

The AI/ML Interview Book

From Scratch to Pro — Theory, Code, and Interview Q&A

Part II — Classical Machine Learning

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Preface

If you are preparing for a machine learning interview in the current market, you are facing a strange paradox. The field has never been more accessible — free courses, open models, and endless tutorials — and yet interviews have never been harder to prepare for. The bar has moved. A decade ago you could pass an ML interview by explaining the bias-variance tradeoff and implementing k-means. Today you may be asked to derive backpropagation on a whiteboard at 10 a.m., design a retrieval-augmented generation system at 1 p.m., and debug a training loss curve at 4 p.m. — all for the same job.

This book exists to carry you across that entire range. It is written for three kinds of readers at once. The **complete beginner** who has decided to enter the field and wants a single, sequential path from "what is a matrix?" to "how does speculative decoding work?". The **practitioner** who uses scikit-learn or PyTorch daily but has never had to justify *why* Adam differs from RMSprop out loud, under pressure. And the **interview-ready candidate** who needs dense, honest model answers rather than hand-waving.

Three principles govern every chapter:

Nothing is assumed. Every topic begins in plain language, as if you have never met it before, and then climbs — intuition first, then formal definitions, then derivations with every symbol defined, then a small worked numeric example you can check with a pencil.

All code runs. Every code listing in this book was executed before it was printed. Where output is shown, it is real output. You can type any listing into a Python file and run it.

Interviews are the destination. Every chapter ends with real interview-style questions and model answers written at the depth an interviewer actually expects — including notes on what the interviewer is listening for.

How to use this book

The book has eight parts and is designed to be read two different ways.

The scratch-to-pro path (8-16 weeks). Read sequentially, Part I through Part VIII. Part I builds the mathematical and programming foundation; nothing later will assume more than it teaches. Do the worked examples by hand. Type out and run the code. This path turns a motivated beginner into a credible candidate.

The interview-in-two-weeks path. If you already work in the field, go straight to where roughly 80% of interview questions concentrate: Part II (classical ML), Part III (deep learning), and Part IV (NLP and LLMs), plus Chapter 34 (implement-from-scratch — the single most requested interview practice topic). Skim the Q&A sections of every other chapter, and read Part VIII the weekend before your loop.

Whichever path you take, treat the end-of-chapter Q&A as active practice, not reading material: cover the answer, say your own answer out loud, then compare. Speaking an answer is a different skill from recognizing one.

A note on the appendices: Appendix A (formula cheat sheets) and Appendix C (100 rapid-fire one-liners) are designed for the final 48 hours before an interview. Appendix E (SQL and Python quick reference) covers the data-manipulation screens that increasingly precede ML rounds.

Notation

The same symbols mean the same things in every chapter of this book. When a chapter introduces a symbol not listed here, it defines it at first use.

Symbol	Meaning
x	a single input (feature vector), $x \in \mathbb{R}^d$
X	the design matrix of all inputs, shape $n \times d$ (rows are examples)
y	the true label or target
$\hat{y}$	the model's prediction
n	number of examples; d — number of features
w, b	weight vector and bias of a single model or layer
θ	the set of all learnable parameters of a model
η	learning rate
$\mathcal{L}$	loss function (to be minimized)
$\nabla_{\theta} \mathcal{L}$	gradient of the loss with respect to parameters
$\sigma(\cdot)$	the sigmoid function $\sigma(z) = 1/(1 + e^{-z})$
$\mathbb{E}[\cdot]$	expectation; $\text{Var}(\cdot)$ — variance
$P(A B)$	conditional probability of A given B
$\mathcal{N}(\mu, \sigma^2)$	Gaussian distribution with mean μ and variance σ^2
$D_{KL}(P \ Q)$	Kullback–Leibler divergence from Q to P
$H(P)$	entropy of distribution P
$\ v\ $	Euclidean (L2) norm of vector v
A^T	transpose of matrix A ; A^{-1} — inverse
λ	regularization strength (or an eigenvalue — context makes it clear)

Vectors are column vectors unless stated otherwise. Code follows Python conventions: 0-indexed, row-major NumPy arrays where $\mathbf{x}[i]$ is the i -th example.

Part II — Classical Machine Learning

Chapter 4: ML Fundamentals

Every ML interview loop contains at least one conversation that is really this chapter. Whatever the surface question — "your model does great offline and terrible in production, why?", "when would you prefer a simpler model?", "how do you split this dataset?" — the interviewer is checking whether you own the conceptual machinery of generalization: what it means to learn from data, why models fail in the two opposite directions, and how to measure performance without fooling yourself. Candidates who can recite algorithms but wobble on bias-variance or data leakage fail loops; the reverse almost never happens.

This chapter covers the framework everything later hangs on: the learning paradigms (supervised, unsupervised, semi-supervised, self-supervised, reinforcement), the bias-variance tradeoff, overfitting and underfitting and the regularization that manages them, evaluation protocol (train/validation/test splits and cross-validation in its k-fold, stratified, and time-series forms), data leakage — the most expensive practical mistake in applied ML — the curse of dimensionality, and the parametric/non-parametric distinction. Each section gives the intuition first, then the formal statement, then the interview framing.

The learning paradigms

Supervised learning learns a mapping from inputs to labeled outputs: given pairs (x, y) drawn from some distribution, find a function f such that $f(x)$ predicts y on *new* draws. Classification (discrete y) and regression (continuous y) are its two faces. The defining cost is labels: someone — an annotator, a logging system, the passage of time — must supply y , and label acquisition is usually the budget item that shapes the whole project. Formally, supervised learning minimizes expected loss $E[L(f(x), y)]$ over the data distribution, estimated by the average loss on a sample; the gap between the sample estimate and the true expectation is the entire subject of this chapter.

Unsupervised learning gets only x and must find structure: clusters (k-means, Chapter 8), density (mixture models), lower-dimensional organization (PCA, embeddings), anomalies (points the structure fails to explain). There is no ground truth to score against, which makes evaluation notoriously slippery — a fact interviewers like to probe ("how would you know your clustering is good?").

Semi-supervised learning uses a small labeled set plus a large unlabeled pool — the practical regime of most industrial problems, where data is abundant and labels are expensive. Techniques: pseudo-labeling (train on labels, predict on the pool, keep confident predictions as new labels), consistency regularization (a model should predict the same thing for perturbed versions of an input), and label propagation over a similarity graph (Chapter 3's BFS flood-fill, weighted).

Self-supervised learning manufactures labels from the data itself: predict the masked word (BERT), predict the next token (GPT), predict whether two augmented views come from the same image (contrastive learning, SimCLR). The trick converts unlabeled data into supervised training signal at unlimited scale, and it is the paradigm that built modern foundation models — Chapters 20–21 are its story. The interview distinction to keep sharp: semi-supervised *mixes* human labels with unlabeled data; self-supervised *needs no human labels at all* for pretraining (they return at fine-tuning time).

Reinforcement learning (Chapter 32) learns from *evaluative* feedback rather than *instructive* feedback: an agent acts, the environment returns a reward, and no one says what the correct action was. Credit assignment across time and the exploration-exploitation dilemma are its defining difficulties. It reappears in the LLM era as RLHF (Chapter 21).

Paradigm	Signal	Canonical tasks	Label cost
Supervised	Human/logged labels y	Classification, regression	High — the budget item
Unsupervised	None (structure in x)	Clustering, density, dim. reduction	Zero
Semi-supervised	Few labels + large pool	Pseudo-labeling, consistency	Low
Self-supervised	Manufactured from x	Masked/next-token prediction, contrastive	Zero for pretraining
Reinforcement	Delayed scalar reward	Control, games, RLHF	Reward design, not labels

The boundaries blur in practice — recommendation systems mix supervised click prediction with bandit exploration; LLM training stacks self-supervised pretraining, supervised fine-tuning, and RL alignment — and saying so, with one example, reads as experience.

The bias-variance tradeoff

Take a model family, train it on a random training sample, and measure its error on a fixed test point. Repeat with a fresh training sample. The prediction *changes* — the trained model is itself a random object, because it depends on the random draw of training data. The bias-variance decomposition splits the expected squared error at a point into three sources:

$$\mathbb{E}[(y - \hat{f}(x))^2] = (f(x) - \mathbb{E}[\hat{f}(x)])^2 + \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2] + \sigma^2$$

The three terms, in order: **bias**², **variance**, and **irreducible noise**.

Bias is systematic error: the gap between the *average* model (averaged over training samples) and the truth. It comes from the model family being too rigid to represent the target — a line fit to a curve is biased at almost every point, no matter how much data you have. **Variance** is sensitivity to the sample: how much the trained model wobbles around its own average as the training set is redrawn. It comes from the family being flexible enough to chase noise. **Irreducible noise** σ^2 is the part of y that x simply does not determine; no model touches it, and recognizing when you have hit it ("the label itself is stochastic — two identical users, different clicks") is a senior-level observation.

The tradeoff: within a model family, adding capacity (higher polynomial degree, deeper tree, more features, smaller regularization) lowers bias and raises variance; removing capacity does the reverse. Total error is U-shaped in capacity, and the minimum — the sweet spot — depends on how much data you have, because variance shrinks with sample size while bias does not. This is why "more data" is the cure for variance problems and a placebo for bias problems, and why the same model can be over-parameterized for 1,000 rows and under-parameterized for 10 million. Listing 1 makes all of this concrete by *measuring* bias² and variance empirically for polynomial fits.

Diagnosis in practice reads the two gaps: training error far above acceptable → high bias (underfitting) — add capacity, add features, train longer, reduce regularization. Training error fine but validation error far above it → high variance (overfitting) — more data, regularization, simpler model, ensembling (bagging exists precisely to cut variance, Chapter 7). The two prescriptions are opposites, which is why the diagnosis must come first — the most common junior mistake is reaching for more regularization when the model is *underfitting*.

Learning curves extend the two-number diagnosis into a picture: plot training and validation error against training-set size. High-bias signature: the two curves converge quickly to a common plateau *above* the target — more data is visibly useless because the curves have already met (Listing 8's flat linear column, drawn out). High-variance signature: a persistent gap between a low training curve and a high validation curve that narrows as n grows — more data is visibly working, and extrapolating the gap tells you roughly how much more you need. Interviewers like this tool because it converts "should we buy more labels?" — a budget question — into a plot you can actually produce, and because quoting it signals you diagnose empirically rather than by vibes.

Two caveats earn bonus points. The clean decomposition is specific to squared loss (classification analogues exist but are messier). And modern deep networks complicate the classical U-shape: heavily over-parameterized models can interpolate the training data yet still generalize, with test error descending again past the interpolation point ("double descent") — the classical tradeoff still governs the underparameterized regime and the intuition (variance from chasing noise) survives, but "always reduce capacity to fix overfitting" is no longer the whole story (Chapter 13 returns to this).

Overfitting, underfitting, and regularization

Overfitting is memorizing the sample instead of learning the distribution: the model fits signal *and* noise, so training error keeps falling while generalization error rises.

Underfitting is failing to fit even the signal: both errors are high and close together. The operational test is always the same pair of numbers — training error and validation error — read as (level, gap): high level + small gap = underfit; low training level + large gap = overfit. Listing 2 traces both curves across model capacity and shows the characteristic divergence.

Regularization is any modification that trades a little training fit for better generalization — in bias-variance terms, deliberately adding bias to remove more variance. The canonical form penalizes weight magnitude in the loss:

$$\mathcal{L}_{reg}(w) = \mathcal{L}(w) + \lambda\Omega(w)$$

L2 (ridge), $\Omega = ||w||_2^2$, shrinks all weights smoothly toward zero but rarely *to* zero: the penalty's gradient $2\lambda w$ is proportional to the weight, so pressure fades as weights shrink. Geometrically, the constraint region is a sphere — the loss contours touch it anywhere. Effects: handles correlated features by splitting weight among them, always has a unique solution (it makes $X'X + \lambda I$ invertible even with more features than rows), and has a Bayesian reading as a Gaussian prior on weights (MAP estimation, Chapter 1).

L1 (lasso), $\Omega = ||w||_1$, produces exact zeros — automatic feature selection. The gradient of $|w|$ has constant magnitude λ regardless of weight size, so small weights get pushed all the way to zero; geometrically the constraint region is a diamond whose *corners* (where coordinates vanish) are where the loss contours preferentially touch. Bayesian reading: Laplace prior. Costs: among correlated features it arbitrarily picks one and zeroes the rest (unstable selections), and the non-differentiability at zero requires coordinate descent or proximal methods rather than plain gradient descent.

Elastic Net mixes both, $\Omega = \alpha||w||_1 + (1-\alpha)||w||_2^2$, to get sparsity *and* grouped handling of correlated features. λ (and α) are hyperparameters tuned on validation data — never on test. Listing 3 fits all three on data with known sparse truth and shows L1 recovering the true zeros while L2 merely shrinks. One practical prerequisite that interviewers probe: penalties compare weight magnitudes across features, so features must be on comparable scales — standardize before regularized fits, and never penalize the intercept.

Regularization is broader than norm penalties, and listing the family signals range: early stopping (halt when validation error turns up — gradient descent visits simple models first), dropout and weight decay in deep nets (Chapter 13), pruning trees (Chapter 6), data augmentation (enlarging the effective sample), and even "more data" itself — all are variance-reduction levers. λ 's dial is a pure bias-variance knob: $\lambda \rightarrow 0$ recovers the unregularized fit (variance), $\lambda \rightarrow \infty$ crushes weights to zero (bias); the U-shaped validation curve over λ is Listing 2's complexity curve mirrored.

Splits and cross-validation

The cardinal rule of evaluation: a model must be scored on data that played *no role* in producing it — not in fitting weights, not in choosing hyperparameters, not in selecting features, not in deciding when to stop. Every violation inflates the score, and the inflation is invisible until production.

Train/validation/test. Three roles, three sets: train fits parameters; validation compares models and tunes hyperparameters; test — touched *once, at the end* — estimates final performance. The reason validation and test must be separate is subtle enough to be an interview question: after you pick the best of fifty configurations *by validation score*, that winning score is biased upward (you selected the luckiest), so an untouched set is needed for an honest estimate. Typical splits 60/20/20 or 80/10/10; big data regimes can afford tiny fractions because absolute set sizes are what control estimate noise.

K-fold cross-validation. With modest data, a single validation split wastes data and yields a noisy estimate that depends on the luck of the split. K-fold: partition into k folds; train on $k-1$, validate on the held-out fold; rotate; average the k scores. Every point is validated on exactly once, the average is far more stable than any single split, and the spread across folds is a free error bar (report mean $\pm$ std, and say so in interviews). Cost: k fits. $k = 5$ or 10 is standard — larger k means more training data per fit (less pessimistic bias) but more compute and higher variance between the highly-overlapping training sets; leave-one-out ($k = n$) is the extreme, cheap only for models with closed-form refits. Crucially, CV *selects* models and hyperparameters; after selection, retrain on all training data with the winning configuration. Listing 4 implements k-fold from scratch and shows the fold-to-fold spread a single split hides.

Stratified k-fold preserves class proportions in every fold (Chapter 3's stratified split, applied per fold). With imbalance — fraud at 1%, a 5-fold split of 1,000 rows — plain random folds can contain a handful or zero positives, making per-fold metrics garbage. Stratify by default for classification.

Time-series CV. Random splits are *wrong* whenever data has temporal order and the deployment task is forecasting: shuffling lets the model train on the future and validate on the past, and autocorrelation means even innocent-looking random folds leak (adjacent, near-duplicate observations straddle the split). Correct protocol: walk-forward validation — train on an expanding (or rolling) window, validate on the next block, slide; sklearn's `TimeSeriesSplit`. A purge gap between train and validation blocks guards against features that smear information across the boundary (rolling averages, lagged labels). Listing 6 shows a shuffled split flattering a model that walk-forward correctly grades much worse. Grouped data has the same disease in another organ: multiple rows per user/patient must be split *by group* (`GroupKFold`), or the model recognizes individuals across the boundary.

Data leakage

Leakage is any path by which information unavailable at prediction time reaches the model during training or evaluation. It is the most expensive practical failure in ML because it produces *great offline metrics* — the model genuinely has extra signal — and collapses only in production, after the launch decision. "Offline AUC 0.95, production disaster" is the interview scenario, and leakage should be your first hypothesis.

Target leakage: a feature encodes the label because it is generated *after* (or by) the outcome — days-in-hospital as a feature for readmission risk, "charged-back" flags for fraud, aggregate statistics computed over a window that includes the outcome. The tell: one feature with implausibly dominant importance; the test: could this value exist, with this content, at the moment of prediction?

Train-test contamination: preprocessing fit on all data before splitting. Standardizing with the global mean/std, selecting features by correlation with the label on the full dataset, fitting imputation or target encodings globally — each lets test-set statistics shape the training pipeline. The fix is mechanical: *every* fitted step happens inside the training fold only, then transforms validation/test — which is exactly what sklearn's Pipeline inside `cross_val_score` automates and why "wrap it in a Pipeline" is the expected answer (Chapter 2's fit/transform contract, now load-bearing). Listing 5 demonstrates the classic version: feature selection on the full dataset before CV produces a spectacular score on *pure noise*.

Temporal and group leakage: covered above — future into past, same entity on both sides of the split. Duplicate rows (a data-engineering artifact) that land on both sides are the degenerate case; deduplicate before splitting.

Leakage checklist worth reciting: know each feature's availability timestamp; split *first* (by time/group where applicable), preprocess inside folds; be suspicious of scores that beat the plausible ceiling; and validate the deployed feature pipeline computes the same values the training pipeline did (training-serving skew, Chapter 27).

The curse of dimensionality

High-dimensional space behaves in ways that break low-dimensional intuition, and several ML pathologies follow directly.

Data sparsity. To cover the unit cube in d dimensions at resolution 0.1 per axis takes 10^d cells — with $d = 20$, more cells than you will ever have rows. Any fixed dataset becomes vanishingly sparse as d grows, so methods that rely on *local* neighborhoods (KNN, kernel smoothing, histogram density estimation) run out of neighbors: the nearest "neighbor" is no longer near.

Distance concentration. As d grows with independent-ish features, the ratio between the nearest and farthest pairwise distances approaches 1 — everyone is roughly equidistant, so "nearest" carries little information and distance-based methods degrade. This is also why KD-trees stop pruning (Chapter 3) and why high-dimensional similarity search moves to approximate methods and *learned* low-dimensional embeddings (Chapter 24). Listing 7 measures the effect directly: the nearest/farthest ratio climbing toward 1 as d goes $2 \rightarrow 1000$.

Geometry gets weird. Nearly all the volume of a high- d ball sits in a thin shell near its surface; nearly all of a Gaussian's mass sits at radius $\approx \sigma\sqrt{d}$, not near the mode. Sampled points are all "far out" and all about equally far.

Overfitting risk. More dimensions = more parameters and more directions to fit noise; with d comparable to n , a linear model can interpolate anything (variance, again). Rules of thumb about samples-per-feature exist but the honest statement is: required data grows fast with *intrinsic* dimension.

The escape hatches: real data usually lies near a low-dimensional manifold (images of faces do not fill pixel space) — which is why learned representations work at all; dimensionality reduction (PCA, Chapter 8; embeddings, Chapter 19+); feature selection (Chapter 9); regularization; and models whose inductive biases match the structure (convolutions assume locality, Chapter 14). "The blessing of non-uniformity" — data concentrating near structure — is the phrase that shows you have read around.

Parametric vs non-parametric models

A **parametric** model commits to a fixed-size parameter vector before seeing data: linear/logistic regression, Naive Bayes, neural networks of fixed architecture. Capacity is set by the form; data refines the parameters. Consequences: training compresses the data into θ (prediction is $O(|\theta|)$, independent of n — fast serving, small memory); strong assumptions = high bias, low variance, better in small-data regimes; and if the assumed form is wrong, no amount of data fixes it (the bias floor).

A **non-parametric** model lets complexity grow with the data — "parameters" scale with n : KNN (the training set *is* the model), kernel density estimation, kernel SVMs (support vectors scale with data), decision trees grown to fit (structure chosen by data), Gaussian processes. Consequences: weak assumptions = low bias, high variance, data-hungry; prediction cost and memory grow with n (KNN's $O(nd)$ per query, Chapter 3); with enough data they approximate anything.

The name is a historical misnomer worth defusing in one sentence: non-parametric models have parameters — often many — what they lack is a *fixed, finite* parameterization chosen in advance. Interview edge cases: k in KNN and tree depth are *hyperparameters*, not parameters; a neural network is parametric (fixed weight count) even though it is a universal approximator; gradient-boosted trees blur the line

(ensemble size fixed by you, tree structures data-driven). The practical selection heuristic: small data or strong domain knowledge → parametric (you need the bias); large data, unknown form, latency budget permitting → non-parametric or high-capacity parametric. Listing 8 races linear regression against KNN regression as n grows on a nonlinear target: the parametric model hits its bias floor early; KNN keeps improving because its capacity grows with the data.

Code implementations

Every listing was executed as shown; outputs are real. These are simulations worth internalizing — each one turns an abstract claim from this chapter into a number.

Listing 1 — Measuring bias² and variance empirically

The decomposition is usually asserted; here it is *measured* — 300 retrainings on fresh samples per degree. Degree 1: bias dominates. Degree 3: sweet spot. Degree 9: bias near zero, variance catastrophic. The sum tracks total expected error (noise $\sigma^2 = 0.09$).

```
"""Listing 1 -- Measuring bias^2 and variance empirically."""
import numpy as np

rng = np.random.default_rng(0)
f = lambda x: np.sin(2 * np.pi * x)          # true function
sigma = 0.3                                  # irreducible noise std
x_test = np.linspace(0.05, 0.95, 25)         # fixed test points

def fit_poly(degree, n=30):
    """Train one polynomial model on a fresh random sample; predict at x_test."""
    x = rng.uniform(0, 1, n)
    y = f(x) + rng.normal(0, sigma, n)
    coeffs = np.polyfit(x, y, degree)
    return np.polyval(coeffs, x_test)

def bias_variance(degree, trials=300):
    """Retrain `trials` times; decompose average error at the test points."""
    preds = np.array([fit_poly(degree) for _ in range(trials)]) # (trials, 25)
    avg_pred = preds.mean(axis=0)                               # E[f_hat(x)] per test point
    bias2 = ((avg_pred - f(x_test)) ** 2).mean()                # (E[f_hat] - f)^2
    variance = preds.var(axis=0).mean()                         # E[(f_hat - E[f_hat])^2]
    return bias2, variance

print(f"{'degree':>6} {'bias^2':>8} {'variance':>9} {'sum+noise':>10}")
for d in [1, 3, 6, 9]:
    b2, v = bias_variance(d)
    print(f"{'d':>6} {'b2':>8.4f} {'v':>9.4f} {'b2 + v + sigma**2:>10.4f}")
```

Output:

degree	bias^2	variance	sum+noise
1	0.1610	0.0218	0.2729
3	0.0029	0.0132	0.1061
6	0.0000	0.0993	0.1893
9	0.0271	8.0816	8.1987

Listing 2 — Train vs validation error across capacity

The two-number diagnosis in action. Watch the gap: degrees 1-2 can't reach the noise floor (0.09) even on train — bias. Past degree 9, train error keeps improving while validation turns up — variance. The diagnosis column applies the (level, gap) rule.

```

"""Listing 2 -- Train vs validation error across model capacity."""
import warnings
import numpy as np
warnings.simplefilter("ignore")  # silence polyfit conditioning warnings at high
degree

rng = np.random.default_rng(1)
f = lambda x: np.sin(2 * np.pi * x)
x_tr = rng.uniform(0, 1, 40); y_tr = f(x_tr) + rng.normal(0, 0.3, 40)
x_va = rng.uniform(0, 1, 200); y_va = f(x_va) + rng.normal(0, 0.3, 200)

print(f'{"degree":>6} {"train MSE":>10} {"val MSE":>9}  diagnosis')
for d in [1, 2, 3, 5, 9, 15, 25]:
    c = np.polyfit(x_tr, y_tr, d)
    tr = ((np.polyval(c, x_tr) - y_tr) ** 2).mean()
    va = ((np.polyval(c, x_va) - y_va) ** 2).mean()
    # noise floor: sigma^2 = 0.09 is the best achievable MSE
    diag = ("underfit" if tr > 0.13 else  # can't even fit train past the floor
           "overfit" if va > 0.11 else "good")  # val above the floor while train
    dips below
    print(f'{"d":>6} {"tr":>10.3f} {"va":>9.3f}  {"diag}')
```

Output:

degree	train MSE	val MSE	diagnosis
1	0.149	0.284	underfit
2	0.149	0.284	underfit
3	0.040	0.094	good
5	0.035	0.090	good
9	0.034	0.091	good
15	0.031	0.118	overfit
25	0.029	0.640	overfit

Listing 3 — L1 vs L2 on a known sparse truth

Ground truth has 3 real features and 9 zeros. OLS and Ridge keep all 12 coefficients nonzero (Ridge only shrinks); Lasso recovers near-exact sparsity — 4 survivors of 12. This is the diamond-corner geometry made visible.

```

"""Listing 3 -- L1 vs L2: sparsity vs shrinkage on a known sparse truth."""
import numpy as np
from sklearn.linear_model import LinearRegression, Ridge, Lasso

rng = np.random.default_rng(2)
n, d = 60, 12
X = rng.normal(0, 1, (n, d))
w_true = np.zeros(d); w_true[:3] = [3.0, -2.0, 1.5]  # only 3 real features
y = X @ w_true + rng.normal(0, 1.0, n)

models = {"OLS": LinearRegression(),
          "Ridge(a=10)": Ridge(alpha=10.0),
          "Lasso(a=0.3)": Lasso(alpha=0.3)}
```

```
print("true w:", w_true[:5].round(1), "... rest exactly 0")
for name, m in models.items():
    m.fit(X, y)
    w = m.coef_
    nz = (np.abs(w) > 1e-6).sum()
    print(f"{name:<12} nonzero={nz:>2} w[:5]={np.round(w[:5], 2)}")
```

Output:

```
true w: [ 3.  -2.  1.5  0.  0. ] ... rest exactly 0
OLS      nonzero=12 w[:5]=[ 2.9 -1.88  1.58  0.08  0.02]
Ridge(a=10) nonzero=12 w[:5]=[ 2.36 -1.54  1.18  0.08  0.08]
Lasso(a=0.3) nonzero= 4 w[:5]=[ 2.52 -1.49  1.21  0.  0. ]
```

Listing 4 — K-fold cross-validation from scratch

The mechanics are ten lines; the payload is the last line — a single random split could have reported anything between the min and max fold, a $3\times$ range. The mean $\pm$ std is the honest report.

```
"""Listing 4 -- k-fold CV from scratch; the spread a single split hides."""
import numpy as np
from sklearn.linear_model import Ridge

def kfold_indices(n, k, seed=0):
    """Shuffle indices, cut into k folds; yield (train_idx, val_idx)."""
    idx = np.random.default_rng(seed).permutation(n)
    folds = np.array_split(idx, k)
    for i in range(k):
        val = folds[i]
        train = np.concatenate([folds[j] for j in range(k) if j != i])
        yield train, val

rng = np.random.default_rng(3)
n, d = 100, 5
X = rng.normal(0, 1, (n, d))
y = X @ np.array([1.0, -1.0, 0.5, 0.0, 2.0]) + rng.normal(0, 1.0, n)

scores = []
for tr, va in kfold_indices(n, k=5):
    m = Ridge(alpha=1.0).fit(X[tr], y[tr])
    mse = ((m.predict(X[va]) - y[va]) ** 2).mean()
    scores.append(mse)
scores = np.array(scores)
print("per-fold MSE:", scores.round(3))
print(f"CV estimate: {scores.mean():.3f} +/- {scores.std():.3f}")
print(f"single-split illusion: could report anywhere from "
      f"{scores.min():.3f} to {scores.max():.3f}")
```

Output:

```
per-fold MSE: [0.736 0.537 1.294 0.915 1.455]
CV estimate: 0.987 +/- 0.342
single-split illusion: could report anywhere from 0.537 to 1.455
```


Listing 5 — Leakage demo: selection before the split

Features and labels are *pure noise* — the true accuracy ceiling is 0.5. Selecting the 20 most label-correlated features on the full dataset before CV yields 90% accuracy. Moving selection inside the folds (Pipeline) returns the truth. This exact bug ships in real papers.

```
"""Listing 5 -- Leakage demo: feature selection before the split, on pure noise."""
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score

rng = np.random.default_rng(4)
n, d = 100, 5000
X = rng.normal(0, 1, (n, d))          # pure noise features
y = rng.integers(0, 2, n)             # random labels: true accuracy ceiling = 0.5

# WRONG: select the 20 features most correlated with y using ALL rows...
corr = np.abs((X - X.mean(0)).T @ (y - y.mean())) / n
top = np.argsort(corr)[-20:]
leaky = cross_val_score(LogisticRegression(max_iter=1000), X[:, top], y, cv=5)

# RIGHT: selection must happen inside each training fold
from sklearn.pipeline import Pipeline
from sklearn.feature_selection import SelectKBest, f_classif
pipe = Pipeline([("select", SelectKBest(f_classif, k=20)),
                  ("clf", LogisticRegression(max_iter=1000))])
clean = cross_val_score(pipe, X, y, cv=5) # SelectKBest refit per fold

print(f"leaky CV accuracy: {leaky.mean():.2f} <- 'discovers' signal in pure noise")
print(f"clean CV accuracy: {clean.mean():.2f} <- the truth: coin-flip")
```

Output:

```
leaky CV accuracy: 0.90 <- 'discovers' signal in pure noise
clean CV accuracy: 0.45 <- the truth: coin-flip
```

Listing 6 — Random split vs walk-forward on autocorrelated data

Adjacent observations of a random walk are near-duplicates, so a shuffled split gives every test point a twin in training — a $131\times$ optimism factor versus the walk-forward protocol that matches deployment.

```
"""Listing 6 -- Random split vs walk-forward on autocorrelated data."""
import numpy as np
from sklearn.neighbors import KNeighborsRegressor

rng = np.random.default_rng(5)
n = 500
# slow random walk + noise: adjacent points are near-duplicates
signal = np.cumsum(rng.normal(0, 0.1, n))
t = np.arange(n)
X = np.c_[np.sin(t / 20), np.cos(t / 20), t / n]
y = signal + rng.normal(0, 0.05, n)

model = KNeighborsRegressor(n_neighbors=3)

# WRONG: shuffled split -- each test point has near-twin neighbors in train
idx = rng.permutation(n)
```

```
tr, te = idx[:400], idx[400:]
mse_shuffled = ((model.fit(X[tr], y[tr]).predict(X[te]) - y[te]) ** 2).mean()

# RIGHT: walk-forward -- train on the past, test on the future
tr, te = t[:400], t[400:]
mse_forward = ((model.fit(X[tr], y[tr]).predict(X[te]) - y[te]) ** 2).mean()

print(f"shuffled-split MSE:      {mse_shuffled:.3f} <- flattering illusion")
print(f"walk-forward MSE:       {mse_forward:.3f} <- deployment reality")
print(f"optimism factor:        {mse_forward / mse_shuffled:.0f}x")
```

Output:

```
shuffled-split MSE:    0.010 <- flattering illusion
walk-forward MSE:     1.295 <- deployment reality
optimism factor:      131x
```

Listing 7 — Distance concentration as dimension grows

The nearest/farthest distance ratio climbs from 0.005 (d=2: 'near' means something) to 0.88 (d=1000: everyone is equidistant). This one table explains why KNN, kernels, and KD-trees all degrade in high dimensions.

```
"""Listing 7 -- Distance concentration as dimension grows."""
import numpy as np

rng = np.random.default_rng(6)
n = 500
print(f"{'dim':>5} {'nearest/farthest':>16} {'mean dist':>10}")
for d in [2, 10, 100, 1000]:
    X = rng.uniform(0, 1, (n, d))
    # pairwise distances of a random subset of pairs
    i, j = rng.integers(0, n, 2000), rng.integers(0, n, 2000)
    keep = i != j
    dist = np.sqrt(((X[i[keep]] - X[j[keep]]) ** 2).sum(axis=1))
    print(f"{'d':>5} {'dist.min() / dist.max()':>16.3f} {'dist.mean()':>10.2f}")
```

Output:

dim	nearest/farthest	mean dist
2	0.005	0.51
10	0.257	1.26
100	0.685	4.08
1000	0.881	12.91

Listing 8 — Parametric bias floor vs non-parametric growth

A linear model on a nonlinear target stalls at $\text{MSE} \approx 0.51$ forever — its bias floor; no data volume helps. KNN keeps absorbing data, converging toward the noise floor (0.04). The tradeoff runs the other way at $n=30$, where KNN's variance would hurt on a harder problem.

```
"""Listing 8 -- Parametric bias floor vs non-parametric growth with n."""
import numpy as np
```

```

from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor

rng = np.random.default_rng(7)
f = lambda x: np.sin(3 * x) + 0.5 * x          # nonlinear truth
def sample(n):
    x = rng.uniform(0, 3, (n, 1))
    return x, f(x).ravel() + rng.normal(0, 0.2, n)

x_te, y_te = sample(2000)
print(f"{'n_train':>8} {'linear MSE':>11} {'KNN(k=5) MSE':>13}")
for n in [30, 100, 1000, 10000]:
    x_tr, y_tr = sample(n)
    lin = ((LinearRegression().fit(x_tr, y_tr).predict(x_te) - y_te) ** 2).mean()
    knn = ((KNeighborsRegressor(5).fit(x_tr, y_tr).predict(x_te) - y_te) ** 2).mean()
    print(f"{'n':>8} {'lin':>11.3f} {'knn':>13.3f}")
print("noise floor (sigma^2): 0.040")

```

Output:

```

n_train  linear MSE  KNN(k=5) MSE
    30         0.530         0.074
   100         0.553         0.057
  1000         0.516         0.048
 10000         0.511         0.048
noise floor (sigma^2): 0.040

```

Pitfalls, comparisons and practical tips

The diagnosis-before-prescription rule. Underfitting and overfitting have opposite cures, so compute train and validation error *first*. The table interviewers expect you to have internalized:

Train error	Val error	Diagnosis	Levers
High	High (close)	High bias / underfit	More capacity, more/better features, less regularization, train longer
Low	High (large gap)	High variance / overfit	More data, regularization, simpler model, bagging, augmentation
Low	Low	Healthy	Ship it — after checking for leakage
High	Lower than train	Protocol bug	Regularization applied only at train time is normal; otherwise suspect a split/metric bug

Suspiciously good = leakage until proven otherwise. If validation performance beats your plausible ceiling — or one feature dominates importance — audit before celebrating. The three-question audit: is any feature computed *after* the label event? Was any fitted transform (scaler, imputer, encoder, selector) fit outside the training fold? Do time or group structure straddle the split? Listing 5's noise-to-90%-accuracy shows how convincing the illusion is.

Split hygiene mistakes, ranked by frequency. (1) Preprocessing before splitting — fit scalers/encoders inside folds via Pipeline. (2) Tuning on test — once test data influences any decision, it is validation data; you no longer have a test set. (3) Random splits on temporal data — walk-forward or nothing. (4) Ignoring groups — the same user on both sides of a split is a memorization channel. (5) Reporting single-split numbers with no error bar — Listing 4's $3\times$ fold spread is typical at modest n .

Regularization mistakes. Penalizing unstandardized features (the penalty then arbitrates by units, not importance); penalizing the intercept; tuning λ on training loss (it will always choose $\lambda=0$); expecting L1 to make stable feature selections under correlated features (it picks one of each correlated group arbitrarily — refit on bootstrap samples to see the instability); forgetting that "more data" is also a variance cure and cheaper than architecture surgery when available.

Bias-variance verbal traps. "High variance" describes the *estimator across resamples*, not the variance of predictions on one set. Bagging reduces variance, boosting primarily reduces bias (Chapter 7) — reversing these is a classic error. σ^2 is irreducible only *given the features*: better features move error out of the "irreducible" bucket. And regularization does not "reduce error" — it trades bias for variance; whether total error falls depends on which side of the U you are on.

Curse-of-dimensionality nuance. Adding features can hurt even when each is weakly informative — the variance cost can exceed the signal gain (this is why feature selection exists, Chapter 9). But dimensionality per se is not fatal: what matters is intrinsic dimension and whether your model's inductive bias matches the structure. Text has $\sim 10^5$ sparse dimensions and linear models thrive on it; the pathology bites *distance-based* methods hardest.

CV subtleties that read as senior. After CV selects hyperparameters, retrain on all training data — CV scores select, they are not the final model. Nested CV (an outer loop for evaluation, inner for tuning) is the honest protocol when you must both tune and report from the same modest dataset; without it, reported scores inherit selection optimism. For imbalanced classification, stratify; for time series, purge a gap between train and validation blocks; for grouped data, GroupKFold. And k -fold's $k=5$ vs $k=10$ matters less than getting the fold *structure* right.

Interview questions and answers

Paradigms

Q1. Classify these as supervised, unsupervised, semi-supervised, self-supervised, or RL: spam filtering, customer segmentation, BERT pretraining, RLHF, fraud detection with 100 labeled cases and 1M unlabeled transactions.

Spam filtering: supervised classification (labels from user reports). Customer segmentation: unsupervised clustering. BERT pretraining: self-supervised — masked-token labels are manufactured from raw text. RLHF: reinforcement learning (reward model provides evaluative feedback; the reward model itself is trained supervised on human preferences — saying both parts is the strong answer). Fraud with 100 labels + 1M unlabeled: semi-supervised (pseudo-labeling, or anomaly detection as an unsupervised assist).

Q2. What exactly distinguishes self-supervised from unsupervised learning, given neither uses human labels?

Self-supervised learning constructs a *supervised* task from raw data — mask a word and predict it, remove a patch and reconstruct it — so training uses a standard supervised loss with manufactured targets; the goal is usually representations for downstream tasks. Unsupervised learning has no prediction target at all: it models structure directly (clusters, densities, components). The boundary is fuzzy (autoencoders sit on it), and acknowledging that while giving the crisp version above is the right register.

Q3. Why is evaluation fundamentally harder for unsupervised learning, and what do people do about it?

No ground truth means no error to measure: any score must appeal to internal criteria (silhouette, inertia — which reward the objective the algorithm already optimized), stability under resampling, or downstream utility (do the clusters improve a supervised task or a business decision?). The practical answers: internal metrics for sanity, human evaluation of samples, and — most convincing — downstream task performance. Flagging that internal metrics can't compare across k or across algorithms with different objectives is the depth marker (Chapter 8 details).

Bias-variance**Q4. State the bias-variance decomposition and define each term operationally — what experiment would measure them?**

Expected squared error at x = $\text{bias}^2 + \text{variance} + \sigma^2$. Operationally (Listing 1): draw many training sets, train a model on each, predict at x . The average prediction's gap from truth, squared, is bias^2 ; the spread of predictions around their own average is variance; σ^2 is the label noise no model touches. The word "expected" is doing work: the decomposition averages over training-set draws, which is why one trained model has no measurable "variance" by itself.

Q5. Your model's training error is 25% against a 5% target. Your colleague suggests more training data. Evaluate.

Won't work. Training error far above target is a bias problem — the model cannot even fit the data it has seen; more of the same data leaves the bias floor where it is (Listing 8's flat linear column is the picture). Cures: higher-capacity model, better features, longer training, less regularization. More data is the prescription for the *opposite* diagnosis — low train error, high validation error. *Interviewers listen for: diagnosis from the train-error level before any prescription.*

Q6. Why does variance shrink with more data but bias doesn't?

Variance is estimation noise: with more samples, the fit depends less on any particular draw — averages stabilize (law of large numbers), so the trained model converges toward the family's best-possible member. Bias is the gap between that best member and the truth — a property of the model family, not the sample. Infinite data gives you the best line; if the truth is a curve, the line's error remains. Corollary: optimal model complexity grows with n , which is why big-data regimes tolerate (and reward) enormous models.

Q7. How do bagging and boosting map onto bias and variance?

Bagging trains high-variance/low-bias learners (deep trees) on bootstrap resamples and averages: averaging cancels the resample-driven wobble — variance drops, bias roughly unchanged. Boosting builds an additive model of high-bias/low-variance learners (stumps), each fitting the residual errors of the ensemble so far: capacity grows additively — bias drops, and unchecked boosting can eventually overfit (variance returns). One sentence each direction is exactly the expected answer; Chapter 7 has the machinery.

Regularization**Q8. Why does L1 produce exact zeros while L2 doesn't? Give two explanations.**

Gradient view: L2's penalty gradient $2\lambda w$ shrinks proportionally — pressure vanishes as w approaches 0, so weights approach but never reach it; L1's gradient is a constant $\pm\lambda$, which can outmuscle a small data-gradient entirely and pin the weight at exactly 0. Geometric view: the constraint ball for L1 is a diamond with corners on the axes; loss contours typically first touch a corner — where coordinates are exactly zero — while L2's sphere has no corners, so tangency lands at generic (nonzero) points. Listing 3 shows both behaviors on known sparse truth.

Q9. When would you prefer Ridge over Lasso, and what does Elastic Net buy?

Ridge: when most features plausibly matter (dense truth), under strong feature correlation (Ridge shares weight across a correlated group; Lasso picks one arbitrarily — unstable), and when you want a unique, well-conditioned solution ($X'X + \lambda I$ always invertible). Lasso: when you believe the truth is sparse or need feature selection for interpretability/serving cost. Elastic Net: sparse selection *with* grouped handling of correlated features, and better behavior when $d \gg n$ (Lasso alone selects at most n features). All three: standardize first, don't penalize the intercept.

Q10. Interpret L2 regularization as a prior. What λ does a stronger prior correspond to?

MAP estimation with likelihood $\times$ prior: maximizing log-posterior = minimizing loss – log prior. A zero-mean Gaussian prior $N(0, \tau^2)$ on each weight contributes $\|w\|^2 / (2\tau^2)$ — L2 with $\lambda \propto \sigma^2 / \tau^2$. Smaller τ (tighter prior, stronger belief that weights are small) = larger λ . Lasso is the same story with a Laplace prior (sharp peak at zero — mass on exact sparsity). This connects Chapter 1's MLE-vs-MAP directly: regularization is the prior.

Q11. Explain early stopping as regularization. What's the practical protocol?

Gradient descent from small initial weights explores simple functions first and adds effective complexity as training proceeds; halting early caps that complexity — an implicit constraint on how far weights travel from initialization (closely related to L2 in linear models). Protocol: monitor validation loss each epoch; stop when it hasn't improved for a patience window; restore the best-validation checkpoint. It's free (no extra training), which is why it's universal in deep learning — and the validation set it consumes is one more reason validation $\neq$ test.

Splits, CV, leakage

Q12. Why do you need both a validation set and a test set?

Because selection biases scores. Compare 50 hyperparameter configurations by validation score and pick the best: that winning number is partly luck — the maximum of 50 noisy estimates overestimates true performance. The test set, used once after all decisions are frozen, gives an unbiased final estimate. The moment test data influences any choice — even "which model do we ship" — it has become validation data. *Interviewers listen for: the max-of-noisy-estimates argument, not just "to avoid overfitting."*

Q13. When is k-fold CV the wrong tool?

Temporal data (train-on-future leakage — use walk-forward); grouped data (same entity across folds — GroupKFold); very large datasets (a single split is statistically sufficient and $k \times$ training cost buys nothing); non-stationary distributions (past folds don't represent deployment); and expensive models (LLM fine-tuning — a single split or fewer folds). Also: k-fold estimates *procedure* performance, not the performance of one specific trained artifact — subtle but occasionally the point of the question.

Q14. Your fraud model shows 0.98 AUC offline and fails in production. Walk through your investigation.

Leakage first: (1) feature availability audit — any feature derived from post-transaction information (chargeback flags, account-closure status, aggregates over windows including the outcome)? (2) protocol audit — preprocessing/selection fit before the split? random split across time (fraud is temporal — must be walk-forward)? same card/account on both sides? duplicates? (3) then non-leakage causes: distribution shift between offline data and production traffic, label-definition drift, training-serving skew in feature computation (Chapter 27). The ranked order — leakage before drift — is the experience signal; 0.98 on fraud is past the plausible ceiling.

Q15. You standardize features using the full dataset's mean and std, then split and cross-validate. How bad is this really?

It's real leakage but usually mild: fold statistics differ from global statistics by $O(1/\sqrt{n})$ — with large n and benign distributions the inflation is small. It becomes serious when: n is small, features are heavy-tailed (a test-set outlier shifts the global std), or the "preprocessing" is supervised — target encoding, feature selection by label correlation, imputation using the label — where inflation can be catastrophic (Listing 5: $0.50 \rightarrow 0.90$). The honest answer grades the severity rather than reciting "never do this"; the fix (Pipeline inside CV) costs nothing, so do it regardless.

Q16. Design the evaluation protocol for a model predicting 30-day hospital readmission, trained on 5 years of multi-visit patient records.

Two structures to respect: patients repeat (group) and medicine drifts (time). Split by *patient* so no individual straddles train/test (else the model memorizes patients, not risk factors); split by *time* so evaluation mimics deployment — train on years 1–4, test on year 5, ideally both (grouped, time-based). Features must be computable at discharge time — no post-discharge labs, no future visits. Stratify metrics by subgroup, use PR-AUC given imbalance (Chapter 10), and state the deployment claim the protocol licenses: "trained on the past, evaluated on later patients it never saw."

Q17. What is nested cross-validation and when is it worth the cost?

Two loops: the inner CV tunes hyperparameters; the outer CV evaluates the *entire tuned procedure* on folds the tuning never saw. It removes the optimism of reporting the inner loop's winning score (max-of-noisy-estimates again). Worth it when data is too small to spare a real test set and you must both tune and publish an estimate — typical in medical/scientific settings. Cost: $k_{\text{outer}} \times k_{\text{inner}}$ fits. With abundant data, a plain train/val/test split achieves the same honesty cheaper.

Curse of dimensionality, parametric vs non-parametric

Q18. Explain distance concentration and its consequence for KNN.

In high dimensions, pairwise distances cluster tightly around their mean — the nearest/farthest ratio approaches 1 (Listing 7: 0.005 at $d=2$, 0.88 at $d=1000$). Intuition: distance² sums d independent per-coordinate terms, so its relative spread shrinks like $1/\sqrt{d}$ (CLT). Consequence: "nearest" neighbors are barely nearer than random points, so KNN's vote carries little signal, and KD-tree pruning bounds never exclude anything — search degenerates to brute force. Escape: learned low-dimensional embeddings, where distance is meaningful again (Chapter 24).

Q19. "Nearly all the volume of a high-dimensional orange is in the peel." Make this precise and give an ML consequence.

The fraction of a d -ball's volume within thickness ε of the surface is $1 - (1-\varepsilon)^d \rightarrow 1$ as d grows: at $d=100$, 63% of the volume is in the outer 1%. Consequences: uniform samples all sit near the boundary (extrapolation, not interpolation, becomes the norm); Gaussian mass concentrates on a shell at radius $\sigma\sqrt{d}$, so "typical" points are far from the mode — which is why high-dimensional density estimation is hard and why naive "sample near the mean" intuitions fail for generative models (Chapter 18's latent spaces).

Q20. Is a neural network parametric or non-parametric? Defend your answer against the obvious objection.

Parametric: the architecture fixes a finite parameter count before data arrives, and prediction cost is independent of n . Objection: "but it can approximate anything — isn't that non-parametric behavior?" Response: universal approximation is a statement about the *family across widths*; any given network is a fixed-capacity parametric model. The honest refinement: in practice we scale architecture with data (bigger corpus $\rightarrow$ bigger model), so the *workflow* is non-parametric even though each model is parametric — that distinction usually ends the debate favorably.

Q21. Your team must choose between logistic regression and KNN for a 50k-row tabular problem with 30 features and a 10ms serving budget. Argue it.

Serving: logistic regression predicts in $O(d)$ — microseconds; KNN is $O(nd)$ per query ($50k \times 30$ distance ops), likely fine raw but needing an index to be safe, and memory holds the whole training set. Statistics: at 50k rows both are viable; if boundaries are roughly linear (or good features make them so), LR's bias is harmless and its calibrated probabilities and coefficients are free wins; KNN handles local nonlinear structure but suffers at $d=30$ without feature weighting (distance concentration begins to bite). Default: LR first as the baseline; escalate to gradient-boosted trees — not KNN — if it underfits (Chapter 7 is the tabular workhorse). The meta-answer interviewers want: constraints first, model second.

Q22. Why do non-parametric methods need more data, in bias-variance language?

Weak assumptions mean the data must supply the structure the model refuses to assume: low bias, high variance. Variance is paid down by samples — locally, a KNN estimate at x averages the k neighbors *near* x , so you need enough data that "near x " contains k relevant points everywhere in feature space (exponentially many in dimension — the curse). A parametric model pools *all* data into a few parameters — high bias if the form is wrong, but tiny variance per parameter. Listing 8: at $n=30$ the gap is modest; at $n=10,000$ KNN has bought its way to the noise floor while the linear model still pays its bias.

Scenarios and synthesis**Q23. Training loss decreasing, validation loss decreasing, but both far above target. What do you do?**

This is underfitting in progress, not overfitting — the gap is small and both curves are still falling. First: keep training (the curves haven't plateaued). If they plateau high: add capacity, improve features, reduce regularization, check for label noise or a broken feature pipeline (garbage features make an unfittable problem that masquerades as bias), and sanity-check the target metric is achievable — estimate the noise floor with a human baseline or duplicate-input disagreement. Reaching for dropout or more data here would be treating the wrong disease.

Q24. Your colleague reports 5-fold CV accuracy of 94.2% and wants to publish "94% accuracy." Critique.

Three critiques, escalating: (1) no error bar — report mean $\pm$ std across folds; if folds ranged 89–97%, "94%" overstates certainty (Listing 4). (2) If any tuning used those same folds, 94.2% carries selection optimism — nested CV or an untouched test set gives the honest number. (3) Accuracy may be the wrong metric entirely (class balance? cost asymmetry? — Chapter 10). Bonus: CV measures the procedure; the shipped model retrained on all data is a different artifact — usually slightly better, but unmeasured.

Q25. A feature engineer adds 500 new features and CV score improves from 0.81 to 0.83. List everything you'd check before believing it.

Leakage: any of the 500 computed with future/label information, or engineered on the full dataset (target encodings, aggregates) outside the folds? Selection optimism: was 0.83 the best of many feature-set attempts scored on the same folds — informal overfitting to the validation protocol? Stability: is $+0.02$ within fold-to-fold noise (compare to the CV std; a paired per-fold comparison is stronger)? Cost: 500 features raise variance, serving latency, and training-serving skew surface. Verdict protocol: fresh holdout or time-forward test before shipping. *Interviewers listen for: treating the validation protocol itself as an overfittable resource.*

Q26. When is the train/validation/test framework itself insufficient, no matter how carefully applied?

When deployment data differs from any split you can make: distribution shift (new users, new fraud patterns, seasonality your window missed), feedback loops (the model's own decisions change future data — recommendations, credit), adversarial adaptation, and training-serving skew (the pipeline computes different features live). Offline splits certify generalization *to the same distribution*; production needs monitoring, A/B tests, and periodic retraining (Chapters 26–27). Naming the boundary of the tool — "IID in, IID out" — is a senior close.

Q27. Give a scenario where adding regularization hurts, and the general principle.

Any high-bias regime: a linear model already underfitting a nonlinear target — more λ shrinks the few useful coefficients and raises both train and validation error. Also: very large n relative to capacity (variance is already tiny; λ adds pure bias), and penalizing unstandardized features can specifically crush the most useful large-scale feature. Principle: regularization moves you along the bias-variance curve in one direction — it helps iff you start on the variance side of the minimum. The train/val table tells you which side you're on.

Q28. Interviewer: "Explain overfitting to a product manager in two sentences, then to a statistician in two sentences."

PM: "The model memorized quirks of the historical data instead of learning the real pattern, like a student who memorizes past exam answers. It looks great in our tests but will miss on new customers, so the impressive number won't survive launch."

Statistician: "The estimator has excess variance — it fits noise components of the sample, so empirical risk diverges from expected risk; the generalization gap grows with capacity-to-sample ratio. Standard remedies constrain the hypothesis space or its effective complexity: penalties, early stopping, averaging." *Interviewers listen for: register control — same concept, two audiences, no jargon leaking into the first.*

Chapter 5: Regression

Regression is where interviewers go to find out whether your ML knowledge has foundations or just a surface. Linear regression looks like the easiest model in the book — one line of sklearn — which is exactly why the questions about it are hard: "derive the normal equation", "what assumptions are you making, and which ones actually matter for prediction?", "why does Lasso produce zeros and Ridge doesn't?", "why is logistic regression called regression when it classifies?". Each of these has a crisp, complete answer, and interviewers use them to separate candidates who have derived things once from candidates who have only imported them. Regression also carries an outsized share of the field's conceptual weight: gradient descent, regularization, maximum likelihood, and the link between loss functions and probabilistic assumptions all appear here first and reappear everywhere — the last layer of nearly every neural network in this book is a linear or logistic regression wearing learned features.

This chapter builds the full ladder: the linear model and its assumptions (separated, carefully, into what prediction needs versus what inference needs), the two ways to fit it — the closed-form normal equation and gradient descent — and exactly when each wins, polynomial regression as the gateway to capacity control, the regularized family (Ridge, Lasso, Elastic Net) with the geometry that explains their different behavior, logistic regression from log-odds to decision boundary to cross-entropy gradient, and finally Generalized Linear Models, the framework that reveals linear, logistic, and Poisson regression as one idea with three link functions.

Linear regression: the model and its assumptions

Linear regression predicts a continuous target as a weighted sum of features:

$$y = Xw + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Here X is the $n \times d$ design matrix (one row per example, one column per feature, plus a column of ones for the intercept), w the weight vector, and ε the noise — everything about y that the features do not determine. The model's claim is modest and precise: the *conditional mean* of the target is a linear function of the features, $\mathbb{E}[y | x] = w^\top x$. Each weight w_j reads as "the expected change in y per unit change in x_j , holding the other features fixed" — a sentence worth rehearsing, because the *holding others fixed* clause is where interview traps live (it is exactly the clause that multicollinearity destroys).

"Linear" means linear *in the parameters*, not in the raw inputs. $y = w_1x + w_2x^2 + w_3\log x$ is still linear regression — the features are nonlinear transforms, but the model is a linear combination of them, and everything in this chapter applies unchanged. This one observation is the entire mechanism of polynomial regression, splines, and interaction

features, and stating it early in an interview signals you know where the model's actual boundary is: it cannot learn which *transforms* to apply; it can only weight the ones you give it. (Neural networks, Chapter 12, are the machine that learns the transforms too.)

The classical assumptions. Textbooks list them as a block; strong candidates split them by what they are *for*. The four:

1. **Linearity:** $\mathbb{E}[y | x]$ really is linear in the supplied features. If it isn't, the model is biased no matter how much data arrives (Chapter 4's high-bias regime).
2. **Independence of errors:** the ε_i are independent across observations. Violated by time series (autocorrelation), grouped data (multiple rows per user), and spatial data.
3. **Homoscedasticity:** constant error variance σ^2 at every x . Violated when spread grows with the prediction — incomes, prices, counts.
4. **Normality of errors:** $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

The split that earns interview points: **for pure prediction, only linearity truly matters** (plus enough independence that your train/test protocol is honest — Chapter 4's leakage discussion). OLS still produces the best linear approximation to the conditional mean under heteroscedastic or non-normal errors; the *point predictions* are fine. What breaks under violations 2–4 is **inference**: standard errors, confidence intervals, p-values, and any "is this coefficient significant?" claim. Under heteroscedasticity the textbook standard-error formula is simply wrong — Listing 8 measures a nominal 95% confidence interval covering the truth only 83% of the time — and under correlated errors it is usually far too optimistic, because n correlated observations carry the information of far fewer independent ones. Normality is the least important in practice: for large n the CLT (Chapter 1) rescues coefficient inference anyway; it matters mainly for prediction *intervals* and small samples.

There is a fifth, often-unstated assumption: **no perfect multicollinearity** — no feature is an exact linear combination of others, which $X^\top X$ needs to be invertible. Near-collinearity is legal but toxic to interpretation, and gets its own treatment below and in Listing 7.

Diagnostics are how the assumptions are checked in practice, and naming them makes answers concrete: plot residuals against fitted values (a curve means nonlinearity; a funnel means heteroscedasticity); plot residuals in time order for autocorrelation (or Durbin-Watson); a Q-Q plot of residuals for normality. The universal message of residual plots: a well-specified model's residuals should look like structureless noise — any pattern left in them is signal the model missed.

Fitting the model: OLS, the normal equation, and gradient descent

Ordinary least squares chooses the weights minimizing the mean of squared residuals:

$$\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|_2^2$$

Why squared error, and not absolute error or something else? Three converging justifications, any of which an interviewer may probe. *Statistical*: under the Gaussian-noise model, maximizing the likelihood of the data is exactly minimizing squared error — write the Gaussian density for each y_i , take logs, and the exponent delivers the sum of squares (MLE, Chapter 1). Squared loss is not an arbitrary choice; it is the Gaussian assumption in disguise, which is also why it is outlier-sensitive: a Gaussian says huge errors are astronomically rare, so the fit contorts itself to shrink them. *Analytical*: the loss is smooth and convex with a closed-form minimizer. *Decision-theoretic*: the function minimizing expected squared error is the conditional mean $\mathbb{E}[y | x]$ — squared loss is what you use when you want the mean. (Absolute loss targets the conditional *median* — more robust, no closed form; that trade is a classic question.)

The normal equation. Set the gradient to zero. Expanding $\mathcal{L}(w) = \frac{1}{n} (y - Xw)^\top (y - Xw)$ and differentiating with respect to w (matrix calculus from Chapter 1):

$$\begin{aligned}\nabla_w \mathcal{L} &= -\frac{2}{n} X^\top (y - Xw) = 0 \Rightarrow X^\top Xw = X^\top y \\ \hat{w} &= (X^\top X)^{-1} X^\top y\end{aligned}$$

Convexity guarantees this stationary point is the global minimum. The geometry is worth one interview sentence: $X\hat{w}$ is the orthogonal projection of y onto the column space of X — the fitted values are the closest point to y reachable by any linear combination of the features, and the residual vector is perpendicular to every feature column ($X^\top (y - X\hat{w}) = 0$ is literally the orthogonality statement).

A tiny worked example makes the machinery concrete. Points $(0, 1), (1, 3), (2, 5)$; model $y = b + wx$. With the intercept column included, $X^\top X$ is the 2×2 matrix with rows $(3, 3)$ and $(3, 5)$, and $X^\top y = (9, 13)$. The normal equation is then the pair $3b + 3w = 9$ and $3b + 5w = 13$, giving $w = 2, b = 1$ — the exact line through the points, residuals zero, as expected for collinear data.

Normal equation vs gradient descent is a standard compare-and-contrast, and the answer has real numbers in it. Solving the normal equation costs $O(nd^2 + d^3)$ — forming $X^\top X$ dominates at $O(nd^2)$, solving the $d \times d$ system adds $O(d^3)$. Gradient descent costs $O(nd)$ per step:

$$w \leftarrow w - \eta \nabla_w \mathcal{L}$$

So: for small-to-moderate d (thousands), the closed form is exact, deterministic, and effectively instant — use it (sklearn's LinearRegression does, via SVD-based least squares, which is the numerically stable route: it never forms $X^\top X$, whose condition number is the *square* of X 's). For very large d , or data too big for memory, or streaming settings, or the moment the loss stops being squared error (logistic regression has no closed form at all), gradient descent and its stochastic variants are the only game — which is why the entire deep learning stack (Chapter 13) runs on them. The honest

summary: the normal equation is a special-case luxury of squared loss; gradient descent is the general-purpose engine.

Gradient descent brings its own obligations. The learning rate η must respect the curvature: convergence requires $\eta < 2/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of the Hessian $\frac{2}{\eta}X^\top X$. And the *ratio* $\lambda_{\max}/\lambda_{\min}$ — the condition number — controls the speed: gradient descent zigzags down ill-conditioned (elongated) loss bowls, making progress at the rate of the *worst* direction. Features on wildly different scales produce exactly this elongation, which is the real reason for standardization before gradient-based fitting. Listing 2 measures it: raw features with a $100\times$ scale gap give a condition number of 8,605 and either divergence or a crawl; standardized, the condition number is 1.0 and the same problem converges $10\times$ faster with a $1,000\times$ larger learning rate. Note what scaling does *not* affect: the closed-form solution — OLS predictions are identical under any affine rescaling of features; only the *optimizer* cares.

Polynomial regression

Polynomial regression is linear regression on manufactured features: replace x with $(x, x^2, \dots, x^k)$ and fit exactly as before. Nothing about the fitting changes — same normal equation, same gradients — because the model is still linear in w . What changes is *capacity*, and polynomial degree is the cleanest capacity dial in ML, which is why it is the canonical demonstration of Chapter 4's bias-variance tradeoff and a favorite interview sketch: underfit line, good cubic, wild degree-15 wiggle through every noise point.

Listing 3 runs the sweep on a known cubic truth with noise of variance 9: degree 1 underfits (validation MSE 23.4), degree 3 hits the truth (validation MSE 9.1 — almost exactly the irreducible noise floor, the best any model could do), and degree 15 overfits (training MSE keeps falling to 3.1, validation rises to 12.5). Two readings worth saying out loud in an interview: the validation minimum sitting *at the noise floor* is what "the model has learned everything learnable" looks like; and training error below the noise floor ($3.1 < 9$) is the smoking gun of memorized noise.

Practical mechanics that distinguish candidates who have actually fit one:

- **Scale explosion:** with $x \in [-3, 3]$, the degree-15 feature spans $\pm 14,000,000$; raw polynomial features produce catastrophically ill-conditioned $X^\top X$. Standardize the expanded features (as Listing 3's pipeline does) or use an orthogonal polynomial basis.
- **Multivariate blowup:** the number of polynomial terms in d features of degree k grows combinatorially — degree-3 in 100 features is already 176,851 terms. Polynomial expansion is a univariate/low- d tool; in high dimensions you want kernels (Chapter 6) or learned features (Chapter 12).
- **Extrapolation:** polynomials shoot to $\pm\infty$ outside the training range, and high-degree fits do so violently. Never trust a polynomial model outside the data's support.

- **Degree selection** is hyperparameter tuning: choose by cross-validation (Chapter 4), never by training error, which is monotonically decreasing in degree.

Ridge, Lasso, and Elastic Net

Chapter 4 introduced penalized loss $\mathcal{L}_{reg}(w) = \mathcal{L}(w) + \lambda\Omega(w)$ as a variance-reduction lever. Applied to linear regression, the three canonical penalties define three named models, and the interview questions concentrate on *why their behaviors differ*.

Ridge regression ($\Omega = \|w\|_2^2$) has a closed form — add λ to the diagonal and the normal equation becomes

$$\hat{w}_{ridge} = (X^\top X + \lambda I)^{-1} X^\top y$$

That $+\lambda I$ is doing three jobs at once, and naming all three is a strong answer.

Numerically, it makes the matrix invertible even when features are collinear or $d > n$ — Ridge always has a unique solution where OLS may have none. *Statistically*, it shrinks coefficients toward zero, trading a little bias for a large variance cut, with the biggest shrinkage applied along the directions of least data variance (in the SVD view, each singular direction is damped by $\sigma_i^2/(\sigma_i^2 + \lambda)$ — small singular values, i.e. ill-determined directions, get crushed; well-determined ones pass nearly untouched). *Bayesianly*, it is MAP estimation under a Gaussian prior $w \sim \mathcal{N}(0, \tau^2 I)$, with $\lambda = \sigma^2/\tau^2$ (Chapter 1's MLE-vs-MAP distinction, made concrete).

Lasso ($\Omega = \|w\|_1$) produces exact zeros — it is simultaneously a fitting procedure and a feature selector. The standard geometric explanation, which interviewers explicitly ask for: picture the constraint view, minimizing squared loss subject to $\Omega(w) \leq t$. The L2 ball is a sphere; the elliptical loss contours expand until they *touch* it, and a sphere's surface has no special points — tangency lands at generic (nonzero) coordinates. The L1 ball is a diamond with corners *on the axes*, where coordinates are exactly zero, and expanding ellipses hit corners with high probability. Same story in gradient language: the L2 penalty's pull toward zero is proportional to w_j — it fades as the weight shrinks, so weights approach but never reach zero. The L1 penalty's pull is a constant λ regardless of size; any weight whose data-gradient can't sustain λ of force gets pinned to exactly zero. (Formally, the soft-thresholding update: for orthonormal features,

$\hat{w}_j = \text{sign}(w_j^{OLS}) \cdot \max(|w_j^{OLS}| - \lambda, 0)$ — small OLS coefficients are zeroed outright.)

Lasso's Bayesian reading is a Laplace prior, sharply peaked at zero. Its costs: no closed form (the kink at zero defeats plain gradient descent; solvers use coordinate descent or proximal methods), at most n nonzero coefficients when $d > n$, and unstable selection among correlated features — from a group of near-duplicates it picks one nearly arbitrarily and zeros the rest, and the pick can flip with a re-sampled dataset.

Elastic Net mixes the two, with sklearn's parameterization

$$\mathcal{L}_{enet}(w) = \frac{1}{2n} \|y - Xw\|_2^2 + \lambda \left(\alpha \|w\|_1 + \frac{1-\alpha}{2} \|w\|_2^2 \right)$$

The L1 part gives sparsity; the L2 part stabilizes it, and in particular fixes the correlated-features pathology: near-duplicate features get *shared* weight (the "grouping effect") instead of an arbitrary winner-take-all. Elastic Net is the default recommendation when you want sparsity and features are correlated — which in real data is almost always.

Listing 4 stages all of this on a 20-feature problem with 5 true nonzeros and one nearly-duplicated feature pair: OLS splits the duplicate pair's weight absurdly (0.05 and 6.07 for a true 3-and-3), Ridge splits it evenly (2.86, 2.89) but zeros nothing, Lasso finds 11 of the 15 true zeros but splits the pair unevenly, and Elastic Net gets both properties at once.

Operational rules that interviews reward: **standardize features first** — the penalty compares coefficient magnitudes across features, which is meaningless if features have different units (a coefficient on "meters" is 1000× the same effect on "kilometers", and the penalty would punish the wrong one); **never penalize the intercept** (it encodes the target's baseline level, not model complexity — penalizing it just biases predictions toward zero); and **tune λ by cross-validation** on a logarithmic grid, retraining on all training data at the winner. When to reach for which: Ridge when most features plausibly matter (dense truth) or as the default variance-control; Lasso when you believe few features matter and want the model to say which; Elastic Net when you want Lasso's selection but features correlate; and remember $\lambda \rightarrow 0$ recovers OLS while $\lambda \rightarrow \infty$ drives all (penalized) weights to zero — the full bias-variance dial of Chapter 4.

Logistic regression

Why "regression" for a classifier? Because logistic regression *is* a regression — on the log-odds. It models a continuous quantity, $\log \frac{p}{1-p}$, as a linear function of the features; classification only happens when you threshold the resulting probability. The name is historically accurate and conceptually helpful, and the one-line interview answer is: "it regresses the log-odds linearly; the class label is a decision rule bolted on afterward."

Build it from the need. We want $p(y = 1 \mid x)$ — a probability in $(0, 1)$ — from a linear score $z = w^\top x + b$ that lives in $(-\infty, \infty)$. The bridge is the **log-odds** (logit): odds $\frac{p}{1-p}$ map $(0, 1)$ to $(0, \infty)$, and the log maps that to all of $\mathbb{R}$. Setting

$$\log \frac{p}{1-p} = w^\top x + b$$

and inverting gives the **sigmoid**:

$$p(y = 1 \mid x) = \sigma(w^\top x + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid facts worth having at recall speed: $\sigma(0) = 0.5$; $\sigma(-z) = 1 - \sigma(z)$; its derivative is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, maximal at 0.25 when $z = 0$ and vanishing in the tails (the saturation behind vanishing gradients, Chapter 12); and it turns weight w_j into an **odds ratio**: one unit of x_j multiplies the odds by e^{w_j} — the interpretation used across medicine and credit scoring, and verified numerically in Listing 5.

The decision boundary is linear. Predicting class 1 when $p \geq 0.5$ is predicting when $z \geq 0$, and $w^\top x + b = 0$ is a hyperplane. The sigmoid is monotonic, so it never bends the boundary — it only calibrates distances from the hyperplane into probabilities. Logistic regression is a linear classifier; if the true boundary is a circle, it fails exactly the way linear regression fails on a curve, and the fix is the same (feature engineering — add x_1^2, x_2^2 and the "linear" boundary in the expanded space is a circle in the original). The threshold 0.5 is not sacred: it minimizes error only under symmetric costs, and moving it trades precision against recall (Chapter 10); with class imbalance or asymmetric costs, tuning the threshold on validation data is standard practice.

The loss. Squared error on probabilities is doubly wrong for classification: it is non-convex when composed with the sigmoid (bad optimization), and it barely punishes confident mistakes (bad statistics). Maximum likelihood gives the right loss. Each label is Bernoulli with parameter $\hat{p}_i = \sigma(w^\top x_i + b)$; the likelihood of the dataset is $\prod_i \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1 - y_i}$, and the negative log-likelihood, averaged, is **binary cross-entropy** (log loss):

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log (1 - \hat{p}_i)]$$

Cross-entropy punishes confident wrongness without bound — $\hat{p} \rightarrow 0$ on a true positive costs $-\log \hat{p} \rightarrow \infty$ — which is exactly the incentive a probability model should face. It is convex in w (a sum of log-sum-exp terms), so gradient descent finds the global optimum; there is **no closed form** (the sigmoid makes the stationarity condition transcendental), which is the standard answer to "why does linear regression have a normal equation and logistic doesn't?".

The gradient is the punchline. Differentiating through the sigmoid (chain rule, using $\sigma' = \sigma(1 - \sigma)$), almost everything cancels:

$$\nabla_w \mathcal{L} = \frac{1}{n} X^\top (\hat{p} - y)$$

— *identical in form* to linear regression's gradient with $\hat{p}$ in place of Xw : residual times features. Deriving this cancellation on a whiteboard is a top-five ML interview request, and the cleanliness is not an accident — it recurs for every GLM with its canonical link (next section) and for softmax + cross-entropy (Chapter 12). Listing 5 implements exactly this update and matches sklearn to three decimals.

Practical notes interviewers listen for: sklearn's LogisticRegression **applies L2 regularization by default** (parameter $C = 1/\lambda$, so *larger* C = *less* regularization — a perennial gotcha, and why Listing 5 sets $C = 10^{10}$ to compare against the unregularized scratch version). On **perfectly separable data**, unregularized weights diverge to infinity — the sigmoid can only reach its 0/1 targets in the limit $\|w\| \rightarrow \infty$, so regularization is not optional there. Probabilities from a well-fit logistic regression are typically decently **calibrated** (log loss is a proper scoring rule); heavily regularized or class-weighted fits

shift this, and Chapter 10 covers recalibration. Multiclass extension is **softmax regression** (multinomial logistic): K score vectors, softmax normalization, categorical cross-entropy — Chapter 12 treats it as the output layer of every classifier network.

Generalized Linear Models

Linear regression, logistic regression, and Poisson regression look like three models; GLMs reveal them as one recipe with three settings. The recipe has three components:

1. **A random component:** the target's conditional distribution, drawn from the exponential family — Gaussian, Bernoulli, Poisson, Gamma, and relatives, all sharing the form $f_Y(y; \theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)\right)$ where θ is the natural parameter.
2. **A systematic component:** the linear predictor $z = w^\top x + b$.
3. **A link function g** connecting them: $g(E[y | x]) = z$, i.e. $E[y | x] = g^{-1}(w^\top x)$.

The link is chosen to map the mean's legal range onto all of $\mathbb{R}$. Gaussian mean lives in $\mathbb{R}$: identity link, and the GLM is linear regression. Bernoulli mean is a probability in $(0, 1)$: logit link, logistic regression — the log-odds construction above is exactly this step. Poisson mean is a positive rate: log link, Poisson regression, with the pleasant multiplicative reading that one unit of x_j *multiplies* the expected count by e^{w_j} .

Target	Distribution	Link $g(\mu)$	Inverse link	Model
Continuous, symmetric noise	Gaussian	identity	identity	Linear regression
Binary	Bernoulli	$\log \frac{\mu}{1-\mu}$	sigmoid	Logistic regression
Counts	Poisson	$\log \mu$	e^z	Poisson regression
Positive, right-skewed	Gamma	$\log \mu$ (common)	e^z	Gamma regression

Each listed link (identity, logit, log) is the distribution's **canonical link** — the one making the natural parameter itself linear in x — and canonical links are why the gradient miracle repeats: for every one of these models, the negative log-likelihood gradient is $\frac{1}{n} X^\top (\hat{\mu} - y)$, residual times features, with only the inverse link changing how $\hat{\mu}$ is computed. Listing 6 exploits this: the Poisson fitting loop differs from the logistic one by a single line (`np.exp` in place of `sigmoid`) and recovers the true parameters to three decimals, matching `sklearn`'s `PoissonRegressor`.

Why interviewers care, beyond taxonomy: GLMs are the right answer to "how would you model counts / rates / strictly positive targets?" — questions where naive linear regression embarrasses itself. Listing 6 shows linear regression on Poisson counts

predicting -5.34 events in a low-rate region; the GLM predicts 0.048 , a legal rate. GLMs also encode variance structure — Poisson variance *equals* the mean, so heteroscedasticity is built into the model rather than violating it (when real counts show variance exceeding the mean — "overdispersion" — negative binomial regression is the standard upgrade, a detail that reads as field experience). Fitting is by maximum likelihood, classically via iteratively reweighted least squares (IRLS — Newton's method that solves a weighted OLS problem per iteration), or plain gradient descent as in the listings. In production ML, GLMs survive as the giant sparse logistic regressions behind ad click-through prediction, insurance pricing (Poisson frequency $\times$ Gamma severity), and any setting where calibrated probabilities, convex training, and per-feature interpretability beat raw accuracy.

Code implementations

Every listing was executed as shown; outputs are real. Together they turn each claim in this chapter into a number: the normal equation agreeing with sklearn to machine precision, scaling deciding whether gradient descent converges at all, the validation minimum landing on the noise floor, Lasso's zeros, the logistic gradient's cancellation, the GLM recipe changing one line between models, and the two failure modes — collinearity and heteroscedasticity — measured rather than asserted.

Listing 1 — OLS three ways: normal equation, pseudoinverse, sklearn

Three routes to the same coefficients. The normal equation solves the linear system (never invert explicitly — `np.linalg.solve` is faster and more stable than computing an inverse); the pseudoinverse route runs on the SVD and survives singular designs; sklearn does SVD-based least squares internally. Agreement to $8.9\text{e-}16$ is machine epsilon: these are the same estimator.

```
"""Listing 1: OLS three ways -- normal equation, pseudoinverse, sklearn."""
import numpy as np
from sklearn.linear_model import LinearRegression

rng = np.random.default_rng(42)
n, d = 200, 3
X = rng.normal(size=(n, d))
true_w, true_b = np.array([2.0, -1.0, 0.5]), 4.0
y = X @ true_w + true_b + rng.normal(scale=0.5, size=n) # noisy linear truth

# Normal equation:  $w = (X'X)^{-1} X'y$  (solve, never invert explicitly)
Xb = np.hstack([np.ones((n, 1)), X]) # prepend bias column
w_ne = np.linalg.solve(Xb.T @ Xb, Xb.T @ y)

# SVD pseudoinverse: numerically stable even when  $X'X$  is near-singular
w_pinv = np.linalg.pinv(Xb) @ y

lr = LinearRegression().fit(X, y)
w_sk = np.r_[lr.intercept_, lr.coef_]

print("truth          :", np.r_[true_b, true_w])
print("normal equation:", np.round(w_ne, 4))
print("pseudoinverse   :", np.round(w_pinv, 4))
```

```
print("sklearn      :", np.round(w_sk, 4))
print("max |diff|   :", f"{np.abs(w_ne - w_sk).max():.2e}")
```

Output:

```
truth      : [ 4.  2. -1.  0.5]
normal equation: [ 3.9815  2.0184 -1.0038  0.4742]
pseudoinverse : [ 3.9815  2.0184 -1.0038  0.4742]
sklearn     : [ 3.9815  2.0184 -1.0038  0.4742]
max |diff|  : 8.88e-16
```

Listing 2 — Gradient descent and the case for feature scaling

One feature has 100× the scale of the other. On raw features, the workable learning rates form a needle's eye: 1e-3 diverges to infinity, 1e-5 crawls (MSE 7.8 after 5,000 steps). Standardizing drops the condition number of the loss surface from 8,605 to 1.0, and the same problem converges to MSE 0.25 in a tenth of the steps at a 1,000× larger learning rate. The closed form, by contrast, is scale-indifferent — this is purely an optimizer problem.

```
"""Listing 2: gradient descent for OLS -- why feature scaling decides convergence."""
import numpy as np

rng = np.random.default_rng(0)
n = 500
x1 = rng.normal(0, 1, n)          # unit-scale feature
x2 = rng.normal(0, 100, n)        # feature 100x larger scale
X = np.c_[x1, x2]
y = 3 * x1 + 0.05 * x2 + rng.normal(0, 0.5, n)

def gd(X, y, lr, steps):
    n, d = X.shape
    w, b = np.zeros(d), 0.0
    for _ in range(steps):
        r = X @ w + b - y          # residuals
        w -= lr * (2 / n) * (X.T @ r) # dMSE/dw
        b -= lr * (2 / n) * r.sum()   # dMSE/db
        if not np.isfinite(w).all(): # diverged
            return w, b, np.inf
    return w, b, np.mean((X @ w + b - y) ** 2)

np.seterr(over="ignore", invalid="ignore") # divergence demo overflows on purpose
for lr_ in [1e-3, 1e-4, 1e-5]:
    w, b, mse = gd(X, y, lr_, 5000)
    print(f"raw      lr={lr_:<6g} MSE={mse:>10.4f}")

mu, sd = X.mean(0), X.std(0)
Xs = (X - mu) / sd                    # standardize
w, b, mse = gd(Xs, y, 0.1, 500)
print(f"scaled lr=0.1 MSE={mse:>10.4f} (500 steps, not 5000)")

# The culprit: condition number of X'X
for name, M in [("raw", X), ("scaled", Xs)]:
    ev = np.linalg.eigvalsh(M.T @ M / n)
    print(f"condition number ({name:6s}): {ev.max()/ev.min():.1f}")
```

Output:


```

raw      lr=0.001  MSE=      inf
raw      lr=0.0001 MSE=      1.4491
raw      lr=1e-05  MSE=      7.8486
scaled   lr=0.1    MSE=      0.2548 (500 steps, not 5000)
condition number (raw ): 8604.6
condition number (scaled): 1.0

```

Listing 3 — Polynomial degree sweep on a cubic truth

Noise variance is 9, so 9 is the unbeatable validation floor. Degree 3 essentially reaches it (9.13). Training error dropping *below* 9 (degrees 9–15) is noise memorization by definition — there is no legitimate signal left to fit. The pipeline standardizes the expanded features, without which high degrees would be numerically hopeless.

```

"""Listing 3: polynomial regression -- capacity sweep on a cubic truth."""
import numpy as np
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

rng = np.random.default_rng(7)
def f(x): return 0.5 * x**3 - x**2 + 2 * x          # cubic ground truth
x_tr = rng.uniform(-3, 3, 40); y_tr = f(x_tr) + rng.normal(0, 3, 40)
x_va = rng.uniform(-3, 3, 200); y_va = f(x_va) + rng.normal(0, 3, 200)

print(f'{"deg":>3} {"train MSE":>10} {"val MSE":>10}')
for deg in [1, 2, 3, 5, 9, 15]:
    model = make_pipeline(PolynomialFeatures(deg), StandardScaler(),
                           LinearRegression())
    model.fit(x_tr[:, None], y_tr)
    tr = mean_squared_error(y_tr, model.predict(x_tr[:, None]))
    va = mean_squared_error(y_va, model.predict(x_va[:, None]))
    print(f'{"deg":>3} {"tr":>10.2f} {"va":>10.2f}')

```

Output:

deg	train MSE	val MSE
1	26.44	23.36
2	15.10	15.24
3	4.75	9.13
5	4.64	9.12
9	4.18	11.83
15	3.11	12.48

Listing 4 — Ridge, Lasso, Elastic Net on a sparse truth

Twenty features, five truly nonzero, and features 0 and 1 are near-duplicates (both with true weight 3). OLS handles the duplicate pair catastrophically — 0.05 and 6.07, an arbitrary split with the right sum. Ridge shares the weight evenly but keeps all 20 features. Lasso zeros 11 of the 15 true zeros but splits the correlated pair unevenly (2.62 / 3.36 — and the winner can flip on a re-drawn sample). Elastic Net delivers sparsity *and* an even split: the grouping effect, live.

```

"""Listing 4: Ridge vs Lasso vs Elastic Net on a sparse truth with correlated
features."""
import numpy as np
from sklearn.linear_model import Ridge, Lasso, ElasticNet, LinearRegression
from sklearn.preprocessing import StandardScaler

rng = np.random.default_rng(1)
n, d = 100, 20
X = rng.normal(size=(n, d))
X[:, 1] = X[:, 0] + rng.normal(0, 0.05, n)      # feature 1 ~ duplicate of feature 0
true_w = np.zeros(d); true_w[[0, 1, 2, 3, 4]] = [3, 3, -2, 1.5, 1]  # 15 true zeros
y = X @ true_w + rng.normal(0, 1, n)
Xs = StandardScaler().fit_transform(X)

models = {"OLS": LinearRegression(),
          "Ridge(a=10)": Ridge(alpha=10),
          "Lasso(a=0.1)": Lasso(alpha=0.1),
          "ENet(a=0.1)": ElasticNet(alpha=0.1, l1_ratio=0.5)}
print(f"{'model':<13}{'zeros':>6}{'w0':>8}{'w1':>8} (truth: w0=w1=3, 15 zeros)")
for name, m in models.items():
    m.fit(Xs, y)
    w = m.coef_
    print(f"{'name':<13}{np.sum(np.abs(w) < 1e-6):>6}{w[0]:>8.2f}{w[1]:>8.2f}")

```

Output:

model	zeros	w0	w1	(truth: w0=w1=3, 15 zeros)
OLS	0	0.05	6.07	
Ridge(a=10)	0	2.86	2.89	
Lasso(a=0.1)	11	2.62	3.36	
ENet(a=0.1)	7	2.92	2.96	

Listing 5 — Logistic regression from scratch

The whole training loop is the clean gradient: p minus y , times features. It matches sklearn (run nearly unregularized via $C=1e10$) to three decimals on both accuracy and coefficients. The last line checks the odds-ratio interpretation numerically.

```

"""Listing 5: logistic regression from scratch -- gradient descent on cross-entropy."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(3)
X, y = make_classification(n_samples=1000, n_features=5, n_informative=3,
                           n_redundant=0, random_state=3)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, random_state=3)

def sigmoid(z):
    return 1.0 / (1.0 + np.exp(-np.clip(z, -500, 500)))

def fit_logistic(X, y, lr=0.1, steps=3000):
    n, d = X.shape
    w, b = np.zeros(d), 0.0
    for _ in range(steps):
        p = sigmoid(X @ w + b)                # predicted P(y=1|x)
        g = p - y                             # dL/dz -- the famous clean gradient
        w -= lr * (X.T @ g) / n
        b -= lr * g.mean()
    return w, b

```

```

w, b = fit_logistic(X_tr, y_tr)
p_te = sigmoid(X_te @ w + b)
acc_scratch = ((p_te >= 0.5) == y_te).mean()

sk = LogisticRegression(C=1e10, max_iter=5000).fit(X_tr, y_tr) # ~no regularization
acc_sk = sk.score(X_te, y_te)

print(f"scratch  acc={acc_scratch:.4f}  w={np.round(w, 3)}")
print(f"sklearn  acc={acc_sk:.4f}  w={np.round(sk.coef_[0], 3)}")
# log-odds sanity check: moving one unit along x0 multiplies the odds by e^{w0}
print(f"odds ratio for x0: e^{w[0]:.2f} = {np.exp(w[0]):.2f}")

```

Output:

```

scratch  acc=0.8833  w=[-0.253  2.174 -0.07  0.85  0.377]
sklearn  acc=0.8833  w=[-0.253  2.174 -0.069  0.851  0.377]
odds ratio for x0: e^{-0.25} = 0.78

```

Listing 6 — Poisson regression: the GLM recipe, one changed line

Compare `fit_poisson` here to `fit_logistic` above: the only substantive difference is `np.exp` where `sigmoid` was — the inverse link. Same clean gradient, same loop, different distribution. The scratch fit recovers the truth and matches `sklearn`'s `PoissonRegressor` to three decimals. The closer: linear regression on the same counts predicts -5.34 events for a low-rate input; rates cannot be negative, and the GLM's 0.048 is the kind of answer the problem actually admits.

```

"""Listing 6: GLMs -- Poisson regression from scratch vs linear regression on
counts."""
import numpy as np
from sklearn.linear_model import PoissonRegressor, LinearRegression

rng = np.random.default_rng(5)
n = 2000
X = rng.normal(size=(n, 2))
true_w, true_b = np.array([0.8, -0.5]), 0.3
lam = np.exp(X @ true_w + true_b) # log link: log E[y] = Xw + b
y = rng.poisson(lam) # count target

def fit_poisson(X, y, lr=0.05, steps=4000):
    n, d = X.shape
    w, b = np.zeros(d), 0.0
    for _ in range(steps):
        mu = np.exp(X @ w + b) # inverse link
        g = mu - y # same clean gradient form as logistic/OLS!
        w -= lr * (X.T @ g) / n
        b -= lr * g.mean()
    return w, b

w, b = fit_poisson(X, y)
print("truth      :", np.r_[true_b, true_w])
print("scratch GLM :", np.round(np.r_[b, w], 3))
sk = PoissonRegressor(alpha=0).fit(X, y)
print("sklearn     :", np.round(np.r_[sk.intercept_, sk.coef_], 3))

# Why not just linear regression? It happily predicts negative counts.
lin = LinearRegression().fit(X, y)
x_new = np.array([[-2.5, 2.5]]) # low-rate region

```

```
print(f"linear regression predicts {lin.predict(x_new)[0]:.2f} events (impossible)")
print(f"Poisson GLM predicts      {np.exp(x_new @ w + b)[0]:.3f} events")
```

Output:

```
truth      : [ 0.3  0.8 -0.5]
scratch GLM : [ 0.298  0.805 -0.531]
sklearn     : [ 0.299  0.805 -0.531]
linear regression predicts -5.34 events (impossible)
Poisson GLM predicts      0.048 events
```

Listing 7 — Multicollinearity: coefficient instability, measured

True weights are 1 and 1 throughout. As the correlation between the two features rises from 0 to 0.999, the OLS coefficient's standard deviation across re-drawn training sets explodes from 0.08 to 1.81 — at $\rho=0.999$ the fitted "effect of x_1 " routinely comes out negative or above 3, while predictions remain accurate (the *sum* along the shared direction is well determined; the split is not). Ridge caps the instability at every correlation level, at the price of mild shrinkage bias (mean 0.97 rather than 1.0).

```
"""Listing 7: multicollinearity -- coefficient instability, measured; ridge as the
fix."""
import numpy as np
from sklearn.linear_model import LinearRegression, Ridge

rng = np.random.default_rng(11)
n, trials = 150, 500

def coef_spread(rho, model):
    """Std of the first coefficient across re-drawn training sets."""
    w0 = []
    for _ in range(trials):
        x1 = rng.normal(size=n)
        x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=n) # corr(x1,x2)=rho
        X = np.c_[x1, x2]
        y = 1.0 * x1 + 1.0 * x2 + rng.normal(0, 1, n)
        w0.append(model.fit(X, y).coef_[0])
    return np.mean(w0), np.std(w0)

print(f"{'rho':>5} {'OLS mean w1':>12} {'OLS std':>8} {'Ridge mean w1':>14} {'Ridge
std':>10}")
for rho in [0.0, 0.9, 0.99, 0.999]:
    m_o, s_o = coef_spread(rho, LinearRegression())
    m_r, s_r = coef_spread(rho, Ridge(alpha=10))
    print(f"{'rho':>5} {'m_o':>12.3f} {'s_o':>8.3f} {'m_r':>14.3f} {'s_r':>10.3f}")
print("\nPredictions stay fine; it is the *coefficients* (and their story) that
break.")
```

Output:

rho	OLS mean w1	OLS std	Ridge mean w1	Ridge std
0.0	0.995	0.082	0.936	0.079
0.9	0.990	0.184	0.969	0.118
0.99	0.989	0.609	0.969	0.086
0.999	0.985	1.813	0.969	0.048

Predictions stay fine; it is the *coefficients* (and their story) that break.

Listing 8 — When OLS inference breaks: heteroscedasticity and coverage

The estimator stays unbiased under heteroscedasticity (slope mean 1.997 for a true 2) — that is not the problem. The problem is the *error bars*: the textbook standard-error formula assumes constant variance, and when error spread grows with x , the "95%" confidence interval covers the truth only 83% of the time — $3.4\times$ the promised error rate, silently. Fixes: robust (heteroscedasticity-consistent) standard errors, weighted least squares, or modeling the variance (a GLM view).

```
"""Listing 8: when OLS inference breaks -- standard errors under heteroscedasticity."""
import numpy as np

rng = np.random.default_rng(21)
n, sims = 200, 3000

def one_fit(hetero):
    x = rng.uniform(0, 4, n)
    noise_sd = 0.1 * np.exp(x) if hetero else 0.5 # error variance grows with x
    y = 1.0 + 2.0 * x + rng.normal(0, noise_sd, n)
    Xb = np.c_[np.ones(n), x]
    w = np.linalg.solve(Xb.T @ Xb, Xb.T @ y)
    resid = y - Xb @ w
    s2 = resid @ resid / (n - 2) # homoscedastic sigma^2 estimate
    se = np.sqrt(s2 * np.linalg.inv(Xb.T @ Xb)[1, 1]) # textbook SE for slope
    covered = abs(w[1] - 2.0) < 1.96 * se # does the 95% CI contain truth?
    return w[1], covered

for hetero in [False, True]:
    slopes, cover = zip(*[one_fit(hetero) for _ in range(sims)])
    label = "heteroscedastic" if hetero else "homoscedastic "
    print(f"{label}: slope mean={np.mean(slopes):.3f} "
          f"95% CI coverage={np.mean(cover):.1%} (should be 95%)")
```

Output:

```
homoscedastic : slope mean=2.001 95% CI coverage=95.0% (should be 95%)
heteroscedastic: slope mean=1.997 95% CI coverage=83.0% (should be 95%)
```

Pitfalls, comparisons and practical tips

The comparison table interviewers expect you to have internalized:

	OLS	Ridge	Lasso	Elastic Net	Logistic
Penalty	none	L2	L1	L1 + L2	usually L2
Closed form	yes	yes	no	no	no
Exact zeros	no	no	yes	yes	with L1

	OLS	Ridge	Lasso	Elastic Net	Logistic
Correlated features	unstable split	even split	arbitrary pick	grouped	as penalty chooses
Works when $d > n$	no (non-unique)	yes	yes ($\leq n$ nonzero)	yes	with penalty
Bayesian prior	flat	Gaussian	Laplace	mixture	Gaussian (L2)

Normal equation vs gradient descent, the decision in one line each: closed form when d is modest and the loss is squared error — exact, no hyperparameters; gradient descent when d is huge, data streams or exceeds memory, or the loss is anything else. SGD (mini-batches, Chapter 13) is the industrial default for the latter.

Classic traps, and the tell for each:

- **Interpreting coefficients under collinearity.** "Feature X has a negative weight, so it hurts the target" — with correlated features, individual weights are nearly arbitrary (Listing 7's std of 1.81 on a true weight of 1). Check correlations or variance inflation factors before telling coefficient stories; prefer Ridge when you must interpret under correlation.
- **Regularizing unstandardized features.** The penalty then punishes features for their *units*, not their complexity. Standardize first, always; and don't penalize the intercept.
- **Tuning λ on the test set.** λ is a hyperparameter like any other — cross-validate it (Chapter 4's protocol). Reporting the CV-winning score as the final estimate is selection bias; use an untouched test set.
- **R^2 worship.** R^2 rises with every added feature, junk included; adjusted R^2 or held-out error are the honest versions (metrics in depth: Chapter 10). And a high R^2 says nothing about assumption validity — Anscombe's quartet has four identical R^2 values for wildly different data shapes; plot residuals.
- **sklearn's C in LogisticRegression.** It is *inverse* regularization strength, and regularization is *on by default* — a silent difference from statsmodels/textbook logistic regression that has burned many a coefficient comparison.
- **Squared-loss thinking on classification.** Fitting class labels 0/1 with linear regression produces predictions outside [0,1], a loss dominated by easy far-side points, and a boundary dragged by outliers. If asked "why not just linear-regress the labels?", those three failures are the answer, plus the non-convexity of squared loss through a sigmoid.
- **Trusting standard errors under violated assumptions.** Listing 8: unbiased slope, garbage confidence interval. If inference matters, check residuals and use robust standard errors or WLS.
- **Outliers under squared loss.** One point with a residual of 10 contributes as much loss as 100 points with residuals of 1. Options: investigate (it may be a data error —

Chapter 9), robust losses (Huber: quadratic near zero, linear in the tails; or absolute loss = median regression), or RANSAC for gross corruption.

- **Perfect separation in logistic regression.** Weights diverge, probabilities saturate to 0/1, and solvers warn or run to `max_iter`. Regularization fixes it; noticing it in the wild (a feature that perfectly encodes the label) is often a leakage alarm (Chapter 4).

Quick sanity habits. Fit an intercept-only baseline first (predicting the mean is $R^2 = 0$; anything worse than baseline means a bug). Compare train vs validation error before interpreting anything (Chapter 4's two-number diagnosis). For counts, rates, and strictly positive targets, reach for the matching GLM rather than transforming and hoping — and if you do log-transform the target, remember predictions back-transform with a bias correction (the mean of a log-normal is not the exponential of the mean of the logs).

Interview questions and answers

Q1. Derive the normal equation for linear regression.

Loss: $L(w) = (1/n)\|y - Xw\|^2$. Expand: $(1/n)(y^T y - 2w^T X^T y + w^T X^T X w)$. Gradient with respect to w : $(2/n)(X^T X w - X^T y)$. Set to zero: $X^T X w = X^T y$, so $\hat{w} = (X^T X)^{-1} X^T y$.

Convexity (the Hessian $(2/n)X^T X$ is positive semi-definite) makes the stationary point a global minimum. *Interviewers listen for: defining each matrix dimension, and mentioning the invertibility condition — $X^T X$ must be full rank, i.e. no perfectly collinear features.*

Q2. List the assumptions of linear regression and say which matter for prediction vs inference.

Linearity of the conditional mean, independent errors, homoscedasticity (constant error variance), normal errors — plus no perfect multicollinearity. For point prediction, linearity is the load-bearing one: OLS still gives the best linear approximation under violations of the rest. Independence, homoscedasticity, and normality matter for *inference* — standard errors, confidence intervals, p-values — which become miscalibrated when they fail (a nominal 95% CI covering 83% of the time in this chapter's Listing 8). Normality is the least critical: CLT rescues coefficient inference at large n .

Q3. Normal equation vs gradient descent — when would you use each, and what are the costs?

Normal equation: $O(nd^2 + d^3)$ — exact, hyperparameter-free, the right choice for modest d (up to a few thousand) with squared loss. Gradient descent: $O(nd)$ per iteration — the choice when d is very large, data doesn't fit memory or streams, or the loss has no closed form (logistic, most everything else). Practical additions: sklearn actually uses SVD-based least squares (more stable than forming $X^T X$, whose condition number is the square of X 's), and industrial scale uses SGD on mini-batches.

Q4. Why do we minimize squared error rather than absolute error? When would you prefer absolute error?

Squared error is the Gaussian MLE: assuming $y = w^T x + \varepsilon$ with Gaussian ε , maximizing likelihood is exactly minimizing squared error — plus it's smooth, convex, and admits a closed form. It estimates the conditional *mean*. Absolute error estimates the conditional *median*, has no closed form, and is robust to outliers — prefer it (or Huber loss, quadratic center with linear tails) when the target has heavy tails or gross errors, or when the median is the business-relevant summary. The deep version of the answer: choosing a loss is choosing a noise model and a target summary statistic.

Q5. Does gradient descent need feature scaling? Does the normal equation? Why the difference?

The normal equation is scale-indifferent — rescaling features rescales coefficients inversely and predictions are identical. Gradient descent's convergence rate is governed by the condition number of $X^T X$: wildly different feature scales elongate the loss bowl, forcing a learning rate small enough for the steepest direction and therefore a crawl along the shallowest. Listing 2 measures it: condition number 8,605 raw vs 1.0 standardized; divergence-or-crawl vs fast convergence. Regularized models add a second, independent reason to scale: the penalty compares coefficient magnitudes across features.

Q6. Why does Lasso produce exactly zero coefficients while Ridge only shrinks them?

Geometric answer: minimizing loss subject to a norm ball constraint — the L1 ball is a diamond with corners on the coordinate axes, and expanding elliptical loss contours preferentially touch corners, where coordinates are exactly zero; the L2 ball is a sphere with no special points, so tangency lands at generic nonzero coordinates. Gradient answer: L2's pull toward zero is proportional to the weight ($2\lambda w$ — fades as w shrinks, never finishes the job); L1's pull is constant λ regardless of size, so any weight whose data-gradient can't sustain λ of opposing force is pinned at exactly zero (soft thresholding). *Interviewers listen for: either picture told precisely; both is a strong answer.*

Q7. State the Bayesian interpretation of Ridge and Lasso.

Both are MAP estimation: the penalized loss is the negative log posterior with the penalty as negative log prior. Ridge = Gaussian prior $w \sim N(0, \tau^2 I)$ — smoothly discourages large weights, with $\lambda = \sigma^2/\tau^2$ connecting noise variance and prior width. Lasso = Laplace prior — sharply peaked at zero with heavier tails, encoding "most weights are exactly or nearly zero, a few may be large". This is Chapter 1's MLE-vs-MAP distinction realized: OLS is the flat-prior MLE, regularization is prior information.

Q8. You have 10,000 features, 500 samples, and you suspect only a few dozen features matter. Which regression do you fit and why?

OLS is unavailable ($d > n$ means $X^T X$ is singular; infinitely many perfect fits, none trustworthy). The sparsity belief points to L1: Lasso, or better Elastic Net since with 10,000 features many will be correlated and pure Lasso picks arbitrarily among correlated groups (and can select at most $n = 500$ features). Standardize features, tune λ (and the L1 ratio) by cross-validation — stratified/grouped as the data demands, per Chapter 4. Mentioning that the selected support is unstable under resampling — so downstream "the model chose these genes/features" stories need stability checks (e.g. selection frequency across bootstraps) — reads as real experience.

Q9. Why is logistic regression called regression if it's a classifier?

Because it literally is a regression — on the log-odds: $\log(p/(1-p)) = \mathbf{w}^T \mathbf{x} + b$ models a continuous quantity linearly. The output is a probability; classification appears only when you threshold it, and the threshold is a separate decision-theoretic choice (0.5 only under symmetric costs). Bonus depth: it's the Bernoulli GLM with the canonical logit link, sitting in the same family as linear (Gaussian, identity link) and Poisson (log link) regression.

Q10. Derive the gradient of logistic regression's loss and comment on its form.

Per-example cross-entropy: $L = -[y \log p + (1-y) \log(1-p)]$ with $p = \sigma(z)$, $z = \mathbf{w}^T \mathbf{x} + b$. Chain rule: $dL/dp = -y/p + (1-y)/(1-p)$; $dp/dz = p(1-p)$. Multiply: $dL/dz = p(1-p) \cdot [(p-y)/(p(1-p))] = p - y$. So $\nabla_{\mathbf{w}} L = (p - y)\mathbf{x}$ — residual times features, the identical form as linear regression's gradient. The cancellation is the signature of a canonical link paired with its exponential-family likelihood, and it recurs for softmax + cross-entropy. *Interviewers listen for: the clean execution of the cancellation; fumbling $d\sigma/dz = \sigma(1-\sigma)$ is the common stumble.*

Q11. Why not train logistic regression with squared error on the probabilities?

Two failures. Optimization: squared error composed with the sigmoid is non-convex — gradient descent can stall in flat regions (a confidently wrong prediction sits in the sigmoid's saturated tail where the squared-loss gradient is tiny). Statistics: squared loss caps the penalty for confident wrongness at 1, while cross-entropy sends it to infinity — the correct incentive for a probability forecaster, and the reason cross-entropy is the Bernoulli MLE while squared error corresponds to a Gaussian noise story that's false for binary outcomes.

Q12. What happens to unregularized logistic regression on linearly separable data?

Weights diverge: any separating hyperplane's loss keeps decreasing as $\|w\|$ grows (probabilities push toward 0/1 asymptotically, never arriving), so the MLE doesn't exist. Symptoms: solver max-iteration warnings, enormous coefficients, probabilities saturated at 0 and 1. Fix: any regularization makes the optimum finite. The field note that scores points: perfect separation on real data is often a leakage alarm — some feature encodes the label (Chapter 4).

Q13. Interpret a logistic regression coefficient of 0.7 on feature x_1 .

One unit increase in x_1 adds 0.7 to the log-odds, i.e. multiplies the odds of the positive class by $e^{0.7} \approx 2.01$ — doubled odds, holding other features fixed. Careful additions: this is *odds*, not probability (the probability change depends on where you start — biggest near $p = 0.5$, tiny in the tails); "holding others fixed" is unreliable under collinearity; and if features were standardized, the unit is one standard deviation.

Q14. What are the three components of a GLM? Map linear, logistic, and Poisson regression onto them.

(1) Random component: an exponential-family conditional distribution for y . (2) Systematic component: linear predictor $z = w^T x + b$. (3) Link function g with $g(E[y|x]) = z$. Linear regression: Gaussian + identity link. Logistic: Bernoulli + logit link. Poisson: Poisson + log link. The link maps the mean's legal range onto $\mathbb{R}$ (probabilities via logit, positive rates via log), and the listed links are canonical — which is why all three share the gradient form $X^T(\hat{\mu} - y)$.

Q15. You're predicting daily support-ticket counts. A colleague fits linear regression. Critique and propose better.

Counts are non-negative integers with variance that typically grows with the mean; linear regression can predict negative counts (Listing 6: -5.34 events), assumes constant variance, and models the wrong error distribution. Propose Poisson regression: log link keeps rates positive, coefficients read multiplicatively (e^w per unit feature), variance = mean is built in. Check for overdispersion (variance $>$ mean in residuals — common with bursty tickets); if present, negative binomial. If many exact zeros beyond what the rate explains (e.g. closed days), zero-inflated models. That escalation path — Poisson $\rightarrow$ overdispersion check $\rightarrow$ negative binomial — is the experienced answer.

Q16. Your linear regression's coefficients change wildly when you add a new feature, but validation error barely moves. Explain.

The new feature correlates strongly with existing ones. Along a shared direction, only the coefficient *sum* is well-determined by the data; the split among correlated features is nearly arbitrary and flips with small perturbations (Listing 7: coefficient std of 1.81 on a true weight of 1 at $\rho = 0.999$). Predictions live on the well-determined sum, so validation error is stable. Implications: don't tell causal or importance stories from these coefficients; use Ridge (stabilizes the split by preferring even sharing), drop redundant features, or examine VIFs. *Interviewers listen for: the distinction between prediction (fine) and interpretation (broken).*

Q17. What is polynomial regression, and why is it still "linear"?

Regression on expanded features ($x, x^2, \dots, x^k$) — the model remains linear *in the parameters*, so the normal equation, gradients, and all linear-model theory apply verbatim; only the feature map changed. It is the cleanest capacity dial: degree sweeps trace the full bias-variance curve (Listing 3: validation error falls to the noise floor at the true degree, then rises). Practical cautions: standardize expanded features (scale explosion), never extrapolate (polynomials diverge outside the data), and select degree by cross-validation, never training error.

Q18. How would you decide between Ridge, Lasso, and Elastic Net for a given problem?

By your belief about the true coefficient structure and your need for selection. Dense truth (many small effects — e.g. genomic polygenic scores, pixel weights): Ridge. Sparse truth, want interpretable feature selection: Lasso. Sparse-ish truth with correlated features (the common real case): Elastic Net — L1 for zeros, L2 for the grouping effect. Then let validation decide: cross-validate all three over λ grids; the metric differences are often small, and saying "I'd let CV arbitrate but expect X because Y" shows both priors and empiricism.

Q19. Why must features be standardized before regularized regression but not before OLS?

OLS is equivariant to affine feature rescaling: coefficients absorb the scale change, predictions unchanged. Penalties break this: $\lambda \Sigma w^2$ compares raw coefficient magnitudes, and a feature measured in grams gets a $1000\times$ smaller coefficient than the same feature in kilograms — so the penalty punishes the kilogram version $10^6\times$ less for identical predictive content. Standardizing puts all coefficients in per-standard-deviation units, making the penalty compare like with like. Related: never penalize the intercept — it encodes baseline level, not complexity.

Q20. Walk through what happens to Ridge solutions as λ goes from 0 to infinity. What about the bias-variance profile?

$\lambda = 0$: OLS — minimum-bias, maximum-variance member of the family. As λ grows, coefficients shrink monotonically toward zero (in the SVD basis, each component damped by $\sigma_i^2/(\sigma_i^2 + \lambda)$, weakest-data directions first); bias rises smoothly, variance falls. $\lambda \rightarrow \infty$: all penalized weights $\rightarrow 0$, model collapses to the intercept (prediction = mean). Validation error over λ is U-shaped, and the minimum shifts toward smaller λ as n grows — with more data you need less variance control. This is Chapter 4's regularization dial with an exact mechanism attached.

Q21. Implement one gradient-descent step for linear regression, by hand, on paper: X has rows (1, 0) and (1, 2) (intercept included), $y = (1, 3)$, w starts at (0, 0), $\eta = 0.1$.

Predictions: $Xw = (0, 0)$. Residuals $r = Xw - y = (-1, -3)$. Gradient $= (2/n)X^Tr = (2/2) \cdot ((1 \cdot -1 + 1 \cdot -3), (0 \cdot -1 + 2 \cdot -3)) = (-4, -6)$. Update: $w \leftarrow (0, 0) - 0.1 \cdot (-4, -6) = (0.4, 0.6)$. Sanity: both coordinates move toward the least-squares solution (1, 1). Interviewers listen for: correct residual direction (prediction minus target) and the transpose in X^Tr — the two classic sign/shape errors.

Q22. Your logistic regression outputs probabilities clustered near 0.5 with poor accuracy, but you know the features are informative. List plausible causes.

Ordered by likelihood: (1) over-regularization — sklearn's default $C=1.0$ can be strong for small or unscaled data; weights crushed toward zero push all probabilities toward 0.5; (2) unscaled features plus regularization — the penalty disproportionately kills large-scale features' coefficients; (3) the informative signal is nonlinear — a linear boundary through a circular pattern predicts ~ 0.5 everywhere; engineer interaction/quadratic features or move to a nonlinear model; (4) not converged — check `max_iter` warnings; (5) label noise or class overlap making 0.5 genuinely correct — check the Bayes rate with a strong nonlinear baseline.

Q23. What is heteroscedasticity, how do you detect it, and what does it break?

Non-constant error variance across the feature space — spread growing with predicted value is the classic pattern (incomes, prices). Detect: residuals-vs-fitted plot showing a funnel; formal tests (Breusch-Pagan). It does not bias coefficients (Listing 8: slope mean 1.997 for truth 2) — it breaks *inference*: textbook standard errors assume constant variance and become miscalibrated (measured 83% coverage for a "95%" CI). Fixes: heteroscedasticity-robust (sandwich) standard errors, weighted least squares if you can model the variance, or a GLM whose variance structure matches (Poisson/Gamma), or transform the target.

Q24. Why does sklearn's LogisticRegression sometimes give different coefficients than statsmodels' Logit on the same data?

sklearn regularizes by default (L2, $C = 1.0$; C is *inverse* strength), while statsmodels fits the unpenalized MLE. Shrunk coefficients differ from MLE coefficients — sometimes drastically for small n or unscaled features. To match: set C very large (or `penalty=None` in recent versions). Also worth naming: solver differences don't change the optimum (the loss is convex) but non-convergence does — check warnings. This is a real-world debugging classic disguised as an API question.

Q25. When $d > n$, what exactly goes wrong with OLS, and how does each regularizer resolve it?

$X^T X$ is at most rank $n < d$: singular, so the normal equation has infinitely many solutions — many fit training data perfectly and generalize arbitrarily badly. Ridge: $X^T X + \lambda I$ is full rank for any $\lambda > 0$ — unique solution, the minimum-norm-flavored one. Lasso: selects at most n features, yielding a unique sparse solution under mild conditions. Both encode the extra information (small or sparse weights) that the data alone cannot supply. Modern aside if invited: minimum-norm interpolation and double descent (Chapter 13) complicate the " $d > n$ is hopeless" story, but the classical answer stands for the classical regime.

Q26. The decision boundary of logistic regression is where $p = 0.5$. Show it's a hyperplane and explain when you'd move the threshold.

$p = \sigma(z) = 0.5$ exactly when $z = 0$, i.e. $w^T x + b = 0$ — a hyperplane, because σ is monotonic and only rescales distances from it into probabilities. Move the threshold under asymmetric costs or imbalance: threshold t corresponds to predicting positive when the odds exceed $t/(1-t)$, and the cost-optimal threshold is $C_{FP}/(C_{FP} + C_{FN})$ for misclassification costs C . Fraud with a 100:1 miss-to-false-alarm cost ratio wants a threshold near 0.01, not 0.5. Tune on validation data against the business metric (Chapter 10's precision-recall machinery).

Q27. Write the update loop for Poisson regression from scratch, and say what changes relative to logistic regression.

Loop: `mu = np.exp(X @ w + b); g = mu - y; w -= lr * X.T @ g / n; b -= lr * g.mean()`.

The only substantive change from logistic is the inverse link — `np.exp` where `sigmoid` was — because both are canonical-link GLMs sharing the gradient form $X^T(\hat{\mu} - y)$ (Listing 6 verifies against `sklearn` to three decimals). Numerical care: the exponential can overflow early in training — standardize features, start weights at zero, keep the learning rate modest or clip the linear predictor.

Q28. R^2 on train is 0.92; on validation it's 0.31. A teammate proposes adding polynomial features. React.

Oppose, with the diagnosis: a huge train-validation gap is high variance (Chapter 4's two-number rule) — the model already overfits, and polynomial features add capacity, making it worse. Prescriptions run the other direction: regularization (Ridge/Lasso with CV-tuned λ), fewer/better features, more data. Also audit for leakage (Chapter 4) — gaps this large in tabular problems are often a leaking feature rather than honest overfitting, and checking feature provenance costs nothing. Polynomial features become the right move only in the opposite regime: both scores low and close (high bias). *Interviewers listen for: diagnosis before prescription, and the leakage instinct.*

Chapter 6: Classification Algorithms

The four algorithms in this chapter — K-Nearest Neighbors, Naive Bayes, Support Vector Machines, and Decision Trees — are interview fixtures for a reason that has little to do with production usage numbers: each one isolates a fundamental idea in its purest form. KNN is pure memory: no training, no parameters, just distances — the cleanest possible probe of whether you understand bias-variance, the curse of dimensionality, and why scaling matters. Naive Bayes is Bayes' theorem turned into a classifier by one heroic assumption, and the discussion of when that assumption's wrongness matters is a probability interview disguised as an ML one. SVMs carry the margin story and the kernel trick — the most elegant idea in classical ML, and a perennial "explain it precisely" request. Decision trees are greedy recursive partitioning with information theory as the split judge, and they are the atoms from which random forests and gradient boosting (Chapter 7) are built — you cannot understand the ensembles that dominate tabular ML without understanding the tree's specific failure mode they were designed to fix.

For each algorithm this chapter follows the same arc: the mechanism from zero, the mathematics with every symbol defined, what the hyperparameters really control (always, in the end, bias versus variance — Chapter 4's dial wearing four costumes), where the algorithm breaks, and what interviewers probe. Logistic regression, the fifth member of the classical-classifier canon, has its own treatment in Chapter 5.

K-Nearest Neighbors

KNN classifies a point by plurality vote of its k closest training examples. That single sentence is the entire algorithm: there is no training phase beyond storing the data, no parameters are fit, and all computation is deferred to prediction time — the textbook example of a **lazy** (instance-based), **non-parametric** learner (Chapter 4's taxonomy). Its inductive bias is bare smoothness: points near each other probably share a label, and nothing more.

Formally: given query x , distance metric $d(\cdot, \cdot)$, and the set $N_k(x)$ of the k training points closest to x , predict

$$\hat{y}(x) = \operatorname{argmax}_c \sum_{i \in N_k(x)} \mathbf{1}(y_i = c)$$

— optionally weighting each vote by $1/d(x, x_i)$ so nearer neighbors count more (this also breaks ties gracefully and softens the choice of k). Probability estimates fall out as vote fractions, though they are coarse for small k . The regression variant averages neighbor targets instead of voting; everything below applies to it unchanged.

k is the bias-variance dial, exactly. $k = 1$ memorizes: training accuracy is 1.0 by construction (every point is its own nearest neighbor) and the decision boundary is a jagged Voronoi mosaic that traces every noise point — maximal variance. $k = n$ predicts the global majority class everywhere — maximal bias. In between, larger k averages over more neighbors, smoothing the boundary: variance falls, bias rises. Listing 1 walks the dial on two-moons data: $k=1$ gives 1.000 train / 0.892 test (the overfit gap), $k=15$ the best test accuracy at 0.896, and $k=201$ sags to 0.821 as the vote pools the two moons together. Choose k by cross-validation; odd k avoids ties in binary problems; and a rule-of-thumb starting point is $k \approx \sqrt{n}$, said with the caveat that CV decides.

Distance is the model. Euclidean (ℓ_2) is the default; Manhattan (ℓ_1) is less dominated by any single coordinate; cosine similarity serves high-dimensional sparse data like text (Chapter 19); Hamming handles categorical vectors. Because raw distances add feature contributions in squared units, **KNN is defenseless against scale**: a feature measured in grams contributes a million times the squared difference of the same feature in kilograms, silently deciding every neighborhood by itself. Listing 1 stages this — multiplying one coordinate by 1,000 drops accuracy from 0.887 to 0.792, and standardization restores it. Scale features, always, before any distance-based method (the same imperative as Chapter 5's gradient descent and regularization, for a different reason: there it was the optimizer and the penalty; here the scale-sensitivity *is the model*).

The curse of dimensionality bites KNN first and hardest. Chapter 4 established the geometry: in high dimensions, distances concentrate — the nearest and farthest neighbors become nearly equidistant, and "nearest" stops carrying information. KNN's entire signal is the distance ranking, so it degrades before almost anything else. Practical mitigations: dimensionality reduction first (PCA, Chapter 8; learned embeddings, Chapters 12+), feature selection (Chapter 9), or metric learning. The modern redemption arc is worth one interview sentence: KNN over *learned embeddings* is exactly how retrieval systems, vector databases, and RAG (Chapter 24) work — the deep network fixes the representation so that distance means something, then KNN does what it always did, at billion-point scale via approximate indexes (HNSW, IVF — Chapter 24).

Cost profile. Training $O(1)$ beyond storage; prediction $O(nd)$ per query, brute force — backwards from most learners, and disqualifying for latency-critical serving at scale unless you accept approximate neighbors (KD-trees and ball trees help only in low dimension; past a few dozen dimensions they degenerate to linear scans). Memory: the entire training set, forever. Imbalance: the majority class wins votes just by density — distance weighting and stratified sampling help. KNN's honest niche: small-to-medium tabular data, strong local structure, zero training budget, and as the algorithmic core of embedding retrieval.

Naive Bayes

Naive Bayes classifies with Bayes' theorem (Chapter 1), choosing the class with the highest posterior:

$$p(y = c \mid x) \propto p(y = c) \prod_{j=1}^d p(x_j \mid y = c)$$

The product is the "naive" step: it assumes features are **conditionally independent given the class** — knowing the class, seeing one feature tells you nothing further about another. This is false essentially always ("free" and "cash" co-occur in spam far beyond independence), and the algorithm works anyway; resolving that tension is the standard interview probe, and the resolution is precise: classification needs only the *argmax* over classes, not correct probabilities. Double-counting correlated evidence inflates the winning posterior's confidence, but often leaves the *ranking* of classes intact — decisions survive, calibration dies. So: use the labels, distrust the probabilities (recalibrate — Platt scaling or isotonic regression, Chapter 10 — if you need them).

Training is one counting pass: estimate the prior $p(y = c)$ as class frequency, and each $p(x_j \mid y = c)$ from the class's data alone — a per-feature, per-class estimate that never touches feature interactions. That is why NB is fast (one pass, $O(nd)$), why it tolerates high dimension (it never estimates a joint), and why it works from shockingly little data (d univariate estimates need far fewer samples than one d -variate estimate — a bias-variance argument: NB is a high-bias, low-variance model that wins exactly when data is scarce relative to dimension).

The three variants differ only in the likelihood model for $p(x_j \mid y = c)$, keyed to feature type:

- **Gaussian NB** — continuous features: fit a per-class mean and variance per feature, evaluate the normal density. Listing 2 implements it from scratch in ~20 lines (means, variances, log-densities, argmax) and matches sklearn to the fourth decimal.
- **Multinomial NB** — count features, the text-classification workhorse: $p(w \mid c)$ is the smoothed fraction of class- c word occurrences that are word w ; a document's log-score is the count-weighted sum of word log-probabilities plus the log prior.
- **Bernoulli NB** — binary presence/absence features: models each feature as a coin flip per class, and — the distinction interviewers fish for — **explicitly penalizes absent words** (the $(1 - p)$ factors), where Multinomial simply ignores them. Bernoulli can beat Multinomial on short documents where absence is informative; Multinomial wins on longer text where counts carry signal.

Numerics and smoothing are where implementations fail. A product of hundreds of small probabilities underflows to zero — always work in log space, turning products into sums (Listing 3 does, as does every serious implementation). And an unseen word-class

pair gives $p = 0$, whose log is $-\infty$, vetoing the entire class on one missing count.

Laplace (add-one) smoothing fixes it:

$$\hat{p}(w | c) = \frac{\text{count}(w, c) + \alpha}{\sum_{w'} \text{count}(w', c) + \alpha V}$$

with $\alpha = 1$ classic, α tunable, and V the vocabulary size — a Dirichlet-prior MAP estimate in disguise (Chapter 1 again). Listing 3 builds the whole spam filter from scratch — vectorize, count, smooth, sum logs — agrees with sklearn, and shows the un-smoothed zero that would have killed it.

Where NB earns its keep in 2026: text baselines that take seconds and embarrass slower models more often than anyone admits, spam filtering, streaming/online settings (counts update incrementally), and any tiny-data regime. Its known lies: correlated features double-counted (two copies of the same feature literally squares its evidence), and probabilities pushed toward 0/1 — overconfident by construction.

Support Vector Machines

A linear classifier's decision boundary is a hyperplane $w^\top x + b = 0$; among all hyperplanes that separate the classes, which one? The perceptron takes any; logistic regression takes the one maximizing likelihood. The SVM's answer: the one that **maximizes the margin** — the distance to the nearest training point of either class. Wide margins are a variance-control argument before they are anything else: a boundary with clearance survives perturbations of the data that a grazing boundary does not, and the formal statement (margin-based generalization bounds independent of dimension) is why SVMs dominated the 2000s.

Hard margin. Scale w, b so the closest points satisfy $|w^\top x + b| = 1$ (the canonical form). The margin — distance between the two supporting hyperplanes $w^\top x + b = \pm 1$ — is $2/\|w\|$, so maximizing margin is minimizing $\|w\|^2$ subject to every point classified with clearance:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s. t.} \quad y_i(w^\top x_i + b) \geq 1 \quad \forall i$$

with labels $y_i \in \{-1, +1\}$. A convex quadratic program: unique solution, no local minima. The points that achieve equality — the ones touching the margin — are the **support vectors**; the solution is a weighted combination of them alone, and deleting every other training point changes nothing. That sparsity is the SVM's signature fact.

Soft margin. Real data is not separable, and hard margins are hostage to their single closest point. Introduce slack $\xi_i \geq 0$ measuring each point's violation and charge for it:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \quad \text{s. t.} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i$$

C is the bias-variance dial: large C punishes violations severely — narrow margin, boundary contorted around individual points, high variance; small C tolerates violations for width — smoother boundary, higher bias. (Note the inversion relative to λ in Chapter 5: C multiplies the *loss*, not the penalty, so $C \approx 1/\lambda$ — the same gotcha as sklearn's LogisticRegression.) Listing 4 makes it tangible: C from 0.01 to 100 shrinks the geometric margin from 3.88 to 1.04 and the support-vector count from 48 to 7. Equivalent unconstrained view for the loss-function fluent: soft-margin SVM minimizes **hinge loss** $\max(0, 1 - y_i(w^\top x_i + b))$ plus L2 — zero loss for points beyond the margin (hence sparsity and robustness to easy points), linear loss for violators (hence robustness to outliers relative to squared losses); compare cross-entropy, which never reaches exactly zero and keeps pulling on well-classified points.

The kernel trick. In the dual formulation of the QP, training data enters *only through inner products* $x_i^\top x_j$, and prediction only through $x^\top x_i$ against support vectors. So replace every inner product with a **kernel** $K(a, b) = \phi(a)^\top \phi(b)$ — the inner product in some feature space ϕ maps to — and you have fit a maximum-margin hyperplane in that space *without ever computing* ϕ . Listing 5 shows the books balancing exactly: for the degree-2 kernel $K(a, b) = (a^\top b)^2$ in 2-D, the explicit map is $\phi(a) = (a_1^2, a_2^2, \sqrt{2}a_1a_2)$, and $(a^\top b)^2 = \phi(a)^\top \phi(b)$ to machine precision. The economics: the RBF kernel $K(a, b) = \exp(-\gamma\|a - b\|^2)$ corresponds to an *infinite-dimensional* feature space, computed in $O(d)$ per pair. A hyperplane in that space is a smooth nonlinear surface in the original one — Listing 5's concentric circles go from 0.510 test accuracy (linear: a hyperplane cannot enclose a disk) to 0.995 (poly or RBF). Validity: any symmetric positive semi-definite function is a legal kernel (Mercer's condition); sums and products of kernels are kernels.

Kernel choice and γ . Linear for high-dimensional sparse data (text — often already separable; use LinearSVC/liblinear, which scales far better); RBF as the default nonlinear choice; polynomial when interaction structure of known degree is plausible. RBF's γ is a *second* variance dial: it sets the kernel's reach ($\gamma = 1/(2\sigma^2)$ in bandwidth terms), so large γ means each support vector influences only its immediate vicinity — boundaries shrink-wrap the data, overfitting; small γ approaches a linear-ish smooth boundary. C and γ interact; tune them jointly on a log grid. And scale features first — the RBF kernel is a distance computation, inheriting KNN's scale-sensitivity verbatim.

Practical profile. Training is between $O(n^2)$ and $O(n^3)$ for kernel SVMs — past $\sim 100k$ points, use linear SVMs or different models entirely (this scaling, plus deep learning eating the perception domains and boosting eating tabular, explains the SVM's fall from default status). No native probabilities (Platt scaling bolts them on, at the cost of a held-out fit); multiclass via one-vs-rest or one-vs-one (libsvm's choice); memory is the support-vector set at prediction time. Their enduring niches: medium-sized clean datasets, high-dimensional/low-sample regimes (genomics), and interviews — because margin + duality + kernels remains the densest concentration of beautiful, testable theory in classical ML.

Decision Trees

A decision tree classifies by asking a sequence of feature questions — is income $\leq 55k$? is default_history = 1? — routing each example down branches to a leaf, and predicting the leaf's majority class (or class proportions). It is the only model in this chapter whose decision process a non-technical stakeholder can read directly, and the only one indifferent to feature scaling and monotone transforms (splits compare against thresholds; only the *order* of values matters — log-transforming a feature changes nothing).

Growing the tree is greedy recursive partitioning. At each node, try every feature and every threshold; score each candidate split by how much it purifies the labels; take the best; recurse on the two children; stop when leaves are pure or a limit is hit. Exhaustive search over tree structures is NP-hard; greedy top-down is the universal compromise, and its myopia is real — a split useless now may enable a great one later (XOR is the classic construction: neither feature alone reduces impurity, so a greedy tree can't start), which foreshadows why ensembles of trees beat single trees.

Impurity measures make "purifies" precise. For a node with class proportions p_c :

$$\text{Entropy} = -\sum_c p_c \log_2 p_c \quad \text{Gini} = 1 - \sum_c p_c^2$$

Both are zero for a pure node and maximal for a uniform mix (entropy 1 bit, Gini 0.5 for balanced binary). Gini reads as "probability of misclassifying a random node member if labeled by the node's own class distribution"; entropy is Chapter 1's information measure. **Information gain** is the impurity drop from a split — parent impurity minus the size-weighted average of child impurities:

$$\text{Gain} = I(\text{parent}) - \frac{n_L}{n} I(\text{left}) - \frac{n_R}{n} I(\text{right})$$

Listing 6 computes everything by hand on a 12-row loan dataset: parent entropy exactly 1 bit (6/6 split), and both criteria select the same split (default_history) with gains of 0.655 bits and 0.357 Gini. **Gini vs entropy in practice:** they disagree rarely and unsystematically — Listing 8 measures it across 40 datasets: gini "wins" 22, entropy 18, mean absolute accuracy difference 0.019, identical fit times. Gini is the sklearn default (no logarithm to compute); the honest interview answer is that the criterion is a rounding error next to depth control. One real distinction worth knowing: information gain is biased toward high-cardinality features (an ID column splits into pure singletons for spectacular fake gain), which C4.5 corrects with **gain ratio** (gain normalized by the split's own entropy); continuous features are handled by sorting values and testing boundary thresholds, categorical ones by subset or one-hot splits depending on the implementation.

Trees overfit by default, and pruning is the cure. Grown to purity, a tree memorizes — one leaf per hard example, boundaries tracing noise. Listing 7, with 15% label noise: the full tree has 90 leaves, train accuracy 1.000, test 0.798; depth-capped versions do

better; and **cost-complexity pruning** — grow full, then collapse subtrees whose accuracy contribution doesn't justify their leaves, choosing the penalty α by cross-validation — lands at 11 leaves and test 0.842, beating every fixed depth. Pre-pruning alternatives (`max_depth`, `min_samples_leaf`, `min_impurity_decrease`) are cheaper and tunable the same way; post-pruning sees the whole tree before deciding, dodging the greedy horizon problem. Either way, the tree's variance problem is structural: deep trees are wildly unstable — resample the data and the top split flips, cascading into a different tree entirely. That instability is precisely the property bagging exploits (Chapter 7: average many high-variance trees into one low-variance forest), which is why single trees are today a teaching tool and an ensemble ingredient more than a production model.

Boundary geometry and blind spots. Every split is axis-aligned, so tree boundaries are staircases; a diagonal linear boundary costs a deep staircase of splits that a linear model gets for free — worth naming as the dual of the SVM's weakness on rectangle-shaped truths. Trees extrapolate as constants (a leaf's prediction holds however far out you go), handle missing values via surrogate splits (CART) or learned default directions (XGBoost, Chapter 7), and their "feature importance" (total impurity reduction per feature) inherits the cardinality bias — permutation importance (Chapter 11) is the more honest read.

Code implementations

Every listing was executed as shown; outputs are real. The set covers the four algorithms' defining behaviors: KNN's k -dial and scale fragility, Gaussian NB from scratch matching sklearn, a complete smoothed spam filter, the SVM margin shrinking as C grows, the kernel identity verified to machine precision, tree splits computed by hand, pruning beating every fixed depth, and the Gini-vs-entropy question settled empirically.

Listing 1 — KNN: the k dial and the scaling trap

$k=1$ posts perfect training accuracy by construction — each point is its own nearest neighbor — with the overfit gap to prove it. $k=201$ pools the two moons into mush. The second half multiplies one feature by 1,000: neighborhoods are then decided by that feature alone, and accuracy drops nine points until standardization restores it.

```
"""Listing 1: KNN -- the k dial is a bias-variance dial, and scaling is not
optional."""
import numpy as np
from sklearn.datasets import make_moons
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler

rng = np.random.default_rng(0)
X, y = make_moons(n_samples=600, noise=0.3, random_state=0)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=0)

print(f"{'k':>4} {'train acc':>10} {'test acc':>9}")
for k in [1, 5, 15, 51, 201]:
    m = KNeighborsClassifier(n_neighbors=k).fit(X_tr, y_tr)
    print(f"{'k':>4} {m.score(X_tr, y_tr):>10.3f} {m.score(X_te, y_te):>9.3f}")
```

```
# Scaling: put one feature in "grams" (x1000) and watch distances break
X2 = X.copy(); X2[:, 1] *= 1000
X2_tr, X2_te, y2_tr, y2_te = train_test_split(X2, y, test_size=0.4, random_state=0)
raw = KNeighborsClassifier(15).fit(X2_tr, y2_tr).score(X2_te, y2_te)
sc = StandardScaler().fit(X2_tr)
scaled = KNeighborsClassifier(15).fit(sc.transform(X2_tr),
y2_tr).score(sc.transform(X2_te), y2_te)
print(f"\nfeature 2 in x1000 units: raw acc={raw:.3f} scaled acc={scaled:.3f}")
```

Output:

k	train acc	test acc
1	1.000	0.892
5	0.939	0.887
15	0.917	0.896
51	0.914	0.879
201	0.825	0.821

```
feature 2 in x1000 units: raw acc=0.792 scaled acc=0.887
```

Listing 2 — Gaussian Naive Bayes from scratch

The whole classifier is twenty lines: per-class priors, per-feature means and variances, log-densities summed under the independence assumption, argmax. Exact agreement with sklearn. Note the log-space computation and the variance floor ($1e-9$) — the two numerics habits that keep NB implementations alive.

```
"""Listing 2: Gaussian Naive Bayes from scratch -- Bayes' rule plus one big
assumption."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

X, y = make_classification(n_samples=1500, n_features=6, n_informative=4,
                           n_redundant=0, random_state=4)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, random_state=4)

class ScratchGNB:
    def fit(self, X, y):
        self.classes = np.unique(y)
        self.prior, self.mu, self.var = {}, {}, {}
        for c in self.classes:
            Xc = X[y == c]
            self.prior[c] = len(Xc) / len(X)           # P(y=c)
            self.mu[c] = Xc.mean(axis=0)               # per-feature mean
            self.var[c] = Xc.var(axis=0) + 1e-9         # per-feature variance
        return self

    def _log_joint(self, X, c):
        # log P(y=c) + sum_j log N(x_j; mu_cj, var_cj) -- the naive independence step
        ll = -0.5 * (np.log(2 * np.pi * self.var[c]) + (X - self.mu[c])**2 /
self.var[c])
        return np.log(self.prior[c]) + ll.sum(axis=1)

    def predict(self, X):
        joint = np.column_stack([self._log_joint(X, c) for c in self.classes])
        return self.classes[joint.argmax(axis=1)]

acc_scratch = (ScratchGNB().fit(X_tr, y_tr).predict(X_te) == y_te).mean()
```



```
acc_sklearn = GaussianNB().fit(X_tr, y_tr).score(X_te, y_te)
print(f"scratch GNB: {acc_scratch:.4f}    sklearn GNB: {acc_sklearn:.4f}")
```

Output:

```
scratch GNB: 0.8022    sklearn GNB: 0.8022
```

Listing 3 — Multinomial NB spam filter with Laplace smoothing

A complete text classifier from scratch: count vectorization, smoothed per-class word probabilities, log-space scoring, and agreement with sklearn. The last line shows the bug smoothing prevents: "deadline" never appears in spam, so its unsmoothed probability is zero, and one $\log(0)$ would veto the spam class for any document containing it.

```
"""Listing 3: Multinomial NB for text -- word counts, log-space, Laplace smoothing."""
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

train_docs = ["win cash prize now", "claim your free prize", "win free cash now",
              "meeting at noon tomorrow", "project deadline next week",
              "lunch meeting with the team"]
train_y = np.array([1, 1, 1, 0, 0, 0]) # 1 = spam
test_docs = ["free cash prize meeting", "team project lunch tomorrow"]

vec = CountVectorizer()
Xtr = vec.fit_transform(train_docs).toarray()
Xte = vec.transform(test_docs).toarray()
vocab = np.array(vec.get_feature_names_out())
V = len(vocab)

# From scratch, with Laplace smoothing (alpha=1)
logp_w = {}
for c in [0, 1]:
    counts = Xtr[train_y == c].sum(axis=0)
    logp_w[c] = np.log((counts + 1) / (counts.sum() + V)) # P(word|class), smoothed
log_prior = {c: np.log((train_y == c).mean()) for c in [0, 1]}

for doc, x in zip(test_docs, Xte):
    scores = {c: log_prior[c] + (x * logp_w[c]).sum() for c in [0, 1]}
    pred = max(scores, key=scores.get)
    print(f"{doc!r}: spam-score={scores[1]:.2f} ham-score={scores[0]:.2f} -> {'SPAM'
    if pred else 'ham'}")

sk = MultinomialNB(alpha=1.0).fit(Xtr, train_y)
print("sklearn agrees:", sk.predict(Xte))
```

Output:

```
'free cash prize meeting': spam-score=-11.13 ham-score=-13.46 -> SPAM
'team project lunch tomorrow': spam-score=-14.43 ham-score=-11.78 -> ham
sklearn agrees: [1 0]
```

Listing 4 — Soft-margin SVM: what C actually does

Same data, three values of C. The geometric margin ($2/\|w\|$) shrinks from 3.88 to 1.04 as C rises, and the support-vector count falls from 48 to 7 — a wide tolerant boundary leaning on many points versus a narrow strict one leaning on few. Test accuracy barely moves here because the blobs are nearly separable; on noisier data, large C is where overfitting lives.

```
"""Listing 4: SVM soft margin -- C controls the margin/violation trade, support vectors
tell the story."""
import numpy as np
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

X, y = make_blobs(n_samples=400, centers=2, cluster_std=1.6, random_state=6)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=6)

print(f"{'C':>8} {'support vecs':>13} {'margin width':>13} {'train acc':>10} {'test
acc':>9}")
for C in [0.01, 1, 100]:
    m = SVC(kernel="linear", C=C).fit(X_tr, y_tr)
    margin = 2 / np.linalg.norm(m.coef_) # geometric margin = 2/||w||
    print(f"{'C':>8g} {'m.n_support_.sum():>13} {'margin:>13.3f} "
          f"{'m.score(X_tr, y_tr):>10.3f} {'m.score(X_te, y_te):>9.3f}")
```

Output:

C	support vecs	margin width	train acc	test acc
0.01	48	3.882	0.988	0.975
1	10	1.347	0.992	0.969
100	7	1.036	0.992	0.969

Listing 5 — The kernel trick, verified to machine precision

Concentric circles: a linear SVM scores 0.510 — coin-flip, since no hyperplane encloses a disk — while degree-2 polynomial and RBF kernels both reach 0.995. Then the trick itself, with the books balanced explicitly: the degree-2 kernel $(a \cdot b)^2$ equals the inner product of the explicit feature maps $\phi(a) \cdot \phi(b)$, computed without ever materializing ϕ .

```
"""Listing 5: the kernel trick -- circles data, and a kernel computed two ways."""
import numpy as np
from sklearn.datasets import make_circles
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

X, y = make_circles(n_samples=500, factor=0.5, noise=0.1, random_state=8)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=8)

for kernel in ["linear", "poly", "rbf"]:
    m = SVC(kernel=kernel, degree=2, C=1).fit(X_tr, y_tr)
    print(f"{'kernel:>7} kernel: test acc = {m.score(X_te, y_te):.3f}")

# The trick itself: K(a,b) = (a.b)^2 equals a dot product in feature space
# phi(a) = (a1^2, a2^2, sqrt(2) a1 a2) -- explicit degree-2 map for 2D input
a, b = X[0], X[1]
```

```
phi = lambda v: np.array([v[0]**2, v[1]**2, np.sqrt(2) * v[0] * v[1]])
lhs = (a @ b) ** 2 # kernel: O(d) work
rhs = phi(a) @ phi(b) # explicit map: O(d^2) features, more for higher degree
print(f"\n(a.b)^2 = {lhs:.6f}    phi(a).phi(b) = {rhs:.6f}    equal: {np.isclose(lhs, rhs)}")
```

Output:

```
linear kernel: test acc = 0.510
poly kernel: test acc = 0.995
rbf kernel: test acc = 0.995

(a.b)^2 = 0.000293    phi(a).phi(b) = 0.000293    equal: True
```

Listing 6 — Decision tree splitting by hand

A 12-row loan dataset, both impurity criteria implemented from scratch, and exhaustive search over every feature and threshold — exactly what a tree does at every node. Parent entropy is exactly 1 bit (six approvals, six rejections). Both criteria choose the same split, on different scales: the default-history feature, gaining 0.655 bits (entropy) or 0.357 (Gini).

```
"""Listing 6: decision tree splitting from scratch -- information gain and Gini, by hand."""
import numpy as np

# Toy loan dataset: (income_k, has_default_history) -> approved
X = np.array([[25, 1], [35, 1], [45, 0], [20, 0], [65, 0], [80, 0], [30, 1], [90, 0],
              [55, 1], [70, 0], [40, 0], [85, 1]])
y = np.array([0, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0])

def entropy(y):
    if len(y) == 0: return 0.0
    p = np.bincount(y, minlength=2) / len(y)
    p = p[p > 0]
    return -(p * np.log2(p)).sum()

def gini(y):
    if len(y) == 0: return 0.0
    p = np.bincount(y, minlength=2) / len(y)
    return 1 - (p ** 2).sum()

def best_split(X, y, impurity):
    base = impurity(y)
    best = (None, None, -1) # feature, threshold, gain
    for j in range(X.shape[1]):
        for t in np.unique(X[:, j]):
            left, right = y[X[:, j] <= t], y[X[:, j] > t]
            if len(left) == 0 or len(right) == 0: continue
            child = (len(left) * impurity(left) + len(right) * impurity(right)) / len(y)
            gain = base - child # information gain (weighted impurity drop)
            if gain > best[2]:
                best = (j, t, gain)
    return best

print(f"parent entropy = {entropy(y):.4f}, parent gini = {gini(y):.4f}")
for name, imp in [("entropy", entropy), ("gini", gini)]:
```

```
j, t, g = best_split(X, y, imp)
feat = ["income", "default_history"][j]
print(f"best split by {name:>7}: {feat} <= {t}    gain = {g:.4f}")
```

Output:

```
parent entropy = 1.0000, parent gini = 0.5000
best split by entropy: default_history <= 0    gain = 0.6549
best split by    gini: default_history <= 0    gain = 0.3571
```

Listing 7 — Tree overfitting and cost-complexity pruning

With 15% label noise, the unrestricted tree grows 90 leaves and memorizes (train 1.000, test 0.798). Cost-complexity pruning — grow full, prune by cross-validated α — lands at 11 leaves and test 0.842, beating every fixed depth cap. The lesson: let the tree see everything, then charge rent per leaf.

```
"""Listing 7: tree overfitting and cost-complexity pruning."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier

X, y = make_classification(n_samples=1200, n_features=10, n_informative=5,
                          flip_y=0.15, random_state=9) # 15% label noise
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=9)

print(f"{'max_depth':>9} {'leaves':>7} {'train acc':>10} {'test acc':>9}")
for depth in [2, 4, 8, None]:
    t = DecisionTreeClassifier(max_depth=depth, random_state=9).fit(X_tr, y_tr)
    print(f"{'str(depth):>9} {t.get_n_leaves():>7} {t.score(X_tr, y_tr):>10.3f} "
          f"{t.score(X_te, y_te):>9.3f}")

# Cost-complexity pruning: grow full, then prune by CV over alpha
path = DecisionTreeClassifier(random_state=9).cost_complexity_pruning_path(X_tr, y_tr)
best_alpha, best_acc = 0, 0
for a in np.unique(path.ccp_alphas)[:3]:
    acc = cross_val_score(DecisionTreeClassifier(ccp_alpha=a, random_state=9), X_tr,
                          y_tr, cv=5).mean()
    if acc > best_acc: best_alpha, best_acc = a, acc
pruned = DecisionTreeClassifier(ccp_alpha=best_alpha, random_state=9).fit(X_tr, y_tr)
print(f"\npruned (alpha={best_alpha:.5f}): {pruned.get_n_leaves()} leaves, "
      f"train {pruned.score(X_tr, y_tr):.3f}, test {pruned.score(X_te, y_te):.3f}")
```

Output:

max_depth	leaves	train acc	test acc
2	4	0.735	0.725
4	15	0.847	0.810
8	57	0.958	0.810
None	90	1.000	0.798

pruned (alpha=0.00802): 11 leaves, train 0.854, test 0.842

Listing 8 — Gini vs entropy: measured, not argued

Forty datasets, identical trees except for the criterion. Neither wins systematically (22–18), the mean accuracy difference is under two points with no consistent direction, and fit times are equal. The empirical answer to a classic question: pick either, spend your attention on depth and pruning.

```

"""Listing 8: Gini vs entropy -- does the criterion actually matter? Measured."""
import time
import numpy as np
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

rng = np.random.default_rng(13)
same_acc, gini_wins, ent_wins, t_gini, t_ent = 0, 0, 0, 0.0, 0.0
diffs = []
trials = 40
for i in range(trials):
    X, y = make_classification(n_samples=800, n_features=12, n_informative=6,
                              flip_y=0.1, random_state=i)
    X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=i)
    t0 = time.perf_counter()
    ag = DecisionTreeClassifier(criterion="gini", max_depth=6, random_state=0)
    ag.fit(X_tr, y_tr).score(X_te, y_te)
    t_gini += time.perf_counter() - t0
    t0 = time.perf_counter()
    ae = DecisionTreeClassifier(criterion="entropy", max_depth=6, random_state=0)
    ae.fit(X_tr, y_tr).score(X_te, y_te)
    t_ent += time.perf_counter() - t0
    diffs.append(abs(ag - ae))
    if abs(ag - ae) < 1e-12: same_acc += 1
    elif ag > ae: gini_wins += 1
    else: ent_wins += 1

print(f"over {trials} datasets: identical acc {same_acc}, gini better {gini_wins},
entropy better {ent_wins}")
print(f"mean |accuracy difference| = {np.mean(diffs):.4f}")
print(f"total fit time: gini {t_gini:.2f}s, entropy {t_ent:.2f}s")

```

Output:

```

over 40 datasets: identical acc 0, gini better 22, entropy better 18
mean |accuracy difference| = 0.0194
total fit time: gini 0.16s, entropy 0.15s

```

Pitfalls, comparisons and practical tips

The four algorithms, side by side:

	KNN	Naive Bayes	SVM (kernel)	Decision Tree
Training cost	$O(1)$	$O(nd)$, one pass	$O(n^2)$ – $O(n^3)$	$O(n \, d \, \log n)$
Prediction cost	$O(nd)$ per query	$O(cd)$	$O(\#SV \cdot d)$	$O(\text{depth})$

	KNN	Naive Bayes	SVM (kernel)	Decision Tree
Needs scaling	yes — critical	no	yes (RBF/poly)	no
Native probabilities	vote fractions (coarse)	yes (miscalibrated)	no (Platt bolt-on)	leaf fractions (overconfident)
Decision boundary	local, jagged	quadratic (GNB) / linear (MNB)	max-margin, kernel-shaped	axis-aligned staircase
Key dial(s)	k, metric	α smoothing	C, γ , kernel	depth / pruning α
Curse of dimensionality	fatal	tolerant	tolerant (margin bounds)	greedy search dilutes
Interpretability	by example	per-feature log-evidence	low (kernel)	highest

When-to-reach-for-which, one line each. KNN: small data, local structure, or retrieval over embeddings. Naive Bayes: text baselines, tiny data, streaming counts, need for speed. SVM: medium clean datasets, high-d/low-n, when maximum-margin robustness matters. Tree: when the stakeholder must read the model — otherwise its ensemble descendants (Chapter 7) take its place.

Classic traps:

- **Forgetting to scale KNN or RBF-SVM features.** Both are distance machines; an ill-scaled feature silently owns every distance (Listing 1's nine-point drop). Trees and NB are immune — knowing *which* models need scaling and why is a favorite screening question.
- **Reading NB probabilities as calibrated.** The independence violation inflates confidence toward 0/1; ranking survives, probabilities don't. Recalibrate (Chapter 10) or use labels only.
- **Skipping smoothing in Multinomial NB.** One unseen word-class pair vetoes the class with $\log(0)$. Laplace smoothing is not optional (Listing 3).
- **Tuning C without γ (or vice versa) in RBF SVMs.** They interact — a good C at one γ is terrible at another; grid or Bayesian-search them jointly, on log scales.
- **Believing a deep tree's training accuracy.** 1.000 train / 0.798 test (Listing 7) is the default outcome, not an anomaly; always prune or cap, always by validation.
- **Trusting impurity-based feature importance.** Biased toward high-cardinality and continuous features; an ID column looks maximally informative. Permutation importance (Chapter 11) before conclusions.
- **Fitting kernel SVMs on 500k rows.** The quadratic-plus kernel matrix will not fit and the QP will not finish. Linear SVM (liblinear/SGD) or a different model family.
- **Using k=1 "because it's most flexible".** It is also the maximum-variance member of the family, with an expected error approaching twice the Bayes rate asymptotically — a theory fact interviewers enjoy: 1-NN's asymptotic error is

bounded by $2 \times$ Bayes error, so even the crudest rule is within a factor of two of optimal given infinite data.

- **One-hot exploding categorical features for trees.** Trees can split subsets natively (implementation permitting); one-hot fragments a categorical's signal across many shallow binary splits and biases against it. Know your library's categorical handling (Chapter 7's LightGBM/CatBoost treat this as a headline feature).

Threshold and imbalance notes, shared across all four. Every classifier here yields scores; the 0.5-equivalent default threshold assumes symmetric costs (Chapter 5's logistic discussion applies verbatim). Under heavy imbalance: stratify splits (Chapter 4), weight classes (`class_weight` in sklearn's SVM/tree; priors in NB do it natively), and evaluate with PR-AUC rather than accuracy (Chapter 10).

Interview questions and answers

Q1. Why is KNN called a "lazy" and "non-parametric" learner, and what are the consequences of each label?

Lazy: no training computation — the model is the stored dataset, and all work happens at query time. Consequence: $O(1)$ training but $O(nd)$ per prediction, the inverse of the usual profile, which disqualifies brute-force KNN from low-latency serving at scale. Non-parametric: no fixed-size parameter vector — model complexity grows with the data (Chapter 4's distinction). Consequence: capacity adapts to any boundary shape given enough data, but memory holds the full training set forever and generalization leans entirely on the smoothness assumption that nearby points share labels.

Q2. How does k in KNN relate to bias and variance? What happens at $k=1$ and $k=n$?

k is a smoothing dial: small k means small neighborhoods — flexible, jagged boundaries, high variance; large k means votes pooled over wide regions — smooth boundaries, high bias. $k=1$ memorizes (train accuracy 1.0 by construction; the boundary is the Voronoi diagram of the training set, tracing every noise point). $k=n$ predicts the global majority class everywhere, ignoring x entirely. Listing 1 traces the arc: 1.000/0.892 train/test at $k=1$, best test at moderate k , sagging accuracy by $k=201$. Select k by cross-validation. Bonus theory: 1-NN's asymptotic error is at most twice the Bayes error.

Q3. Why does feature scaling matter enormously for KNN and RBF-SVMs but not at all for decision trees or Naive Bayes?

KNN and the RBF kernel are built on distances, and squared differences add across features in raw units — a feature with $1000\times$ the scale contributes $10^6\times$ the squared distance, deciding neighborhoods alone (Listing 1: accuracy $0.887 \rightarrow 0.792$ unscaled). Trees only compare each feature against its own thresholds — any monotone transform leaves the split structure unchanged. NB models each feature independently in its own units; scale is absorbed into the per-feature densities/counts. The pattern worth stating: scaling matters wherever features are *combined* in a common metric (distances, penalties, gradient steps) and not where they are treated per-feature.

Q4. State the Naive Bayes assumption precisely. It's false in practice — why does the classifier still work?

Conditional independence: features are independent *given the class*, so the joint likelihood factorizes into per-feature terms. (Not marginal independence — features may correlate through the class.) It survives its own falseness because prediction needs only the argmax over classes: double-counting correlated evidence inflates the winner's posterior but often preserves which class wins. The casualty is calibration — probabilities are pushed toward 0/1 and shouldn't be trusted without recalibration. The crisp phrasing: the decision is robust, the probabilities are not.

Q5. When would you use Gaussian vs Multinomial vs Bernoulli Naive Bayes?

By feature type, because the variants differ only in the likelihood model. Gaussian: continuous features (per-class mean/variance per feature). Multinomial: counts — the text default with bag-of-words or TF vectors, where a document's score is a count-weighted sum of word log-probabilities. Bernoulli: binary presence/absence — and unlike Multinomial it also scores *absent* features via the $(1-p)$ factors, which can win on short documents where a word's absence is informative. Multinomial usually wins on longer text where repetition carries signal.

Q6. Why is Laplace smoothing necessary in Multinomial NB, and what is it Bayesianly?

An unseen word-class pair has count 0, hence estimated probability 0, hence log-probability $-\infty$ — a single novel word vetoes the entire class regardless of all other evidence (Listing 3's 'deadline' example). Add- α smoothing replaces count/total with $(\text{count} + \alpha) / (\text{total} + \alpha V)$, guaranteeing every word positive probability. Bayesianly it is MAP estimation under a Dirichlet(α) prior on the word distribution — pseudo-counts as prior belief, Chapter 1's MLE-vs-MAP story again. α tunes shrinkage toward uniform: larger α = more smoothing = more bias, less variance.

Q7. Why must NB implementations compute in log space?

The likelihood is a product of many per-feature probabilities, each possibly tiny; a few hundred factors underflow float64 to exactly zero, making all classes tie at 0. Logs turn the product into a sum of manageable negative numbers and turn argmax of products into argmax of sums (log is monotone). Same discipline appears anywhere likelihoods multiply — HMMs, language model scoring (Chapter 20). If actual probabilities are needed, normalize with the log-sum-exp trick, subtracting the max before exponentiating.

Q8. Define the margin of an SVM and derive why maximizing it means minimizing $\|w\|^2$.

The margin is the distance from the separating hyperplane $w^T x + b = 0$ to the closest training point. A point's distance to the hyperplane is $|w^T x + b| / \|w\|$. Fix the scale ambiguity (w, b can be multiplied by any constant) by canonicalizing: the closest points satisfy $|w^T x + b| = 1$. Then the closest distance is $1 / \|w\|$ and the full margin between the two supporting hyperplanes is $2 / \|w\|$. Maximizing $2 / \|w\| =$ minimizing $\|w\| =$ minimizing $\frac{1}{2} \|w\|^2$ (squared for differentiability, halved for a clean gradient), subject to $y_i(w^T x_i + b) \geq 1$. *Interviewers listen for: the canonicalization step — without it "maximize the margin" has no well-defined objective.*

Q9. What is a support vector, and why is the SVM solution sparse in them?

A training point with a nonzero dual coefficient — geometrically, one lying on the margin (or violating it, in soft-margin). The optimal w is a weighted sum of support vectors only; every point comfortably beyond the margin has zero hinge loss and zero gradient pull, so removing it changes nothing. Consequences: prediction cost scales with the number of SVs, not n ; the SV count is a capacity diagnostic (Listing 4: 48 SVs at $C=0.01$ vs 7 at $C=100$); and a very high SV fraction signals an overlapping or misspecified problem.

Q10. Hard margin vs soft margin — what problem does soft margin solve, and what does C control?

Hard margin requires perfect separation: it has no solution on non-separable data and is hostage to its single closest point (one mislabeled example relocates the boundary). Soft margin adds slack $\xi_i \geq 0$ per point — the amount by which it may violate the margin — and charges C per unit of total slack. C is the bias-variance dial: large C approximates hard margin (narrow, contorted, high variance), small C buys width by tolerating violations (smooth, high bias). Note C multiplies the loss term, so it behaves like $1/\lambda$ — the inverse of ridge's λ , a sign-convention trap shared with sklearn's LogisticRegression.

Q11. Explain the kernel trick precisely — what property of the SVM makes it possible?

In the dual, both training and prediction touch data *only through inner products* $\mathbf{x}_i^T \mathbf{x}_j$. Replace every inner product with $K(a,b) = \varphi(a)^T \varphi(b)$ for some feature map φ , and the algorithm runs identically — fitting a max-margin hyperplane in φ -space without ever computing φ . The payoff: K is often far cheaper than φ (Listing 5: $(a^T b)^2$ in $O(d)$ equals the inner product of an $O(d^2)$ -dimensional explicit map, verified to machine precision); for RBF, φ -space is infinite-dimensional yet K costs $O(d)$. Any symmetric positive semi-definite function is valid (Mercer). Applies to any algorithm expressible in inner products — kernelized ridge, PCA, and friends.

Q12. What does the RBF kernel's γ control? Describe overfitting with large γ .

$K(a,b) = \exp(-\gamma\|a-b\|^2)$: γ is inverse bandwidth — the reach of each support vector's influence. Small γ : broad influence, near-linear smooth boundaries, high bias. Large γ : influence shrinks to tiny neighborhoods, the boundary shrink-wraps individual training points — islands around each example, training accuracy ~ 1 , terrible generalization; in the $\gamma \rightarrow \infty$ limit behavior approaches 1-NN memorization. γ and C interact (both push flexibility), so tune jointly on a log grid, with scaled features — the kernel is a distance and inherits distance's scale-sensitivity.

Q13. Compare hinge loss and cross-entropy as classification losses.

Hinge: $\max(0, 1 - y \cdot z)$ — exactly zero beyond the margin, so correctly-and-confidently classified points exert no pull; produces sparse solutions (support vectors) and stops caring once clearance is achieved. Linear growth for violators makes it more outlier-tolerant than squared losses. Not differentiable at the kink (subgradients handle it); no probabilistic reading. Cross-entropy: never exactly zero — every point keeps pulling, weights keep growing without regularization on separable data (Chapter 5); smooth, convex, and is the Bernoulli MLE, so it yields calibrated probabilities. Rule of thumb: hinge when you want a decision with margin sparsity, cross-entropy when you want probabilities.

Q14. Why do kernel SVMs struggle at large n , and what are the practical outs?

The kernel matrix is $n \times n$ — a million points means 10^{12} entries — and QP solvers run $O(n^2)$ – $O(n^3)$; prediction also scales with the support-vector count, which grows with n on noisy data. Outs: linear SVMs via liblinear or SGD with hinge loss (text and other high-d sparse data is often near-separable, making the kernel unnecessary); kernel approximations mapping to explicit low-dimensional features (random Fourier features, Nyström) then fitting linear; subsampling; or conceding the regime to boosted trees / neural nets. Knowing the crossover ($\sim 10^4$ – 10^5 points) is the practical marker.

Q15. Walk through how a decision tree chooses a split, including the exact quantity optimized.

At each node, for every feature and every candidate threshold (midpoints between sorted unique values for continuous features): partition the node's examples, compute the impurity of each child (Gini or entropy), take the size-weighted average, and subtract from the parent's impurity — the information gain. Choose the (feature, threshold) with maximum gain; recurse. Listing 6 does it by hand: parent entropy 1.0 bit, best split gains 0.655 bits. Greedy and local — each split maximizes immediate purification with no lookahead, which is why XOR-structured signals defeat single trees.

Q16. Gini vs entropy — differences in theory and practice?

Theory: entropy = $-\sum p \log_2 p$ (information-theoretic, from Chapter 1); Gini = $1 - \sum p^2$ (probability of misclassifying by the node's own label distribution). Both zero when pure, maximal when uniform; entropy penalizes near-purity slightly differently and is marginally more expensive (a log per class). Practice: they choose the same splits the vast majority of the time and neither wins systematically — measured across 40 datasets (Listing 8): 22–18 with mean accuracy difference 0.019 and identical fit time. Strong answer: name the formulas, then say the criterion is a rounding error next to depth control and pruning — and know that *gain ratio* (C4.5) exists to fix information gain's bias toward high-cardinality features.

Q17. Why do unconstrained decision trees overfit, and how do pre-pruning and post-pruning differ?

Grown to purity, a tree can dedicate a leaf to every hard example — with label noise, that means memorizing noise (Listing 7: 90 leaves, 1.000 train, 0.798 test on 15%-noise data). Pre-pruning stops growth early via caps: `max_depth`, `min_samples_leaf`, `min_impurity_decrease` — cheap, but can stop before a split that would have enabled a good deeper one (the greedy horizon problem). Post-pruning grows the full tree, then collapses subtrees that don't pay for their complexity: cost-complexity pruning penalizes accuracy – $\alpha \cdot (\# \text{leaves})$, sweeping α by cross-validation (Listing 7: 11 leaves, 0.842 test — best of all variants). Post-pruning sees the whole tree before judging, at the cost of growing it first.

Q18. Why are single decision trees said to have high variance, and what exploits that property?

The greedy structure compounds instability: a small data perturbation flips the top split, which changes every descendant's data, which changes their splits — entire subtrees reorganize. Deep trees are therefore low-bias, high-variance estimators whose predictions vary wildly across resamples. Bagging (Chapter 7) exploits exactly this: averaging many high-variance, low-bias trees trained on bootstrap resamples cancels the variance while keeping the low bias — random forests are the institutionalized fix, adding feature subsampling to decorrelate the trees further. The interview arc worth telling: the tree's flaw is the forest's fuel.

Q19. A tree's impurity-based feature importance ranks a user-ID-like column first. Diagnose.

Cardinality bias: a feature with many unique values offers many candidate thresholds, so it can carve data into small pure groups by chance — an ID column can achieve perfect training splits with zero generalizable signal. Impurity-based importance sums these fake gains. Fixes: drop identifier-like columns (also a leakage check — Chapter 4), use permutation importance on held-out data (Chapter 11), or gain-ratio-style corrections. Follow-up worth volunteering: the same bias inflates continuous features relative to low-cardinality categoricals.

Q20. Your KNN model is slow at prediction time in production. Options?

Ordered by invasiveness: reduce n — prototype selection/condensing, or cluster centroids as the reference set; reduce d — PCA or feature selection (also helps accuracy via the curse); exact tree indexes — KD-tree/ball tree, effective only up to a few dozen dimensions; approximate nearest neighbors — HNSW graphs, IVF/product quantization (the vector-database stack, Chapter 24), trading tiny recall loss for orders-of-magnitude speedups; or switch families — fit a parametric model (logistic, tree ensemble) if latency dominates. Naming ANN indexes as the industrial answer signals current practice.

Q21. Concentric-circles data: which of the four classifiers work out of the box, and why?

KNN: yes — purely local voting, shape-agnostic. Kernel SVM (RBF or degree-2 poly): yes — the boundary is linear in feature space, circular in input space (Listing 5: 0.995). Gaussian NB: yes, by luck of the geometry — per-class Gaussians with different variances yield a quadratic boundary, which can be an ellipse. Decision tree: yes but ugly — axis-aligned staircase approximating a circle, needing depth. Linear SVM / logistic regression: no (0.510 in Listing 5) — no hyperplane encloses a disk — unless you engineer features (add x_1^2 , x_2^2 , and the problem becomes linearly separable). The question tests whether you can map model class to boundary geometry.

Q22. Why does Naive Bayes often beat more expressive models on small text datasets?

Bias-variance at high dimension: text has d in the tens of thousands and small n . NB estimates only d univariate distributions per class — its errors are dominated by (stable, predictable) bias. An expressive model estimating interactions has variance that small n cannot support. NB's strong-but-wrong independence assumption acts as extreme regularization, and with argmax-only decisions its miscalibration doesn't hurt labels. Empirically NB also benefits from redundant evidence in text: many weakly informative words vote, and vote-counting is robust. As n grows the ranking flips — linear SVMs and logistic regression overtake it, then neural approaches.

Q23. How do you get probabilities out of an SVM, and what's the caveat?

SVMs output margins (signed distances), not probabilities — hinge loss has no likelihood interpretation. Platt scaling fits a one-dimensional logistic regression $\sigma(a \cdot z + b)$ mapping margin z to probability, on held-out data (sklearn's `probability=True` does this via internal CV). Caveats: it adds an expensive extra fit; probabilities can be inconsistent with the raw decision function near the boundary (predict vs `predict_proba` can disagree); and calibration quality depends on the held-out fit — verify with reliability curves (Chapter 10). Isotonic regression is the nonparametric alternative given enough data.

Q24. Two spam features are near-duplicates ("free" and "FREE" as separate tokens). What happens to NB, and to logistic regression, and why the difference?

NB double-counts: the independence assumption treats the duplicate evidence as fresh, effectively squaring that feature's likelihood contribution — confidence inflates, and with enough duplication the decision itself can tip. Logistic regression, trained discriminatively, sees the redundancy and splits the weight between the copies (Chapter 5's collinearity: coefficient split arbitrary, *sum* — and hence predictions — well-determined). The general principle: generative models estimating $p(x|y)$ per feature can't notice inter-feature redundancy; discriminative models optimizing $p(y|x)$ automatically discount it. That's also a lens on why logistic regression usually edges out NB with sufficient data.

Q25. Sketch KNN, SVM-RBF, and decision-tree decision boundaries on the same 2-D dataset. What do the shapes reveal?

KNN: locally jagged, following data density; jaggedness decreasing with k . SVM-RBF: smooth curved contours with a margin corridor, complexity set by γ and C . Tree: rectangles — axis-parallel line segments meeting at right angles. The shapes are the models' inductive biases made visible: KNN assumes local smoothness, the RBF SVM assumes smooth boundaries with maximal clearance, the tree assumes axis-aligned, hierarchical structure. Diagonal or rotated structure embarrasses the tree; enclosed regions embarrass linear models; fine local texture with small n embarrasses everything but KNN (which then overfits it). *Interviewers listen for: connecting boundary geometry to inductive bias, not just describing pictures.*

Q26. You have 1,000 labeled emails and need a working spam filter today, with per-decision explanations. Pick a model and defend it.

Multinomial NB with Laplace smoothing on bag-of-words. Defense: trains in milliseconds on 1,000 examples; small-n/high-d is NB's winning regime (Q22); explanations are native — each word contributes an additive log-evidence term, so "flagged because: 'prize' (+2.1), 'free' (+1.8)" falls out of the model; and it updates online as new labels arrive by incrementing counts. State the caveats to show judgment: probabilities need recalibration before being shown as confidence scores, and a linear SVM or logistic regression should take over as labels accumulate. *Interviewers listen for: matching constraints (time, data size, explainability) to model properties rather than reaching for the fanciest option.*

Q27. Implement the decision rule of a 3-NN classifier in one line of NumPy, given distance matrix D (queries × train) and labels y.

For binary 0/1 labels: `preds = (y[np.argsort(D, axis=1)[:3]].mean(axis=1) >= 0.5).astype(int)` — argsort each query's distances, take the three nearest indices, look up labels, majority = mean ≥ 0.5 . Points to volunteer: argsort is $O(n \log n)$ per query while `np.argpartition(D, 3, axis=1)` is $O(n)$ and the right call at scale; ties are impossible with odd k in binary problems; and for multiclass replace the mean with a `bincount-argmax` per row.

Q28. For each of the four algorithms, name its single most important hyperparameter and what happens at each extreme.

KNN — k : $k=1$ memorizes (max variance), $k=n$ predicts the majority class (max bias). NB — smoothing α : $\alpha \rightarrow 0$ lets zero counts veto classes (brittle), $\alpha \rightarrow \infty$ flattens likelihoods toward uniform so only priors matter. SVM — C (with γ close behind for RBF): $C \rightarrow \infty$ is hard margin, contorted by every point; $C \rightarrow 0$ ignores the data for width, collapsing toward the majority side. Tree — depth/pruning α : unlimited depth memorizes (Listing 7), depth 1 is a decision stump. The unifying observation that makes the answer land: every one of these is the same bias-variance dial from Chapter 4, wearing different notation.

Chapter 7: Ensemble Methods

If a hiring loop touches tabular data — fraud, credit, churn, pricing, ranking features — this chapter is where the production questions live, because gradient-boosted trees have won essentially every tabular benchmark and Kaggle leaderboard for a decade, and random forests remain the strongest "works on the first try" baseline in the field. The interview coverage is correspondingly dense: "bagging vs boosting" is a top-ten screening question; "why does averaging models help?" tests whether your bias-variance understanding is operational or decorative; and "XGBoost vs LightGBM vs CatBoost" checks that your experience extends past sklearn defaults.

The chapter's through-line is one idea seen from three angles: **combining models beats single models exactly when their errors are imperfectly correlated**, and the three ensemble families differ in how they manufacture that decorrelation. Bagging trains the same learner on resampled data *in parallel* and averages — attacking variance. Boosting trains learners *sequentially*, each focused on its predecessors' mistakes — attacking bias. Stacking trains *different* learners and learns how to combine them — attacking whatever's left. Chapter 6 ended with the observation that a deep decision tree is a low-bias, high-variance learner whose instability looks like a flaw; this chapter is the story of turning that flaw into the engine of the strongest classical models we have.

Why ensembles work

Start with the arithmetic that makes everything else believable. Take k classifiers, each wrong with probability ε , and suppose — unrealistically, for the moment — their errors are independent. A majority vote is wrong only when more than half err simultaneously, a binomial tail that collapses fast: for $\varepsilon = 0.3$ and $k = 21$, the majority errs with probability ≈ 0.026 . Three weak-ish models voting beat one decent model. The catch carries the whole subject: **independence is a fiction** — models trained on the same data see the same noise, share the same blind spots, and err together. Averaging k estimators each with variance σ^2 and pairwise correlation ρ gives ensemble variance

$$\text{Var} = \rho\sigma^2 + \frac{1-\rho}{k}\sigma^2$$

The second term dies as k grows; the first does not. **ρ is the floor**, and every ensemble method is, at heart, a scheme for pushing ρ down without individually ruining the members: bagging resamples the data, random forests additionally resample features, boosting reweights so each member literally solves a different problem, stacking mixes model families. When an interviewer asks "why not just train the same model twice and average?", this equation is the answer: identical models have $\rho = 1$ and averaging changes nothing.

The three families in one table, worth reproducing from memory:

	Bagging	Boosting	Stacking
Training	parallel, independent	sequential, each fixes the last	parallel base + meta-learner
Attacks	variance	bias (then variance if unchecked)	both, via model diversity
Base learner	low-bias/high-variance (deep trees)	high-bias/low-variance (stumps, shallow trees)	diverse, dissimilar families
Combination	average / majority vote	weighted sum, learned weights	learned combiner on out-of-fold predictions
Parallelizable	fully	no (sequential by construction)	base layer yes
Canonical form	Random Forest	AdaBoost, XGBoost/LightGBM/CatBoost	Kaggle-style stacks

The base-learner row is the depth marker: bagging *needs* high-variance members (averaging stable models accomplishes nothing — bagged linear regressions are a waste of CPU), while boosting *needs* weak, biased members (boosting already-strong learners overfits almost immediately, since each round fits the previous rounds' residual noise).

Bagging and Random Forests

Bootstrap aggregating (bagging): draw B bootstrap samples (n rows with replacement — each sample omits $\sim 36.8\%$ of rows, since the chance a given row is never picked is $(1 - 1/n)^n \rightarrow e^{-1}$), train one model per sample, and combine by majority vote (classification) or average (regression). The resampling manufactures the diversity: each tree sees a different noise realization, so their individual overfittings partially cancel. Listing 1 measures the effect directly — retraining a single deep tree on fresh samples produces predictions that disagree with each other on 25.2% of test points; a 100-tree bagged ensemble's disagreement drops to 9.1%, and mean test accuracy rises from 0.779 to 0.852. Nothing about the base learner changed; only the variance did. That is the entire theory of bagging, verified in twelve lines.

Random Forest = bagging + feature subsampling. Bagged trees still correlate: a dominant feature wins the root split in every tree, making the trees structurally similar (high ρ , the floor in the variance formula). The forest's fix is brutal and effective — at *every split*, restrict the candidate features to a random subset of size `max_features` ($\sqrt{d}$ is the classification default). Strong features get benched in many splits, forcing trees to discover different structure; ρ falls; averaging bites deeper. Listing 2 sweeps the dial: `max_features=1` (too random — individual trees too weak) scores 0.878, $\sqrt{d}$ and 0.5d score 0.903, and all-features (plain bagging) drops to 0.893. The U-shape is the

decorrelation-vs-member-strength trade made visible, and `max_features` is the forest's most underrated hyperparameter.

Out-of-bag (OOB) evaluation is the forest's free lunch: each tree never saw $\sim 37\%$ of the rows, so those rows can be scored by that tree as genuine held-out data. Aggregating each row's predictions over only the trees that didn't train on it yields the OOB score — an honest generalization estimate *without a validation split*, essentially free cross-validation. Listing 2: OOB accuracy 0.9035 vs actual test accuracy 0.9030. In small-data regimes where every row is precious, OOB is the answer to "how do you validate without giving up data?"

Practicalities interviewers probe. Forests barely overfit with more trees — `B` is not a complexity parameter; predictions converge as `B` grows (the variance formula again: more `k`, same ρ) — so "more trees" costs compute, never accuracy; tune tree depth/`min_samples_leaf` if the forest overfits. Forests train embarrassingly parallel. Their **feature importances** (mean impurity reduction) inherit Chapter 6's cardinality bias, and add a subtlety: correlated features split the credit, understating each — permutation importance (Chapter 11) before conclusions. And forests inherit trees' inability to extrapolate: leaf averages cannot exceed observed targets, which bites time-trending regression targets.

Boosting I: AdaBoost

Boosting's premise inverts bagging's: take a **weak learner** — barely better than chance, canonically a decision stump — and build a strong one by training a *sequence*, each member focused on what the sequence so far gets wrong. AdaBoost (1997) made it concrete with sample weights:

1. Start with uniform weights $w_i = 1/n$.
2. Fit a weak learner to the weighted data; compute its weighted error $\epsilon_t = \sum_{i: \text{wrong}} w_i$.
3. Give it a say proportional to its quality: $\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}$ — zero say at coin-flip ($\epsilon_t = 0.5$), large say as $\epsilon_t \rightarrow 0$.
4. Reweight: multiply each sample's weight by $e^{-\alpha_t y_i h_t(x_i)}$ — mistakes up, hits down — and renormalize. The next learner faces a distribution concentrated on the hard cases.
5. Final prediction: $\text{sign}\left(\sum_t \alpha_t h_t(x)\right)$, a weighted vote.

Listing 3 implements the loop in ~ 20 lines and matches sklearn exactly: a single stump scores 0.750; fifty reweighted stumps score 0.860, with the earned α 's decaying (0.561, 0.370, 0.236 — each successive stump faces a harder, noisier distribution and earns less say). Two theory notes that upgrade an answer: AdaBoost is provably minimizing **exponential loss** $\sum_i e^{-y_i F(x_i)}$ by greedy stagewise additive fitting — the reweighting scheme falls out of the math rather than being a heuristic — and that same exponential

loss explains its known weakness: mislabeled points get exponentially up-weighted until the ensemble contorts around them, so AdaBoost is notoriously sensitive to label noise (gradient boosting with robust losses handles this better).

Boosting II: Gradient Boosting

Gradient boosting generalizes AdaBoost's "fix your predecessor" into calculus: it is **gradient descent in function space**. Maintain an additive model $F_m(x)$; at each round, compute the loss's negative gradient with respect to the current predictions — for squared loss this is literally the residual $y - F_m(x)$ — fit a small regression tree to those residuals, and take a short step:

$$F_{m+1}(x) = F_m(x) + \eta \cdot h_m(x)$$

Chapter 5's gradient descent updated a weight vector by the loss gradient; gradient boosting updates a *function* by adding a tree that approximates the loss gradient. Same idea, different space — and the framing explains the framework's generality: swap the loss and the same machinery does robust regression (absolute/Huber loss — gradients are signs, not raw residuals, so outliers stop dominating), classification (log loss — residuals become $y - \hat{p}$, Chapter 5's logistic gradient again), and ranking (pairwise losses; LambdaMART). Listing 4 builds the squared-loss version from scratch — constant model, fit residuals, shrink, repeat — and watches test MSE fall $23.6 \rightarrow 20.8 \rightarrow 9.9 \rightarrow 3.3 \rightarrow 1.9$ across 200 trees, landing within noise of sklearn's implementation.

The learning rate η (shrinkage) is boosting's most important knob and the standard interview follow-up. Small steps deliberately under-use each tree, leaving signal for later trees to refine — empirically, many small steps generalize better than few large ones, at the cost of needing more rounds. Listing 5 runs the race on noisy data: $\eta=1.0$ peaks at 8 trees (0.811) then decays; $\eta=0.1$ peaks later and higher (0.828 at 26 trees). Both eventually overfit — boosting, unlike bagging, keeps driving training loss toward zero, and with label noise that means memorizing it — hence the standard recipe: **small η + many rounds + early stopping on a validation set**, letting the data pick the stopping point. The contrast with forests is a mandatory talking point: *more trees never hurt a forest; more rounds eventually hurt boosting*. Additional variance controls: shallow trees (depth 3–8 — each tree a weak learner capturing limited interaction order), row subsampling per round ("stochastic gradient boosting", typically 0.5–0.8), column subsampling (borrowed from forests), and L1/L2 penalties on leaf values in the modern libraries.

XGBoost, LightGBM, CatBoost

The three industrial implementations share the algorithm above and differ in engineering and a few real algorithmic ideas — the differences are a standard "have you actually used these?" interview probe.

XGBoost (2014) established the modern template: **second-order optimization** (uses the loss's Hessian as well as gradient, yielding closed-form optimal leaf values and a principled gain formula), **explicit regularization** in the objective (γ per leaf, λ on leaf values — the gain formula's γ threshold is exactly cost-complexity pruning built into growth), **sparsity-aware splits** with a learned default direction per node (missing values handled natively — no imputation), histogram-based approximate split finding, and column/row subsampling. Trees grow **level-wise** (breadth-first) by default.

LightGBM (2017, Microsoft) is built for speed on big data: aggressive **histogram binning** (features discretized to ~ 255 bins — split search over bins, not values), **leaf-wise growth** (always split the leaf with the largest gain, wherever it is — deeper, more lopsided, more accurate trees for the same leaf budget, with `num_leaves` replacing `max_depth` as the capacity knob and overfitting faster on small data), **GOSS** (gradient-based one-side sampling — keep all large-gradient rows, subsample the well-fit small-gradient ones), and **EFB** (bundling mutually-exclusive sparse features). Native categorical support without one-hot.

CatBoost (2017, Yandex) leads with categorical features and leakage-resistance: **ordered target statistics** — encoding categories by target means computed *only from earlier rows in a random permutation*, so a row's own label never leaks into its encoding (Chapter 4's target-leakage discipline, built into the library; contrast naive target encoding, which leaks by construction) — plus **ordered boosting** (per-round models trained on prefixes, countering the subtle bias where residuals are computed on rows the model already fit), and **symmetric (oblivious) trees** — the same split at every node of a level, giving extremely fast inference and built-in regularization. Strongest defaults of the three; least tuning to reach par.

Rules of thumb for "which one?": LightGBM for large datasets where training speed dominates; CatBoost for category-heavy tabular data or when tuning budget is minimal; XGBoost as the battle-tested default with the largest ecosystem. Honest coda: on most mid-size problems, all three land within noise of each other after tuning — Listing 6 measures XGBoost, LightGBM, and sklearn's HistGradientBoosting on the same 20k-row task at 0.9197 / 0.9233 / 0.9165, a spread smaller than a hyperparameter wiggle. Saying that out loud, then naming the tiebreakers (categoricals $\rightarrow$ CatBoost, scale $\rightarrow$ LightGBM, ecosystem $\rightarrow$ XGBoost), is the experienced answer.

Voting and Stacking

Voting classifiers combine *different* model families trained on the *same* data. Hard voting takes the majority label; **soft voting averages predicted probabilities** — usually better, since it uses confidence, but only meaningful if members' probabilities are comparable (calibrate first — Chapters 6 and 10). The catch Listing 7 documents honestly: with members of unequal strength (0.746–0.864), the equal-weight vote scores 0.837 — *below* the best member. Democracy among unequals drags the ensemble toward

mediocrity; hand-set weights (1,1,3,3) claw back to 0.863, still shy of just using the forest. Voting works when members are comparably strong and genuinely diverse; otherwise it's a way to average away your best model.

Stacking replaces guessed weights with a learned combiner. Train the base models; collect their **out-of-fold predictions** (each row's meta-features come from models that never trained on that row — the anti-leakage step, and the detail interviews test: stacking on in-sample predictions lets the meta-learner learn "trust the most overfit member", which collapses in production); train a simple **meta-learner** (logistic regression, classically) on those predictions. Listing 8 closes the arc from Listing 7: same four members, stacked with 5-fold out-of-fold probabilities, scores 0.868 — beating every member and every voting variant — and the meta-learner's coefficients (−0.09, −1.14, 3.74, 3.44) show it learned to lean on the SVM and forest and to *use the weak NB as a corrective signal* (negative weight), which no fixed voting scheme could express. Deep stacks (multiple meta-levels) win Kaggle margins; production mostly stops at one level — each layer multiplies serving cost, latency, and monitoring surface for shrinking returns (Chapter 27's serving-cost lens).

Code implementations

Every listing was executed as shown; outputs are real. The sequence tells the chapter's story in numbers: bagging's variance cut measured directly, the forest's OOB score matching a real test set, AdaBoost and gradient boosting built from scratch and matching sklearn, the learning-rate race, the big-three head-to-head, and voting's honest failure redeemed by stacking.

Listing 1 — Bagging measurably cuts variance

Twenty-five retrainings on fresh samples. The single deep tree's predictions disagree with each other on 25.2% of test points — that instability is Chapter 6's high-variance diagnosis, quantified. Bagging 100 such trees cuts disagreement to 9.1% and lifts mean accuracy 0.779 → 0.852. Same base learner, same data; only the variance changed.

```
"""Listing 1: bagging measurably cuts variance -- the whole point, in numbers."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

rng = np.random.default_rng(0)
X_pool, y_pool = make_classification(n_samples=6000, n_features=12, n_informative=6,
                                   flip_y=0.1, random_state=0)
X_test, y_test = X_pool[5000:], y_pool[5000:] # fixed test set
X_pool, y_pool = X_pool[:5000], y_pool[:5000]

def spread_and_acc(model, trials=25, n=800):
    """Train on `trials` fresh samples; measure prediction disagreement and
    accuracy."""
    preds, accs = [], []
    for _ in range(trials):
```

```

        idx = rng.choice(len(X_pool), n, replace=False)
        m = model().fit(X_pool[idx], y_pool[idx])
        p = m.predict(X_test)
        preds.append(p); accs.append((p == y_test).mean())
    preds = np.array(preds)
    # variance proxy: how often two retrained models disagree on a test point
    disagree = np.mean([np.mean(preds[i] != preds[j])
                        for i in range(trials) for j in range(i + 1, trials)])
    return disagree, np.mean(accs)

tree = lambda: DecisionTreeClassifier(random_state=0)
bag = lambda: BaggingClassifier(DecisionTreeClassifier(), n_estimators=100,
                               random_state=0, n_jobs=-1)

d1, a1 = spread_and_acc(tree)
d2, a2 = spread_and_acc(bag)
print(f"single tree : disagreement {d1:.3f} mean test acc {a1:.3f}")
print(f"bagged x100 : disagreement {d2:.3f} mean test acc {a2:.3f}")

```

Output:

```

single tree : disagreement 0.252 mean test acc 0.779
bagged x100 : disagreement 0.091 mean test acc 0.852

```

Listing 2 — Random forest: OOB for free, max_features as the decorrelation dial

The OOB score (0.9035) predicts the held-out test score (0.9030) to three decimals — validation without a validation set. The max_features sweep shows the U: 1 is too random (weak members), all-features is plain bagging (correlated members), $\sqrt{d}$ and 0.5 sit at the sweet spot.

```

"""Listing 2: random forest -- OOB error as a free validation set, and the max_features
dial."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

X, y = make_classification(n_samples=3000, n_features=20, n_informative=8,
                           flip_y=0.1, random_state=1)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.33, random_state=1)

# OOB: each tree sees ~63.2% of rows; the rest score it for free
rf = RandomForestClassifier(n_estimators=300, oob_score=True, random_state=1,
                           n_jobs=-1).fit(X_tr, y_tr)
print(f"OOB accuracy : {rf.oob_score_:.4f}")
print(f"test accuracy: {rf.score(X_te, y_te):.4f} (OOB predicted it, no split
needed)")

# max_features: the decorrelation dial
print(f"\n{'max_features':>12} {'test acc':>9}")
for mf in [1, "sqrt", 0.5, None]: # None = all features = plain bagging
    m = RandomForestClassifier(n_estimators=300, max_features=mf,
                              random_state=1, n_jobs=-1).fit(X_tr, y_tr)
    print(f"{str(mf):>12} {m.score(X_te, y_te):>9.4f}")

top = np.argsort(rf.feature_importances_)[::-1][:3]
print(f"\ntop-3 impurity importances: features {top}, "
      f"weights {np.round(rf.feature_importances_[top], 3)}")

```


Output:

```

00B accuracy : 0.9035
test accuracy: 0.9030    (00B predicted it, no split needed)

max_features  test acc
      1      0.8778
     sqrt    0.9030
      0.5    0.9030
     None    0.8929

top-3 impurity importances: features [ 6 13  3], weights [0.177 0.117 0.091]

```

Listing 3 — AdaBoost from scratch

The full algorithm in twenty lines: weighted stump, weighted error, α as the stump's earned say, exponential reweighting of mistakes. One stump scores 0.750; fifty boosted stumps score 0.860, exactly matching sklearn. The decaying α sequence (0.561, 0.370, 0.236) shows each successive stump facing a harder residual problem and earning less voice.

```

"""Listing 3: AdaBoost from scratch -- reweight what you got wrong."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import AdaBoostClassifier
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X, y01 = make_classification(n_samples=1000, n_features=8, n_informative=5,
                             flip_y=0.05, random_state=2)
y = 2 * y01 - 1 # AdaBoost wants labels in {-1,+1}
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, random_state=2)

def adaboost(X, y, rounds=50):
    n = len(y)
    w = np.full(n, 1 / n) # uniform sample weights
    stumps, alphas = [], []
    for _ in range(rounds):
        stump = DecisionTreeClassifier(max_depth=1)
        stump.fit(X, y, sample_weight=w)
        pred = stump.predict(X)
        err = w[pred != y].sum() # weighted error
        alpha = 0.5 * np.log((1 - err) / (err + 1e-12)) # stump's say
        w *= np.exp(-alpha * y * pred) # up-weight mistakes, down-weight
    hits
        w /= w.sum()
        stumps.append(stump); alphas.append(alpha)
    return stumps, alphas

def predict(stumps, alphas, X):
    agg = sum(a * s.predict(X) for s, a in zip(stumps, alphas))
    return np.sign(agg)

stumps, alphas = adaboost(X_tr, y_tr)
acc = (predict(stumps, alphas, X_te) == y_te).mean()
stump_acc = DecisionTreeClassifier(max_depth=1).fit(X_tr, y_tr).score(X_te, y_te)
sk = AdaBoostClassifier(n_estimators=50, random_state=2).fit(X_tr, y_tr)
print(f"one stump      : {stump_acc:.4f}")
print(f"scratch AdaBoost : {acc:.4f} (50 stumps)")
print(f"sklearn AdaBoost  : {sk.score(X_te, y_te):.4f}")

```



```
print(f"\nfirst 3 alphas: {np.round(alphas[:3], 3)} (better stumps earn a bigger say)")
```

Output:

```
one stump      : 0.7500
scratch AdaBoost : 0.8600 (50 stumps)
sklearn AdaBoost : 0.8600

first 3 alphas: [0.561 0.37  0.236] (better stumps earn a bigger say)
```

Listing 4 — Gradient boosting from scratch

Constant model, fit a shallow tree to the residuals, add a shrunk step, repeat — gradient descent in function space, for squared loss. Test MSE falls $23.6 \rightarrow 1.89$ over 200 trees, within noise of sklearn's 1.88. Note where the learning rate appears twice: in training (each tree fits residuals of the *shrunk* running sum) and in prediction.

```
"""Listing 4: gradient boosting from scratch -- fit the residuals, shrink, repeat."""
import numpy as np
from sklearn.datasets import make_friedman1
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor

X, y = make_friedman1(n_samples=1500, noise=1.0, random_state=3)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, random_state=3)

def gbm_fit(X, y, rounds=200, lr=0.1, depth=3):
    f0 = y.mean() # start from the constant model
    pred = np.full_like(y, f0)
    trees = []
    for _ in range(rounds):
        resid = y - pred # negative gradient of squared loss
        t = DecisionTreeRegressor(max_depth=depth).fit(X, resid)
        pred += lr * t.predict(X) # small step toward the residuals
        trees.append(t)
    return f0, trees

def gbm_predict(f0, trees, X, lr=0.1):
    return f0 + lr * sum(t.predict(X) for t in trees)

def mse(a, b): return np.mean((a - b) ** 2)

f0, trees = gbm_fit(X_tr, y_tr)
print(f"constant model MSE : {mse(y_te, y_tr.mean()):.3f}")
for k in [1, 10, 50, 200]:
    print(f"after {k:>3} trees : {mse(y_te, gbm_predict(f0, trees[:k], X_te)):.3f}")

sk = GradientBoostingRegressor(n_estimators=200, learning_rate=0.1, max_depth=3,
                              random_state=3).fit(X_tr, y_tr)
print(f"sklearn GBM : {mse(y_te, sk.predict(X_te)):.3f}")
```

Output:

```
constant model MSE : 23.561
after  1 trees      : 20.825
after 10 trees      :  9.937
```

```

after 50 trees      : 3.319
after 200 trees     : 1.887
sklearn GBM        : 1.880

```

Listing 5 — Learning rate vs rounds: where boosting overfits

Deliberately noisy labels (20% flipped) to make overfitting visible. $\eta=1.0$ peaks at 8 trees then decays; $\eta=0.1$ peaks later and higher. Unlike a forest, boosting keeps optimizing training loss into the noise — the staged test-accuracy curve turning downward is the signature, and early stopping on validation is the cure.

```

"""Listing 5: the learning-rate / n_estimators trade, and where boosting overfits."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import train_test_split

X, y = make_classification(n_samples=2000, n_features=15, n_informative=6,
                           flip_y=0.2, random_state=4) # noisy: overfittable
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=4)

for lr in [1.0, 0.1]:
    m = GradientBoostingClassifier(n_estimators=500, learning_rate=lr,
                                   max_depth=3, random_state=4).fit(X_tr, y_tr)
    accs = [ (p == y_te).mean() for p in m.staged_predict(X_te) ]
    best = int(np.argmax(accs)) + 1
    print(f"lr={lr:<4}: best test acc {max(accs):.3f} at {best:>3} trees; "
          f"acc at 500 trees {accs[-1]:.3f}")

```

Output:

```

lr=1.0 : best test acc 0.811 at 8 trees; acc at 500 trees 0.800
lr=0.1 : best test acc 0.828 at 26 trees; acc at 500 trees 0.804

```

Listing 6 — XGBoost vs LightGBM vs sklearn HistGB

Same 20k-row problem, comparable settings. The accuracy spread (0.9165–0.9233) is smaller than a typical hyperparameter wiggle — the honest headline. Timing differences here reflect this dataset and default threading more than fundamentals; LightGBM's speed advantages show at much larger scale, where histogram binning and GOSS pay off.

```

"""Listing 6: XGBoost vs LightGBM vs sklearn HistGB -- same job, different engineering."""
import time, warnings
import numpy as np
warnings.filterwarnings("ignore", category=UserWarning)
from sklearn.datasets import make_classification
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.model_selection import train_test_split
import xgboost as xgb
import lightgbm as lgb

X, y = make_classification(n_samples=20000, n_features=30, n_informative=12,
                           flip_y=0.1, random_state=5)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, random_state=5)

```

```
models = {
    "XGBoost": xgb.XGBClassifier(n_estimators=300, learning_rate=0.1, max_depth=6,
                                tree_method="hist", verbosity=0, random_state=5),
    "LightGBM": lgb.LGBMClassifier(n_estimators=300, learning_rate=0.1,
                                   num_leaves=63, verbose=-1, random_state=5),
    "sklearn HistGB": HistGradientBoostingClassifier(max_iter=300, learning_rate=0.1,
                                                      random_state=5),
}
print(f"{'model':<15} {'fit seconds':>11} {'test acc':>9}")
for name, m in models.items():
    t0 = time.perf_counter()
    m.fit(X_tr, y_tr)
    dt = time.perf_counter() - t0
    print(f"{'name':<15} {'dt':>11.2f} {'m.score(X_te, y_te):>9.4f}")
```

Output:

model	fit seconds	test acc
XGBoost	0.74	0.9197
LightGBM	2.04	0.9233
sklearn HistGB	0.50	0.9165

Listing 7 — Voting: the honest failure mode

Four members of unequal strength. The equal-weight vote (hard 0.837, soft 0.834) lands *below* the best member (0.864) — the weak members drag it down. Skill-weighted soft voting recovers to 0.863, still not beating the forest. Voting is not magic; it needs comparably strong, diverse members — or a learned combiner (Listing 8).

```
"""Listing 7: voting classifiers -- hard vs soft, and why diversity is the fuel."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import VotingClassifier, RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

X, y = make_classification(n_samples=2500, n_features=15, n_informative=6,
                          flip_y=0.15, random_state=6)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=6)

members = [
    ("logreg", make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000))),
    ("nb", GaussianNB()),
    ("svm", make_pipeline(StandardScaler(), SVC(probability=True, random_state=6))),
    ("rf", RandomForestClassifier(n_estimators=200, random_state=6)),
]
for name, m in members:
    print(f"{'name':<7}: {m.fit(X_tr, y_tr).score(X_te, y_te):.4f}")

hard = VotingClassifier(members, voting="hard").fit(X_tr, y_tr)
soft = VotingClassifier(members, voting="soft").fit(X_tr, y_tr)
print(f"\nhard vote: {hard.score(X_te, y_te):.4f} (majority of labels)")
print(f"soft vote: {soft.score(X_te, y_te):.4f} (average of probabilities)")

# Equal votes let weak members drag the ensemble below its best member.
# Weight by (rough) skill and the ensemble recovers -- and edges out everyone.
weighted = VotingClassifier(members, voting="soft",
```

```
weights=[1, 1, 3, 3]).fit(X_tr, y_tr)
print(f"weighted : {weighted.score(X_te, y_te):.4f}    (soft, weights 1,1,3,3)")
```

Output:

```
logreg : 0.7630
nb      : 0.7460
svm     : 0.8620
rf      : 0.8640

hard vote: 0.8370    (majority of labels)
soft vote: 0.8340    (average of probabilities)
weighted : 0.8630    (soft, weights 1,1,3,3)
```

Listing 8 — Stacking: learn who to trust

Same four members as Listing 7, but the combiner is learned on out-of-fold probabilities (cv=5 — no member's meta-feature was predicted on rows it trained on). Result: 0.868, beating every member and every voting variant. The meta-learner's coefficients are the story: near-zero on logistic regression, negative on NB (used as a corrective signal), heavy on SVM and forest — weights no fixed vote could guess.

```
"""Listing 8: stacking -- let a meta-learner learn who to trust, out-of-fold."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import StackingClassifier, RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

X, y = make_classification(n_samples=2500, n_features=15, n_informative=6,
                          flip_y=0.15, random_state=6)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.4, random_state=6)

members = [
    ("logreg", make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000))),
    ("nb", GaussianNB()),
    ("svm", make_pipeline(StandardScaler(), SVC(probability=True, random_state=6))),
    ("rf", RandomForestClassifier(n_estimators=200, random_state=6)),
]
# cv=5: each member's meta-features are OUT-OF-FOLD predictions -- the anti-leakage
# step
stack = StackingClassifier(members, final_estimator=LogisticRegression(max_iter=2000),
                          cv=5, stack_method="predict_proba").fit(X_tr, y_tr)
print(f"best single member : 0.8640 (random forest, Listing 7)")
print(f"stacking           : {stack.score(X_te, y_te):.4f}")
w = stack.final_estimator_.coef_[0]
print(f"meta-learner weights on (logreg, nb, svm, rf) probs: {np.round(w, 2)}")
```

Output:

```
best single member : 0.8640 (random forest, Listing 7)
stacking           : 0.8680
meta-learner weights on (logreg, nb, svm, rf) probs: [-0.09 -1.14  3.74  3.44]
```

Pitfalls, comparisons and practical tips

Random forest vs gradient boosting — the decision most often faced in practice:

	Random Forest	Gradient Boosting (XGB/LGBM/Cat)
Trees	deep, independent, parallel	shallow, sequential, each fixes the last
Attacks	variance	bias, then variance via shrinkage/stopping
More trees	never hurts (converges)	eventually overfits — early stop
Tuning burden	low (works near-default)	real (lr, rounds, depth/leaves, subsampling, regularization)
Training	embarrassingly parallel	sequential rounds (parallel within round)
Typical ceiling	strong baseline	state of the art on tabular
Robustness to noise/outliers	high	lower (chases residuals) — use robust losses
Free validation	OOB score	needs a validation set (for early stopping anyway)

Classic traps:

- **Tuning `n_estimators` for a forest by grid search.** It's not a complexity knob; accuracy is monotone-ish in it. Set it as large as compute allows and tune depth/min_samples_leaf/max_features instead. For boosting, `n_estimators` is *the* overfitting axis — don't grid it, early-stop it.
- **Comparing boosting configs at a fixed round count.** A lower learning rate always looks worse at 50 rounds; give each config its early-stopped best. The lr/rounds pair is one knob, not two.
- **Forgetting boosting's noise sensitivity.** 20% label noise turned Listing 5's curves downward within dozens of rounds. Suspect label quality before adding rounds; consider Huber loss for regression; AdaBoost specifically amplifies mislabeled points exponentially.
- **Stacking on in-sample predictions.** The meta-learner learns "trust whoever overfit hardest." Out-of-fold predictions or nothing — this is Chapter 4's leakage discipline applied to ensembles, and it is the first thing a reviewer checks in a stacking pipeline.
- **Naive target encoding of categoricals for boosted trees.** Encoding a category by its mean target computed over all rows leaks each row's own label (Chapter 4). Use out-of-fold encoding or CatBoost's ordered statistics, which exist precisely for this.

- **Soft-voting uncalibrated members.** Averaging an overconfident NB's 0.99 with a calibrated forest's 0.6 lets the worst-calibrated member shout loudest. Calibrate (Chapter 10) before soft voting; or stack, which learns to discount.
- **Reading forest feature importances as causal or even reliable.** Impurity importances inherit cardinality bias and split credit across correlated features. Permutation importance on held-out data (Chapter 11), and even that splits credit under correlation.
- **Deploying a 5-layer stack for a 0.3% offline gain.** Serving cost, latency, retraining orchestration, and monitoring surface all multiply per layer (Chapter 27). Production ensembles are usually one boosted model, or boosting + one linear/NN partner, or a single-level stack.
- **Assuming trees/ensembles extrapolate.** All tree ensembles predict within the convex hull of observed leaf values — a time-trending target walks straight out of range. Detrend, or use a model with a linear component.

Defaults that survive contact with reality. Forest: `n_estimators` 300–500, `max_features='sqrt'`, tune `min_samples_leaf` $\in \{1, 5, 20\}$. Boosting: `learning_rate` 0.05–0.1, early stopping on 10–20% validation with patience ~ 50 rounds, `max_depth` 4–8 (XGBoost) or `num_leaves` 31–127 (LightGBM), `subsample` and `colsample` ~ 0.8 , then regularization (λ , γ / `min_child_weight`) if the val curve says overfit. And always benchmark against the boring baseline first: a forest at defaults tells you what the fancy model must beat.

Interview questions and answers

Q1. Bagging vs boosting — give the two-minute answer that covers mechanism, target, and base learner.

Bagging trains the same learner on bootstrap resamples in parallel and averages/votes; it attacks *variance*, so it wants low-bias, high-variance members — deep trees. Boosting trains a sequence, each member fitted to the current ensemble's mistakes (reweighted samples in AdaBoost, loss gradients in gradient boosting), combined as a weighted sum; it attacks *bias*, so it wants weak, high-bias members — stumps or shallow trees. Consequences that show depth: bagging parallelizes and more members never hurt; boosting is sequential and more rounds eventually overfit, so it needs early stopping. Listing 1 measures bagging's variance cut (disagreement 0.25→0.09); Listing 4 shows boosting's bias destruction (MSE 23.6→1.9).

Q2. Why does averaging models reduce error at all? When does it fail?

Ensemble variance = $\rho\sigma^2 + (1-\rho)\sigma^2/k$ for k members with variance σ^2 and pairwise correlation ρ : the uncorrelated part averages away as $1/k$; the correlated part $\rho\sigma^2$ is an irreducible floor. Averaging helps exactly when errors are imperfectly correlated and members are comparably competent. It fails when $\rho \approx 1$ (same model retrained — nothing to cancel), when members are unequally strong (Listing 7: equal votes scored 0.837 vs the best member's 0.864 — democracy among unequals), or when the members share a systematic bias (averaging many models that all miss the same interaction still misses it — variance reduction cannot fix bias).

Q3. What fraction of the data does each bootstrap sample miss, and why does that number matter?

$P(\text{a row is never drawn in } n \text{ draws with replacement}) = (1 - 1/n)^n \rightarrow e^{-1} \approx 36.8\%$. It matters twice: it's the diversity engine (each tree trains on a different $\sim 63\%$ of rows), and it's free validation — the out-of-bag rows are genuine held-out data for that tree, and aggregating OOB predictions across trees gives an honest generalization estimate with no split (Listing 2: OOB 0.9035 vs test 0.9030). Deriving the e^{-1} limit on request is a favorite probability crossover question (Chapter 1).

Q4. What does a random forest add over plain bagged trees, and why is it needed?

Feature subsampling per split: only a random `max_features`-sized subset competes at each node. Bagged trees stay correlated because dominant features win the top splits in every tree — and ρ is the floor in the variance formula, so bagging alone stalls. Benching strong features forces structurally different trees, cutting ρ . Listing 2's sweep shows the trade: `max_features=1` weakens members too much (0.878), all features = plain bagging keeps them correlated (0.893), $\sqrt{d}$ hits the balance (0.903). One sentence for the mechanism, one for the U-shape — that's the complete answer.

Q5. Why doesn't a random forest overfit as you add trees, while boosting does as you add rounds?

Forest trees are i.i.d. draws from a fixed distribution of trees; averaging more of them converges to the expectation of that distribution — variance falls monotonically toward the $\rho\sigma^2$ floor and the model stops changing. Complexity is set by the individual trees, not their count. Boosting's rounds are not exchangeable draws — each round further optimizes training loss, so the ensemble's effective capacity grows with every round; with noisy labels it eventually fits the noise (Listing 5: test accuracy peaks at 8-26 rounds and decays). Hence: forests — max out $n_{\text{estimators}}$; boosting — early stop on validation.

Q6. Walk through AdaBoost's update equations and explain what each accomplishes.

Weighted error $\epsilon_t = \sum w_i$ over mistakes measures the stump against the *current* distribution. Its sign $\alpha_t = \frac{1}{2} \log((1 - \epsilon_t)/\epsilon_t)$: zero at coin-flip, growing as error falls, negative if worse than chance (flip its vote). Weight update $w_i \leftarrow w_i \cdot \exp(-\alpha_t y_i h_t(x_i))$ then renormalize: mistakes multiplied up, hits down, so the next stump faces a distribution concentrated on the survivors' failures. Prediction: $\text{sign}(\sum \alpha_t h_t)$. The depth marker: this is exactly greedy stagewise minimization of exponential loss $\sum \exp(-yF(x))$ — the updates fall out of the derivation rather than being heuristics — and exponential loss is also why AdaBoost is brittle to label noise: a mislabeled point's weight grows exponentially until the ensemble contorts around it.

Q7. In what sense is gradient boosting "gradient descent in function space"?

Ordinary gradient descent updates a parameter vector: $w \leftarrow w - \eta \nabla L$. Gradient boosting updates a *function*: at each round compute the negative gradient of the loss with respect to current predictions (one number per training point — for squared loss it's the residual $y - F(x)$; for log loss, $y - \hat{p}$), fit a small tree to approximate that gradient vector, and step: $F \leftarrow F + \eta \cdot \text{tree}$. The tree is the "direction", η the step size. The framing's payoff is generality — swap the loss, keep the machinery: Huber for robust regression, log loss for classification, pairwise losses for ranking (LambdaMART). *Interviewers listen for: the residual = negative gradient identification, and that it holds only for squared loss.*

Q8. Why does a small learning rate generalize better in boosting, and what does it cost?

Shrinkage deliberately under-commits to each tree: each round corrects only a fraction of the visible residual, so early trees don't lock in coarse, noise-contaminated corrections, and later trees refine with more context — a regularization effect empirically worth real accuracy (Listing 5: $\eta=0.1$ peaks at 0.828 vs $\eta=1.0$'s 0.811). Cost: proportionally more rounds to reach the same training fit, so more compute and a bigger model. The operational recipe: fix η small (0.05–0.1), set rounds high, early-stop on validation — treating η and $n_estimators$ as one joint knob, never comparing configs at a fixed round count.

Q9. Name the headline differences among XGBoost, LightGBM, and CatBoost, and give your when-to-use rules.

XGBoost: second-order (gradient+Hessian) objective with explicit regularization (γ per leaf, λ on leaf values), sparsity-aware splits with learned default directions for missing values, level-wise growth, the largest ecosystem. LightGBM: histogram binning everywhere, leaf-wise growth (num_leaves is the capacity knob; deeper lopsided trees, faster accuracy per leaf, faster overfitting on small data), GOSS row sampling and EFB feature bundling — built for scale. CatBoost: ordered target statistics for categoricals (leakage-free encoding by construction), ordered boosting, symmetric trees (fast inference), strongest defaults. Rules: big data/speed → LightGBM; heavy categoricals or low tuning budget → CatBoost; default/ecosystem → XGBoost. Honest coda (Listing 6): after tuning, mid-size results usually land within noise of each other.

Q10. What problem do CatBoost's "ordered target statistics" solve, exactly?

Naive target encoding replaces category c with the mean target over all rows of category c — including the row being encoded, so each row's own label leaks into its feature (Chapter 4's target leakage; rare categories are worst: a singleton category's encoding is its label). Ordered statistics fix it structurally: draw a random permutation, encode each row using only rows that precede it — its own label and all "future" labels are excluded, mimicking a stream where you can only know the past. CatBoost also applies the same trick to boosting itself (ordered boosting), countering the bias of computing residuals on rows the model already fit. Naming target leakage as the underlying disease, not just describing the mechanism, is what scores.

Q11. Hard voting vs soft voting — and when does voting actively hurt?

Hard: majority of predicted labels. Soft: average predicted probabilities, then `argmax` — usually better because confidence information survives, but only if members' probabilities are comparable (calibrate overconfident members first, or they shout loudest). Voting hurts when members are unequally strong — equal weights drag the ensemble below its best member (Listing 7: 0.837 vs 0.864) — or insufficiently diverse, where it adds cost without cancellation. Fixes in order of principle: weight by validation skill; or stop guessing weights and stack (Listing 8 learned weights of -0.09 , -1.14 , 3.74 , 3.44 — including using a weak member as a negative signal, inexpressible by voting).

Q12. Describe stacking's data flow, and the leakage trap in it.

Level 0: train diverse base models. Meta-features: each training row's base-model predictions, produced *out-of-fold* — split into k folds, predict each fold with models trained on the other $k-1$. Level 1: a simple meta-learner (logistic regression classically) trained on those meta-features; at inference, base models (refit on all data) predict, meta-learner combines. The trap: using in-sample predictions as meta-features. The most overfit base model then has the most accurate-looking training predictions, so the meta-learner learns "trust the overfitter" — great offline, collapses live. Out-of-fold generation is not an optimization; it is the correctness condition. (Chapter 4's leakage taxonomy, ensemble edition.)

Q13. Your gradient-boosted model's validation error started rising at round 400 of 2,000. What do you do, and what do you not do?

Do: stop at (or roll back to) the validation minimum — that's early stopping working as designed; most libraries automate it (`early_stopping_rounds` / callbacks with a patience window). Then, if you want more headroom: lower the learning rate and rerun with more rounds, add row/column subsampling, shallower trees, or stronger leaf regularization — all push the minimum later and lower. Don't: keep training ("it might come back" — it won't; the curve is fitting label noise), tune `n_estimators` by grid as if it were independent of `lr`, or evaluate the final model on the same validation set that chose the stopping point without noting the mild selection bias (Chapter 4: the val-selected score is optimistic; confirm on untouched test).

Q14. Why are shallow trees (depth 3-8) the standard boosting base learner rather than stumps or deep trees?

Tree depth bounds interaction order: a depth- d tree can express interactions among at most d features along a path. Stumps (depth 1) model purely additive effects — fine when the truth is additive, blind to interactions. Deep trees are strong learners; boosting them means each round can fit residual noise wholesale, overfitting in few rounds and wasting the sequential correction structure. Depth 3-8 buys low-order interactions while keeping each member weak enough that shrinkage + many rounds does the fitting gradually. The tell in tuning: if best depth trends high, you may be missing engineered interaction features; if stumps win, the signal is additive.

Q15. How does XGBoost handle missing values, and why is that better than imputation for trees?

Sparsity-aware split finding: at each node, during training, the gain of sending all missing-valued rows left vs right is evaluated, and the better "default direction" is stored per node. Missingness thus becomes signal the tree can exploit — informative missingness (a blank income field correlates with outcome) is captured for free, whereas mean-imputation actively destroys it by blending missing rows into the average-value population. It also removes an entire preprocessing step and its leakage surface (imputers fit on full data before splitting — a Chapter 4 classic). LightGBM and CatBoost have equivalents; sklearn's HistGB too.

Q16. A random forest's feature importance ranks a nearly-random high-cardinality ID-like feature in the top 3. Explain and fix.

Impurity-based importance sums each feature's impurity reductions over all splits; high-cardinality features offer many candidate thresholds and can carve chance purity, accumulating fake credit (Chapter 6's cardinality bias, now averaged over trees — subtler but intact). Correlated features add a second distortion: credit split arbitrarily among near-duplicates understates each. Fixes: permutation importance on held-out data (shuffle a feature, measure the score drop — Chapter 11), drop identifier-like columns after a leakage audit (an ID that predicts the target often proxies time or batch — Chapter 4), and for correlated groups, permute the group jointly or cluster features first.

Q17. When would you choose a random forest over gradient boosting in production, even expecting slightly lower accuracy?

When the operational profile matters more than the last accuracy point: forests are near-tuning-free (a wrong default rarely disasters), robust to label noise and outliers (no residual-chasing), embarrassingly parallel to train and easy to retrain on a schedule, give OOB validation without a split, and degrade gracefully. Boosting's edge costs tuning discipline, early-stopping infrastructure, and noise sensitivity. Concretely: small team, noisy labels, frequent automated retrains, or a baseline needed this week — forest. Stable pipeline, clean-ish labels, accuracy-critical margin — boosted trees. Stating it as an operations decision, not an accuracy decision, is what reads senior.

Q18. Estimate the prediction-time cost of a 500-tree forest vs a 500-round boosted model with depth-6 trees. Same? Different?

Per query, both walk ~ 500 trees of depth ≤ 6 — about 500×6 comparisons, so raw traversal cost is comparable and small (microseconds). Differences that matter at scale: boosted trees are usually shallower and symmetric in CatBoost's case (vectorizable into table lookups — its inference speed claim); forests parallelize trivially across trees; both pale next to a single linear model, which is why extreme-latency ranking systems distill ensembles into simpler models (Chapter 21's distillation, tabular edition). If asked for one number: both are $O(\text{trees} \times \text{depth})$ per row, and the real cost battle is memory locality, not arithmetic.

Q19. Why does boosting reduce bias? Be precise about the mechanism.

Each round adds a function fitted to the current ensemble's systematic errors — by construction, whatever consistent pattern the ensemble misses becomes the next member's target. A single weak learner has high bias (a stump can't represent much), but sums of many weak learners form a far richer function class: 500 depth-3 trees can represent complex surfaces no individual member can. Boosting greedily builds toward the best function in that additive class, driving approximation error (bias) down each round. The flip side: as bias falls, the remaining "signal" in residuals is increasingly noise, which is where the variance risk and early stopping enter. Bagging never does this — averaging can't represent anything its members can't.

Q20. Design an ensemble for a fraud model where labels are noisy and 0.5% positive. What do you pick and why?

Gradient boosting (LightGBM/XGBoost) as the core — tabular, interaction-heavy, state of the art — with noise-and-imbalance discipline: class weights or $\text{scale_pos_weight} \approx \text{neg/pos}$ rather than aggressive oversampling (Chapter 9); PR-AUC or recall-at-precision as the metric, never accuracy (Chapter 10); early stopping on a stratified validation split; modest depth and strong subsampling because label noise + boosting is the known bad marriage (Listing 5); avoid AdaBoost outright (exponential loss amplifies mislabels). Consider a forest sanity-check model — its noise robustness makes disagreement with the boosted model a good label-quality alarm. Calibrate scores (Chapter 10) because fraud thresholds are cost-driven. Every clause maps to a chapter concept, which is the point of the question.

Q21. What is stochastic gradient boosting, and why does subsampling help a method that isn't variance-limited?

Fit each round's tree on a random row subsample (0.5–0.8 typical; column subsampling analogously). Two benefits: regularization — each tree sees a perturbed view, decorrelating consecutive trees' errors and behaving like bagging grafted onto boosting, measurably delaying the overfitting turn; and compute — smaller fits per round. Why it helps despite boosting being bias-oriented: by mid-training, bias is largely destroyed and the marginal rounds are fitting noise — precisely a variance regime, where resampling's decorrelation works. LightGBM's GOSS is the refinement: keep all high-gradient (poorly fit) rows, subsample the well-fit ones — biased sampling toward the informative examples with a correction factor.

Q22. Your stacked ensemble scores 0.91 in offline CV but 0.79 in the online A/B. The base models alone score 0.86 offline / 0.84 online. Diagnose.

The base models transfer (small offline-online gap); the stack does not — the meta-layer is the suspect. Prime hypothesis: meta-features were generated in-sample (not out-of-fold), so the meta-learner learned to trust overfit base predictions — inflating offline scores and collapsing live (Q12's trap). Second: the meta-learner was tuned/selected on the same CV folds that generated meta-features (nested leakage — selection bias on top). Third: distribution shift interacting with the stack — base-model calibration drifted online, and a meta-learner keyed to offline probability levels misfires (compare online score distributions to offline). Verify in that order: regenerate meta-features strictly out-of-fold, hold out a truly untouched test period, check probability histograms offline vs online.

Q23. Why is a weak learner "weak" on purpose in boosting? What goes wrong with strong ones?

Boosting's error-correction structure assumes each member leaves something for the next to fix. A strong learner (deep tree) fits most of the signal *and a chunk of noise* in round one; subsequent rounds then fit mostly noise, and the exponential/gradient reweighting concentrates on unfixable points (mislabels, inherent overlap). Capacity also compounds: the additive ensemble of strong learners reaches interpolation in few rounds, skipping the gradual regularized path that shrinkage is designed to walk. Weak learners keep per-round capacity small so the ensemble's capacity grows slowly and controllably — complexity added in η -sized increments, stopped by validation. It's the same philosophy as small learning rates in SGD (Chapter 13), one level up.

Q24. Implement one boosting round for squared loss in four lines of NumPy/sklearn, given current predictions pred, targets y, and learning rate lr.

```
resid = y - pred; tree = DecisionTreeRegressor(max_depth=3).fit(X, resid); pred = pred + lr * tree.predict(X); trees.append(tree).
```

Worth narrating as you write: the residual is the negative gradient of $\frac{1}{2}(y-F)^2$, so this is one function-space gradient step; for log loss the only change is `resid = y - sigmoid(F)` (Chapter 5's GLM gradient again); and the shrunk step is why prediction must apply the same `lr` (Listing 4's `gbm_predict`). Four lines plus that narration is a complete, correct answer.

Q25. Level-wise vs leaf-wise tree growth — mechanics and consequences.

Level-wise (XGBoost default): expand all nodes at the current depth before going deeper — balanced trees, capacity capped by `max_depth`, conservative. Leaf-wise (LightGBM): always split the single leaf with the largest gain, wherever it sits — lopsided trees that spend their leaf budget where the loss says it matters, achieving lower loss per leaf count. Consequences: leaf-wise reaches higher accuracy faster on large data but overfits faster on small data (a deep chain can isolate tiny groups — hence `num_leaves` and `min_data_in_leaf` as the controlling knobs, and the standard advice `num_leaves < 2^max_depth`). Knowing that `num_leaves`, not `max_depth`, is LightGBM's real capacity dial is the practitioner tell.

Q26. Can ensembling ever hurt? Give three concrete scenarios.

(1) Averaging in weaker members: equal-weight voting with unequal members lands below the best one (Listing 7, 0.837 vs 0.864). (2) Correlated members: same architecture retrained with different seeds has ρ near 1 — cost multiplies, variance barely moves; ensembling only pays for genuine diversity. (3) Calibration and interpretability casualties: averaging distorts probability calibration (recalibrate after), and a stakeholder-readable single tree becomes an unexplainable committee — sometimes a regulatory non-starter (Chapter 11/33). Bonus fourth: boosting a noisy-label dataset past the validation minimum makes the ensemble strictly worse than a shorter version of itself.

Q27. Why do gradient-boosted trees still beat neural networks on most tabular problems?

Tabular data's structure fits trees' inductive bias: heterogeneous feature types and scales (trees are scale-indifferent; nets need careful normalization/embeddings), abundant uninformative features (greedy split selection is implicit feature selection; nets must learn to ignore), sharp threshold-like decision rules common in business data (axis-aligned splits express them natively; smooth activations approximate them awkwardly), and typically modest n where the net's capacity advantage can't cash in. Add operational maturity — fast CPU training, early stopping, native missing/categorical handling. The honest boundary: at very large n with heavy feature interactions, or with multimodal inputs (text/image columns), neural and hybrid approaches close the gap and win (Chapters 12+). Citing benchmarks both ways, rather than tribal loyalty, is the strong finish.

Q28. You may deploy exactly one model artifact but can train anything. How do you capture ensemble gains in a single model?

Knowledge distillation (Chapter 21's idea, tabular edition): train the big ensemble/stack as a teacher, then train one compact student — a single boosted model or small net — on the teacher's *soft predictions* (probabilities, not hard labels), often on augmented/unlabeled data the teacher pseudo-labels. The soft targets carry the teacher's learned inter-class structure and smoothing, so the student typically lands closer to the teacher than to a same-size model trained on raw labels. Alternatives worth naming: pick the ensemble's strongest member and retune it hard; or weight-averaging tricks that approximate ensembling inside one artifact — stochastic weight averaging (SWA) and snapshot ensembles. Distillation is the canonical answer; naming SWA shows breadth.

Chapter 8: Unsupervised Learning

Unsupervised learning is where ML meets data with no answer key: only x , no y , and the job is to find structure — groups, densities, low-dimensional organization, exceptions. Interviewers use it for two things. First, the algorithms themselves: k-means and PCA are among the most-asked derivations in ML interviews, EM is a rite of passage, and "how would you choose the number of clusters?" is a screening staple. Second — and this is what separates candidates — the *epistemics*: without labels there is no accuracy to compute, so every claim ("these are the customer segments") requires justifying an evaluation strategy, and "how would you know your clustering is good?" has ended more interviews than any derivation. Chapter 4 flagged this slipperiness; here it gets teeth.

The chapter covers clustering in four flavors (k-means's centroid geometry, hierarchical's merge tree, DBSCAN's density reachability, GMM's probabilistic mixtures — each with a different implicit definition of "cluster", which is the real lesson), the EM algorithm underneath mixtures, dimensionality reduction from linear (PCA, LDA) to neighbor-preserving (t-SNE, UMAP), association rule mining with its support/confidence/lift triple, and anomaly detection. Everything is implemented and measured in the listings — including the honest failures.

K-Means

K-means partitions n points into k clusters by choosing centroids $\mu_1, \dots, \mu_k$ minimizing **inertia** — within-cluster sum of squared distances:

$$J = \sum_{i=1}^n \min_j \|x_i - \mu_j\|^2$$

Exact minimization is NP-hard; **Lloyd's algorithm** is the universal heuristic: (1) assign each point to its nearest centroid; (2) move each centroid to the mean of its assigned points; repeat until nothing moves. Both steps provably decrease J — the assignment step is optimal given centroids, the update step is optimal given assignments (the mean minimizes summed squared distance, a Chapter 1 fact) — so the algorithm always converges. But only to a *local* minimum, and which one depends entirely on initialization: Listing 1's twenty random-restart runs land anywhere in a $1.24\times$ inertia spread. The standard fixes are restarts (keep the best of n_{init} runs) and **k-means++ initialization** — seed the first centroid uniformly, then pick each next centroid with probability proportional to its squared distance from the nearest existing seed, spreading the seeds across the data with a provable $O(\log k)$ approximation guarantee. Listing 1's k-means++ run matches the best of twenty random restarts on the first try.

Choosing k — the inevitable follow-up — has no oracle, only heuristics. The **elbow method** plots inertia against k and looks for the bend: inertia decreases monotonically in k (more centroids can only shorten distances — at $k = n$ it is zero, uselessly), so what carries information is the *rate* slowing. Listing 2 shows drops of ~44% per increment up to the true $k = 6$, collapsing to ~6% after — an unusually clean elbow; real elbows are mushier, which is the criticism to volunteer. The **silhouette score** is sturdier: for each point, with a = mean distance to its own cluster and b = mean distance to the nearest other cluster, $s = (b - a) / \max(a, b) \in [-1, 1]$; average over points. It rewards tight, well-separated clusters and *can decrease* — Listing 2's silhouette peaks exactly at $k = 6$ (0.662). Honest caveats: silhouette presumes compact-ish geometry (it will prefer 2 super-clusters when groups overlap — a scale-of-analysis ambiguity, not a bug), costs $O(n^2)$ distances, and neither method beats domain knowledge about how many segments the business can act on. Gap statistic and BIC-via-GMM are the other names worth having.

Limitations are the richest k-means interview territory, because every one traces back to the objective: squared Euclidean distance to a single centroid presumes clusters are **spherical, similar in size and density, and convex**. Listing 3 breaks each assumption and measures the damage (adjusted Rand index against truth): two moons — k-means 0.249, because no centroid pair can represent interlocking crescents (DBSCAN: 1.000); sheared anisotropic Gaussians — 0.430, because "nearest centroid" draws boundaries perpendicular to the wrong axes (GMM with full covariance: 1.000); one big diffuse cluster plus two small tight ones — 0.252, because splitting the big cluster lowers inertia more than respecting the small ones (GMM: 0.985). Further standing issues: sensitivity to feature scale (it's a distance method — standardize, Chapter 6's rule), sensitivity to outliers (means get dragged; k-medoids uses actual points as centers), hard assignments (no "60% cluster A" — GMM's opening), and the requirement to pick k up front. Say the summary sentence: *k-means doesn't find your clusters; it finds the best spherical-Voronoi story it can tell, whether or not your data is one.*

Two closers interviewers like: k-means is exactly a GMM with equal spherical covariances and hard assignments (the EM connection below makes this precise), and at scale it runs as mini-batch k-means (stochastic centroid updates — the SGD of clustering) and powers vector-quantization workhorses from codebooks to product-quantized vector indexes (Chapter 24).

Hierarchical clustering

Hierarchical clustering returns not one partition but a **dendrogram** — a binary merge tree over the data, cuttable at any height into any number of clusters. **Agglomerative** (bottom-up: start with n singletons, repeatedly merge the two closest clusters) is the standard; **divisive** (top-down recursive splitting) is rarer and mostly conceptual. No k is needed to build the tree; k is chosen afterward by cutting — and a large jump in merge

distance marks a natural cut (Listing 4: merge distances $13.0 \rightarrow 85.1 \rightarrow 95.3$; the 13→85 jump says "three clusters", and cutting there recovers the true sizes exactly).

Everything hinges on the **linkage** — the definition of distance between clusters: **single** (closest pair — follows chains and filaments, can "leak" through noise bridges), **complete** (farthest pair — compact, similar-diameter clusters, outlier-sensitive), **average** (mean over all pairs — the compromise, UPGMA), and **ward** (merge the pair whose union least increases within-cluster variance — inertia-greedy, k-means's sibling, the usual default). The choice is a prior about cluster shape, and Listing 4 stages the clean double dissociation: on compact blobs, ward/average/complete score ARI 0.985–0.993 while single scores 0.571 (chaining through inter-cluster noise); on two moons, single scores a perfect 1.000 (chaining along the crescents is exactly right) while ward scores 0.162 (compactness is exactly wrong). Neither is "better" — they define "cluster" differently, and saying that is the answer.

Practicalities: $O(n^2)$ memory for the distance matrix and $O(n^2 \log n)$ -ish time bound it to tens of thousands of points; it inherits distance's scaling requirements; it is deterministic (no init lottery); and the dendrogram itself is often the deliverable — taxonomies, gene-expression heatmap orderings, topic hierarchies — a reason to choose it even when a flat partition would score the same.

DBSCAN and HDBSCAN

DBSCAN defines clusters by **density, not distance to a center**: a **core point** has $\geq \text{min_samples}$ neighbors within radius ϵ ; points within ϵ of a core point are **density-reachable**; clusters are maximal chains of density-reachable points; everything else is **noise** — a first-class output, not a failure. The consequences interviewers probe: no k required (the data's density structure decides), arbitrary cluster shapes (moons, rings, filaments — anything density-connected; Listing 5: ARI 0.970 on noisy moons where k-means got 0.249), and native outlier flagging (the injected 10% uniform noise lands in the -1 label).

The cost is ϵ , and it is a cliff, not a dial: Listing 5's sweep shows $\epsilon = 0.05$ shattering the data into 34 fragments with 275 "noise" points ($\text{ARI} \approx 0$), $\epsilon = 0.1$ – 0.2 nearly perfect, $\epsilon = 0.4$ fusing everything into one blob ($\text{ARI} 0.123$). The k -distance plot (sorted distance to each point's k -th neighbor; look for the knee) is the standard ϵ -picking heuristic. The deeper failure mode: **one global ϵ cannot serve clusters of different densities** — tight enough for the dense cluster shatters the sparse one; loose enough for the sparse one fuses the dense pair. That is precisely what **HDBSCAN** fixes: it builds a hierarchy over *all* density levels (conceptually, DBSCAN at every ϵ at once), then extracts the most stable clusters across the hierarchy — no ϵ at all, just `min_cluster_size`. Listing 5: HDBSCAN matches tuned DBSCAN ($\text{ARI} 0.967$) with nothing to tune, which is why it has become the practical default for density clustering. Remaining shared caveats: distance-

based (scale features; degrades in high dimension like KNN — Chapter 6), and border-point assignments can vary between runs in some implementations.

Gaussian Mixture Models and EM

A GMM says the data was generated by a two-stage story: pick component j with probability π_j , then draw $x \sim \mathcal{N}(\mu_j, \Sigma_j)$. The density is a weighted sum of Gaussians:

$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \Sigma_j)$$

Clustering becomes posterior inference — the **responsibility** $r_{ij} = p(\text{component } j \mid x_i)$ by Bayes' rule — and assignments are **soft**: a point between two components is 60/40, not forced to choose. Full covariance matrices let components be ellipses at any orientation, which is exactly what rescued Listing 3's anisotropic case (ARI 1.000 vs k-means 0.430). Covariance structure is the capacity dial: spherical $\subset$ diag $\subset$ tied $\subset$ full, trading flexibility against parameters-per-component (full costs $O(d^2)$ each — in high d , regularize or restrict). And because a GMM is a *generative density model*, you can sample new points, score likelihoods ($\rightarrow$ anomaly detection below), and select k with honest information criteria: BIC/AIC penalize the likelihood by parameter count, something inertia can never do.

Direct maximum likelihood is intractable — the log of a *sum* doesn't decompose — because the component labels are **latent variables**. **EM (Expectation-Maximization)** is the standard escape, and it alternates two readable steps. **E-step**: with parameters fixed, compute responsibilities r_{ij} (Bayes' rule — "which component probably made this point?"). **M-step**: with responsibilities fixed, update parameters by responsibility-weighted MLE — π_j = average responsibility, μ_j = weighted mean, Σ_j = weighted covariance; each formula is Chapter 1's Gaussian MLE with fractional counts. The guarantee that makes EM famous: **each iteration cannot decrease the observed-data log-likelihood** (it maximizes a tight lower bound — Jensen's inequality — that touches the likelihood at the current parameters). Listing 6 implements the full loop from scratch in ~25 lines with the monotonicity asserted programmatically: log-likelihood climbs $-7746 \rightarrow -2213 \rightarrow -2161$, converges in 21 steps, and recovers the true mixture (π, μ, σ all within noise, matching sklearn). The k-means connection completes the circuit: freeze all covariances to ϵI , let $\epsilon \rightarrow 0$, and responsibilities harden to nearest-centroid assignments — EM degenerates into exactly Lloyd's algorithm. K-means is a GMM that stopped believing in uncertainty.

EM's caveats are interview follow-ups: local maxima (initialize from k-means, use restarts); **singularities** — a component can collapse onto one point, $\sigma \rightarrow 0$, likelihood $\rightarrow \infty$, which is why implementations floor the covariance (reg_covar); label switching (component identities are arbitrary — sort before comparing, as Listing 6 does); and convergence can be slow when components overlap. EM itself is general far beyond

GMMs — missing data, hidden Markov models, any latent-variable likelihood — and "derive EM for a mixture" is a canonical hard-round request that Listing 6 is designed to rehearse.

Dimensionality reduction: PCA, LDA, t-SNE, UMAP

PCA finds the orthogonal directions of maximal variance and projects onto the top k . Equivalent formulations worth being able to swap between: (1) maximize projected variance; (2) minimize squared reconstruction error (the best rank- k linear approximation — the two objectives are the same by Pythagoras); (3) eigendecompose the covariance matrix — or, numerically better, **SVD the centered data**: $X_c = USV^\top$, principal axes = rows of $V^\top$, component variances = $S^2/(n-1)$. Centering is mandatory (uncentered "PCA" finds the direction to the mean, not of the spread); scaling matters exactly as in every distance method — standardize unless features share units, or the largest-variance raw feature *is* the first PC by default. **Explained variance ratio** guides k : Listing 7 on 64-pixel digits — 10 components keep 73.8%, 40 keep 98.8% — and the from-scratch SVD matches sklearn exactly, with reconstruction demonstrated (project to 10-D, map back, pixel MSE 4.91). Uses beyond visualization: decorrelation/whitening, noise reduction, compression before distance methods (KNN, k-means — Chapter 6's curse mitigation), and multicollinearity removal for regression (Chapter 5). Limits: linear (a Swiss-roll manifold defeats it), variance $\neq$ importance (the discriminative direction may be PC 40 — see LDA), and components are combinations of everything, costing interpretability.

LDA (Linear Discriminant Analysis) is the supervised counterpoint: it uses *labels* to find $\leq$ (classes – 1) directions maximizing between-class variance over within-class variance — separation, not spread. Listing 8 makes the contrast quantitative on digits: 2-D PCA supports 15-NN accuracy 0.641; 2-D LDA, 0.687 — better, because its two axes were chosen to separate the ten classes, not to preserve pixel variance. LDA belongs in this chapter as the reminder that "dimensionality reduction" is not inherently unsupervised — when labels exist and the goal is class structure, use them.

t-SNE and UMAP abandon linearity and global fidelity to preserve *neighborhoods*. t-SNE converts pairwise distances into neighbor probabilities (bandwidths set per point by **perplexity**, effectively "how many neighbors count" — the main knob, typically 5–50), then arranges points in 2-D so the low-dimensional neighbor distribution (heavy-tailed Student-t, which is what prevents the crowding problem of squeezing high-D neighborhoods into 2-D) matches via KL-divergence minimization. UMAP builds a fuzzy k-NN graph and optimizes a cross-entropy layout — similar spirit, better preserved global structure, faster on large n , and it can transform *new* points (t-SNE classically cannot — a production-relevant difference). The payoff is dramatic: Listing 8's digits go from 0.64 (PCA) to 0.976/0.978 (t-SNE/UMAP) in 2-D 15-NN accuracy — local class structure almost perfectly preserved. The mandatory warnings, straight from interview scripts: **cluster sizes and inter-cluster distances in a t-SNE plot are not meaningful** (the algorithm equalizes densities; distances between islands are artifacts), different

perplexities tell different stories (always show more than one), it will happily draw "clusters" in pure noise, and axes have no interpretation. These are *visualization and neighbor-structure* tools — cluster on them with care (HDBSCAN-on-UMAP is a popular, effective, and theoretically awkward pipeline; know both facts), and never feed t-SNE coordinates into downstream models as features.

Association rule mining

Market-basket analysis mines co-occurrence rules $A \rightarrow B$ from transaction sets. The three metrics, on baskets containing itemsets:

- **Support**($A \rightarrow B$) = $P(A \cup B \text{ in basket})$ — how common the pattern is; filters rare noise.
- **Confidence**($A \rightarrow B$) = $P(B | A) = \text{support}(A \cup B) / \text{support}(A)$ — how reliable the implication is.
- **Lift**($A \rightarrow B$) = $\text{confidence} / \text{support}(B) = P(A \cup B) / (P(A)P(B))$ — how much more than *chance* the pair co-occurs. Lift > 1: complements; = 1: independent; < 1: substitutes. Lift is the honesty check: Listing 9's beer $\rightarrow$ milk rule has confidence 0.80 (looks strong!) and lift 1.00 — milk is just in 80% of baskets; the rule says nothing. Meanwhile beer $\rightarrow$ diapers, confidence 0.80, lift 1.33 — a real association. Confidence-without-lift is the classic rookie reading, and catching it is the entire point of the question.

Apriori finds all frequent itemsets level-wise using the **downward-closure property**: every subset of a frequent itemset is frequent, so k-itemsets need only be generated from frequent (k−1)-itemsets, pruning the exponential lattice (Listing 9 implements the first two levels from scratch: 5 frequent items $\rightarrow$ 5 frequent pairs at 30% support, then scores all rules). **FP-Growth** reaches the same answer without candidate generation: compress transactions into a prefix tree (FP-tree) sharing common prefixes, then mine it recursively — typically much faster on large sparse baskets. Modern framing to volunteer: association rules survive in retail analytics and rule-based recommendation baselines, and lift's logic (observed co-occurrence over independence expectation) is the same idea as PMI in NLP (Chapter 19).

Anomaly detection

Anomaly detection asks "which points don't belong?" — usually unsupervised, because anomalies are rare, unlabeled, and unlike each other (fraud, sensor faults, intrusions). The three classical detectors triangulate different definitions of "doesn't belong", and Listing 10 scores them on blobs + scattered outliers:

- **Isolation Forest** (0.82/0.82 precision/recall): anomalies are *easy to isolate* — random axis-aligned splits strand an outlier in few cuts, so short average path

length across random trees = anomalous. $O(n \log n)$, no distances (scale-tolerant, high-d-tolerant), the industrial default.

- **One-Class SVM** (0.71/0.70): learns a boundary around the "normal" region in kernel space; ν bounds the outlier fraction. Elegant, but kernel-scaling and γ -sensitivity make it the fussiest of the three.
- **LOF (Local Outlier Factor)** (0.84/0.84): compares each point's local density to its neighbors' — a point in a sparse spot *relative to its neighborhood* is anomalous. Catches local anomalies global methods miss (a point at a dense cluster's edge); inherits KNN's scaling/dimension issues; classically has no natural predict-on-new-data mode (novelty=True changes the contract).

Two more families complete the answer: **GMM/density scoring** — fit the density, flag low-likelihood points (natural when you already have a generative model), and **autoencoders** — train a bottlenecked reconstructor on (mostly) normal data; anomalies reconstruct poorly, and reconstruction error is the score (Chapter 18 builds them; the idea works because the bottleneck learns the normal manifold and has no capacity budget for rarities). The shared operational truths: **contamination is a chosen alarm rate**, not a discovered fact — you are picking a point on the precision/recall curve (Listing 10's knob), and thresholds should be set against the cost of missed-vs-false alarms with whatever labeled examples exist (Chapter 10's machinery); evaluation without labels falls back on injection studies (plant synthetic anomalies, measure recovery — exactly Listing 10's design) and human triage of the top-k alarms.

Code implementations

Every listing was executed as shown; outputs are real. The set implements the chapter: Lloyd's algorithm and the init lottery, elbow vs silhouette on known k , three staged k -means failures with the model that fixes each, the linkage double dissociation, DBSCAN's ϵ cliff and HDBSCAN's escape from it, EM from scratch with monotonicity asserted, PCA via SVD with reconstruction, the four-way embedding bake-off, Apriori with the lift lesson, and three anomaly detectors scored against injected truth.

Listing 1 — K-means from scratch and the initialization lottery

Lloyd's two-step alternation in twelve lines. Twenty random initializations spread across a $1.24\times$ inertia range — same data, same algorithm, different local minima. `k-means++` hits the best observed value in one seeded try.

```
"""Listing 1: k-means from scratch (Lloyd's algorithm), and why initialization
matters."""
import numpy as np
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans

X, _ = make_blobs(n_samples=1500, centers=6, cluster_std=1.2, random_state=0)

def kmeans(X, k, init_centers, iters=100):
```

```

C = init_centers.copy()
for _ in range(iters):
    # Assignment step: each point to its nearest center
    d = ((X[:, None, :] - C[None, :, :]) ** 2).sum(-1)      # (n, k) sq-distances
    labels = d.argmin(1)
    # Update step: each center to the mean of its points
    newC = np.array([X[labels == j].mean(0) if (labels == j).any() else C[j]
                     for j in range(k)])
    if np.allclose(newC, C): break
    C = newC
inertia = ((X - C[labels]) ** 2).sum()
return labels, C, inertia

rng = np.random.default_rng(1)
# Random init, 20 restarts: the spread shows the local-minimum lottery
inertias = []
for _ in range(20):
    init = X[rng.choice(len(X), 6, replace=False)]
    _, inertia = kmeans(X, 6, init)
    inertias.append(inertia)
print(f"random init, 20 runs: best {min(inertias):.0f} worst {max(inertias):.0f} "
      f"({max(inertias)/min(inertias):.2f}x spread)")

sk = KMeans(n_clusters=6, init="k-means++", n_init=10, random_state=1).fit(X)
print(f"k-means++ (sklearn) : inertia {sk.inertia_:.0f}")

```

Output:

```

random init, 20 runs: best 3518 worst 4359 (1.24x spread)
k-means++ (sklearn) : inertia 3518

```

Listing 2 — Choosing k: elbow vs silhouette on known k=6

Inertia falls forever; its *percentage drop* collapses from ~44% to ~6% right after the true k — the elbow. Silhouette, which can decrease, peaks exactly at k=6. On real data the elbow is rarely this crisp; the silhouette peak is the sturdier signal.

```

"""Listing 2: choosing k -- elbow (inertia) vs silhouette, on data with known k=6."""
import numpy as np
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

X, _ = make_blobs(n_samples=1500, centers=6, cluster_std=1.0,
                  center_box=(-15, 15), random_state=0)

print(f"{'k':>3} {'inertia':>9} {'drop %':>7} {'silhouette':>11}")
prev = None
for k in range(2, 11):
    km = KMeans(n_clusters=k, n_init=10, random_state=0).fit(X)
    sil = silhouette_score(X, km.labels_)
    drop = f"{100*(prev-km.inertia_)/prev:>6.1f}%" if prev else " - "
    print(f"{'k':>3} {'km.inertia_':>9.0f} {drop} {sil:>11.3f}")
    prev = km.inertia_

```

Output:

```

k    inertia  drop %  silhouette
2    27760    -      0.589

```


3	15334	44.8%	0.552
4	8861	42.2%	0.566
5	4953	44.1%	0.621
6	2785	43.8%	0.662
7	2615	6.1%	0.591
8	2438	6.8%	0.549
9	2291	6.0%	0.472
10	2130	7.0%	0.420

Listing 3 — Three staged k-means failures, each with its cure

Adjusted Rand index against known truth. Moons: no two centroids can represent interlocking crescents — DBSCAN's density chains can. Anisotropic: nearest-centroid boundaries cut across the stretched clusters — full-covariance GMM tilts its ellipses to match. Unequal sizes: splitting the big diffuse cluster buys more inertia than respecting the small tight ones — GMM's per-component variances fix it.

```

"""Listing 3: k-means limitations -- when the spherical assumption is the wrong
prior."""
import numpy as np
from sklearn.datasets import make_moons, make_blobs
from sklearn.cluster import KMeans, DBSCAN
from sklearn.mixture import GaussianMixture
from sklearn.metrics import adjusted_rand_score as ari

rng = np.random.default_rng(2)

# Case 1: non-convex shapes (two moons)
Xm, ym = make_moons(n_samples=600, noise=0.06, random_state=2)
km = KMeans(2, n_init=10, random_state=2).fit_predict(Xm)
db = DBSCAN(eps=0.2, min_samples=5).fit_predict(Xm)
print("two moons      : k-means ARI", f"{ari(ym, km):.3f}",
      "  DBSCAN ARI", f"{ari(ym, db):.3f}")

# Case 2: anisotropic (stretched) Gaussians
Xb, yb = make_blobs(n_samples=600, centers=3, random_state=3)
Xa = Xb @ np.array([[0.6, -0.64], [-0.4, 0.85]]) # shear the space
km = KMeans(3, n_init=10, random_state=3).fit_predict(Xa)
gm = GaussianMixture(3, covariance_type="full", n_init=10,
random_state=3).fit_predict(Xa)
print("anisotropic   : k-means ARI", f"{ari(yb, km):.3f}",
      "  GMM(full) ARI", f"{ari(yb, gm):.3f}")

# Case 3: unequal cluster sizes/densities
Xu, yu = make_blobs(n_samples=[500, 60, 40], centers=[[0,0],[5,5],[9,0]],
                    cluster_std=[2.2, 0.5, 0.5], random_state=4)
km = KMeans(3, n_init=10, random_state=4).fit_predict(Xu)
gm = GaussianMixture(3, covariance_type="full", n_init=10,
random_state=4).fit_predict(Xu)
print("unequal sizes : k-means ARI", f"{ari(yu, km):.3f}",
      "  GMM(full) ARI", f"{ari(yu, gm):.3f}")

```

Output:

```

two moons      : k-means ARI 0.249  DBSCAN ARI 1.000
anisotropic    : k-means ARI 0.430  GMM(full) ARI 1.000
unequal sizes  : k-means ARI 0.252  GMM(full) ARI 0.985

```

Listing 4 — Linkage choices: a clean double dissociation

On compact blobs, ward/average/complete are near-perfect and single linkage chains through the gaps (0.571). On moons, single linkage is *perfect* (chaining along a crescent is exactly right) and ward is worst (0.162 — compactness is exactly wrong). The linkage is a prior about cluster shape. The dendrogram's merge distances then locate the natural cut: 13.0 → 85.1 is the jump, so cut into three.

```
"""Listing 4: agglomerative clustering -- linkage choices are different definitions of
'close'."""
import numpy as np
from scipy.cluster.hierarchy import linkage, fcluster
from sklearn.datasets import make_blobs, make_moons
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import adjusted_rand_score as ari

# Compact blobs: everything works; ward is the default for a reason
Xb, yb = make_blobs(n_samples=400, centers=[[0, 0], [7, 4], [1, 8]], cluster_std=1.0,
                    random_state=5)
# Elongated chains (moons): single linkage's chaining is a feature here, ward's
compactness a bug
Xm, ym = make_moons(n_samples=400, noise=0.05, random_state=5)

print(f"{'linkage':>9} {'blobs ARI':>10} {'moons ARI':>10}")
for link in ["single", "complete", "average", "ward"]:
    ab = AgglomerativeClustering(3, linkage=link).fit_predict(Xb)
    am = AgglomerativeClustering(2, linkage=link).fit_predict(Xm)
    print(f"{'link':>9} {'ari(yb, ab):>10.3f} {'ari(ym, am):>10.3f}")

# The dendrogram is the real output: cut it anywhere for any number of clusters
Z = linkage(Xb, method="ward") # (n-1) merges: [idx_a, idx_b, distance, size]
print("\nlast 3 merges (ward), distances:", np.round(Z[-3:, 2], 1))
for k in [2, 3, 4]:
    labels = fcluster(Z, t=k, criterion="maxclust")
    print(f"cut at k={k}: cluster sizes {np.bincount(labels)[1:]}")
```

Output:

```
linkage blobs ARI moons ARI
single      0.571      1.000
complete    0.985      0.389
average     0.993      0.259
ward        0.993      0.162

last 3 merges (ward), distances: [13.  85.1 95.3]
cut at k=2: cluster sizes [134 266]
cut at k=3: cluster sizes [134 132 134]
cut at k=4: cluster sizes [134 132 39 95]
```

Listing 5 — DBSCAN's ϵ cliff, and HDBSCAN's escape

Moons plus 10% uniform noise. $\epsilon=0.05$ shatters (34 fragments, $\text{ARI} \approx 0$); $\epsilon=0.1$ – 0.2 nails it and flags the noise; $\epsilon=0.4$ fuses everything. HDBSCAN gets the same quality with no ϵ — `min_cluster_size` is its only real knob.

```
"""Listing 5: DBSCAN -- density clustering with noise, its eps cliff, and HDBSCAN's
fix."""
```

```

import numpy as np
from sklearn.cluster import DBSCAN, HDBSCAN
from sklearn.datasets import make_moons
from sklearn.metrics import adjusted_rand_score as ari

rng = np.random.default_rng(6)
Xm, ym = make_moons(n_samples=600, noise=0.06, random_state=6)
X = np.vstack([Xm, rng.uniform(-1.5, 2.5, (60, 2))]) # add 10% uniform noise
y = np.hstack([ym, np.full(60, -1)]) # noise gets label -1

print(f"{'eps':>5} {'clusters':>9} {'noise pts':>10} {'ARI':>6}")
for eps in [0.05, 0.1, 0.2, 0.4]:
    lab = DBSCAN(eps=eps, min_samples=5).fit_predict(X)
    n_clusters = len(set(lab)) - (1 if -1 in lab else 0)
    print(f"{'eps':>5} {'n_clusters':>9} {(lab == -1).sum():>10} {ari(y, lab):>6.3f}")

hdb = HDBSCAN(min_cluster_size=15).fit_predict(X)
n_clusters = len(set(hdb)) - (1 if -1 in hdb else 0)
print(f"\nHDBSCAN: {n_clusters} clusters, {(hdb == -1).sum()} noise pts, "
      f"ARI {ari(y, hdb):.3f} -- no eps to tune")

```

Output:

eps	clusters	noise pts	ARI
0.05	34	275	-0.016
0.1	2	59	0.970
0.2	2	44	0.948
0.4	1	32	0.123

HDBSCAN: 2 clusters, 52 noise pts, ARI 0.967 -- no eps to tune

Listing 6 — EM for a Gaussian mixture, from scratch

The complete E-step (responsibilities via Bayes) and M-step (responsibility-weighted MLE), with EM's defining guarantee asserted in code: the log-likelihood never decreases. Converges in 21 steps and recovers the true mixture, matching sklearn.

```

"""Listing 6: EM for a two-component 1-D Gaussian mixture, from scratch."""
import numpy as np
from sklearn.mixture import GaussianMixture

rng = np.random.default_rng(7)
# True mixture: 40% N(-2, 0.8^2) + 60% N(3, 1.5^2)
x = np.hstack([rng.normal(-2, 0.8, 400), rng.normal(3, 1.5, 600)])

def norm_pdf(x, mu, sd):
    return np.exp(-0.5 * ((x - mu) / sd) ** 2) / (sd * np.sqrt(2 * np.pi))

# init: crude split at the extremes
pi, mu, sd = np.array([0.5, 0.5]), np.array([x.min(), x.max()]), np.array([1.0, 1.0])
prev_ll = -np.inf
for step in range(200):
    # E-step: responsibilities -- P(component | point), Bayes' rule
    dens = np.stack([pi[k] * norm_pdf(x, mu[k], sd[k]) for k in range(2)]) # (2, n)
    r = dens / dens.sum(0)
    # M-step: weighted MLE updates
    Nk = r.sum(1)
    pi = Nk / len(x)
    mu = (r * x).sum(1) / Nk
    sd = np.sqrt((r * (x - mu[:, None]) ** 2).sum(1) / Nk)
    ll = np.log(dens.sum(0)).sum()

```

```

if step in (0, 1, 4, 20):
    print(f"step {step:>3}: log-likelihood {ll:>10.2f}")
    assert ll >= prev_ll - 1e-9, "EM must never decrease the log-likelihood"
    if ll - prev_ll < 1e-8: break
    prev_ll = ll

print(f"converged at step {step}: ll {ll:.2f}")
print(f"scratch : pi={np.round(pi,3)} mu={np.round(mu,2)} sd={np.round(sd,2)}")
sk = GaussianMixture(2, random_state=7).fit(x[:, None])
order = np.argsort(sk.means_.ravel())
print(f"sklearn : pi={np.round(sk.weights_[order],3)} mu={np.round(sk.means_.ravel()
[order],2)} "
      f"sd={np.round(np.sqrt(sk.covariances_.ravel()[order]),2)}")
print(f"truth   : pi=[0.4 0.6] mu=[-2.  3.] sd=[0.8 1.5]")

```

Output:

```

step   0: log-likelihood   -7746.58
step   1: log-likelihood   -2213.23
step   4: log-likelihood   -2163.52
step  20: log-likelihood   -2160.60
converged at step 21: ll -2160.60
scratch : pi=[0.399 0.601] mu=[-2.08  2.92] sd=[0.74 1.44]
sklearn : pi=[0.401 0.599] mu=[-2.08  2.93] sd=[0.75 1.43]
truth   : pi=[0.4 0.6] mu=[-2.  3.] sd=[0.8 1.5]

```

Listing 7 — PCA from scratch via SVD

Center, SVD, read off axes and variances — and the explained-variance ratios match sklearn boolean-exactly. The reconstruction round trip (project to 10-D, map back to 64-D) is the "PCA as best rank-k approximation" claim, measured.

```

"""Listing 7: PCA from scratch via SVD -- variance explained, reconstruction, sklearn
parity."""
import numpy as np
from sklearn.datasets import load_digits
from sklearn.decomposition import PCA

X, _ = load_digits(return_X_y=True)          # 1797 digits, 64 pixels
Xc = X - X.mean(0)                          # center -- mandatory for PCA

# SVD of the centered data: principal axes = rows of Vt, scores = U*S
U, S, Vt = np.linalg.svd(Xc, full_matrices=False)
var = S**2 / (len(X) - 1)                   # eigenvalues of the covariance matrix
evr = var / var.sum()                      # explained variance ratio
print("explained variance ratio, first 5 PCs:", np.round(evr[:5], 3))
for k in [2, 10, 20, 40]:
    print(f" top {k:>2} PCs keep {evr[:k].sum():.1%} of variance")

# Reconstruction: project to k dims, map back, measure pixel error
k = 10
Z = Xc @ Vt[:k].T                          # (n, k) compressed representation
X_rec = Z @ Vt[:k] + X.mean(0)             # back to 64-D
mse = np.mean((X - X_rec) ** 2)
print(f"\nreconstruction MSE with k=10: {mse:.2f} (pixel values 0-16)")

sk = PCA(n_components=5).fit(X)
print("sklearn EVR matches:", np.allclose(sk.explained_variance_ratio_, evr[:5]))

```

Output:

```

explained variance ratio, first 5 PCs: [0.149 0.136 0.118 0.084 0.058]
top 2 PCs keep 28.5% of variance
top 10 PCs keep 73.8% of variance
top 20 PCs keep 89.4% of variance
top 40 PCs keep 98.8% of variance

reconstruction MSE with k=10: 4.91 (pixel values 0-16)
sklearn EVR matches: True

```

Listing 8 — Four ways to 2-D: PCA vs LDA vs t-SNE vs UMAP

The quality metric is 15-NN accuracy *in the 2-D embedding* — how much class-relevant neighborhood structure survived the trip down from 64-D. Linear methods keep ~0.64–0.69; the neighbor-preserving nonlinear methods keep ~0.98. The price: t-SNE/UMAP coordinates are for looking at, not for measuring distances or feeding models.

```

"""Listing 8: PCA vs LDA vs t-SNE vs UMAP -- what survives the trip to 2-D."""
import warnings, time
warnings.filterwarnings("ignore")
import numpy as np
from sklearn.datasets import load_digits
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
from sklearn.manifold import TSNE
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
import umap

X, y = load_digits(return_X_y=True)

def knn_quality(Z, y):
    """How well do 2-D neighborhoods preserve class structure? 15-NN accuracy."""
    Z_tr, Z_te, y_tr, y_te = train_test_split(Z, y, test_size=0.3, random_state=8)
    return KNeighborsClassifier(15).fit(Z_tr, y_tr).score(Z_te, y_te)

methods = {
    "PCA (2)": lambda: PCA(2).fit_transform(X),
    "LDA (2)": lambda: LDA(n_components=2).fit_transform(X, y), # supervised!
    "t-SNE": lambda: TSNE(2, perplexity=30, random_state=8).fit_transform(X),
    "UMAP": lambda: umap.UMAP(n_components=2, random_state=8).fit_transform(X),
}
print(f"{'method':<8} {'seconds':>8} {'2-D 15-NN acc':>14}")
for name, fn in methods.items():
    t0 = time.perf_counter()
    Z = fn()
    print(f"{'name':<8} {'time.perf_counter()-t0:>8.1f} {'knn_quality(Z, y):>14.3f}")

```

Output:

method	seconds	2-D 15-NN acc
PCA (2)	0.0	0.641
LDA (2)	0.0	0.687
t-SNE	6.5	0.976
UMAP	14.6	0.978

Listing 9 — Association rules from scratch: the lift lesson

Apriori's level-wise search using downward closure, then every rule scored. The teaching moment is in the table: beer → milk has confidence 0.80 but lift 1.00 — milk is simply everywhere, the rule is noise. beer → diapers, same confidence, lift 1.33 — a real association. Never read confidence without lift.

```

"""Listing 9: association rules from scratch -- support, confidence, lift on
baskets."""
from itertools import combinations

baskets = [
    {"bread", "milk"}, {"bread", "diapers", "beer", "eggs"},
    {"milk", "diapers", "beer", "cola"}, {"bread", "milk", "diapers", "beer"},
    {"bread", "milk", "diapers", "cola"}, {"bread", "milk"},
    {"milk", "diapers", "beer"}, {"bread", "cola"}, {"bread", "milk", "diapers"},
    {"milk", "beer"},
]
n = len(baskets)
def support(itemset):
    return sum(itemset <= b for b in baskets) / n

# Apriori level-wise search: an itemset can be frequent only if all subsets are
min_sup = 0.3
items = sorted({i for b in baskets for i in b})
L1 = [frozenset([i]) for i in items if support(frozenset([i])) >= min_sup]
L2 = [frozenset(c) for c in combinations(sorted({i for s in L1 for i in s}), 2)
      if support(frozenset(c)) >= min_sup]
print("frequent single items:", sorted(sorted(s)[0] for s in L1))
print("frequent pairs      :", [tuple(sorted(s)) for s in L2])

print(f"\n{'rule':<22} {'support':>8} {'confidence':>11} {'lift':>6}")
for pair in L2:
    for a in pair:
        A, B = frozenset([a]), pair - frozenset([a])
        sup, conf = support(pair), support(pair) / support(A)
        lift = conf / support(B)
        b = list(B)[0]
        print(f"{a:>8} -> {b:<10} {sup:>8.2f} {conf:>11.2f} {lift:>6.2f}")

```

Output:

```

frequent single items: ['beer', 'bread', 'cola', 'diapers', 'milk']
frequent pairs      : [('beer', 'diapers'), ('beer', 'milk'), ('bread', 'diapers'),
('bread', 'milk'), ('diapers', 'milk')]

```

rule	support	confidence	lift
beer -> diapers	0.40	0.80	1.33
diapers -> beer	0.40	0.67	1.33
beer -> milk	0.40	0.80	1.00
milk -> beer	0.40	0.50	1.00
bread -> diapers	0.40	0.57	0.95
diapers -> bread	0.40	0.67	0.95
bread -> milk	0.50	0.71	0.89
milk -> bread	0.50	0.62	0.89
milk -> diapers	0.50	0.62	1.04
diapers -> milk	0.50	0.83	1.04

Listing 10 — Anomaly detection: three detectors, scored against planted truth

An injection study: 950 normal points, 50 planted outliers, no labels used in fitting. All three detectors take a contamination/ ν knob — the chosen alarm rate — and precision/recall land accordingly. LOF and Isolation Forest lead here; One-Class SVM trails, true to its reputation as the fussiest.

```

"""Listing 10: anomaly detection -- Isolation Forest vs One-Class SVM vs LOF,
scored."""
import numpy as np
from sklearn.datasets import make_blobs
from sklearn.ensemble import IsolationForest
from sklearn.neighbors import LocalOutlierFactor
from sklearn.svm import OneClassSVM
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import precision_score, recall_score

rng = np.random.default_rng(9)
X_norm, _ = make_blobs(n_samples=950, centers=3, cluster_std=1.0, random_state=9)
X_anom = rng.uniform(X_norm.min(0) - 4, X_norm.max(0) + 4, (50, 2)) # scattered
outliers
X = StandardScaler().fit_transform(np.vstack([X_norm, X_anom]))
y = np.hstack([np.zeros(950), np.ones(50)]) # 1 = anomaly

detectors = {
    "IsolationForest": IsolationForest(contamination=0.05, random_state=9),
    "One-Class SVM": OneClassSVM(nu=0.05, gamma="scale"),
    "LOF": LocalOutlierFactor(n_neighbors=25, contamination=0.05),
}
print(f'{"detector":<16} {"precision":>10} {"recall":>7}')
for name, det in detectors.items():
    pred = det.fit_predict(X) # -1 = anomaly in sklearn convention
    flag = (pred == -1).astype(int)
    print(f'{"name":<16} {"precision_score(y, flag):>10.2f} {"recall_score(y,
flag):>7.2f}')

```

Output:

detector	precision	recall
IsolationForest	0.82	0.82
One-Class SVM	0.71	0.70
LOF	0.84	0.84

Pitfalls, comparisons and practical tips

The clustering algorithms, side by side:

	k-means	Hierarchical	DBSCAN/ HDBSCAN	GMM
Cluster definition	near a centroid	merge-tree cut	density-connected region	Gaussian component
Needs k	yes	no (cut later)	no	yes (BIC helps)

	k-means	Hierarchical	DBSCAN/ HDBSCAN	GMM
Shapes	spherical/ convex	linkage-dependent	arbitrary	ellipsoidal
Noise/outliers	dragged means	outlier-sensitive (except single)	native noise label	soft, but singularities
Assignment	hard	hard	hard + noise	soft (responsibilities)
Scale (n)	excellent (mini-batch)	$O(n^2)$ — tens of thousands	good (index- accelerated)	moderate
Determinism	init lottery	deterministic	deterministic-ish	init lottery
Key risk	wrong-shape fiction	wrong linkage	ϵ cliff / density variation	singularities, local maxima

Classic traps:

- **Clustering unscaled data.** Every method here except the tree-free ones runs on distances; one wide-range feature owns the geometry silently (Chapter 6's lesson, third appearance — it keeps appearing because people keep doing it).
- **Trusting the elbow.** Real elbows are mush; inertia comparisons across k are structurally rigged toward more clusters. Silhouette, gap statistic, BIC (GMM), stability under resampling — and the business question "how many segments can we act on?" — before declaring k .
- **Validating with the objective you optimized.** "Inertia is low" praises k -means for being k -means. Use *external* checks: downstream task lift, label agreement where partial labels exist (ARI, as the listings do), human inspection of samples, stability across resamples.
- **Reading t-SNE geometry.** Island sizes and gaps are artifacts; perplexity changes the story; noise makes clusters. Show multiple perplexities, never measure distances on the plot, never feed the coordinates to models.
- **PCA without centering — or on unstandardized mixed units.** Uncentered SVD finds the mean direction, not variance; unstandardized PCA elects the largest-unit feature as PC1. Also: variance $\neq$ relevance — the label-discriminative direction can hide in late PCs (LDA exists for that).
- **DBSCAN with one ϵ on varying densities.** Structurally impossible fit; the k -distance plot picks ϵ for *one* density regime only. HDBSCAN is the modern default answer.
- **GMM singularities.** A component collapsing onto a point sends likelihood to ∞ — floor the covariances (reg_covar), sanity-check fitted σ 's before believing a "great" likelihood.
- **Confidence without lift.** Listing 9's beer→milk: confidence 0.80, lift 1.00, information zero. Popular consequents make every rule look confident.

- **Anomaly contamination as gospel.** `contamination=0.05` defines a 5% alarm rate; it doesn't discover the true anomaly fraction. Set it from triage capacity and cost asymmetry, and validate by injection or labeled samples.
- **Forgetting the pipeline order.** Reduce-then-cluster (PCA→k-means, UMAP→HDBSCAN) changes results and their meaning; the reduction is part of the model, so tune and report it as such — and fit it on training data only when any downstream supervised step exists (Chapter 4's leakage rule applies to *unsupervised* preprocessing too).

Default reaches. Segmenting compact numeric data: k-means++ with silhouette-guided k. Unknown shapes, noise expected: HDBSCAN. Need soft memberships, densities, or generative scoring: GMM with BIC over k and covariance types. Need a taxonomy: agglomerative + dendrogram. Visualization: UMAP (and a PCA beside it for the linear-truth check). Anomalies at scale: Isolation Forest first.

Interview questions and answers

Q1. Walk through Lloyd's algorithm and prove it converges. To what does it converge?

Alternate: (1) assign each point to its nearest centroid; (2) move each centroid to its cluster's mean. Convergence: both steps weakly decrease inertia — assignment is optimal given centroids by definition of "nearest"; the mean minimizes summed squared distance given assignments (differentiate $\sum \|x_i - \mu\|^2$, set to zero, get $\mu = \text{mean}$). Inertia is bounded below and there are finitely many partitions, so the descent terminates. But only at a *local* minimum — which one depends on initialization (Listing 1: $1.24 \times$ inertia spread over 20 random inits). Hence restarts and k-means++. *Interviewers listen for: the two "each step is optimal given the other" arguments — that's the proof.*

Q2. What is k-means++ and why does it work?

An initialization scheme: pick the first centroid uniformly at random; pick each subsequent centroid from the data with probability proportional to $D(x)^2$ — squared distance to the nearest already-chosen centroid. Far-from-covered regions are likely to get the next seed, so seeds spread across the data's structure instead of clumping (random init's failure: two seeds in one true cluster, zero in another, and Lloyd's local descent can't recover). It carries a provable guarantee — expected inertia within $O(\log k)$ of optimal before Lloyd's even runs — and costs one pass per seed. sklearn's default for good reason.

Q3. Your stakeholder asks "how many customer segments are there?" — describe your actual workflow.

Triangulate, don't oracle: scale features; run k over a range with silhouette (peak) and inertia (elbow, held loosely — Listing 2 shows why silhouette is sturdier); check BIC with a GMM as a likelihood-honest second opinion; test stability (recluster on bootstrap resamples — real segments recur, artifacts don't); and crucially invert the question — segments are for *acting on*, so 4 actionable segments beat 11 statistically-pretty ones. Present 2-3 candidate resolutions with profiles, not one number. Saying "the data may be hierarchical — 2 super-segments containing 6 sub-segments, both true" signals maturity, since cluster count is scale-dependent.

Q4. Name k-means's implicit assumptions and what breaks when each fails.

Squared-Euclidean-to-centroid implies: spherical clusters (fails on stretched/anisotropic data — boundaries cut across the grain; Listing 3: ARI 0.430), similar sizes and densities (fails with one big diffuse + small tight clusters — splitting the big one buys more inertia; ARI 0.252), convexity (fails on moons/rings — no centroid represents a crescent; ARI 0.249), and comparable feature scales. Fixes map one-to-one: GMM with full covariance for anisotropy and unequal spread, DBSCAN/spectral for non-convexity, standardization for scale. The strong-answer close: k-means always returns k clusters, whether or not they exist — it reports its assumptions, not your data.

Q5. Define the silhouette score and its failure modes.

Per point: a = mean intra-cluster distance, b = mean distance to the nearest other cluster; $s = (b-a)/\max(a,b) \in [-1,1]$; report the mean (and per-cluster means — one bad cluster hides in a global average). Near 1: tight and separated; near 0: on a boundary; negative: likely misassigned. Failures: presumes compact geometry, so density-chain clusters (moons) score poorly even when perfect; prefers coarse resolutions when groups overlap (can peak at 2 super-clusters); $O(n^2)$ cost bites at scale (subsample); and comparing silhouettes across different distance metrics or preprocessing is meaningless. Use it to compare k under one fixed pipeline, not as an absolute quality certificate.

Q6. Agglomerative linkages — describe each and give a case where single linkage is the best choice and one where it's the worst.

Single: cluster distance = closest pair (chains, filament-friendly, noise-bridge-vulnerable). Complete: farthest pair (compact, outlier-sensitive). Average: mean pairwise. Ward: merge minimizing the increase in within-cluster variance (k-means's sibling; the default). Listing 4's dissociation is the answer template: on two moons single linkage is *perfect* (1.000 — crescents are chains) while ward is worst (0.162); on compact blobs with scattered noise, single chains through the gaps (0.571) while ward/average are near-perfect (0.993). Linkage = your prior about cluster shape; pick it from the geometry you expect.

Q7. How do you read a dendrogram, and how do you choose where to cut it?

Leaves are points; each merge is a horizontal join at a height equal to the inter-cluster distance at merge time; cutting at any height yields the clusters below the cut. Choose the cut at a large gap in merge heights — many cheap merges then a sudden expensive one means "these groups resisted merging" (Listing 4: distances 13.0 → 85.1; cut in the gap for $k=3$, recovering exact true sizes). Also legitimate: cut for a fixed k the business needs, or by minimum cluster size. Mention cophenetic correlation as the "does the tree respect the original distances?" check if asked how to validate the hierarchy itself.

Q8. Explain DBSCAN's core/border/noise taxonomy and its two parameters.

Core point: $\geq \text{min_samples}$ points within radius ϵ (density above threshold). Border: within ϵ of a core point but not core itself — joins the cluster, extends nothing. Noise: neither — labeled -1 , a first-class output. Clusters are maximal sets connected through core points (density-reachability chains). min_samples sets the density bar (rule of thumb $\geq d+1$, larger for noisy data); ϵ sets the scale and is the sensitive one — Listing 5's sweep goes shatter (34 fragments) → perfect → single blob across 0.05→0.4. Pick ϵ from the knee of the sorted k -distance plot; or skip the problem with HDBSCAN.

Q9. What problem does HDBSCAN solve that DBSCAN structurally cannot?

Clusters of different densities under one global ε : an ε tight enough for the dense cluster labels the sparse cluster noise; loose enough for the sparse one merges the dense pair. HDBSCAN examines all density levels at once — conceptually the full hierarchy of DBSCAN solutions over ε — and extracts clusters that persist across the widest range of densities (stability-based selection). Result: no ε ; `min_cluster_size` is the main knob and it's interpretable ("smallest group I care about"). Listing 5: tuned-DBSCAN quality (ARI 0.967) with nothing tuned. Bonus: it yields per-point membership strengths and an outlier score, and pairs famously with UMAP embeddings.

Q10. Derive the E and M steps for a GMM and state EM's convergence guarantee precisely.

E-step: responsibilities $r_{ij} = \pi_j N(\mathbf{x}_i | \mu_j, \Sigma_j) / \sum_l \pi_l N(\mathbf{x}_i | \mu_l, \Sigma_l)$ — Bayes' rule for "which component made this point", with parameters frozen. M-step: with $N_j = \sum_i r_{ij}$ — $\pi_j = N_j/n$, $\mu_j = \sum_i r_{ij} \mathbf{x}_i / N_j$, $\Sigma_j = \sum_i r_{ij} (\mathbf{x}_i - \mu_j)(\mathbf{x}_i - \mu_j)^T / N_j$ — Gaussian MLE with fractional counts. Guarantee: each full iteration cannot decrease the observed-data log-likelihood (the E-step builds a lower bound tight at current parameters via Jensen's inequality; the M-step maximizes it), so EM converges monotonically — to a local maximum or saddle, not necessarily global (Listing 6 asserts the monotonicity in code and watches $-7746 \rightarrow -2161$). Follow-ups to expect: singularities (floor Σ), init from k-means, restarts.

Q11. In what precise sense is k-means a special case of GMM?

Constrain the GMM: equal weights, shared spherical covariance εI , and take $\varepsilon \rightarrow 0$. The responsibilities $r_{ij} \propto \exp(-\|\mathbf{x}_i - \mu_j\|^2 / 2\varepsilon)$ then concentrate all mass on the nearest centroid — soft assignments harden into Lloyd's assignment step — and the M-step's weighted mean becomes the plain cluster mean, Lloyd's update step. So k-means is EM on a degenerate GMM with hard assignments and no covariance learning. The practical reading: choosing k-means over GMM is asserting spherical, equal-variance clusters and refusing uncertainty — sometimes exactly right (speed, simplicity), but it should be a choice, not a default (Listing 3 shows the price when it's wrong).

Q12. Why can't you fit a GMM by just differentiating the log-likelihood and using gradient descent?

You can, actually — the objection is subtler and stating it well scores points. The log-likelihood $\log \sum_j \pi_j N(\mathbf{x}|\theta_j)$ is differentiable; but the sum inside the log couples all components (no per-component decomposition), constraints must be maintained (π on the simplex, Σ positive-definite — requiring reparameterization), and the surface has the same local maxima *plus* the singularity problem in raw form. EM handles all of this structurally: the E-step decouples the components (given responsibilities, each component's M-step is an independent, closed-form, constraint-respecting weighted MLE), no learning rate exists, and monotone improvement is guaranteed. Gradient/second-order methods do get used (and can converge faster near optima); EM wins on robustness, simplicity, and closed-form steps.

Q13. Derive PCA as a variance-maximization problem. Why do eigenvectors appear?

Seek unit vector $\mathbf{w}$ maximizing the variance of the projection: $\text{Var}(X\mathbf{w}) = \mathbf{w}^T \Sigma \mathbf{w}$ with Σ the covariance matrix. Lagrangian for the constraint $\|\mathbf{w}\|=1$: $\mathbf{w}^T \Sigma \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{w} - 1)$; stationarity gives $\Sigma \mathbf{w} = \lambda \mathbf{w}$ — an eigenvector equation. The variance achieved is $\mathbf{w}^T \Sigma \mathbf{w} = \lambda$, so the maximizer is the eigenvector with the largest eigenvalue; subsequent components repeat the argument orthogonal to those already chosen. Equivalent view: minimizing squared reconstruction error yields the same subspace (Pythagoras splits total variance into kept + lost). In practice compute by SVD of centered data — never form Σ explicitly for conditioning reasons (Listing 7 matches sklearn exactly this way).

Q14. Why must data be centered before PCA, and when should it also be standardized?

PCA analyzes the covariance structure — spread around the mean. Uncentered, the "first component" of X points at the data's mean (the dominant direction of raw second moments), not along its variance; results are geometrically meaningless. Standardize (unit variance per feature) when features have different units or scales: variance is unit-dependent, and a feature in centimeters has 10^4 times the variance of the same feature in meters — raw-scale PCA elects loud-unit features as PCs. Equivalent statement: standardized PCA = eigendecomposition of the correlation matrix. Keep raw scales only when units are shared and magnitude differences are genuinely meaningful (e.g. spectra, pixel intensities).

Q15. PCA vs LDA — objectives, constraints, when each wins.

PCA: unsupervised, finds directions of maximal total variance; up to d components; ignores labels entirely. LDA: supervised, finds directions maximizing between-class scatter over within-class scatter; at most ($\text{classes} - 1$) components (the between-class scatter matrix's rank). When labels exist and the goal is class structure, LDA's axes are chosen for exactly that (Listing 8: 0.687 vs 0.641 in 2-D on digits); PCA can bury the discriminative direction in a late component since variance $\neq$ relevance. PCA wins when there are no labels, when you need more components than classes allow, as generic decorrelation/compression, or when LDA's Gaussian-equal-covariance assumptions are badly violated. Common pipeline: PCA to denoise, then LDA.

Q16. Your colleague measures the distance between two clusters on a t-SNE plot to argue they're "more different" than another pair. Respond.

The measurement is invalid. t-SNE preserves local neighborhoods and deliberately sacrifices global geometry: inter-cluster distances and cluster sizes in the embedding are artifacts of the optimization (the heavy-tailed low-D kernel and per-point bandwidths equalize densities and repel non-neighbors indiscriminately). The same data at different perplexities rearranges the islands. What the plot does support: points in the same island are mutual neighbors in high-D. For the colleague's actual question, compute distances in the original (or PCA) space, or use a method with better global fidelity (UMAP is better but not exempt) and confirm quantitatively. Listing 8's framing helps: t-SNE's 0.976 is *neighborhood* preservation, nothing more.

Q17. Compare t-SNE and UMAP for a practitioner.

Both build neighbor graphs in high-D and lay them out in low-D; differences that matter: UMAP is typically faster at scale, preserves global structure somewhat better, and — decisive for production — has a transform() for new points, while classic t-SNE must re-fit (out-of-sample extensions exist but aren't standard). t-SNE's perplexity $\leftrightarrow$ UMAP's n_neighbors (local-global dial); UMAP adds min_dist (visual packing). Both: stochastic (seed-dependent layouts), axes meaningless, densities distorted, "clusters" appear in noise. Listing 8: equal neighborhood quality (0.976 vs 0.978) on digits. Default: UMAP for routine work and pipelines; t-SNE remains standard in some fields (single-cell bio) and for small careful visualizations.

Q18. Define support, confidence, and lift. Why is high confidence alone misleading?

For rule $A \rightarrow B$ over baskets: support = $P(A, B)$ — pattern frequency; confidence = $P(B|A)$ — implication reliability; lift = $P(A, B)/(P(A)P(B))$ — co-occurrence relative to independence. Confidence inherits the consequent's base rate: if B is in 80% of baskets, almost *any* A gives confidence ≈ 0.8 (Listing 9: beer $\rightarrow$ milk, confidence 0.80, lift 1.00 — pure base rate; beer $\rightarrow$ diapers, same confidence, lift 1.33 — real signal). Lift corrects by dividing out $P(B)$: >1 complements, 1 independent, <1 substitutes. Caveats worth adding: lift is symmetric (loses direction), unstable at tiny supports (screen by support first), and none of the three implies causation.

Q19. Explain the Apriori property and how FP-Growth improves on Apriori.

Apriori (downward closure): every subset of a frequent itemset is frequent — equivalently, adding items never raises support. This prunes the exponential itemset lattice: candidate k-itemsets need only be built from frequent (k-1)-itemsets, and any candidate with an infrequent subset is dead on arrival (Listing 9's level-wise construction). Apriori's costs: multiple database scans and huge candidate sets at low support. FP-Growth removes candidate generation: compress transactions into an FP-tree (prefix tree with counts; frequent items sorted so prefixes share), then recursively mine conditional subtrees — two scans total, typically much faster on large sparse data. Same output, different search.

Q20. How does Isolation Forest decide something is anomalous, and why is it the industrial default?

Inverted logic: instead of modeling "normal", it measures how easy a point is to *isolate* — grow trees with random feature/threshold splits; a point's anomaly score comes from its average path length to isolation across trees (outliers strand alone in few splits; inliers need many). Default-worthy properties: $O(n \log n)$ with small constants; no distance computations (no scaling sensitivity, tolerates high dimension and irrelevant features better than KNN-family methods); handles mixed magnitudes; trivially parallel; subsampled trees make it robust. Weaknesses: axis-aligned splits miss some structured anomalies; scores need calibration to costs (contamination = chosen alarm rate — Listing 10). Interview bonus: it's an ensemble (Chapter 7) whose members are deliberately *random*, not accurate — diversity without skill, because the signal is structural.

Q21. LOF vs Isolation Forest — when does the local method matter?

LOF scores each point by the ratio of its neighbors' local density to its own — anomalous means "sparse relative to *its own neighborhood*". That relativity is the point: a reading at the edge of a tight cluster can be globally unremarkable (denser than the sparse cluster's core) yet locally anomalous; global methods (Isolation Forest, GMM likelihood) miss it, LOF catches it. Costs: KNN machinery — $O(n^2)$ -ish without indexes, scale-sensitive, curse-of-dimensionality-fragile, and the classic version doesn't score unseen points (novelty=True refits the contract). Practical rule: Isolation Forest first for scale and robustness; LOF when clusters of very different densities are known to exist and local context defines "anomalous" (Listing 10: 0.84/0.84 — best in that geometry).

Q22. How do autoencoders detect anomalies, and what silently breaks the approach?

Train a bottlenecked encoder-decoder to reconstruct (mostly normal) data; the bottleneck forces learning the normal manifold, so unseen anomalies reconstruct badly — reconstruction error is the anomaly score (details in Chapter 18). Silent breakers: anomalies contaminating training (the AE learns to reconstruct them too — clean or robustify the training set); over-capacity (a too-wide/deep AE reconstructs *everything*, including anomalies — the bottleneck must actually bottleneck); anomalies lying *on* the normal manifold (fraudulent transactions crafted to look typical — reconstruction can't see intent); and threshold drift as normal behavior evolves (monitor the error distribution, Chapter 27). Fair summary: powerful for high-dimensional structured data (images, sequences) where classical detectors' distances fail; overkill for 10 tabular features.

Q23. How do you evaluate a clustering when no labels exist — and when partial labels exist?

No labels: internal metrics (silhouette, Davies-Bouldin) — knowing they reward the geometry the algorithm already optimized; stability — recluster under resampling/perturbation, measure agreement (ARI between runs), since real structure recurs; downstream utility — does adding cluster-ID as a feature lift a supervised task, do segments move a business metric in A/B; and human triage of samples per cluster. Partial labels: external indices — ARI (chance-corrected pair agreement; the listings' metric), NMI (information-theoretic), purity — computed on the labeled subset. The interview frame: internal metrics are sanity checks, stability is evidence, downstream utility is proof; and any evaluation must hold the preprocessing/distance fixed or it compares pipelines, not clusterings.

Q24. Why does high dimensionality hurt clustering, and what are the standard mitigations?

Distance concentration (Chapter 4): as d grows with independent-ish features, all pairwise distances converge to the same value, so "nearest" and "densest" lose contrast — k-means boundaries become arbitrary, DBSCAN finds one blob or all noise (density thresholds are distance-based), silhouettes flatten. Mitigations: reduce first — PCA to the elbow of explained variance, or UMAP for nonlinear structure (with the caveat that the reduction now shapes the answer); feature-select with domain knowledge; use cosine distance on sparse/directional data (text — Chapter 19); subspace/spectral clustering when clusters live in different low-D subspaces; and for mixtures, restrict covariances (diagonal) so parameters don't explode. The honest note: if informative structure is confined to a few dimensions, finding *those* is the clustering problem.

Q25. Design the unsupervised half of a fraud pipeline: you have 10M transactions, no labels yet, and analysts who can review 200 cases/day.

Scope by the review budget: 200/day over 10M/period fixes the alarm rate — that's the contamination knob, chosen not discovered (Listing 10's lesson). Detector: Isolation Forest on engineered features (amounts, velocities, novelty counters — Chapter 9) for scale; add a GMM/density score per customer segment (cluster first — behavior is heterogeneous, and "anomalous for a student account" $\neq$ "for a business account": LOF's local logic at the segment level). Rank, don't threshold: send the top-200 to analysts daily. Their verdicts become labels — the pipeline's real product — feeding a supervised model (Chapter 7's boosted trees) that gradually takes over; keep the unsupervised layer for novel-pattern detection forever. Monitor score drift (Chapter 27). Every design choice here traces to a chapter concept, which is why the question gets asked.

Q26. A GMM fit reports spectacular log-likelihood, and one component has variance 1e-9. Diagnose.

A singularity, not a discovery: a component has collapsed onto one point (or a few duplicates); as $\sigma \rightarrow 0$ that point's density $\rightarrow \infty$, so the likelihood diverges — the "spectacular" number is the pathology itself. The MLE for mixtures is technically unbounded; EM found the degenerate direction. Fixes: floor the covariance (reg_covar), use MAP-EM with a prior (inverse-Wishart shrinkage — Bayesian regularization, Chapter 1's MAP again), remove duplicate points, reduce k, restart from different inits and prefer solutions with sane σ 's. Then re-select k by BIC computed only over non-degenerate fits. Recognizing "likelihood too good = broken" is the point of the question.

Q27. Implement one k-means assignment step in two lines of NumPy, given X (n×d) and centers C (k×d).

`d2 = ((X[:, None, :] - C[None, :, :]) ** 2).sum(-1)` — broadcasting to an (n, k) matrix of squared distances — then `labels = d2.argmin(1)`. Volunteer the production notes: the broadcast materializes $n \times k \times d$ — for large n use the expansion $\|x\|^2 = 2XC^T + \|c\|^2$ to stay at $n \times k$ (`d2 = (X**2).sum(1)[:, None] - 2*X@C.T + (C**2).sum(1)`); skip the square root (argmin is monotone-invariant); and this exact computation is the inner loop of vector quantization and IVF index training in vector databases (Chapter 24).

Q28. When would you deliberately choose plain k-means over every fancier option in this chapter?

When its assumptions plausibly hold (roundish, similar-scale groups in scaled numeric features) or when its *role* doesn't require them to: as a quantizer (codebooks, IVF cells, minibatch-friendly embedding compression — correctness is "low distortion", not "true clusters"); at scales where $O(n^2)$ methods and full-covariance GMMs are unaffordable; when determinism-enough (k-means++ + n_init) and five-line explainability matter for handoff; as the init for GMM/EM; or as the honest baseline every fancier clustering must beat (Chapter 7's baseline discipline, unsupervised edition). The disciplined phrasing: k-means is a fast biased estimator of cluster structure — ideal when speed matters and the bias is known to be tolerable.

Chapter 9: Feature Engineering & Data Preprocessing

Models get the glory; features decide the outcome. The same gradient-boosted trees that win Kaggle competitions lose them when fed raw columns, and most production wins attributed to "a better model" are actually a better representation of the same data. Interviewers know this, which is why feature engineering questions are less about memorizing transformations and more about *judgment under leakage risk*: nearly every preprocessing step — imputation, encoding, scaling, selection, resampling — involves fitting something to data, and fitting it to the wrong data (test rows, future rows, the labels themselves) manufactures offline scores that evaporate in production. Chapter 4 introduced leakage as a concept; this chapter is where it becomes muscle memory, because feature engineering is where leakage actually happens.

The chapter walks the outline's territory in order: missing values and the mechanism taxonomy (MCAR/MAR/MNAR) that decides which imputation is even valid; categorical encoding from one-hot through target encoding to learned embeddings, with the naive-target-encoding leak staged and measured (+8 points of pure fiction in Listing 2); scaling and why the choice of scaler is a statement about your outliers; feature selection's three families (filter, wrapper, embedded) judged against planted ground truth; class imbalance, where the honest experiment shows resampling is mostly *not* about ranking metrics — and SMOTE-before-split fabricates a 0.999 PR-AUC; outlier detection and treatment, including why univariate rules miss exactly the outliers that do the most damage; the classical nonlinearity-buying transforms (log, binning, interactions), each shown rescuing a linear model from a specific failure; and the high-cardinality problem, where one-hot, hashing, frequency, and smoothed target encoding fight it out on a 900-category feature. Throughout, one rule organizes everything: **fit every transform on training data only, inside the cross-validation loop, ideally inside a Pipeline** — and every listing that violates it does so deliberately, with the damage measured.

Missing values and imputation

Before choosing an imputation strategy, ask *why* the values are missing — the answer determines what is statistically legitimate. The taxonomy (Rubin's) has three levels.

MCAR (missing completely at random): missingness is independent of everything — a sensor dropped packets. Any reasonable imputation is unbiased; you lose only efficiency.

MAR (missing at random): missingness depends on *observed* columns — older customers skip the income field, and age is recorded. Model-based imputation that conditions on the observed columns (iterative, KNN) can recover the structure. **MNAR**

(missing not at random): missingness depends on the *missing value itself* or on

unobserved facts — high earners decline to state income, defaulters hide debts. No imputation computed from observed data can be unbiased, because the observed distribution of that column is systematically censored. What *does* work under MNAR is refusing to pretend: **the missingness itself is a feature**. Listing 1 stages this: when holes are punched in a label-dependent way, mean/median/KNN/iterative imputation all cluster around 0.82–0.83 accuracy, while simply appending a binary missing-indicator column jumps to 0.862 — the indicator *is* the signal — and a model that handles NaN natively hits 0.923.

The strategies, in escalating sophistication. **Deletion** (drop rows or columns): defensible only under MCAR with few holes; under MAR/MNAR it biases the sample toward the kind of rows that answer questions. **Mean/median imputation**: fills with a constant; median resists skew and outliers. Cheap, but it shrinks the column's variance, distorts its correlations with everything else (imputed points sit on a horizontal line in any scatter plot), and under MNAR fills censored holes with the average of the *uncensored* — exactly wrong. **KNN imputation**: fill from the k nearest rows in the observed columns — respects local structure, costs $O(n^2)$ distances, inherits every scaling caveat of Chapter 6. **Iterative (model-based) imputation** (MICE-family): round-robin regress each incomplete column on all others, iterate to convergence — the strongest general-purpose option under MAR, and Listing 1's MCAR winner among imputers (0.809 vs mean's 0.790). **Missing-indicator augmentation**: add the mask as a column *alongside* any fill — nearly free, and the only honest play under suspected MNAR. **Native handling**: modern gradient boosting (XGBoost, LightGBM, HistGradientBoosting) treats NaN as a value, learning per-split which side missing rows go — Listing 1's overall winner in both regimes (0.877 MCAR, 0.923 MNAR), which is why "impute nothing, use HGB" is a legitimate answer, not an evasion.

Two leakage traps close the section. First, **fit imputers on training data only**: a mean computed over all rows includes test rows — mild in practice, but a pipeline violation that generalizes badly to deployment, where the "test set" hasn't arrived yet. Second, **impute after splitting, inside the CV loop** — an imputer fit on the full training set before cross-validation leaks each validation fold into the statistics used to fill it. Both are automatic if imputation lives in a `sklearn Pipeline`; both are hand-rolled bugs otherwise. Interview closer: never silently impute the target — rows with missing y are unlabeled data, not imputation candidates.

Encoding categorical features

Models eat numbers; categories must become numbers without acquiring false arithmetic. The choice of encoding is a choice about *what structure you assert*, and asserting wrong structure is the classic junior mistake.

Label / integer encoding maps categories to arbitrary integers (red=0, green=1, blue=2). For tree models this is merely awkward (a tree can carve $\{0,2\}$ vs $\{1\}$ in two

splits); for linear models and anything distance-based it is wrong — it asserts blue = 2×green, an order and a magnitude that don't exist. Reserve integer codes for **ordinal encoding**, where the order is real (S < M < L < XL; ratings; education levels) and you assign integers *in that order* — asserting equal spacing, which is itself an assumption to flag out loud.

One-hot encoding creates one binary column per category — the default for nominal categories with modest cardinality. Costs and caveats: dimensionality grows linearly with cardinality (the high-cardinality section below is the escape plan); the columns are perfectly multicollinear with the intercept (the "dummy variable trap" — drop one column for unregularized linear models; keep all for regularized or tree models, where it's harmless and drop-one actually distorts regularization symmetry); trees fragment — a 900-category one-hot gives each split a single yes/no on a single rare category, so depth is spent enumerating instead of generalizing; and unseen test categories need an explicit policy (`handle_unknown="ignore"` gives the all-zeros row — decide, don't discover).

Target encoding (mean encoding) replaces each category with the mean of the target over that category's rows — one dense column carrying exactly the signal a model wants. It is simultaneously the most powerful classical encoding and the most dangerous, because the encoding is *computed from labels*. Done naively — category means over all rows, including the row being encoded and the test set — each row's own label contaminates its feature. Listing 2 makes the fraud precise: a 2,000-category column of **pure noise** (three rows per category, zero true signal) plus one real feature. Honest model on the real feature: test 0.688. Add naively-encoded noise categories: test **0.780** — the model looks 8 points better *because test labels leaked into test features through the category means*. Out-of-fold encoding (each row encoded using means computed on other folds; test encoded with train-only means) returns 0.695 — the truth, give or take fold noise. The production forms add **smoothing**: shrink each category's mean toward the global mean in proportion to its count, $\tilde{y}_c = \frac{n_c \bar{y}_c + m \bar{y}}{n_c + m}$, so a 3-row category contributes mostly prior — this is an empirical-Bayes move, and it is what `category_encoders.TargetEncoder` and CatBoost's ordered target statistics implement (CatBoost encodes each row using only rows *before it* in a random permutation — leakage-proof by construction).

Embeddings are the deep-learning answer (Chapter 12 builds them): map each category to a learned dense vector, trained end-to-end with the model. They shine when cardinality is huge *and* categories have latent similarity structure worth learning (users, products, words), and they transfer across tasks. For tabular work with classical models, they're usually overkill — but "entity embeddings of categorical variables" is a name worth dropping, and the conceptual bridge (an embedding layer *is* a one-hot times a learned matrix) is a favorite interview connection. Cheap intermediate options that need no labels: **frequency encoding** (category → its count; surprisingly effective when popularity correlates with the target) and **hashing** (below, in high-cardinality).

Scaling and normalization

Scaling exists because many algorithms read *distance* or *gradient* meaning into raw feature magnitudes. Who cares: KNN, k-means, SVMs (kernel distances), PCA (variance directions), regularized linear models (a penalty $\lambda\|w\|^2$ taxes coefficients of small-scale features harder — Chapter 5), and neural networks (conditioning of the loss surface). Who doesn't: trees and tree ensembles — splits are order-based, and any monotone transform of a feature leaves its split structure unchanged. Listing 3 supplies the disaster number: one feature on a $500\times$ scale drops unscaled KNN to 0.579 (barely above chance) while any scaler restores ~ 0.88 — the big feature owned every distance, and five informative features were reduced to spectators.

The three standard scalers are three different claims about your data. **Standardization** (z-score): subtract mean, divide by standard deviation — the default; produces mean 0, variance 1; assumes no gross outliers, because both mean and especially σ are outlier-sensitive. **Min-max normalization**: map to $[0, 1]$ by the observed range — bounded output (nice for NNs and image-style inputs), but the *range* is the most outlier-fragile statistic there is: one absurd value compresses everyone else into a sliver. **Robust scaling**: subtract median, divide by IQR — immune to anything outside the middle 50%. Listing 3 quantifies the choice: with 2% gross outliers planted in the big feature, standard and min-max scaling leave the inliers' IQR at 0.031 and 0.0036 respectively — the outliers inflated the divisor, crushing the working scale of the real data $\sim 50\text{--}300\times$ below the other features' — while robust scaling holds $\text{IQR} = 1.0$ by construction. Accuracy differences stay modest here because KNN with 15 neighbors is forgiving, but the mechanism is the interview point: *a scaler fit on contaminated statistics silently re-weights features*.

Housekeeping that separates candidates: **fit the scaler on train, transform test with train's parameters** — scaling with test-set statistics is leakage (mild, but it's the same pipeline discipline as everything else); scale *inside* CV via Pipeline; sparse data wants `MaxAbsScaler` or no centering (subtracting a mean densifies sparsity); and **normalization** in the sklearn sense (`Normalizer`) is a different operation entirely — it rescales each *row* to unit norm (for cosine-similarity text vectors), not each column, and confusing the two is a small but telling error.

Feature selection

Feature selection buys three things: generalization (noise features are variance a model must overcome), economics (fewer features = cheaper inference, faster training, less to maintain in a feature store), and interpretability. The three families differ in what they consult. **Filter methods** score each feature against the target *without any model*: correlation, ANOVA F-statistic, mutual information (catches nonlinear dependence that correlation misses), chi-square for categorical pairs. Fast, scalable, embarrassingly parallel — and blind to interactions (a feature useless alone but decisive in combination

scores zero) and to redundancy (ten copies of one signal all score high; the filter happily keeps all ten). **Wrapper methods** search feature subsets by actually training and validating the model: forward selection, backward elimination, and **RFE** (recursive feature elimination: fit, drop the weakest by coefficient or importance, refit, repeat). Interaction-aware, model-specific — and expensive, $O(\text{model fits} \times \text{subsets explored})$, plus a real overfitting risk: enough subset search on one validation set fits the validation set. **Embedded methods** get selection as a fitting side effect: **L1/Lasso** drives coefficients exactly to zero (Chapter 5's geometry); tree-ensemble **feature importances** rank features by total impurity reduction or, better, **permutation importance** (shuffle a column, measure the score drop — model-agnostic, computed on validation data, immune to impurity importance's bias toward high-cardinality features).

Listing 4 judges all three against planted truth — 30 features: 6 informative, 6 redundant linear copies, 18 pure noise. Mutual-information filter, RFE, and random-forest importances each recover 8/8 real features in their top-8. The instructive miss is L1 at strong regularization: 6/10 real — it zeroed some *redundant* copies (correct behavior! Lasso picks one of a correlated group and discards the rest, Chapter 5's known instability) but let four noise features through. The lesson isn't "L1 is bad"; it's that each family has a characteristic failure signature — filters keep redundant copies, Lasso arbitrarily picks among them, importances split credit across them — and naming the signature is what the interviewer wants.

The section's leakage trap outranks the methods: **selecting features on the full dataset before cross-validation is leakage**, and it is the single most common way practitioners fabricate results on high-dimensional data. Select on all rows (test included), then CV: the selected features were chosen partly for their correlation with test labels, and with enough candidate features (genomics: 20,000 genes, 100 patients) you can reach near-perfect "cross-validated" accuracy on pure noise. The fix is mechanical: selection is a fitted transform, so it lives inside the Pipeline, refit per fold. Same rule, same reason, third appearance — it will not be the last.

Handling imbalanced data

With a 3% positive class, accuracy is a broken compass (97% by predicting "no" always — Chapter 10 dwells on metric choice; here we use PR-AUC). The menu: **class weights** (reweight the loss so each positive counts $1/\pi$ as much — no data change, one line: `class_weight="balanced"`); **random undersampling** (throw away majority rows until balanced — fast, loses information); **random oversampling** (duplicate minority rows — no information gained, overfitting risk as trees memorize repeated points); **SMOTE** (synthesize minority points by interpolating between minority nearest neighbors — the famous one); and **threshold moving** (train as-is, then place the decision threshold where the precision/recall trade-off belongs — often the rightest and least-used answer, because the default 0.5 threshold was never sacred).

Listing 5's honest experiment deflates the folklore. On 3.9% positives with a random forest, judged by PR-AUC — a *ranking* metric: baseline 0.768, class weights 0.748, random oversampling 0.749, SMOTE 0.708, undersampling 0.670. **Nothing beats doing nothing.** The mechanism: resampling and reweighting change the *threshold geometry* and probability calibration — they shift where scores concentrate — but a ranking metric only cares about the *order* of scores, and the baseline forest already orders fine; meanwhile undersampling discarded 92% of the majority class (real information, real cost) and SMOTE's linear interpolations manufacture points on segments between true positives, which in a 12-dimensional space with class overlap plants synthetic positives in majority territory. The honest summary for interviews: *imbalance handling mostly buys you a usable default threshold and calibrated-ish hard predictions; if you evaluate with ranking metrics and tune the threshold downstream, the baseline plus threshold moving is very hard to beat.* Class weights remain the cheapest defensible default when hard predictions are needed; focal loss (Chapter 14, born in object detection where imbalance is 1000:1) is the deep-learning-native variant — it down-weights already-easy examples so the loss concentrates on the hard minority.

Then the trap, which is worth more than the menu: **resample inside the CV loop, after splitting, training folds only** — never before. SMOTE before the split interpolates between minority points that will land in *different* sides of the split, so synthetic training points are literally blends of test rows; and the balanced "test set" no longer resembles the 3% world the model will face. Listing 5 measures the fantasy: PR-AUC **0.999** for SMOTE-before-split vs 0.708 done honestly. Any pipeline showing a suspiciously beautiful number on imbalanced data should be audited for exactly this bug first — it is epidemic in published applied-ML work. (`imblearn.pipeline.Pipeline` exists precisely so samplers can sit inside CV and apply only to training folds.)

Outlier detection and treatment

An outlier is a point that some model of "normal" says doesn't belong; the operational questions are *which model of normal* and *what to do about it*. Detection rules escalate in dimension-awareness. **Univariate rules:** z-score ($|z| > 3$ under approximate normality — but mean and σ are themselves outlier-inflated, so gross outliers mask each other; the MAD-based robust z-score fixes this) and **Tukey fences** (outside $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ — the boxplot whiskers; robust and distribution-lite). Both examine one column at a time. **Multivariate detectors** examine joint structure: Mahalanobis distance (elliptical normality), and Chapter 8's trio — Isolation Forest, LOF, one-class SVM — which need no distributional story at all.

Listing 6 stages why the distinction matters: 2% of rows corrupted at **high leverage** — extreme x paired with wrong y . Each corrupted value is unremarkable *marginally* (an x of 8 is just a big x ; a y near 0 is a typical y), so the univariate rules collapse: z-score on y flags 1/40 corrupted rows, IQR fences 2/40. Isolation Forest, seeing the *joint* (x, y) — where these points sit far from the data's line — catches 32/40. This is the general truth:

the outliers that damage models most are precisely the ones only visible jointly, because leverage is a relationship between columns, not a property of one.

The same listing measures treatment. The corruption drags OLS's slope from the true 2.0 to 0.897 — high-leverage points grab the regression line with force proportional to their distance from the x-center (Chapter 5's leverage). Options, with their measured recoveries: **winsorizing** x at the 1st/99th percentiles (clip, don't delete) manages only 0.941 — it pulls the extreme x's inward but *keeps the wrong y's*, now planted at moderate leverage: winsorizing treats a value, not a row, and a corrupted row stays corrupted. **Deleting flagged rows** recovers 1.668; **Huber loss** — Chapter 5's robust regression, quadratic near zero, linear in the tails, so gross residuals get bounded influence — recovers 1.759 with no detection step at all. Neither fully reaches 2.0 (the detector missed 8 corrupted rows; honesty in numbers). The judgment call interviewers probe: deletion is right when flagged points are *errors* (sensor glitches, joined-wrong rows) and dangerous when they're *real but rare* (a fraud model deleting its own fraud cases; the largest customer in a revenue model). When you can't know, prefer treatments that bound influence without destroying data: robust losses, robust scalers, rank/log transforms — and always ask where the outliers came from before choosing.

Binning, log transforms, and interaction features

These are the classical tools for buying nonlinearity without leaving the linear-model world — each rescues a specific, nameable failure, and Listing 7 measures all three. **Log transforms** fix multiplicative structure and right skew: when the truth is log-linear (elasticities, dose-response, anything where effects compound), a linear fit on the raw skewed feature manages R^2 0.603 while the same model on $\log(x)$ reaches 0.973. Rules: log needs positive inputs ($\log1p$ for counts with zeros; signed-log or Yeo-Johnson for real-valued); Box-Cox generalizes to a *fitted* power transform; and the workhorse justification — compressing a heavy right tail so the model isn't dominated by its largest values — applies to incomes, prices, view counts, and almost everything human-generated.

Binning (discretization) converts a numeric feature into interval indicators, letting a linear model fit a step function: on a U-shaped truth, raw-x linear regression achieves R^2 -0.001 — *literally nothing*, since the U's two arms cancel any single slope — while 10 equal-width one-hot bins reach 0.889. Costs: information loss inside bins, arbitrary edges (quantile bins beat equal-width on skewed data), and discontinuities at boundaries; splines are the smooth upgrade. Trees bin implicitly, which is one reason trees need none of this section. **Interaction features** multiply columns so an additive model can see joint effects: on an XOR-style truth (class = whether x_1 and x_2 agree in sign), logistic regression on $[x_1, x_2]$ scores 0.493 — a coin flip, because neither feature *alone* carries any signal, the textbook definition of a pure interaction — while adding the single engineered column $x_1 \cdot x_2$ scores 0.998. `PolynomialFeatures` automates degree-2 expansion at quadratic column cost; domain knowledge (price $\times$ promotion, dose $\times$ age) beats blind expansion; and this is exactly the gap kernels (Chapter 6) and deep networks (Chapter 12) close automatically — a connection worth saying aloud.

High-cardinality categoricals

Zip codes, user IDs, merchant IDs, URLs: thousands to millions of categories, long-tailed, with new values arriving daily. One-hot is arithmetic suicide at this scale (a million columns; each tree split interrogating one rare category), so the alternatives trade off density, leakage risk, and vocabulary maintenance. Listing 8's testbed — 900 categories, ~9 rows each, every category carrying a real effect — keeps the comparison honest.

One-hot (900 columns): 0.727 — workable at 900, dead at 900,000. **Feature hashing** (32 columns): hash each category string into a fixed number of buckets, accept collisions — 0.625, the price of ~28 categories sharing each bucket's identity. The trade is explicit: no vocabulary, no state, no unseen-category problem (new strings hash like old ones), memory fixed in advance — bought with collision noise that grows as buckets shrink. The tool of choice for streaming/online systems and million-way text features. **Frequency encoding** (1 column): 0.619 — here popularity carries no signal by construction, so it merely separates categories into rough count-buckets; on real data where popularity correlates with the target (popular products, active users) it routinely embarrasses fancier options at zero leakage risk. **Smoothed target encoding** (1 column): 0.734 — the winner, matching one-hot's information in 1/900th the width, *because* smoothing shrinks each ~9-row category's mean toward the prior instead of trusting it raw. The ranking to internalize: when categories genuinely predict the target, properly-regularized target encoding is the classical best-in-class; hashing when the vocabulary is unbounded or memory is fixed; frequency as the free lunch to always try; embeddings (Chapter 12) when there's enough data to learn structure; and the **rare-category bucket** (collapse everything below a count threshold into "other") composes with all of the above. Every one of these must handle the unseen-category case explicitly — global mean for target encoding, zero for frequency, its hash for hashing — because in production the categories you never saw are guaranteed to arrive.

Code implementations

Every listing was executed as shown; outputs are real. The set implements the chapter: the imputation bake-off across missingness mechanisms, the naive-target-encoding leak measured against out-of-fold honesty, the unscaled-KNN disaster and the outlier-vs-scaler study, the three selection families judged against planted truth, the imbalance menu deflated by a ranking metric plus the SMOTE-before-split fantasy, leverage outliers invisible to univariate rules, the three nonlinearity-buying transforms each rescuing a linear model, and the four-way high-cardinality shootout.

Listing 1 — Imputation strategies, and why the missingness mechanism decides

Under MCAR the imputers are within a point of each other and iterative wins the imputer race. Under MNAR — holes punched by the label — the missing-indicator column beats every clever fill, and native-NaN gradient boosting beats everything twice.

```

"""Listing 1: imputation strategies -- and why the missingness mechanism decides."""
import warnings
warnings.filterwarnings("ignore")
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.experimental import enable_iterative_imputer # noqa
from sklearn.impute import SimpleImputer, KNNImputer, IterativeImputer
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

rng = np.random.default_rng(0)
X, y = make_classification(n_samples=3000, n_features=8, n_informative=6,
random_state=0)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.33, random_state=0)

def punch_holes(X, y, mnar):
    """MCAR: random 25% holes. MNAR: holes depend on the label -- missingness IS
    signal."""
    Xm = X.copy()
    if mnar:
        p = np.where(y == 1, 0.45, 0.05) # positives lose feature 0 often
        mask = rng.random(len(X)) < p
        Xm[mask, 0] = np.nan
    else:
        mask = rng.random(X.shape) < 0.25
        Xm[mask] = np.nan
    return Xm

for regime, mnar in [("MCAR (random holes)", False), ("MNAR (label-driven)", True)]:
    Xm_tr, Xm_te = punch_holes(X_tr, y_tr, mnar), punch_holes(X_te, y_te, mnar)
    strategies = {
        "mean": SimpleImputer(strategy="mean"),
        "median": SimpleImputer(strategy="median"),
        "KNN": KNNImputer(n_neighbors=5),
        "iterative": IterativeImputer(max_iter=10, random_state=0),
        "mean+indicator": SimpleImputer(strategy="mean", add_indicator=True),
    }
    print(f"\n{regime}")
    for name, imp in strategies.items():
        pipe = make_pipeline(imp, StandardScaler(), LogisticRegression(max_iter=2000))
        pipe.fit(Xm_tr, y_tr)
        print(f"    {name:<15}: {pipe.score(Xm_te, y_te):.4f}")
    hgb = HistGradientBoostingClassifier(random_state=0).fit(Xm_tr, y_tr)
    print(f"    {'native (HGB)':<15}: {hgb.score(Xm_te, y_te):.4f}    (no imputation at
all)")

```

```

MCAR (random holes)
mean          : 0.7899
median        : 0.7980
KNN           : 0.8000
iterative     : 0.8091
mean+indicator : 0.7960
native (HGB)  : 0.8768    (no imputation at all)

MNAR (label-driven)
mean          : 0.8293
median        : 0.8273
KNN           : 0.8232
iterative     : 0.8232
mean+indicator : 0.8616
native (HGB)  : 0.9232    (no imputation at all)

```

Listing 2 — Naive target encoding: the leak, measured

The categorical column is 2,000 categories of pure noise. Naive all-rows encoding 'improves' test accuracy by 8 points over the honest baseline — fiction manufactured by baking test labels into test features. Out-of-fold encoding returns the truth.

```

"""Listing 2: target encoding computed before the split -- an offline mirage,
measured."""
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split, KFold

rng = np.random.default_rng(1)
n = 6000
cat = rng.integers(0, 2000, n)           # ~3 rows per category, NO real signal
x = rng.normal(0, 1, n)                 # the only true signal
y = ((x + rng.normal(0, 1, n)) > 0).astype(int)
df = pd.DataFrame({"cat": cat, "x": x})
tr, te = train_test_split(np.arange(n), test_size=0.33, random_state=1)

def score(X):
    m = RandomForestClassifier(200, random_state=1, n_jobs=-1).fit(X[tr], y[tr])
    return m.score(X[tr], y[tr]), m.score(X[te], y[te])

baseline = score(df[["x"]].values)
print(f"x alone           : train {baseline[0]:.3f} test {baseline[1]:.3f}")

# NAIVE target encoding: category -> mean(y) over ALL its rows (incl. this row's label)
means_all = pd.Series(y).groupby(df["cat"]).mean()
X_naive = np.c_[df["cat"].map(means_all), df["x"]]
trn, ten = score(X_naive)
print(f"target enc (naive) : train {trn:.3f} test {ten:.3f} <- inflated: test
labels leaked")

# OUT-OF-FOLD encoding: each row sees only other folds' means; test sees train means
enc = np.zeros(n)
gm = y[tr].mean()
for f_tr, f_val in KFold(5, shuffle=True, random_state=1).split(tr):
    means = pd.Series(y[tr[f_tr]]).groupby(df["cat"].iloc[tr[f_tr]]).mean()
    enc[tr[f_val]] = df["cat"].iloc[tr[f_val]].map(means).fillna(gm)
means_tr = pd.Series(y[tr]).groupby(df["cat"].iloc[tr]).mean()
enc[te] = df["cat"].iloc[te].map(means_tr).fillna(gm)
tro, teo = score(np.c_[enc, df["x"]])
print(f"target enc (OOF)   : train {tro:.3f} test {teo:.3f} <- honest")
print("\nThe category is pure noise -- yet naive encoding 'beats' the honest 0.695,")
print("because encoding on ALL rows bakes test labels into test features. That +8pt")
print("edge is an offline mirage: in production, new rows have no labels to leak.")

```

```

x alone           : train 1.000 test 0.688
target enc (naive) : train 1.000 test 0.780 <- inflated: test labels leaked
target enc (OOF)   : train 1.000 test 0.695 <- honest

```

The category is pure noise -- yet naive encoding 'beats' the honest 0.695, because encoding on ALL rows bakes test labels into test features. That +8pt edge is an offline mirage: in production, new rows have no labels to leak.

Listing 3 — Scaling: the unscaled disaster, and what outliers do to each scaler

One feature on a 500x scale reduces unscaled KNN to near-chance. With 2% gross outliers planted, standard and min-max scaling crush the inliers' working range (IQR 0.031 and 0.0036) while robust scaling holds it at 1.0 by construction.

```
"""Listing 3: scaling -- the unscaled disaster, and what outliers do to each scaler."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler

rng = np.random.default_rng(2)
X, y = make_classification(n_samples=2000, n_features=6, n_informative=4,
random_state=2)
X[:, 0] *= 500 # one feature on a 500x scale
X[rng.choice(2000, 40), 0] += rng.normal(2e5, 3e4, 40) # plus 2% gross outliers
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.33, random_state=2)

# Other features have IQR ~ 1.3 after their own scaling; compare feature 0's
print(f"{'scaler':<9} {'KNN test acc':>13} {'IQR of scaled f0':>17}")
for name, sc in [("none", None), ("standard", StandardScaler()),
                 ("min-max", MinMaxScaler()), ("robust", RobustScaler())]:
    steps = ([sc] if sc else []) + [KNeighborsClassifier(15)]
    acc = make_pipeline(*steps).fit(X_tr, y_tr).score(X_te, y_te)
    f0 = sc.fit_transform(X_tr[:, 0]) if sc else X_tr[:, 0]
    iqr = np.subtract(*np.percentile(f0, [75, 25]))
    print(f"{'name':<9} {'acc':>13.4f} {'iqr':>17.4f}")
print("\nUnscaled: the 500x feature owns every distance -- accuracy collapses.")
print("Standard and min-max let the 2% outliers inflate the divisor, shrinking the")
print("inliers' working scale ~50x below the other features'; robust (median/IQR)")
print("restores feature 0 to parity by construction (IQR = 1).")
```

scaler	KNN test acc	IQR of scaled f0
none	0.5788	980.2859
standard	0.8833	0.0310
min-max	0.8879	0.0036
robust	0.8727	1.0000

```
Unscaled: the 500x feature owns every distance -- accuracy collapses.
Standard and min-max let the 2% outliers inflate the divisor, shrinking the
inliers' working scale ~50x below the other features'; robust (median/IQR)
restores feature 0 to parity by construction (IQR = 1).
```

Listing 4 — Filter, wrapper, embedded selection vs planted truth

Thirty features: 6 informative, 6 redundant copies, 18 noise. Mutual information, RFE, and forest importances each recover 8/8 real features; L1 at strong regularization shows its signature failure — it discards redundant copies (by design) but admits four noise features.

```
"""Listing 4: feature selection -- filter, wrapper, embedded, judged against known
truth."""
import numpy as np
```



```

from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_selection import (SelectKBest, mutual_info_classif, RFE)
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler

# 30 features: the FIRST 6 are informative, 6 are redundant copies, 18 pure noise
X, y = make_classification(n_samples=2000, n_features=30, n_informative=6,
                          n_redundant=6, shuffle=False, random_state=3)
Xs = StandardScaler().fit_transform(X)
truth = set(range(12))          # informative + their redundant copies

def report(name, chosen):
    chosen = [int(c) for c in chosen]
    hits = len(set(chosen) & truth)
    print(f"{name:<22}: {sorted(chosen)}  ({hits}/{len(chosen)} real)")

# FILTER: score each feature alone (fast, model-free, blind to interactions)
mi = SelectKBest(mutual_info_classif, k=8).fit(Xs, y)
report("filter (mutual info)", np.where(mi.get_support())[0])

# WRAPPER: recursive feature elimination around a model (slow, interaction-aware)
rfe = RFE(LogisticRegression(max_iter=2000), n_features_to_select=8).fit(Xs, y)
report("wrapper (RFE)", np.where(rfe.support_)[0])

# EMBEDDED: selection as a side effect of fitting
l1 = LogisticRegression(penalty="l1", C=0.05, solver="liblinear").fit(Xs, y)
report("embedded (L1)", np.where(np.abs(l1.coef_[0]) > 1e-6)[0])
rf = RandomForestClassifier(300, random_state=3, n_jobs=-1).fit(Xs, y)
report("embedded (RF top-8)", np.argsort(rf.feature_importances_)[::-1][:8])

```

```

filter (mutual info) : [0, 2, 3, 4, 5, 7, 9, 11] (8/8 real)
wrapper (RFE)       : [0, 1, 2, 3, 4, 5, 9, 11] (8/8 real)
embedded (L1)       : [0, 1, 2, 3, 5, 11, 16, 17, 19, 23] (6/10 real)
embedded (RF top-8) : [0, 1, 3, 4, 5, 8, 9, 11] (8/8 real)

```

Listing 5 — Imbalance: weights vs resampling vs SMOTE, and the SMOTE-before-split trap

Judged by PR-AUC — a ranking metric — nothing beats the untouched baseline: resampling reshapes threshold geometry, not ranking. SMOTE applied before the split interpolates test rows into training data and reports a fantasy 0.999.

```

"""Listing 5: class imbalance -- weights vs resampling vs SMOTE, and the SMOTE-before-split trap."""
import warnings
warnings.filterwarnings("ignore")
import numpy as np
from imblearn.over_sampling import SMOTE, RandomOverSampler
from imblearn.under_sampling import RandomUnderSampler
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import average_precision_score
from sklearn.model_selection import train_test_split

X, y = make_classification(n_samples=12000, n_features=12, n_informative=6,
                          weights=[0.97], flip_y=0.02, random_state=4) # 3%
positives
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.33,
                                          stratify=y, random_state=4)

def pr_auc(m, Xt=None, yt=None):

```

```

Xt, yt = (X_te, y_te) if Xt is None else (Xt, yt)
return average_precision_score(yt, m.predict_proba(Xt)[: , 1])

rf = lambda: RandomForestClassifier(300, random_state=4, n_jobs=-1)
print(f"positives: {y_tr.mean():.1%} of train")
print(f"baseline (as-is) : PR-AUC {pr_auc(rf().fit(X_tr, y_tr)):.3f}")
w = RandomForestClassifier(300, class_weight='balanced', random_state=4, n_jobs=-1)
print(f"class_weight=balanced : PR-AUC {pr_auc(w.fit(X_tr, y_tr)):.3f}")
for name, sampler in [("random undersample", RandomUnderSampler(random_state=4)),
                      ("random oversample", RandomOverSampler(random_state=4)),
                      ("SMOTE", SMOTE(random_state=4))]:
    Xr, yr = sampler.fit_resample(X_tr, y_tr)
    print(f"{name:<22}: PR-AUC {pr_auc(rf().fit(Xr, yr)):.3f}")

# THE TRAP: SMOTE before the split -- synthetic points interpolate future test rows
Xs, ys = SMOTE(random_state=4).fit_resample(X, y)
Xs_tr, Xs_te, ys_tr, ys_te = train_test_split(Xs, ys, test_size=0.33, random_state=4)
m = rf().fit(Xs_tr, ys_tr)
print(f"\nSMOTE BEFORE split : PR-AUC {pr_auc(m, Xs_te, ys_te):.3f} <- fantasy
number")

```

```

positives: 3.9% of train
baseline (as-is) : PR-AUC 0.768
class_weight=balanced : PR-AUC 0.748
random undersample : PR-AUC 0.670
random oversample : PR-AUC 0.749
SMOTE : PR-AUC 0.708

SMOTE BEFORE split : PR-AUC 0.999 <- fantasy number

```

Listing 6 — Leverage outliers: invisible marginally, fatal jointly

Corrupted rows pair extreme x with wrong y, so each value is unremarkable alone: univariate rules flag 1-2 of 40; Isolation Forest, seeing the joint distribution, flags 32. The corruption drags the OLS slope from 2.0 to 0.897; winsorizing the value barely helps (the row stays corrupted), while deletion and Huber loss recover most of the truth.

```

"""Listing 6: outliers -- detection rules compared, then treatment options measured."""
import warnings
warnings.filterwarnings("ignore")
import numpy as np
from sklearn.ensemble import IsolationForest
from sklearn.linear_model import LinearRegression, HuberRegressor

rng = np.random.default_rng(7)
n = 2000
x = rng.normal(0, 1, n)
y = 2.0 * x + rng.normal(0, 1, n) # true slope = 2
out = rng.choice(n, 40, replace=False) # 2% corrupted rows...
x[out] = rng.normal(8, 1, 40) # ...at HIGH LEVERAGE (extreme x)
y[out] = rng.normal(0, 3, 40) # ...with wrong y
x_te = rng.normal(0, 1, 1000) # clean test set
y_te = 2.0 * x_te + rng.normal(0, 1, 1000)

# --- detection: three rules, judged against the known corrupted rows ---
z = np.abs((y - y.mean()) / y.std()) # z-score on y alone
q1, q3 = np.percentile(y, [25, 75])
fence = (y < q1 - 1.5 * (q3 - q1)) | (y > q3 + 1.5 * (q3 - q1)) # Tukey, y alone
iso = IsolationForest(contamination=0.02, random_state=7).fit(np.c_[x, y])
flags = {
    "z-score>3 (y only)": np.where(z > 3)[0],
    "1.5*IQR (y only)": np.where(fence)[0],
}

```



```

    "IsolationForest": np.where(iso.predict(np.c_[x, y]) == -1)[0],
}
truth = set(out.tolist())
for name, idx in flags.items():
    tp = len(set(idx.tolist()) & truth)
    print(f"{name:<18}: flagged {len(idx):>3}, real {tp}/40")

# --- treatment: what each choice does to the fitted slope and clean-test RMSE ---
def fit(xs, ys, model=None):
    m = (model or LinearRegression()).fit(xs[:, None], ys)
    rmse = np.sqrt(np.mean((m.predict(x_te[:, None]) - y_te) ** 2))
    return m.coef_[0], rmse

lo, hi = np.percentile(x, [1, 99])
keep = iso.predict(np.c_[x, y]) == 1 # drop IsolationForest flags
for name, (slope, rmse) in {
    "OLS, as-is": fit(x, y),
    "winsorize x 1/99": fit(np.clip(x, lo, hi), y),
    "delete flagged": fit(x[keep], y[keep]),
    "Huber loss": fit(x, y, HuberRegressor()),
}.items():
    print(f"{name:<18}: slope {slope:5.3f} clean-test RMSE {rmse:5.3f}")

```

```

z-score>3 (y only): flagged 5, real 1/40
1.5*IQR (y only) : flagged 13, real 2/40
IsolationForest : flagged 40, real 32/40
OLS, as-is      : slope 0.897 clean-test RMSE 1.493
winsorize x 1/99 : slope 0.941 clean-test RMSE 1.462
delete flagged  : slope 1.668 clean-test RMSE 1.063
Huber loss      : slope 1.759 clean-test RMSE 1.039

```

Listing 7 — Log, bins, interactions: three rescues of a linear model

Each transform targets a nameable failure: log-linear truth (R^2 0.603 raw, 0.973 logged), U-shaped truth (R^2 -0.001 raw — the arms cancel — 0.889 binned), and pure-interaction truth (coin-flip additive, 0.998 with the single product column).

```

"""Listing 7: log transforms, binning, interactions -- buying nonlinearity for linear
models."""
import warnings
warnings.filterwarnings("ignore")
import numpy as np
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import KBinsDiscretizer

rng = np.random.default_rng(5)
n = 4000

# (a) log transform: multiplicative truth, skewed feature
x_skew = rng.lognormal(0, 1, n) # heavy right tail
y_mult = 3 * np.log(x_skew) + rng.normal(0, 0.5, n) # truth is log-linear
tr, te = train_test_split(np.arange(n), test_size=0.33, random_state=5)
def r2(feats, target):
    m = LinearRegression().fit(feats[tr], target[tr])
    return m.score(feats[te], target[te])
print(f"skewed feature, raw : R^2 {r2(x_skew[:, None], y_mult):.3f}")
print(f"skewed feature, log : R^2 {r2(np.log(x_skew[:, None]), y_mult):.3f}")

# (b) binning: U-shaped truth defeats a line; bins buy a step function
x_u = rng.uniform(-3, 3, n)
y_u = x_u**2 + rng.normal(0, 0.7, n)

```

```

print(f"\nU-shaped truth, raw      : R^2 {r2(x_u[:, None], y_u):.3f}")
bins = KBinsDiscretizer(n_bins=10, encode='onehot-dense').fit(x_u[tr, None])
print(f"U-shaped truth, binned: R^2 {r2(bins.transform(x_u[:, None]), y_u):.3f}")

# (c) interactions: XOR-style truth invisible to additive models
x1, x2 = rng.normal(size=n), rng.normal(size=n)
y_xor = ((x1 * x2) > 0).astype(int)          # sign agreement decides the class
Xa = np.c_[x1, x2]
Xi = np.c_[x1, x2, x1 * x2]                  # add the product feature
def acc(X):
    m = LogisticRegression().fit(X[tr], y_xor[tr])
    return m.score(X[te], y_xor[te])
print(f"\nXOR truth, additive      : acc {acc(Xa):.3f} (coin flip)")
print(f"XOR truth, + x1*x2         : acc {acc(Xi):.3f} (one engineered column)")

```

```

skewed feature, raw      : R^2 0.603
skewed feature, log      : R^2 0.973

U-shaped truth, raw      : R^2 -0.001
U-shaped truth, binned: R^2 0.889

XOR truth, additive      : acc 0.493 (coin flip)
XOR truth, + x1*x2       : acc 0.998 (one engineered column)

```

Listing 8 — 900 categories: one-hot vs hashing vs frequency vs smoothed target

Every category carries real signal, ~9 rows each. Smoothed target encoding wins in one column (0.734), matching 900-column one-hot (0.727); hashing pays a measured collision tax (0.625); frequency encoding fails here by construction — popularity was built to carry no signal.

```

"""Listing 8: high-cardinality categoricals -- one-hot vs hashing vs frequency vs
smoothed target."""
import warnings
warnings.filterwarnings("ignore")
import numpy as np
import pandas as pd
from category_encoders import TargetEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_extraction import FeatureHasher
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(6)
n, k = 8000, 900                                # 900 categories, ~9 rows each
cat = rng.integers(0, k, n)
effect = rng.normal(0, 1, k)                      # each category has a real effect
x = rng.normal(0, 1, n)
y = ((effect[cat] + x + rng.normal(0, 1, n)) > 0).astype(int)
cat_str = np.array([f"c{c}" for c in cat])
tr, te = train_test_split(np.arange(n), test_size=0.33, random_state=6)

def score(Xtr, Xte, note=""):
    m = RandomForestClassifier(300, random_state=6, n_jobs=-1).fit(Xtr, y[tr])
    return m.score(Xte, y[te])

# one-hot: 900 columns, ~9 ones each -- wide, sparse, fragmenting
oh = pd.get_dummies(pd.Series(cat_str))
X_oh = np.c_[oh.values, x]
print(f"one-hot ({oh.shape[1]} cols)      : acc {score(X_oh[tr], X_oh[te]):.3f}")

```

```
# hashing trick: fixed 32 columns, no vocabulary, collisions accepted
H = FeatureHasher(n_features=32, input_type="string")
Xh = np.c_[H.transform([[c] for c in cat_str]).toarray(), x]
print(f"hashing (32 cols)      : acc {score(Xh[tr], Xh[te]):.3f}")

# frequency encoding: category -> its count (cheap, target-free)
freq = pd.Series(cat_str[tr]).value_counts()
Xf = np.c_[pd.Series(cat_str).map(freq).fillna(0), x]
print(f"frequency (1 col)      : acc {score(Xf[tr], Xf[te]):.3f}")

# smoothed target encoding (category_encoders: fits on TRAIN only, shrinks rare cats)
tenc = TargetEncoder(cols=[0], smoothing=5.0)
Xt_tr = np.c_[tenc.fit_transform(pd.DataFrame(cat_str[tr]), y[tr]).values, x[tr]]
Xt_te = np.c_[tenc.transform(pd.DataFrame(cat_str[te])).values, x[te]]
print(f"target enc (1 col)      : acc {score(Xt_tr, Xt_te):.3f}")
```

```
one-hot (900 cols)      : acc 0.727
hashing (32 cols)       : acc 0.625
frequency (1 col)       : acc 0.619
target enc (1 col)      : acc 0.734
```

Pitfalls, comparisons and practical tips

The leakage ledger. Every fitted preprocessing step must be fit on training folds only, inside the CV loop. The chapter measured three violations: naive target encoding (+8 points of fiction, Listing 2), SMOTE before the split (0.708 → 0.999, Listing 5), and — from the literature — selection-before-CV, which can certify pure noise as near-perfectly predictive. Milder cousins (imputer/scaler statistics computed over test rows) follow the identical rule. One mechanical habit prevents all of it: transforms live in a `Pipeline`, and the `Pipeline` is what enters `cross_val_score`. If a preprocessing step touches labels — target encoding, SMOTE, supervised selection — treat it as maximally dangerous by default.

Encoding chooser.

Situation	Encoding	Watch for
Nominal, low cardinality (< ~30)	One-hot	Dummy trap for unregularized linear; unseen categories
True order exists	Ordinal (in order)	Equal-spacing assumption
High cardinality, signal in category	Smoothed / OOF target encoding	Leakage; rare categories need shrinkage
Unbounded vocabulary, streaming	Hashing	Collision noise; pick bucket count by memory
Popularity plausibly predictive	Frequency	Free to try, target-free
Huge cardinality + deep model	Learned embeddings	Needs data; Chapter 12

Situation	Encoding	Watch for
Long tail of rare values	Rare-category bucket first	Threshold is a hyperparameter

Scaler chooser. Standard by default; robust when outliers are real and staying; min-max when the downstream wants bounded inputs and the data is clean; MaxAbs/no-centering for sparse. Trees: none needed. Row-wise `Normalizer` is a different animal (unit-norm rows for cosine similarity) — don't confuse it with column scaling.

Imputation quick rules. Diagnose the mechanism before choosing the tool: plot missingness against other columns and against the target (a missingness-vs-target dependence is the MNAR alarm). Under suspected MNAR, add the indicator column — it costs nothing and is the only unbiased move available. Gradient boosting with native NaN handling is a legitimate complete answer. Never impute the target.

Imbalance quick rules. First fix the metric (PR-AUC / recall-at-precision, Chapter 10), then ask whether you need better *ranking* (rarely improvable by resampling — Listing 5) or better *hard predictions at a threshold* (fix by threshold moving or class weights). SMOTE is a legitimate tool that mostly matters for extreme scarcity of minority examples with simple decision boundaries — and it must live inside the CV loop via `imblearn`'s Pipeline.

Outlier judgment. Univariate rules miss leverage (Listing 6); check jointly when columns should covary. Ask provenance before treatment: error → delete or fix upstream; real-but-rare → keep, bound its influence (Huber, robust scaler, log/rank transform); the outlier IS the phenomenon (fraud, failures) → it's your positive class, not your trash. Winsorizing treats a value, not a row — a corrupted row keeps lying through its other columns.

Trees vs linear: who needs what. A useful compression of the whole chapter: tree ensembles are indifferent to monotone transforms, scaling, and binning, handle NaN natively (modern ones), and split on raw categoricals (LightGBM/CatBoost) — their feature engineering need is interactions across many variables (which depth supplies) and signal-bearing encodings for huge cardinalities. Linear models need everything: scaling for the penalty, log/bins/splines for nonlinearity, explicit interaction columns, one-hot or target encodings, and imputation. "Which model are you feeding?" is the first question of any feature-engineering answer.

Interview questions and answers

Q1. Define MCAR, MAR, and MNAR, and explain why the mechanism decides which imputation is even valid.

MCAR (missing completely at random): missingness is independent of everything — a dropped sensor packet. Any reasonable fill is unbiased; you lose only efficiency.

MAR: missingness depends on *observed* columns (older customers skip income, age is recorded), so model-based imputation conditioning on those columns (iterative, KNN) can recover the structure.

MNAR: missingness depends on the *missing value itself* or unobserved facts (high earners decline to state income). No imputation from observed data is unbiased, because the observed distribution of that column is systematically censored. Rubin's taxonomy isn't trivia — it tells you whether a fill can be honest at all, and the diagnostic is plotting missingness against other columns and the target.

Q2. Under suspected MNAR, what is the only unbiased move, and why?

Stop pretending you can fill the hole and instead make the *missingness itself a feature*: append a binary indicator column alongside any fill. Under MNAR the fact that a value is missing carries signal (defaulters hide debts), so the mask often out-predicts the imputed number. Listing 1 measures it: with label-driven holes, every imputer clusters at 0.82–0.83 accuracy while `mean + indicator` jumps to 0.862 — the indicator *is* the signal. It costs one nearly-free column and is the only move that doesn't bake the censoring bias into the fill.

Q3. Compare mean/median, KNN, and iterative (MICE) imputation on cost and assumptions.

Mean/median: fill with one constant (median resists skew); cheap, but it shrinks the column's variance and flattens its correlations — imputed points sit on a line in any scatter — and under MNAR fills censored holes with the uncensored average, exactly wrong. *KNN*: fill from the k nearest rows in the observed columns; respects local structure, costs $O(n^2)$ distances, inherits every scaling caveat. *Iterative / MICE*: round-robin regress each incomplete column on all others to convergence — the strongest general-purpose option under MAR (Listing 1's MCAR winner among imputers, 0.809 vs mean's 0.790). Sophistication buys MAR-validity, not MNAR-validity — nothing computed from observed data escapes censoring.

Q4. Where in the pipeline must imputation live, and what breaks if it doesn't?

Inside a `Pipeline`, fit on training folds only, refit per CV fold. A mean computed over all rows includes test rows; an imputer fit on the full training set before cross-validation leaks each validation fold into the statistics used to fill it. Both are silent optimism that evaporates in deployment, where "test rows" haven't arrived. The rule is mechanical and automatic once the imputer is a pipeline step; it's a hand-rolled bug otherwise — the same fit-on-train-only discipline that governs scaling, selection, and resampling.

Q5. Should you ever impute the target variable?

No. Rows with missing `y` are unlabeled data, not imputation candidates — inventing their labels manufactures signal from nothing and corrupts every downstream metric. Drop them from supervised training (or exploit them via semi-supervised methods that treat them *as* unlabeled), but never fill them with a model's guess and then train on that guess as if it were ground truth.

Q6. When is "don't impute at all, use gradient boosting" a legitimate answer rather than an evasion?

Whenever your model is modern gradient boosting (XGBoost, LightGBM, HistGradientBoosting), which treats NaN as a value and learns per-split which side missing rows go. It handles MCAR and MNAR structure without any fill: Listing 1's overall winner in both regimes (0.877 MCAR, 0.923 MNAR), beating every imputer. It sidesteps the leakage traps entirely because there's no fitted fill statistic to leak. The honest caveat: it only works because the tree can route missingness — a linear model or KNN still needs a real imputation strategy.

Q7. Label vs ordinal vs one-hot encoding — what structure does each assert?

Label/integer (red=0, green=1, blue=2) asserts an order *and* a magnitude — blue = 2×green — that don't exist; fine for trees (they can carve {0,2} vs {1}), wrong for anything linear or distance-based. *Ordinal* assigns integers *in a real order* (S<M<L<XL, education levels), asserting equal spacing — itself an assumption to flag aloud. *One-hot* asserts no order at all, one binary column per category — the correct default for nominal categories, at the cost of dimensionality. The junior mistake is asserting order where there is none; the senior move is naming exactly what arithmetic each encoding smuggles in.

Q8. Explain the dummy-variable trap. When do you drop a one-hot column and when do you keep all of them?

A full one-hot set is perfectly collinear with the intercept (the columns sum to 1), so an unregularized linear model has a non-identifiable coefficient — *drop one column* (the reference level). But keep *all* columns for regularized linear models and trees: with an L2/L1 penalty, drop-one distorts the regularization symmetry (the dropped level gets penalty-free treatment via the intercept), and trees don't care about collinearity at all. So the rule is conditional: drop-one only for unregularized linear; keep-all otherwise.

Q9. What is target encoding, and precisely how does the naive version leak?

Target (mean) encoding replaces each category with the mean of the target over that category's rows — one dense column carrying exactly the signal a model wants, and the strongest classical encoding for high cardinality. Naive computation takes category means over *all* rows, including the row being encoded and the test set, so each row's own label contaminates its feature. Listing 2 stages it on 2,000 categories of *pure noise* (three rows each): honest test AUC 0.688, naive 0.780 — eight points of fiction, because test labels leaked into test features through the category means. In production, new rows have no labels to leak, so the edge is a pure offline mirage.

Q10. How do out-of-fold encoding and smoothing fix target-encoding leakage?

Out-of-fold: encode each training row using means computed on *other* folds, and encode test with train-only means — Listing 2 returns 0.695, the truth give or take fold noise. *Smoothing* shrinks each category's mean toward the global mean in proportion to its count, $y_c = (n_c * \text{mean}_c + m * \text{mean_global}) / (n_c + m)$, so a 3-row category contributes mostly prior — an empirical-Bayes move that stabilizes rare categories. CatBoost's ordered target statistics go further: encode each row using only rows *before it* in a random permutation, leakage-proof by construction. `category_encoders.TargetEncoder` implements the smoothed form and fits on train only.

Q11. When are frequency encoding and the hashing trick the right call, and what do they cost?

Frequency encoding (category → its count) is free, label-free, and surprisingly effective when popularity correlates with the target; it fails when it doesn't — Listing 8's frequencies were built to carry no signal and it lands last at 0.619. *Hashing* maps categories into a fixed number of buckets with no vocabulary — ideal for unbounded/streaming cardinality and tight memory — but pays a measured collision tax (Listing 8: 32 buckets → 0.625, below one-hot's 0.727) because colliding categories share a column and blur their signals. Pick hashing when the vocabulary is open-ended; pick frequency when popularity is plausibly predictive and you want a target-free single column.

Q12. What are entity embeddings, and when are they worth the trouble over target encoding?

An embedding maps each category to a learned dense vector, trained end-to-end with a neural model (Chapter 12). They win when cardinality is huge *and* categories share latent similarity structure worth learning — users, products, words — and they transfer across tasks. For tabular work with classical models they're usually overkill; smoothed/OOF target encoding gets most of the signal in one column with no deep model. The interview bridge worth naming: an embedding layer *is* a one-hot times a learned matrix, so it's the same lookup as one-hot encoding but with a trainable, low-dimensional codebook.

Q13. Which algorithms need feature scaling and which are indifferent — and why?

Scaling matters whenever an algorithm reads *distance* or *gradient* meaning into raw magnitudes: KNN, k-means, SVMs (kernel distances), PCA (variance directions), regularized linear models (an L2 penalty taxes small-scale features' coefficients harder), and neural nets (loss conditioning). Trees and tree ensembles are indifferent — splits are order-based, so any monotone transform leaves the split structure unchanged. Listing 3 supplies the disaster: one feature on a 500× scale drops unscaled KNN to 0.579 (near chance) because that feature owns every distance; any scaler restores ~0.88. The tell is "does the model compare features to each other by magnitude?"

Q14. Standard vs min-max vs robust scaling — what does each assume, and how do outliers affect them?

Standardization (z-score): mean 0, variance 1; the default, but both mean and especially σ are outlier-sensitive. *Min-max*: map to [0,1] by the observed range — bounded output for NNs, but range is the single most outlier-fragile statistic, so one absurd value crushes everyone into a sliver. *Robust*: subtract median, divide by IQR — immune to anything outside the middle 50%. Listing 3 quantifies it: with 2% gross outliers, standard and min-max leave the inliers' IQR at 0.031 and 0.0036 (the outliers inflated the divisor, shrinking the real data's working scale 50–300×), while robust holds IQR = 1.0 by construction. A scaler fit on contaminated statistics silently re-weights features.

Q15. What's the difference between column scaling and sklearn's `Normalizer`?

They operate on different axes. A `StandardScaler`/`MinMaxScaler`/`RobustScaler` rescales each *column* (feature) using statistics across rows. `Normalizer` rescales each *row* to unit norm — used for cosine-similarity text vectors so document length doesn't dominate. Confusing the two is a small but telling error: normalizing rows when you meant to standardize columns destroys per-feature comparability and vice versa. Also mind sparsity: centering (subtracting a mean) densifies sparse data, so sparse inputs want `MaxAbsScaler` or no-centering.

Q16. Filter, wrapper, and embedded feature selection — trade-offs and each family's characteristic failure.

Filter (correlation, ANOVA F, mutual information, chi-square) scores each feature against the target with no model — fast and parallel, but blind to interactions (a feature useless alone but decisive in combination scores zero) and to redundancy (ten copies all score high, it keeps all ten). *Wrapper* (forward/backward, RFE) trains the model on subsets — interaction-aware and model-specific, but expensive and prone to overfitting the validation set through search. *Embedded* (L1/Lasso, tree importances, permutation importance) gets selection as a fitting side effect — cheap and model-aware. Listing 4 (6 informative, 6 redundant, 18 noise): MI-filter, RFE, and RF importances each recover 8/8 real features; the failure signature is what matters — filters keep redundant copies, Lasso arbitrarily picks one of a correlated group, importances split credit across copies.

Q17. In Listing 4, L1 recovered only 6/10 real features. Was L1 wrong?

No — it exposed its known behavior. Lasso picks *one* feature from a correlated group and zeros the rest (Chapter 5's geometry and instability), so it correctly discarded some *redundant* copies — but at strong regularization it also let four noise features slip through while zeroing real-but-redundant ones. The lesson isn't "L1 is bad"; it's that L1 optimizes for a sparse predictive set, not for recovering every ground-truth feature, and among correlated features its choice is arbitrary and unstable across resamples. Naming that signature is the point, not the raw count.

Q18. Why is selecting features on the full dataset before cross-validation the most dangerous leak in high-dimensional work?

Because selection is a fitted transform, and fitting it on all rows (test included) chooses features partly for their correlation with test labels — then CV "validates" on the very labels that guided the choice. With many candidates and few rows (genomics: 20,000 genes, 100 patients) you can reach near-perfect cross-validated accuracy on *pure noise*. The fix is mechanical: selection lives inside the `Pipeline` and is refit per fold, so each fold's features are chosen without seeing that fold's labels. Same rule as imputation, scaling, and resampling — fit on training folds only.

Q19. Permutation importance vs impurity (Gini) importance — why prefer permutation?

Impurity importance ranks features by total split-impurity reduction, but it's biased toward high-cardinality and continuous features (more candidate split points = more chances to reduce impurity by chance) and is computed on training data, so it rewards overfitting. Permutation importance shuffles one column and measures the score drop on *held-out* data — model-agnostic, evaluated on validation, and immune to the cardinality bias. Its cost is one extra scoring pass per feature, and it can under-credit features within a correlated group (shuffling one, the model leans on its twin), so read it alongside the redundancy structure.

Q20. With 3% positives, why is accuracy the wrong metric, and what do you use instead?

A model predicting "negative" always scores 97% accuracy while catching zero positives — accuracy rewards the majority and is blind to the minority you actually care about. Use metrics tied to the positive class: PR-AUC (precision-recall area, which ignores the easy true negatives), recall at a fixed precision, or F-beta at a chosen operating point (Chapter 10). Fixing the metric comes *first* — before touching resampling — because the metric determines whether imbalance handling even helps.

Q21. Class weights, undersampling, oversampling, SMOTE, threshold moving — what does each actually buy?

Class weights reweight the loss (a positive counts $1/\pi$ as much) — one line, no data change, the cheapest defensible default for hard predictions. *Undersampling* drops majority rows — fast, but discards real information. *Oversampling* duplicates minority rows — no new information, overfitting risk. *SMOTE* synthesizes minority points by interpolating between minority neighbors — helps most under extreme scarcity with simple boundaries. *Threshold moving* trains as-is and moves the decision threshold off the never-sacred 0.5 to where the precision/recall trade-off belongs — often the rightest and least-used answer. Listing 5's ranking (PR-AUC): baseline 0.768 beat class weights 0.748, oversample 0.749, SMOTE 0.708, undersample 0.670.

Q22. In Listing 5, doing nothing beat every resampling method. Why?

PR-AUC is a *ranking* metric — it cares only about the order of scores — and the baseline random forest already orders positives above negatives fine. Resampling and reweighting change *threshold geometry* and probability calibration (where scores concentrate), which a ranking metric ignores; meanwhile undersampling threw away 92% of the majority (real cost) and SMOTE's linear interpolations plant synthetic positives on segments between true positives, landing some in majority territory in a 12-D overlapping space. The honest summary: imbalance handling mostly buys a usable default threshold and calibrated hard predictions; if you evaluate by ranking and tune the threshold downstream, baseline + threshold moving is very hard to beat.

Q23. Explain the SMOTE-before-split trap and where SMOTE must actually go.

SMOTE before the train/test split interpolates between minority points that then land on *different* sides of the split, so synthetic training points are literal blends of test rows — and the rebalanced "test set" no longer resembles the 3% world the model will face. Listing 5 measures the fantasy: PR-AUC **0.999** before-split vs 0.708 done honestly. SMOTE (and any sampler) must live *inside* the CV loop, after splitting, applied to training folds only — `imblearn.pipeline.Pipeline` exists precisely so samplers apply to train folds and not to validation. A suspiciously beautiful number on imbalanced data should be audited for this bug first.

Q24. Why do univariate outlier rules miss the outliers that damage models most?

Because the most damaging outliers are *leverage* points — extreme in the joint structure but unremarkable in any single column. Listing 6 corrupts 2% of rows with extreme *x* paired with wrong *y*; each value is marginally ordinary (an *x* of 8 is just a big *x*, a *y* near 0 is a typical *y*), so *z*-score on *y* flags 1/40 and IQR fences 2/40. Isolation Forest, seeing the joint (*x*, *y*) sitting far from the data's line, catches 32/40. Leverage is a *relationship* between columns, not a property of one, so column-at-a-time rules are structurally blind to it — check jointly whenever columns should covary.

Q25. Winsorizing vs deleting vs robust loss — compare, and say when deletion is dangerous.

Listing 6's corruption drags OLS's slope from the true 2.0 to 0.897. *Winsorizing* x at the 1st/99th percentiles reaches only 0.941 — it pulls extreme x 's inward but keeps the wrong y 's, now at moderate leverage: it treats a value, not a row, and a corrupted row keeps lying through its other columns. *Deleting* flagged rows recovers 1.668; *Huber loss* (quadratic near zero, linear in the tails, bounding gross residuals) recovers 1.759 with no detection step at all. Deletion is right when flagged points are *errors* (sensor glitches, bad joins) and dangerous when they're *real but rare* — a fraud model deleting its fraud cases, or dropping the largest customer in a revenue model. When unsure, bound influence instead of destroying data: robust losses, robust scalers, rank/log transforms.

Q26. When do you reach for a log transform, binning, or an interaction feature?

Each buys a linear model a specific nonlinearity (Listing 7). *Log* when the relationship or the feature is multiplicative/right-skewed — it turns multiplicative truth linear (R^2 0.603 $\rightarrow$ 0.973) and compresses heavy tails. *Binning* (discretize into buckets, then one-hot) when the response is non-monotone in a feature — a U-shaped truth invisible to a line (R^2 -0.001) becomes a step function a linear model can fit (0.889), at the cost of lost resolution and a bin-count hyperparameter. *Interactions* (add products like $x_1 \cdot x_2$) when the effect of one feature depends on another — an XOR-style truth is a coin flip additively (0.493) and near-perfect with one product column (0.998). Trees discover all three implicitly; linear models need them handed over.

Q27. A feature has 900 categories, each with real signal. Rank one-hot, hashing, frequency, and smoothed target encoding.

Listing 8 (≈ 9 rows/category): smoothed target encoding wins in a single column (0.734), edging 900-column one-hot (0.727) while being far narrower and not fragmenting trees. Hashing into 32 buckets pays a collision tax (0.625) because distinct categories share columns. Frequency encoding fails here by construction (0.619) — popularity was built to carry no signal. The general lesson: at high cardinality one-hot is wide and fragments trees, hashing trades accuracy for fixed memory, frequency only works if popularity is predictive, and smoothed/OOF target encoding gives the most signal per column — provided it's fit on train only with shrinkage for rare categories.

Q28. "Which model are you feeding?" — summarize how that answer reorganizes all feature engineering.

It's the first question of any feature-engineering answer because tree ensembles and linear models have nearly opposite needs. *Trees* are indifferent to monotone transforms, scaling, and binning, handle NaN natively (modern ones), and can split on raw categoricals (LightGBM/CatBoost) — their real needs are cross-variable interactions (which depth supplies) and signal-bearing encodings for huge cardinalities. *Linear models* need everything: scaling for the penalty, log/bins/splines for nonlinearity, explicit interaction columns, one-hot or target encodings, and imputation. Naming the model first tells you which two-thirds of the preprocessing menu you can skip — and which leakage traps (target encoding, SMOTE, selection) still apply to both regardless.

Chapter 10: Model Evaluation & Metrics

A model is only as trustworthy as the number you judge it by, and most bad deployments trace back to a metric that lied — accuracy on imbalanced data, ROC-AUC where precision was what mattered, an offline R^2 inflated by leakage, or a "2% better" model that was pure sampling noise. Chapter 9 already leaned on this: it evaluated imbalance handling with PR-AUC precisely because accuracy is a broken compass at 3% prevalence. This chapter is where metric choice becomes a first-class skill. The organizing idea is that every metric encodes a value judgment — what counts as a mistake, how much a confident mistake costs, whether ranking or calibration or a point estimate is the deliverable — and choosing the metric *is* choosing what the model optimizes for once it reaches production.

The chapter walks the territory in order: the confusion matrix and the rates built from it (accuracy's failure staged at 6% prevalence, Listing 1); the ROC-vs-PR question, where ROC-AUC stays near 0.95 while PR-AUC falls toward the prevalence line as positives thin out (Listing 2); thresholds as a separate decision from the model, with the default 0.5 shown suboptimal and operating points chosen on validation data (Listing 3); calibration, where a margin-based classifier's probabilities are reshaped by Platt and isotonic and scored by Brier and log loss (Listing 4); proper scoring rules and why accuracy is blind to confidence (Listing 5); regression metrics and their characteristic failure modes — RMSE's outlier sensitivity, MAPE's explosion near zero, R^2 's negativity, adjusted R^2 's penalty for junk features (Listing 6); ranking metrics MRR/MAP/NDCG computed from scratch (Listing 7); the statistics of model comparison — McNemar's test and bootstrap confidence intervals that separate a real gain from noise (Listing 8); and conformal prediction, a distribution-free coverage guarantee (Listing 9). One rule threads through: report the metric that matches the decision, with an honesty interval around it.

The confusion matrix and its rates

Every classification metric for hard predictions is a ratio built from four counts: true positives (TP), false positives (FP), true negatives (TN), false negatives (FN). **Accuracy** = $(TP+TN)/\text{all}$ is the seductive default and the first casualty of imbalance — Listing 1 trains on 6% positives and scores 0.953 accuracy while a model that *always predicts negative* scores 0.938; the 1.5-point "skill" hides that recall is 0.367, i.e. the model misses nearly two-thirds of the positives it exists to catch. **Precision** = $TP/(TP+FP)$ answers "when it says positive, how often is it right?" — the metric you optimize when false alarms are costly (spam that eats real mail, a fraud freeze on a good customer). **Recall** (sensitivity, TPR) = $TP/(TP+FN)$ answers "of the real positives, how many did it catch?" — the metric you optimize when misses are costly (cancer screening, fraud that clears). The two trade

off through the threshold, and naming *which* error is more expensive is the first move in any evaluation answer.

F1 = $2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$ is their harmonic mean — it punishes imbalance between the two (you cannot game it by maximizing one), which is why it beats accuracy on skewed classes; **F β** tilts the balance ($\beta > 1$ weights recall, $\beta < 1$ weights precision). **Balanced accuracy** = $(\text{TPR} + \text{TNR}) / 2$ averages per-class recall, so the always-negative baseline drops to 0.5 (Listing 1: 0.680, honestly mediocre). **Specificity** (TNR) = $\text{TN} / (\text{TN} + \text{FP})$ is recall for the negative class. The interview reflex: when you hear "imbalanced," stop quoting accuracy and switch to precision/recall/F1 or a class-balanced metric, and state the cost asymmetry that picks between them.

ROC-AUC versus PR-AUC

Both summarize a *ranking* of scores across all thresholds, but they answer different questions and disagree sharply under imbalance. The **ROC curve** plots TPR against FPR ($= \text{FP} / (\text{FP} + \text{TN})$); **ROC-AUC** is the probability a random positive outranks a random negative — a clean, prevalence-independent measure of discrimination, and 0.5 is chance. That prevalence-independence is exactly its blind spot: FPR has the giant true-negative pool in its denominator, so on rare positives a model can rack up many false positives while FPR barely moves. Listing 2 holds the model's discrimination roughly fixed and thins positives from 30% to 1.5%: ROC-AUC drifts only $0.951 \rightarrow 0.756$, while **PR-AUC** (average precision, the area under precision-vs-recall) collapses $0.867 \rightarrow 0.266$, tracking toward the no-skill line that for PR is the prevalence.

The rule: **use PR-AUC (or precision/recall at an operating point) when the positive class is rare and it's the class you care about** — fraud, disease, retrieval — because precision has FP against TP, not against the vast TN pool, so it feels every false alarm. Use **ROC-AUC when classes are roughly balanced or you genuinely care about both errors symmetrically**, or when you want a threshold-free discrimination number comparable across datasets of different prevalence. A subtle corollary: ROC-AUC is invariant to prevalence, so a model can look identically good on a balanced offline sample and a 1%-positive production stream while its *precision* — the thing the on-call team actually experiences — quietly craters. PR-AUC would have warned you.

Thresholds and operating points

A probabilistic classifier outputs scores; turning them into decisions needs a threshold, and **the default 0.5 is a convention, not an optimum**. The threshold is a business decision layered on top of the model: it slides you along the precision/recall curve, and where you sit should be set by the cost of each error, not by a library default. Listing 3 makes this concrete on 8% positives — the default 0.5 yields precision 0.974 / recall 0.664; dropping the threshold to the validation-chosen max-F1 point (0.360) trades precision down to 0.893 to buy recall up to 0.749 and a better F1 (0.815); demanding

precision ≥ 0.95 pushes the threshold back up to 0.510, buying precision at recall's expense. Same model, same scores — three different products.

Two disciplines separate candidates. First, **choose the threshold on a validation set, never on the test set** — the threshold is a fitted parameter, and tuning it on test is the same leakage sin as tuning a hyperparameter there (Listing 3 selects on a held-out validation split, then reports on test, which is why the realized test precision doesn't hit the target exactly — an honest reminder that operating points don't transfer perfectly). Second, for problems with an explicit cost matrix, **optimize expected cost directly** rather than F1: threshold where marginal cost of a false positive equals marginal benefit of a true positive. Threshold moving is also the cheapest fix for class imbalance (Chapter 9): train on the natural distribution, then move the threshold — often beating resampling, which mostly just relocates the implicit threshold anyway.

Probability calibration

A model is **calibrated** if, among all cases it scores 0.7, about 70% are actually positive — the scores mean what they say as probabilities. Discrimination (ranking, ROC-AUC) and calibration are independent: a model can rank perfectly yet be badly miscalibrated, and calibration is what you need whenever a downstream decision consumes the *probability itself* — expected-value thresholds, risk pricing, triage queues, or feeding scores into another model. Some learners are calibrated out of the box (logistic regression, by construction); many are not — SVMs output margins, naive Bayes' independence assumption makes it overconfident, and boosted trees push scores toward the extremes. Listing 4 shows an SVM's naive-sigmoid scores tracing a textbook miscalibration: cases it scores 0.15 are positive only 1% of the time, cases it scores 0.85 are positive 98% of the time — an S-shaped reliability curve that a raw Brier score of 0.061 quietly reflects.

The fixes are post-hoc, fit on a held-out calibration set. **Platt scaling** fits a one-parameter logistic (a sigmoid) mapping raw scores to probabilities — cheap, strong when the miscalibration is genuinely sigmoidal, and safe on small calibration sets. **Isotonic regression** fits a free monotonic step function — more flexible, corrects any monotone distortion, but needs more data and can overfit a small calibration set. Listing 4 cuts Brier $0.061 \rightarrow 0.043$ and log loss $0.249 \rightarrow 0.162$ with Platt, isotonic essentially tied here (more data would favor isotonic). Read the reliability curve to *diagnose*, then pick the scaler by data budget: **Platt when calibration data is scarce or the distortion looks sigmoidal; isotonic when you have plenty and the distortion is some arbitrary monotone shape**. Calibrate on data the model didn't train on — calibrating on the training set relearns the training distribution's optimism.

Proper scoring rules: log loss and Brier

A **proper scoring rule** is minimized (in expectation) by reporting your true probabilities — it rewards honesty, so you can't improve your expected score by shading predictions

away from your real belief. The two standard ones grade probabilities, not hard labels. **Log loss** (cross-entropy) = $-\text{mean}(\log p \text{ at the true class})$ punishes confident errors without mercy: a probability approaching 0 on an event that happens sends it toward infinity — Listing 5 clips a ~ 0 probability on a positive to a log loss of 34.5 from that single row. It's unbounded above, which makes it the sharpest instrument and the most sensitive to a single catastrophic overconfidence. **Brier score** = mean squared error between probability and outcome — bounded in $[0,1]$, gentler on outliers, and it decomposes into *reliability* – *resolution* + *uncertainty* (calibration, discrimination, and irreducible base-rate variance).

Listing 5 stages why these beat accuracy: a timid model that always assigns 0.6 to the correct class has perfect hard-label accuracy (1.000) but log loss 0.511, while a bold model that assigns 0.95 and is confidently wrong 8% of the time has *lower* accuracy (0.902) yet *lower* log loss (0.340) — because well-placed confidence, most of the time, pays off under a proper scoring rule even with occasional disasters. The lessons compound: accuracy is a step function blind to confidence, so two models with identical accuracy can have very different log loss; proper scoring rules are the right training and selection objective when you'll consume probabilities; and log loss vs Brier is a sensitivity choice — reach for log loss when a single overconfident miss is unacceptable, Brier when you want a bounded, decomposable, outlier-tolerant number.

Regression metrics

Regression metrics differ in what error they punish and where they break. **MSE** and its rooted, unit-restoring cousin **RMSE** square the residual, so they are dominated by large errors — Listing 6 injects three big misses and RMSE jumps $5.11 \rightarrow 6.75$ while **MAE** (mean absolute error) barely moves $4.08 \rightarrow 4.40$. That is the central trade: RMSE if large errors are disproportionately bad (and if you want the metric your squared-loss model actually optimizes); MAE if every unit of error is equally bad and you want robustness to outliers (MAE is minimized by the median, RMSE by the mean). **MAPE** (mean absolute percentage error) reports scale-free percentages — attractive for cross-series comparison, lethal near zero: Listing 6 sets five targets to 0.1 and MAPE explodes $8.5\% \rightarrow 497\%$ on unchanged predictions, and it asymmetrically punishes over-prediction. Prefer **sMAPE** or **WAPE** when targets approach zero.

$R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ is the fraction of variance explained, benchmarked against predicting the mean; it is unitless and comparable, but it **can go negative** — a model worse than the mean baseline (Listing 6: predicting 0 everywhere scores -24.6), a fact that surprises candidates who think R^2 lives in $[0,1]$. And R^2 *never decreases* when you add features, even pure noise, because in-sample fit can only improve — Listing 6 adds 40 noise columns and raw R^2 ticks up $0.915 \rightarrow 0.919$. **Adjusted R^2** penalizes for parameters, so it falls $0.914 \rightarrow 0.912$ on those junk features — which is why adjusted R^2 (or an out-of-sample R^2) is the honest one for model comparison across different feature counts. The umbrella rule: report an error metric in the target's units (RMSE or MAE)

alongside a normalized one (R^2), and choose squared vs absolute by how much you fear large errors.

Ranking metrics

When the deliverable is an *ordering* — search results, recommendations, retrieval for RAG (Chapter 24) — the metric must reward putting relevant items near the top, which classification and even AUC miss. **MRR** (mean reciprocal rank) scores $1/\text{rank}$ of the *first* relevant hit, averaged over queries — the right metric when the user wants one good answer (a known-item search); Listing 7 gives 1.000 to a list with a relevant item first and 0.333 to one where the first relevant is at rank 3. **MAP** (mean average precision) averages precision at every relevant position — it rewards packing *all* relevants high, not just the first (Listing 7: 0.917 vs 0.411 for the same relevant set reordered). **NDCG** (normalized discounted cumulative gain) is the most general: it supports *graded* relevance (0–3, not just yes/no) and discounts gains logarithmically by position, then normalizes by the ideal ordering so scores are comparable across queries — Listing 7 scores best-first at 1.000 and worst-first at 0.614 on identical graded relevances.

The unifying point Listing 7 makes explicit: a metric that ignores order — $\text{recall}@k$ with k equal to the list length — calls the good and bad rankings *identical*, because they contain the same relevant set. Ranking quality lives in the permutation, so the metric must be permutation-sensitive and position-discounted. Choose by the product: MRR for one-right-answer navigation, MAP for binary relevance where completeness matters, NDCG when relevance is graded or you need cross-query comparability. All three are reported @ k (NDCG@10, MAP@100) because users see a finite window.

Statistical significance of model comparison

Model B scores higher than A on the test set — is the gap real or sampling noise? Two tools answer honestly. **McNemar's test** is the paired test for classifiers on the *same* test set: build the 2×2 table of the rows where they disagree (A-wrong/B-right vs A-right/B-wrong) and test whether the split is lopsided — only the discordant pairs carry information, because rows both models get right or both get wrong say nothing about which is better. Listing 8 finds 184 rows where B rescues A against 54 the other way, $\chi^2 = 58.3$, $p < 0.0001$ — a real difference. **Bootstrap confidence intervals** generalize to any metric: resample the test set with replacement, recompute the metric gap thousands of times, and read the 2.5/97.5 percentiles — Listing 8's AUC gap for B–A is +0.047 with 95% CI [+0.039, +0.055], excluding zero.

The discipline this installs is the antidote to leaderboard theater. Listing 8's second comparison pits two gradient-boosting models differing only by random seed: the AUC gap is +0.0022 with 95% CI [−0.0004, +0.0048] — **spanning zero**, so the "improvement" is noise and shipping on it is superstition. The rules: pair your tests when the models see the same data (McNemar, paired bootstrap, 5×2 cv paired t-test) because

paired tests cancel the variance from *which* rows landed in test; a single train/test split gives one noisy estimate, so cross-validate and report a spread, not a point; and always translate a metric delta into "is the interval clear of zero?" before claiming a win. This is the same anti-optimism instinct as the leakage rules — measure the uncertainty, don't just quote the mean.

Conformal prediction and uncertainty

Point predictions and even calibrated probabilities still don't come with a *guarantee*; **conformal prediction** supplies one that is distribution-free and finite-sample. Split conformal is startlingly simple: hold out a calibration set, define a **nonconformity score** (for classification, $1 - \text{predicted probability of the true class}$), take the appropriate quantile of those calibration scores, and at test time emit the *set* of labels whose score falls under that quantile. The theorem: for a chosen error rate α , the prediction set contains the true label with probability at least $1 - \alpha$, assuming only that the data is exchangeable — no distributional form, no asymptotics. Listing 9 hits 0.904 empirical coverage at a 90% target and 0.951 at 95%, with the finite-sample $(n+1)(1-\alpha)/n$ quantile correction doing the guaranteeing.

The price of the guarantee is **adaptive set size**: where the model is confident the set holds one label; where it's unsure the set grows to two or more, and an empty set can appear when nothing is plausible enough — the set size *is* the honesty, converting vague confidence into a concrete "these are the possibilities at 95%." This is the practitioner's entry to uncertainty quantification, sitting alongside Bayesian posteriors, deep ensembles, and MC-dropout (Chapter 13) — but conformal is the one you can bolt onto *any* trained model for a coverage promise. Interview framing: calibration makes the probabilities honest on average; conformal makes a per-prediction coverage guarantee you can put in a contract, in exchange for sometimes admitting you don't know.

Code implementations

Listing 1 — Confusion matrix: why accuracy lies on imbalanced data

```
"""Listing 1: confusion matrix -- why accuracy lies, and precision/recall/F1 at the
default threshold."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import (confusion_matrix, accuracy_score, precision_score,
                             recall_score, f1_score, balanced_accuracy_score)

X, y = make_classification(n_samples=8000, n_features=20, n_informative=6,
                          weights=[0.94, 0.06], random_state=0)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, random_state=0, stratify=y)
clf = LogisticRegression(max_iter=2000).fit(Xtr, ytr)
pred = clf.predict(Xte)                                     # default 0.5 threshold
```

```

tn, fp, fn, tp = confusion_matrix(yte, pred).ravel()
print(f"prevalence (test) : {yte.mean():.3f}")
print(f"confusion TN {tn} FP {fp} FN {fn} TP {tp}")
print(f"accuracy : {accuracy_score(yte, pred):.3f}")
print(f"always-negative acc : {1 - yte.mean():.3f} (the dumb baseline)")
print(f"precision : {precision_score(yte, pred):.3f}")
print(f"recall (sensitivity): {recall_score(yte, pred):.3f}")
print(f"F1 : {f1_score(yte, pred):.3f}")
print(f"balanced accuracy : {balanced_accuracy_score(yte, pred):.3f}")

```

```

prevalence (test) : 0.062
confusion TN 2233 FP 17 FN 95 TP 55
accuracy : 0.953
always-negative acc : 0.938 (the dumb baseline)
precision : 0.764
recall (sensitivity): 0.367
F1 : 0.495
balanced accuracy : 0.680

```

Listing 2 — ROC-AUC vs PR-AUC as positives get rarer

```

"""Listing 2: ROC-AUC vs PR-AUC -- why ROC flatters a model on rare positives."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, average_precision_score

def evaluate(weight_pos, seed=0):
    X, y = make_classification(n_samples=20000, n_features=20, n_informative=6,
                              weights=[1 - weight_pos, weight_pos], random_state=seed)
    Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.4, random_state=seed,
                                          stratify=y)
    s = LogisticRegression(max_iter=2000).fit(Xtr, ytr).predict_proba(Xte)[ :, 1]
    return yte.mean(), roc_auc_score(yte, s), average_precision_score(yte, s)

print(f"{'prevalence':>10} {'ROC-AUC':>9} {'PR-AUC':>8} {'no-skill PR (=prev)':>20}")
for w in (0.30, 0.10, 0.03, 0.01):
    prev, roc, pr = evaluate(w)
    print(f"{'prev':>10.3f} {'roc':>9.3f} {'pr':>8.3f} {'prev':>20.3f}")
print("\nROC-AUC barely moves as positives get rarer; PR-AUC falls toward the")
print("no-skill line (=prevalence). Same ranking model, same discrimination --")
print("ROC hides that precision is collapsing under a flood of true negatives.")

```

prevalence	ROC-AUC	PR-AUC	no-skill PR (=prev)
0.302	0.951	0.867	0.302
0.104	0.941	0.688	0.104
0.035	0.875	0.417	0.035
0.015	0.756	0.266	0.015

ROC-AUC barely moves as positives get rarer; PR-AUC falls toward the no-skill line (=prevalence). Same ranking model, same discrimination -- ROC hides that precision is collapsing under a flood of true negatives.

Listing 3 — Threshold moving: the default 0.5 is not the operating point

```

"""Listing 3: threshold moving -- the default 0.5 is almost never the right operating point."""

```

```

import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import precision_recall_curve, f1_score, precision_score,
recall_score

X, y = make_classification(n_samples=12000, n_features=20, n_informative=6,
                           weights=[0.92, 0.08], random_state=1)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.4, random_state=1, stratify=y)
# split a validation set to CHOOSE the threshold (never on test)
Xtr2, Xval, ytr2, yval = train_test_split(Xtr, ytr, test_size=0.33, random_state=1,
stratify=ytr)
m = RandomForestClassifier(300, random_state=1, n_jobs=-1).fit(Xtr2, ytr2)
sval = m.predict_proba(Xval)[: , 1]
ste = m.predict_proba(Xte)[: , 1]

def report(t, tag):
    p = (ste >= t).astype(int)
    print(f"{tag:22} thr {t:.3f} P {precision_score(yte,p):.3f} R
{recall_score(yte,p):.3f} F1 {f1_score(yte,p):.3f}")

report(0.5, "default 0.5")
# max-F1 threshold chosen on validation
prec, rec, thr = precision_recall_curve(yval, sval)
f1s = 2*prec*rec/(prec+rec+1e-12)
t_f1 = thr[np.nanargmax(f1s[:-1])]
report(t_f1, "max-F1 (val-chosen)")
# threshold for precision >= 0.95 on validation
ok = np.where(prec[:-1] >= 0.95)[0]
t_p90 = thr[ok[0]] if len(ok) else 0.99
report(t_p90, "precision>=0.95 target")

```

default 0.5	thr 0.500	P 0.974	R 0.664	F1 0.790
max-F1 (val-chosen)	thr 0.360	P 0.893	R 0.749	F1 0.815
precision>=0.95 target	thr 0.510	P 0.974	R 0.659	F1 0.786

Listing 4 — Calibration: reliability, Brier & log loss under Platt and isotonic

```

"""Listing 4: calibration -- reliability, Brier & log loss before/after Platt and
isotonic."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.calibration import CalibratedClassifierCV, calibration_curve
from sklearn.model_selection import train_test_split
from sklearn.metrics import brier_score_loss, log_loss

X, y = make_classification(n_samples=15000, n_features=20, n_informative=8,
                           random_state=2)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.4, random_state=2, stratify=y)

base = SVC(probability=False).fit(Xtr, ytr) # margins -> sigmoid = badly
calibrated
# use decision function pushed through a naive sigmoid as the "raw" probability
raw = 1/(1+np.exp(-base.decision_function(Xte)))
platt = CalibratedClassifierCV(SVC(), method="sigmoid", cv=3).fit(Xtr,
ytr).predict_proba(Xte)[: , 1]
iso = CalibratedClassifierCV(SVC(), method="isotonic", cv=3).fit(Xtr,
ytr).predict_proba(Xte)[: , 1]

```



```

for tag, p in [("raw (naive sigmoid)", raw), ("Platt (sigmoid)", platt), ("isotonic",
iso)]:
    print(f"{tag:22} Brier {brier_score_loss(yte,p):.4f}    log-loss {log_loss(yte,p):.4f}")

# reliability: fraction positive vs mean predicted, 10 bins, for the raw model
frac, mean = calibration_curve(yte, raw, n_bins=10)
print("\nreliability (raw): predicted vs actual by bin")
for mp, fp in zip(mean, frac):
    print(f"  pred {mp:.2f}  ->  actual {fp:.2f}")

```

```

raw (naive sigmoid)    Brier 0.0607    log-loss 0.2491
Platt (sigmoid)       Brier 0.0428    log-loss 0.1621
isotonic              Brier 0.0431    log-loss 0.1675

```

```

reliability (raw): predicted vs actual by bin
  pred 0.07 -> actual 0.01
  pred 0.15 -> actual 0.01
  pred 0.24 -> actual 0.06
  pred 0.34 -> actual 0.12
  pred 0.45 -> actual 0.37
  pred 0.55 -> actual 0.64
  pred 0.65 -> actual 0.81
  pred 0.76 -> actual 0.95
  pred 0.85 -> actual 0.98
  pred 0.93 -> actual 0.99

```

Listing 5 — Proper scoring rules punish confident-wrong; accuracy can't

```

"""Listing 5: proper scoring rules -- log loss and Brier punish confident-wrong;
accuracy can't."""
import numpy as np
from sklearn.metrics import log_loss, brier_score_loss, accuracy_score

rng = np.random.default_rng(3)
n = 2000
y = rng.integers(0, 2, n)
# Model A: hesitant but honest -- correct class gets prob 0.6
# Model B: bold -- correct class gets prob 0.95 ... but 8% of the time it is boldly
WRONG at 0.99
pa = np.where(y == 1, 0.6, 0.4)
pb = np.where(y == 1, 0.95, 0.05).astype(float)
flip = rng.random(n) < 0.08
pb[flip] = 1 - pb[flip]                                # confident and wrong

for tag, p in [("A: hesitant (0.6)", pa), ("B: bold + 8% confident-wrong", pb)]:
    hard = (p >= 0.5).astype(int)
    print(f"{tag:32} acc {accuracy_score(y,hard):.3f}  logloss {log_loss(y,p):.3f}
Brier {brier_score_loss(y,p):.3f}")

# the single-prediction catastrophe: one prob of 0 on a positive -> infinite surprise
(clipped)
p_one = np.array([1e-15]); y_one = np.array([1])
print(f"\none confident-wrong prob (~0 on a positive): log loss {log_loss(y_one,
p_one, labels=[0,1]):.1f}")
print("Accuracy is a step function -- blind to confidence. Log loss is unbounded")
print("above and rewards honesty; Brier is bounded [0,1] and decomposes into")
print("reliability - resolution + uncertainty.")

```

```

A: hesitant (0.6)          acc 1.000  logloss 0.511  Brier 0.160
B: bold + 8% confident-wrong  acc 0.902  logloss 0.340  Brier 0.091

```

one confident-wrong prob (~0 on a positive): log loss 34.5
 Accuracy is a step function -- blind to confidence. Log loss is unbounded above and rewards honesty; Brier is bounded [0,1] and decomposes into reliability - resolution + uncertainty.

Listing 6 — Regression metrics: outliers, near-zero targets, junk features

```
"""Listing 6: regression metrics -- outliers, scale, near-zero targets, and junk
features."""
import numpy as np
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

rng = np.random.default_rng(4)
n = 500
y = rng.normal(50, 10, n)
pred = y + rng.normal(0, 5, n) # honest predictor, RMSE ~5
def row(tag, yt, yp):
    rmse = mean_squared_error(yt, yp) ** 0.5
    mae = mean_absolute_error(yt, yp)
    mape = np.mean(np.abs((yt - yp) / yt)) * 100
    print(f"{tag:20} RMSE {rmse:6.2f} MAE {mae:6.2f} MAPE {mape:6.1f}% R2
{r2_score(yt,yp):.3f}")

row("clean", y, pred)
# inject 3 large errors -> RMSE explodes, MAE barely moves (squared vs absolute)
pred2 = pred.copy(); pred2[:3] += 60
row("3 big errors", y, pred2)
# near-zero targets wreck MAPE
y0 = y.copy(); y0[:5] = 0.1
row("targets near 0", y0, pred) # same preds, tiny denominators

# R2 can go NEGATIVE: predicting a constant worse than the mean baseline
print(f"\nR2 of 'predict 0 for everything' : {r2_score(y, np.zeros(n)):.3f} (worse
than mean => negative)")

# adjusted R2 penalizes useless features
from sklearn.linear_model import LinearRegression
Xr = rng.normal(size=(n, 1)); ytrue = 3*Xr[:,0] + rng.normal(0, 1, n)
Xjunk = np.c_[Xr, rng.normal(size=(n, 40))] # 40 pure-noise columns
def adj(r2, n, p): return 1 - (1 - r2)*(n - 1)/(n - p - 1)
for name, XX in [("1 real feature", Xr), ("40 noise feats", Xjunk)]:
    r2 = LinearRegression().fit(XX, ytrue).score(XX, ytrue)
    print(f"{name:16} R2 {r2:.3f} adjR2 {adj(r2, n, XX.shape[1]):.3f}")
```

clean	RMSE	5.11	MAE	4.08	MAPE	8.5%	R2	0.743
3 big errors	RMSE	6.75	MAE	4.40	MAPE	9.1%	R2	0.552
targets near 0	RMSE	7.14	MAE	4.53	MAPE	497.4%	R2	0.593

R2 of 'predict 0 for everything' : -24.609 (worse than mean => negative)
 1 real feature R2 0.915 adjR2 0.914
 +40 noise feats R2 0.919 adjR2 0.912

Listing 7 — Ranking metrics from scratch: MRR, MAP, NDCG

```
"""Listing 7: ranking metrics from scratch -- MRR, MAP, NDCG reward the TOP of the
list."""
import numpy as np
```



```

def dcg(rels):
    rels = np.asarray(rels, float)
    return np.sum(rels / np.log2(np.arange(2, len(rels) + 2)))
def ndcg(rels):
    ideal = dcg(sorted(rels, reverse=True))
    return dcg(rels) / ideal if ideal > 0 else 0.0
def average_precision(binary):
    binary = np.asarray(binary)
    hits = np.cumsum(binary)
    precs = hits / (np.arange(len(binary)) + 1)
    return (precs * binary).sum() / binary.sum() if binary.sum() else 0.0
def rr(binary):
    idx = np.where(np.asarray(binary) == 1)[0]
    return 1.0 / (idx[0] + 1) if len(idx) else 0.0

# two rankings of the same 6 docs; relevance in {0,1} (graded for NDCG uses same)
good = [1, 1, 0, 1, 0, 0] # relevants near the top
bad = [0, 0, 1, 0, 1, 1] # same # relevant, buried
print(f"{'ranking':8} {'MRR':>6} {'AP':>6} {'NDCG':>6}")
for tag, r in [("good", good), ("bad", bad)]:
    print(f"{'tag':8} {'rr(r)':6.3f} {'average_precision(r)':6.3f} {'ndcg(r)':6.3f}")

# graded relevance (0..3): NDCG rewards putting the MOST relevant first
graded_a = [3, 2, 1, 0]
graded_b = [0, 1, 2, 3]
print(f"\ngraded NDCG best-first {ndcg(graded_a):.3f} worst-first {ndcg(graded_b):.3f}")
print("Same relevant set, different order -> different score. A metric that ignores")
print("order (e.g. recall@k with k=6) would call good and bad identical.")

```

ranking	MRR	AP	NDCG
good	1.000	0.917	0.967
bad	0.333	0.411	0.583

```

graded NDCG best-first 1.000 worst-first 0.614
Same relevant set, different order -> different score. A metric that ignores
order (e.g. recall@k with k=6) would call good and bad identical.

```

Listing 8 — Is B really better? McNemar's test + bootstrap CI

```

"""Listing 8: is model B really better? McNemar's test + bootstrap CI on the metric gap."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, roc_auc_score
from statsmodels.stats.contingency_tables import mcnemar

X, y = make_classification(n_samples=6000, n_features=25, n_informative=8,
                           random_state=5)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.4, random_state=5, stratify=y)
A = LogisticRegression(max_iter=2000).fit(Xtr, ytr)
B = GradientBoostingClassifier(random_state=5, subsample=0.7).fit(Xtr, ytr)
pa, pb = A.predict(Xte), B.predict(Xte)
sa, sb = A.predict_proba(Xte)[:,1], B.predict_proba(Xte)[:,1]
print(f"accuracy A {accuracy_score(yte,pa):.4f} B {accuracy_score(yte,pb):.4f}")
print(f"ROC-AUC A {roc_auc_score(yte,sa):.4f} B {roc_auc_score(yte,sb):.4f}")

# McNemar: only the DISAGREEMENTS count (paired predictions on the same test rows)
b_only = int(((pa != yte) & (pb == yte)).sum()) # A wrong, B right
a_only = int(((pa == yte) & (pb != yte)).sum()) # A right, B wrong
res = mcnemar([[0, a_only], [b_only, 0]], exact=False, correction=True)

```

```

print(f"\nMcNemar discordant: A-wrong/B-right {b_only}, A-right/B-wrong {a_only}")
print(f"McNemar chi2 {res.statistic:.3f} p-value {res.pvalue:.4f}")

# bootstrap 95% CI on the AUC difference (B - A) over resampled test rows
rng = np.random.default_rng(5); diffs = []
for _ in range(2000):
    idx = rng.integers(0, len(yte), len(yte))
    if yte[idx].sum() in (0, len(idx)): continue
    diffs.append(roc_auc_score(yte[idx], sb[idx]) - roc_auc_score(yte[idx], sa[idx]))
lo, hi = np.percentile(diffs, [2.5, 97.5])
print(f"bootstrap AUC gap (B-A): {np.mean(diffs):+.4f} 95% CI [{lo:+.4f}, {hi:+.4f}]")
print("CI excludes 0 => the gap is real, not sampling noise." if lo > 0 else
      "CI includes 0 => cannot claim B beats A.")

# --- contrast: two GBMs differing only by random_state -> tiny gap, within noise ---
B2 = GradientBoostingClassifier(random_state=17, subsample=0.7).fit(Xtr, ytr)
sb2 = B2.predict_proba(Xte)[: ,1]
diffs2 = []
for _ in range(2000):
    idx = rng.integers(0, len(yte), len(yte))
    if yte[idx].sum() in (0, len(idx)): continue
    diffs2.append(roc_auc_score(yte[idx], sb2[idx]) - roc_auc_score(yte[idx], sb[idx]))
lo2, hi2 = np.percentile(diffs2, [2.5, 97.5])
print(f"\ntwo GBMs (seed 17 vs 5): AUC gap {np.mean(diffs2):+.4f} 95% CI [{lo2:+.4f}, {hi2:+.4f}]")
print("CI spans 0 => the seed-to-seed difference is noise; don't ship on it." if lo2
      <= 0 <= hi2
      else "CI excludes 0.")

```

```

accuracy  A 0.8400  B 0.8904
ROC-AUC   A 0.9116  B 0.9585

```

```

McNemar discordant: A-wrong/B-right 184, A-right/B-wrong 63
McNemar chi2 58.300 p-value 0.0000
bootstrap AUC gap (B-A): +0.0470 95% CI [+0.0393, +0.0554]
CI excludes 0 => the gap is real, not sampling noise.

```

```

two GBMs (seed 17 vs 5): AUC gap +0.0022 95% CI [-0.0004, +0.0048]
CI spans 0 => the seed-to-seed difference is noise; don't ship on it.

```

Listing 9 — Split conformal prediction: distribution-free coverage

```

"""Listing 9: split conformal prediction -- distribution-free coverage you can actually
trust."""
import numpy as np
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

X, y = make_classification(n_samples=12000, n_features=20, n_informative=8,
                          n_classes=4, n_clusters_per_class=1, random_state=6)
Xtr, Xrest, ytr, yrest = train_test_split(X, y, test_size=0.5, random_state=6,
                                          stratify=y)
Xcal, Xte, ycal, yte = train_test_split(Xrest, yrest, test_size=0.5, random_state=6,
                                          stratify=yrest)
clf = RandomForestClassifier(300, random_state=6, n_jobs=-1).fit(Xtr, ytr)

# nonconformity = 1 - predicted prob of the TRUE class, scored on calibration set
cal_scores = 1 - clf.predict_proba(Xcal)[np.arange(len(ycal)), ycal]
for alpha in (0.10, 0.05):
    # finite-sample quantile (the (n+1)(1-alpha)/n correction gives the guarantee)
    n = len(cal_scores)
    qhat = np.quantile(cal_scores, np.ceil((n + 1) * (1 - alpha)) / n, method="higher")
    proba = clf.predict_proba(Xte)

```

```

sets = proba >= (1 - qhat)                                # include every class plausible
enough
covered = sets[np.arange(len(yte)), yte].mean()
avg_size = sets.sum(axis=1).mean()
print(f"target coverage {1-alpha:.0%} -> empirical {covered:.3f}    avg set size
{avg_size:.2f}")
print("\nGuarantee holds WITHOUT any distributional assumption: the prediction set")
print("contains the true label >= 1-alpha of the time, in exchange for sometimes")
print("returning more than one class (set size grows where the model is unsure).")

```

```

target coverage 90% -> empirical 0.904    avg set size 0.94
target coverage 95% -> empirical 0.951    avg set size 1.02

```

Guarantee holds WITHOUT any distributional assumption: the prediction set contains the true label $\geq 1 - \alpha$ of the time, in exchange for sometimes returning more than one class (set size grows where the model is unsure).

Pitfalls, comparisons and practical tips

Accuracy on imbalanced data. The evergreen trap: 95% accuracy at 5% prevalence can be worse than the always-negative baseline on the class you care about (Listing 1). The instant an interviewer says "imbalanced," drop accuracy for precision/recall/F1, PR-AUC, or balanced accuracy, and name the cost asymmetry.

ROC-AUC where PR-AUC belongs. ROC-AUC is prevalence-invariant, which is a feature for cross-dataset comparison and a bug for rare-positive problems — it stays high while precision collapses (Listing 2). Rare, valuable positive class $\rightarrow$ PR-AUC and precision/recall at an operating point.

Tuning the threshold on the test set. The decision threshold is a fitted parameter; choose it on validation, report on test (Listing 3). Same rule for choosing among calibration methods and for early-stopping rounds — anything selected by looking at a score is a parameter that can leak.

Confusing discrimination with calibration. A high AUC says the ranking is good, not that the probabilities are trustworthy. If a downstream decision reads the probability (expected value, pricing, triage), check a reliability curve and calibrate (Platt/isotonic) on held-out data (Listing 4). Never calibrate on the training set.

Optimizing accuracy when you'll consume probabilities. Accuracy is a step function blind to confidence; two models with equal accuracy can differ wildly in log loss (Listing 5). Train and select on a proper scoring rule (log loss, Brier) when the output is a probability, and remember log loss is unbounded — one confident-wrong prediction dominates.

MAPE near zero, RMSE with outliers, R^2 assumed bounded. MAPE explodes as targets approach zero (Listing 6) — use sMAPE/WAPE. RMSE is outlier-dominated; report MAE alongside if large errors are rare noise. R^2 can be negative (worse than the mean)

and never falls when features are added — use adjusted or out-of-sample R^2 for model comparison.

Order-blind metrics for ranking problems. Recall@k with k = list length ignores the permutation and calls a top-loaded and a bottom-loaded ranking identical (Listing 7). Use MRR/MAP/NDCG, reported @k, and pick by whether you need one hit, all hits high, or graded relevance.

Claiming a win inside the noise band. A metric delta is an estimate with variance. Pair the test (McNemar, paired bootstrap) and check whether the confidence interval clears zero before shipping — a seed-to-seed gap of 0.002 with a CI spanning zero is not an improvement (Listing 8). Cross-validate and report a spread, not a single split's point estimate.

Data leakage inflating every offline number. The whole chapter's metrics are honest only if the evaluation split is honest. All of Chapter 9's leakage traps (target encoding, SMOTE-before-split, selection-before-CV) manufacture beautiful offline scores that vanish in production — audit the pipeline before trusting the metric.

Metric-decision mismatch. The meta-pitfall: optimizing a metric that isn't the business objective. Precision/recall balance, threshold, and even the choice of ROC vs PR encode a cost model — write the cost model down first, then choose the metric that matches it.

Interview questions and answers

Q1. Define precision, recall, and specificity from the confusion matrix, and give a problem where each is the metric to maximize.

Precision = $TP/(TP+FP)$: of the flagged positives, how many are real — maximize when false alarms are expensive (spam filtering that must not eat real mail, a fraud freeze on good customers). Recall/sensitivity = $TP/(TP+FN)$: of the real positives, how many caught — maximize when misses are expensive (cancer screening, security). Specificity = $TN/(TN+FP)$: recall for the negative class, i.e. how well you clear the truly-negative. The interview move is naming the cost asymmetry first, because it's what picks the metric — precision and recall trade off through the threshold, so you can't max both without moving to F1 or fixing an operating point.

Q2. Why is accuracy misleading on imbalanced data, and what do you report instead?

Because the majority class dominates the count: at 6% positives a model that never fires scores 94% (Listing 1's baseline 0.938), so a "95%" model can be catching almost nothing — its recall was 0.367. Accuracy weights every row equally, which is exactly wrong when the rare class is the point. Report precision/recall/F1, PR-AUC, or balanced accuracy (per-class recall averaged, which drops the trivial baseline to 0.5), and state whether misses or false alarms cost more so the reader knows which of precision/recall to weight.

Q3. What is F1, why the harmonic mean, and when is it the wrong summary?

$F1 = 2PR/(P+R)$, the harmonic mean of precision and recall. The harmonic mean is dominated by the smaller of the two, so you can't inflate F1 by maxing one and tanking the other — it forces balance, which is why it beats accuracy on skewed classes. It's the wrong summary when the two errors have very different costs (use $F\beta$ to tilt, $\beta > 1$ for recall-heavy), when you care about ranking across all thresholds (use AUC), or when the negatives matter too (F1 ignores TN entirely — a fraud model and a random model can share an F1 while differing on specificity).

Q4. Explain ROC-AUC and PR-AUC and when each is appropriate.

ROC-AUC = area under TPR-vs-FPR = $P(\text{a random positive outranks a random negative})$; a threshold-free, prevalence-invariant discrimination score, chance = 0.5. PR-AUC (average precision) = area under precision-vs-recall; its no-skill baseline is the prevalence, not 0.5. Use ROC-AUC when classes are roughly balanced or both errors matter symmetrically, or for cross-dataset comparison. Use PR-AUC when positives are rare and are the class you care about, because precision puts FP against TP (not against the vast TN pool), so it registers false alarms that FPR — with the huge TN denominator — hides.

Q5. A model shows ROC-AUC 0.92 offline but the on-call team complains of endless false alarms in production. What happened?

Prevalence. ROC-AUC is invariant to base rate, so a model that discriminates well on a balanced offline sample keeps its 0.92 when deployed on a 1%-positive stream — but precision is not invariant: with 99% negatives, even a small FPR generates far more false positives than there are true positives, so the alerts the team sees are mostly wrong. Listing 2 measures exactly this drift (ROC-AUC 0.951 $\rightarrow$ 0.756 while PR-AUC 0.867 $\rightarrow$ 0.266). The fix is to evaluate with PR-AUC / precision-at-recall at the production prevalence, then set the threshold for a tolerable alert precision.

Q6. Is the 0.5 threshold optimal? How do you choose an operating point?

0.5 is a convention, not an optimum — it's optimal only under equal error costs and calibrated probabilities. The threshold is a business decision layered on the model: sweep the precision/recall curve on a *validation* set and pick the point that maximizes your objective — max F1, or the highest recall subject to precision $\geq$ target, or minimum expected cost given a cost matrix. Listing 3 shows the same model becoming three products: default 0.5 (P 0.974/R 0.664), max-F1 at 0.360 (P 0.893/R 0.749), precision $\geq$ 0.95 at 0.510. Choose on validation, report on test — and expect slight slippage because operating points don't transfer perfectly.

Q7. What does it mean for a classifier to be calibrated, and how is that different from being accurate?

Calibrated: among cases assigned probability p , a fraction p are actually positive — the numbers mean what they say. This is orthogonal to discrimination/accuracy: a model can rank every positive above every negative (AUC 1.0) yet output 0.51 vs 0.49, badly calibrated; or be well-calibrated but weak at ranking. You need calibration whenever a decision consumes the probability itself — expected-value thresholds, risk pricing, triage, stacking scores into another model. Diagnose with a reliability curve (predicted vs empirical frequency by bin); Listing 4's raw SVM traces an S-curve (pred 0.15 $\rightarrow$ actual 0.01).

Q8. Platt scaling vs isotonic regression — mechanism and when to use each.

Both are post-hoc maps from raw scores to calibrated probabilities, fit on a held-out set. Platt fits a one-parameter sigmoid (logistic regression on the scores) — cheap, low-variance, ideal when the miscalibration is sigmoidal or calibration data is scarce, but it can't fix non-sigmoidal distortions. Isotonic fits a free monotonic step function — corrects any monotone distortion, but is higher-variance and overfits small calibration sets. Rule of thumb: Platt for small data or sigmoidal error, isotonic when you have plenty of calibration data and an arbitrary monotone shape. Listing 4: both cut Brier $0.061 \rightarrow \sim 0.043$; with more data isotonic pulls ahead. Never calibrate on training data.

Q9. What is a proper scoring rule, and why does it matter?

A scoring rule for probabilistic predictions that is optimized (in expectation) by reporting your true beliefs — you cannot lower your expected score by lying about your confidence. Log loss and Brier are proper; accuracy is not (it only sees the hard label, so any probability > 0.5 scores the same). It matters because if you train or select on an improper rule you invite miscalibrated overconfidence, whereas a proper rule rewards being both sharp and honest. When the deliverable is a probability, the selection metric should be proper — otherwise you optimize for the wrong thing.

Q10. Compare log loss and Brier score.

Log loss = $-\text{mean log}(p \text{ at the true class})$: unbounded above, so it punishes confident-wrong catastrophically — a probability near 0 on an event that occurs contributes ~ 34.5 alone (Listing 5). Brier = mean squared error between probability and outcome: bounded $[0,1]$, gentler on a single disaster, and it decomposes into reliability – resolution + uncertainty, cleanly separating calibration from discrimination. Reach for log loss when one overconfident miss is unacceptable and you want maximum sensitivity; reach for Brier when you want a bounded, decomposable, outlier-tolerant number. Both are proper; both grade probabilities, not labels.

Q11. Two models have identical accuracy but very different log loss. How is that possible, and which ship?

Accuracy only checks which side of 0.5 each prediction falls; log loss reads the confidence. A model that's right at 0.51 and a model that's right at 0.95 share accuracy but the confident one has far lower log loss on its hits — and a model that's occasionally confident-and-wrong pays a huge log-loss penalty accuracy never sees (Listing 5: a bold model with *lower* accuracy 0.902 beats a timid always-0.6 model's log loss despite the timid one scoring 1.000 accuracy). Which ship depends on the deliverable: if you consume probabilities, prefer the better log loss; if you only act on the hard label at a fixed threshold, accuracy (at that threshold) is what matters.

Q12. RMSE vs MAE — how do they differ and how do you choose?

RMSE squares residuals before averaging (then roots), so large errors dominate; MAE averages absolute residuals, so every unit counts equally. Consequences: RMSE is outlier-sensitive (Listing 6: three big misses push RMSE $5.11 \rightarrow 6.75$ while MAE moves $4.08 \rightarrow 4.40$), $\text{RMSE} \geq \text{MAE}$ always, and they're minimized by different predictors — RMSE by the conditional mean, MAE by the conditional median. Choose RMSE when large errors are disproportionately costly (and it's the loss your squared-error model optimizes); choose MAE when errors should count linearly and you want robustness to a few bad rows.

Q13. When does MAPE break, and what replaces it?

$\text{MAPE} = \text{mean } |(y - \hat{y})/y|$ divides by the target, so it explodes as targets approach zero — Listing 6 sets five targets to 0.1 and MAPE jumps $8.5\% \rightarrow 497\%$ with unchanged predictions — and it's undefined at exactly zero. It's also asymmetric, penalizing over-prediction more than under-prediction, which biases models that optimize it. Replace it with sMAPE (symmetric, bounded) or WAPE / weighted MAPE (sum of absolute errors over sum of actuals, which dilutes the near-zero denominators). MAPE is only safe when targets are comfortably bounded away from zero and you want a scale-free percentage for cross-series comparison.

Q14. Can R^2 be negative? Why does adding features never decrease it, and what fixes that?

Yes — $R^2 = 1 - \text{SS_res}/\text{SS_tot}$ compares your model to the mean baseline, so any model worse than "predict the mean" is negative (Listing 6: predicting 0 everywhere scores -24.6). And on the training set R^2 never decreases when you add features, even pure noise, because extra parameters can only reduce in-sample residuals (Listing 6: +40 noise columns nudged R^2 $0.915 \rightarrow 0.919$). Adjusted R^2 penalizes for parameter count and so falls on junk features ($0.914 \rightarrow 0.912$); an honest alternative is out-of-sample R^2 on held-out data. Use adjusted or out-of-sample R^2 whenever comparing models with different numbers of features.

Q15. You need a single number for a search/recommendation system. Which metric and why?

A ranking metric, because the deliverable is an ordering and classification metrics ignore position. Pick by the product: MRR (mean reciprocal rank) when the user wants one good answer and only the first hit's rank matters (known-item search); MAP when relevance is binary and packing *all* relevants high matters; NDCG when relevance is graded (0–3) and/or you need cross-query comparability, since it discounts by log position and normalizes by the ideal ranking. Report it @k (NDCG@10) because users see a finite window. Listing 7 shows why order-blind metrics fail — recall@k over the whole list scores a top-loaded and bottom-loaded ranking identically.

Q16. Explain NDCG, including the discount and the normalization.

$\text{DCG} = \sum \text{rel}_i / \log_2(i+1)$: each item's (possibly graded) relevance is discounted by a logarithm of its rank, so a relevant item lower down contributes less — encoding that users pay more attention to the top. NDCG divides DCG by IDCG, the DCG of the ideal ordering (relevances sorted descending), giving a 0–1 score comparable across queries with different numbers of relevant items. Listing 7: best-first graded relevances score 1.000, worst-first 0.614. The two design choices are the position discount (why NDCG rewards ordering, not just presence) and the per-query normalization (why you can average it across heterogeneous queries).

Q17. Model B beats model A by 0.5% accuracy on the test set. Do you ship it?

Not on that number alone — 0.5% may be inside the noise. Test it: McNemar's test on the paired predictions (only the rows where they disagree carry signal), or a bootstrap/cross-validated confidence interval on the metric gap, and ship only if the interval clears zero. Listing 8 shows both outcomes: a real model change gives a CI [+0.039, +0.055] (clearly non-zero) while two models differing only by random seed give +0.0022 with CI [−0.0004, +0.0048] spanning zero — a mirage. A single split is one draw; report a spread, not a point.

Q18. Why McNemar's test rather than comparing two accuracy numbers, and why does it use only disagreements?

Because the two models are evaluated on the *same* test rows, so their errors are correlated — an unpaired comparison of two accuracies ignores that pairing and loses power. McNemar builds the 2×2 table of discordant pairs (A-right/B-wrong vs A-wrong/B-right) and tests whether that split is lopsided under the null of equal error rates. The concordant cells (both right, both wrong) carry no information about *which* model is better — they'd cancel — so only the discordant counts enter the statistic. Listing 8: 184 vs 54 discordant, χ^2 58.3, $p < 0.0001$.

Q19. How does a bootstrap confidence interval on a metric work, and what's its advantage?

Resample the test set with replacement to the same size, recompute the metric (or the gap between two models) on each resample, repeat a few thousand times, and take the 2.5/97.5 percentiles of the resulting distribution as a 95% CI. Its advantage is generality: it works for *any* metric — AUC, F1, NDCG — with no closed-form variance and no distributional assumption, and for model comparison you resample the same indices for both models (paired) to cancel the which-rows-landed-in-test variance. Listing 8 uses it on an AUC gap; the interval clearing zero is the ship/no-ship signal.

Q20. What is conformal prediction, and what exactly does it guarantee?

A wrapper that turns any trained model's scores into prediction *sets* with a finite-sample coverage guarantee. Split conformal: on a held-out calibration set compute nonconformity scores (e.g. $1 - \text{predicted prob of the true class}$), take the $[(n+1)(1-\alpha)]/n$ quantile, and at test time output every label whose nonconformity falls below it. Guarantee: the set contains the true label with probability $\geq 1-\alpha$, assuming only exchangeability — no distributional form, no large-sample asymptotics. Listing 9 hits 0.904 and 0.951 empirical coverage at 90%/95% targets. It's the one uncertainty method you can bolt onto any model for a coverage promise.

Q21. What's the price of conformal's guarantee, and how does calibration differ from it?

The price is variable set size: to keep coverage, the set expands where the model is unsure (two or more labels) and can be empty where nothing is plausible — the size *is* the honesty, an explicit "here are the possibilities at 95%." Calibration is a different promise: it makes the scalar probabilities accurate *on average* (70% of the 0.7-cases are positive) but gives no per-instance guarantee. Conformal gives a marginal coverage guarantee per prediction (a set that provably contains the truth $1-\alpha$ of the time) at the cost of set-valued output. Calibration → trustworthy numbers; conformal → a coverage contract.

Q22. Precision-recall or ROC for early-warning fraud detection at 0.5% prevalence — argue it.

Precision-recall. At 0.5% positives the negative pool is enormous, so ROC's FPR barely registers thousands of false positives while precision — FP against TP — feels every one, and the on-call team lives on precision (how many of my alerts are real) and recall (how much fraud did I catch). ROC-AUC would look reassuringly high and hide an unusable alert precision (Listing 2's mechanism). Report PR-AUC for the ranking, then fix an operating point by precision-at-recall against the fraud team's review capacity, and set the threshold there — the threshold, not AUC, is what ships.

Q23. Micro vs macro vs weighted averaging for multiclass metrics — what's the difference?

Macro averages the per-class metric with equal weight, so rare classes count as much as common ones — use it when minority-class performance matters (macro-F1 is the standard imbalanced-multiclass headline). Micro pools all TP/FP/FN across classes before computing, so it's dominated by frequent classes and, for single-label problems, equals overall accuracy. Weighted averages per-class metrics by class support — a compromise that respects prevalence while still being per-class. The interview point: on imbalanced multiclass, macro exposes the failure on rare classes that micro/accuracy conceal, so state which you're quoting and why.

Q24. How should cross-validation choice change for grouped, imbalanced, or time-series data?

Match the split to the dependence structure. Imbalanced → StratifiedKFold to keep class ratios stable across folds (plain KFold can hand a fold zero positives). Grouped (multiple rows per patient/user/session) → GroupKFold so no group straddles the train/test boundary, else the model memorizes the group and leaks. Time series → forward-chaining / TimeSeriesSplit (train on past, validate on future); a shuffled split lets the model peek at the future and manufactures optimism (Chapter 4's walk-forward vs shuffled gap). The default random KFold is only correct for i.i.d. data with balanced classes.

Q25. Should evaluation metrics ever be computed inside the cross-validation loop, or only at the end?

Compute the metric per fold and aggregate (mean $\pm$ spread), never pool predictions from a single split. Per-fold scoring gives you the variance — the honesty interval — that a single number hides, and it's what lets you tell a real gain from noise. Any step that *fits* — imputation, scaling, encoding, feature selection, resampling, threshold and calibration choice — must live inside the loop and be refit per fold, or its statistics leak the validation fold (Chapter 9's ledger). The metric itself doesn't leak, but everything feeding it can, so the whole preprocessing-plus-scoring stack goes inside the fold.

Q26. What is Cohen's kappa and when would you prefer it over accuracy?

Cohen's kappa = $(p_{\text{observed}} - p_{\text{chance}}) / (1 - p_{\text{chance}})$: agreement corrected for the agreement expected by chance given the class marginals. It's useful when chance agreement is high — heavy imbalance or when comparing to a majority-vote baseline — because it discounts the "free" accuracy a trivial classifier earns; kappa 0 means no better than chance, 1 means perfect. It also generalizes to inter-annotator agreement for label-quality audits, and weighted kappa penalizes ordinal misclassifications by distance. Prefer it (or balanced accuracy/MCC) over raw accuracy whenever the base rates make accuracy look good for free.

Q27. Why is Matthews correlation coefficient (MCC) considered a robust single-number classification metric?

MCC is the correlation between predicted and actual binary labels, computed from all four confusion cells: $(TP \cdot TN - FP \cdot FN) / \sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$, ranging -1 to $+1$. Because it uses TP, TN, FP, and FN symmetrically, it only scores high when the model does well on *both* classes — unlike F1, which ignores TN and can be inflated on imbalanced data. That balance makes it a strong default headline for imbalanced binary problems; a high MCC is hard to fake with a majority-class strategy. Its cost is lower interpretability than precision/recall, so pair it with the operating-point numbers stakeholders act on.

Q28. Walk through how you'd design the evaluation for a new binary classifier end to end.

Start from the decision and its cost asymmetry, which dictates the metric — rare valuable positives → PR-AUC plus precision-at-recall; balanced/both-errors → ROC-AUC or MCC; probabilities consumed downstream → add log loss/Brier and a reliability curve. Build an honest split matching the data's structure (stratified/group/time-series), keep every fitted step inside CV, and reserve a validation slice for the threshold and any calibration. Report the primary metric with a cross-validated spread and a bootstrap CI, choose the operating point on validation, calibrate if probabilities matter, and A/B or McNemar-test against the incumbent so you ship only when the interval clears zero. Optionally wrap conformal for a coverage guarantee. The through-line: the metric mirrors the decision, and every number carries an uncertainty band.

Chapter 11: Model Interpretability & Explainability

A model that predicts well but cannot be explained is a liability: regulators demand reasons for credit denials, doctors will not act on an oracle, debugging a silent failure requires knowing what the model attends to, and stakeholders who cannot interrogate a model will not trust it — often correctly, since Chapter 9's leakage stories showed that a suspiciously strong model is usually strong for the wrong reason. Interpretability is the toolkit for opening the box: measuring which features drive predictions overall (global), why *this* prediction came out the way it did (local), what shape each feature's effect takes, and whether the model's behavior differs across protected groups. Interviewers love this chapter's territory because it separates candidates who can call

`shap.TreeExplainer` from those who know what the numbers mean, when they lie, and which method answers which question.

The chapter follows the outline's arc. First the map: intrinsic versus post-hoc, model-specific versus model-agnostic, global versus local. Then feature importance and its traps — impurity importance's cardinality bias versus permutation importance (Listing 1), and coefficients that mislead when features are unscaled or collinear (Listing 2). Partial dependence and ICE plots come next, with a staged interaction that a PDP averages into a flat line while ICE curves expose it (Listing 3). The heart of the chapter is SHAP: Shapley values computed exactly from the game-theoretic definition and verified against KernelSHAP (Listing 4), then TreeSHAP on a churn-style model with the additivity check and global aggregation (Listing 5). LIME is built from scratch — perturb, weight, fit a local surrogate (Listing 6). A dedicated section stages how correlated features corrupt every importance method (Listing 7). The chapter closes with fairness: demographic parity, equalized odds, disparate impact, and a demonstration that dropping the protected attribute does not remove bias when a proxy carries it (Listing 8).

The interpretability map: global vs local, intrinsic vs post-hoc

Three axes organize every method in this chapter, and stating them is the expected first move in any interpretability interview answer.

Intrinsic versus post-hoc. An *intrinsically interpretable* model is readable by construction: a linear model's coefficients, a shallow decision tree's paths, a rule list. A *post-hoc* method explains an already-trained black box from the outside — permutation importance, PDP, SHAP, LIME all work this way. The classic trade-off is that intrinsic models buy transparency with capacity, though the gap is smaller than folklore suggests:

on tabular data a well-regularized GBM explained with TreeSHAP is often both more accurate and more *usefully* explained than a long linear model with hundreds of interacting coefficients. Rule of thumb: when the stakes are high and the signal is simple, prefer the intrinsic model; when you need the capacity, pair the black box with post-hoc tools and validate the explanations.

Model-specific versus model-agnostic. Coefficients belong to linear models; impurity importance and TreeSHAP belong to trees; saliency maps belong to differentiable networks. Model-agnostic methods — permutation importance, PDP/ICE, KernelSHAP, LIME — treat the model as a function $f(x)$ you can query, which makes them universal but often slower and dependent on how you generate the query points (the source of most of their failure modes, as Listing 7 shows).

Global versus local. A *global* explanation summarizes the model's overall behavior: which features matter on average, what the aggregate effect of a feature looks like. A *local* explanation accounts for one prediction: why did *this* applicant get denied? The two can disagree instructively — a feature that matters enormously for a handful of rows can look minor globally, and Listing 3's interaction shows a feature with a strong effect on every single row that averages to *zero* globally. SHAP bridges the axes: it is defined locally, and its local values aggregate into faithful global summaries (Listing 5).

The vocabulary that recurs downstream: **faithfulness** (does the explanation reflect what the model actually computes?), **stability** (do reruns give the same explanation?), and **plausibility** (does it match human intuition?) — and the trap that an explanation can be plausible without being faithful, which is worse than no explanation because it manufactures unearned trust.

Feature importance methods

Impurity importance (MDI). Tree ensembles ship with a free importance score: each split reduces impurity (Gini, entropy, or variance), and summing the reduction credited to each feature over all splits, weighted by the fraction of samples reaching the node, gives *mean decrease in impurity*. It costs nothing — the numbers were computed during training — and it is the default `feature_importances_` in scikit-learn. It has two well-known biases. First, the **cardinality bias**: features with many possible split points (continuous features, high-cardinality categoricals, ID-like columns) get more chances to produce a spuriously good split, so pure noise with many unique values earns nonzero importance. Listing 1 stages this: a continuous noise column scores MDI 0.074 — level with two *real* informative features and seven times the score of an equally useless binary noise column. Second, MDI is computed on the **training** data structure, so it reflects what the model used to fit, including what it used to overfit — it says nothing about generalization.

Permutation importance fixes both. Shuffle one column of *held-out* data, breaking its relationship with the target while preserving its marginal distribution, and measure how much the test metric drops: importance = performance lost when the feature's

information is destroyed. It is model-agnostic, evaluated on test data, and immune to cardinality bias — in Listing 1 the same continuous noise column falls to +0.002, statistically zero, while the real features keep their ranking. Repeat the shuffle (`n_repeats`) and report the spread. Its costs: one model evaluation pass per feature per repeat, and a serious failure mode under correlated features — permuting one member of a correlated pair creates rows that cannot occur in reality *and* lets the model recover the signal from the surviving twin, deflating both importances (Listing 7 measures the collapse). **Drop-column importance** — retrain without the feature, measure the drop — is the gold standard for "what does this feature contribute *given the others*," but costs a full retrain per feature and answers a subtly different question: a feature can be individually informative yet have zero drop-column importance because a correlated feature covers for it.

Coefficients as importance. For linear models, $|w_j|$ measures importance only after features share a scale: coefficients have units (target change per *unit* of feature), so a feature measured in dollars gets a coefficient 1,500 times smaller than an equally important feature measured in decades. Listing 2 builds two equally predictive features — income in dollars, age in years — and the raw fit reports coefficients 0.000116 versus 0.176: income looks irrelevant. Standardize both and the truth appears: 1.73 versus 1.75, statistically identical. The second trap is collinearity, inherited from Chapter 5: with two near-duplicate features ($\rho \approx 0.999$), bootstrap refits swing the individual coefficients from (+1.34, +0.35) to (+2.09, -0.39) — one even flips sign — while their *sum* stays a stable ~ 1.7 . Only the combined effect is identified; reading individual coefficients as importances under collinearity is reading noise.

Partial dependence and ICE plots

Importance says *how much* a feature matters; **partial dependence** says *what shape* its effect takes. The partial dependence of f on feature j at value v is the average prediction when feature j is forced to v for everyone:

$$PD_j(v) = \frac{1}{n} \sum_{i=1}^n f(x_i^{(-j)}, x_j = v)$$

where $x_i^{(-j)}$ keeps row i 's other features. Sweep v over a grid and plot: the resulting curve reveals linearity, saturation, thresholds, non-monotonicity. Two caveats define the method. First, forcing $x_j = v$ for *everyone* assumes feature j is independent of the rest — with correlated features the average is taken over synthetic rows that cannot exist (a 1.9 m person forced to weigh 45 kg), the same pathology as permutation importance. Second, and the point of Listing 3: **PDP is an average, and averages hide heterogeneity.**

ICE (individual conditional expectation) plots fix the second caveat by not averaging: one curve per row, showing that row's prediction as feature j sweeps the grid — the PDP is exactly the mean of the ICE curves. Listing 3 constructs the canonical failure: $y = 2x_1$

for group B but $y = -2x_1$ for group A. The PDP of x_1 is flat — fitted slope -0.01 , "no effect" — while the ICE curves split into two fans with slopes -1.68 and $+1.69$. A feature with a strong effect on *every single row* averages to nothing because the effects cancel. The practitioner's habit: always plot ICE under (or instead of) the PDP; if the ICE curves are roughly parallel the PDP is a faithful summary, and if they cross or fan out there is an interaction that the PDP is erasing — go find the interacting feature (centered "c-ICE" curves, which pin all curves to zero at the grid's left edge, make the divergence even easier to see).

Shapley values and SHAP

The most principled answer to "how much did each feature contribute to this prediction?" comes from cooperative game theory. Treat the features as players in a coalition game where the payout of a coalition S is the model's expected prediction when only the features in S are known:

$$v(S) = E[f(x_S, X_{\bar{S}})]$$

— features in S fixed to this instance's values, the rest averaged over a background distribution. The **Shapley value** of feature j is its marginal contribution averaged over every possible order in which features could be revealed:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [v(S \cup \{j\}) - v(S)]$$

It is the unique attribution satisfying four axioms: **efficiency** (contributions sum exactly to $f(x) - E[f(X)]$ — the explanation accounts for the whole prediction), **symmetry** (identical contributors get identical credit), **dummy** (a feature that never changes any coalition's value gets zero), and **additivity** (consistent across ensembled models).

Efficiency is the property the other methods lack and the one to name in an interview: SHAP values are a complete decomposition of the prediction's distance from the base rate, not just a ranking. Listing 4 implements the definition literally — all 2^d coalitions, factorial weights — and the additivity check lands exactly: base + contributions = 5.213 = $f(x)$ to the third decimal.

The definition costs $O(2^d)$, hence the approximations. **KernelSHAP** (model-agnostic) exploits the fact that Shapley values are the solution to a specific weighted linear regression on coalition indicators: sample coalitions, evaluate $v(S)$ by marginalizing over a background dataset, and solve the regression with the Shapley kernel weights. Listing 4 confirms the equivalence — with enough samples KernelSHAP matches the exact enumeration to four decimal places (max gap 0.0000). It works on any model but each explanation costs (samples $\times$ background) model evaluations; explaining a large test set can be slower than training. **TreeSHAP** (model-specific) computes *exact* Shapley values for tree ensembles in polynomial time — $O(TLD^2)$ per instance, with T trees, L leaves, depth D — by pushing coalitions through the tree structure analytically. That is the

practical reason SHAP dominates tabular ML: for GBMs and forests you get axiomatic attributions at production speed. Listing 5 runs it on a churn-style GBM: the SHAP matrix for 1,500 test rows arrives in seconds, one customer's prediction decomposes into base -0.012 plus contributions $-4.911 = -4.924$ log-odds $\rightarrow p = 0.007$, matching `predict_proba` exactly, with `contract_len` (-3.614) doing most of the pushing. Note the unit: for classifiers TreeSHAP explains the *margin* (log-odds), not the probability — contributions are additive in log-odds space and only there.

From local to global. Because every row gets an exact decomposition, global importance is just aggregation: mean $|\text{SHAP}|$ per feature. Listing 5 shows the mean- $|\text{SHAP}|$ ranking and the permutation ranking agreeing feature-for-feature — reassuring when it happens, diagnostic when it does not (disagreement usually points at correlated features or at a feature that matters hugely for a few rows). The `shap` library's summary ("beeswarm") plot layers more: each dot a row, x-position the SHAP value, color the feature's value — so you read magnitude, direction, and nonlinearity in one figure. SHAP's caveats: computed against a **background/reference** distribution whose choice changes the values (explaining "versus the average customer" differs from "versus approved customers"); interventional marginalization inherits the impossible-points problem under correlated features (Listing 7); and SHAP explains *the model*, not *the world* — a feature can get large SHAP values because the model leans on it as a proxy, not because it causes anything.

LIME: local surrogate models

LIME (Local Interpretable Model-agnostic Explanations) answers the local question with a different bargain: approximate the black box *near this instance* with a model simple enough to read. Formally it minimizes $\mathcal{L}(f, g, \pi_x) + \Omega(g)$ over a family G of interpretable models — find the surrogate g (usually sparse linear) that best matches f on points weighted by proximity π_x to the instance, penalizing complexity. The algorithm, which Listing 6 implements in ~ 15 lines: (1) sample perturbations around x (Gaussian noise in standardized space for tabular data; masking words or superpixels for text and images); (2) query the black box for each perturbation; (3) weight each perturbed point by an exponential proximity kernel; (4) fit a weighted ridge regression — its coefficients are the explanation.

Listing 6 explains a decision-boundary customer ($p = 0.499$) and gets a readable answer: per standard deviation, `contract_len` pushes -0.278 , `monthly_fee` $+0.190$, with a weighted local fit R^2 of 0.623. Two practical lessons are baked in. First, the *choice of instance matters*: explaining a saturated prediction ($p \approx 0.003$) yields a nearly flat local surface and a meaningless surrogate — LIME explanations are only as informative as the local gradient. Second, *stability must be checked, never assumed*: LIME's perturbations are random, so reruns can reorder features; with 5,000 perturbations the top feature is stable across five seeds here, but the honest workflow always reruns with different seeds and reports whether the story holds. LIME's knobs — kernel width, number of

perturbations, surrogate sparsity — are unprincipled in the sense that no axiom picks them, and the perturbation distribution can wander off the data manifold. Versus SHAP: LIME is faster per explanation than KernelSHAP and more flexible across modalities (its image/text variants are natural), but its attributions satisfy no efficiency axiom — they are local slopes, not a decomposition of the prediction — and are generally less stable. When both are cheap (trees → TreeSHAP), SHAP wins; LIME earns its keep on expensive black boxes and non-tabular inputs.

When explanations mislead: correlated features

Every post-hoc method in this chapter queries the model on synthetic points, and correlated features make those points lie. Listing 7 duplicates a feature ($\rho = 1.000$) and watches the damage. Permutation importance of x_1 collapses from 1.638 to 0.478 — not because x_1 matters less but because when it is shuffled, its twin still carries the signal, so performance barely drops; read naively, "importance 0.478" understates the causal role by 3.4×. SHAP splits the credit instead: mean |SHAP| 1.659 alone becomes 0.864 + 0.797 for the pair — the *sum* is conserved (the efficiency axiom guarantees the total decomposition), but each individual number halves. Neither behavior is a bug; they are different, defensible answers to "who gets credit for shared information" — and both surprise anyone expecting the duplicate to be free. The listing's last lines expose the deeper mechanism: after permuting one twin, the correlation between the two drops from +1.000 to -0.024, i.e. the method is scoring the model on rows where near-identical features disagree wildly — a region the model never saw and reality never produces.

The defenses: cluster correlated features and permute/interpret them as groups; use drop-column importance when you can afford retraining; prefer conditional/observational SHAP variants when the independence assumption is untenable (knowing they trade one subtlety for another — credit then leaks to correlated bystanders); and above all report correlation structure alongside any importance table. The interview sound bite: *feature importance is well-defined only up to the correlation structure of the features* — any single number per feature is a projection of a joint story.

Fairness and bias detection

Interpretability's highest-stakes application is auditing models for discrimination. The vocabulary: a **protected attribute** A (race, gender, age...), a prediction $\hat{Y}$, an outcome Y , and *group fairness criteria* that compare error or selection rates across groups.

Demographic parity demands equal selection rates: $P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$; its ratio form is **disparate impact**, with the legal "80% rule" flagging ratios below 0.8.

Equalized odds demands equal TPR and FPR across groups — errors, not just selections, must be balanced; **equal opportunity** relaxes it to TPR only (qualified candidates get equal chances). **Calibration within groups** demands that a score of 0.7 mean 70% in every group. A foundational impossibility theorem (Kleinberg et al.;

Chouldechova): when base rates differ across groups, calibration and equalized odds cannot hold simultaneously except in degenerate cases — *fairness metrics are choices with trade-offs, not boxes to tick all at once*, and stating this is the strongest single sentence available in a fairness interview.

The most common misconception is **fairness through unawareness** — "we don't use race as a feature, so the model can't discriminate." Listing 8 dismantles it: a hiring model trained *without* the protected attribute, on historical labels where 45% of group-0's qualified candidates were erased (biased past decisions), with a zip-code proxy that strongly encodes group. The model reconstructs the bias through the proxy: selection rates 0.273 vs 0.331, disparate impact 0.82 (hovering at the legal line), and against *true* merit a TPR gap of +0.093 — qualified group-0 candidates are 9 points less likely to be selected. The listing also teaches the subtler lesson: audited against the *historical labels*, the FPR gap (−0.108) looks like the model over-selects group 0 — the audit's ground truth is itself contaminated, and **a fairness audit is only as good as the labels it audits against**. Mitigations come in three families: *pre-processing* (reweigh or repair the training data, remove proxies — hard, since proxies hide), *in-processing* (fairness constraints or adversarial debiasing in the loss), *post-processing* (group-specific thresholds). Listing 8 applies the cheapest, post-processing to equalize opportunity: per-group thresholds chosen to hit TPR ≈ 0.80 on merit in both groups collapse the TPR gap to −0.006 and, here, drag demographic parity to 0.98 as a side effect. SHAP plugs in upstream of all of this: computing SHAP values for the proxy feature by group reveals *how* the model uses the proxy, turning "the metric is bad" into "this feature carries the bias."

Code implementations

Listing 1 — Impurity vs permutation importance: the cardinality trap

```
"""Listing 1: impurity importance vs permutation importance -- the cardinality trap."""
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance
from sklearn.model_selection import train_test_split
from sklearn.datasets import make_classification

rng = np.random.default_rng(0)
X, y = make_classification(n_samples=4000, n_features=5, n_informative=3,
                          n_redundant=0, random_state=0)
df = pd.DataFrame(X, columns=[f"real_{i}" for i in range(5)])
# a pure-noise feature with many unique values (like an ID or a timestamp)
df["noise_hicard"] = rng.normal(size=len(df)) # continuous noise
# a pure-noise feature with only 2 levels
df["noise_binary"] = rng.integers(0, 2, size=len(df)) # binary noise

Xtr, Xte, ytr, yte = train_test_split(df, y, random_state=0)
rf = RandomForestClassifier(n_estimators=300, random_state=0).fit(Xtr, ytr)
print(f"test accuracy: {rf.score(Xte, yte):.3f}")

# 1) impurity-based (Gini) importance -- computed on TRAINING data structure
```

```

imp = pd.Series(rf.feature_importances_, index=df.columns).sort_values(ascending=False)
print("\nimpurity (MDI) importance:")
for k, v in imp.items():
    print(f" {k:<14}{v:+.4f}")

# 2) permutation importance -- computed on HELD-OUT data
pi = permutation_importance(rf, Xte, yte, n_repeats=20, random_state=0)
ps = pd.Series(pi.importances_mean, index=df.columns).sort_values(ascending=False)
print("\npermutation importance (test):")
for k, v in ps.items():
    print(f" {k:<14}{v:+.4f}")

```

Output:

```

test accuracy: 0.849

impurity (MDI) importance:
real_1      0.3766
real_4      0.2100
real_3      0.1788
real_2      0.0763
noise_hicard 0.0740
real_0      0.0739
noise_binary 0.0105

permutation importance (test):
real_1      +0.2762
real_4      +0.1315
real_3      +0.1163
real_0      +0.0054
real_2      +0.0052
noise_binary +0.0024
noise_hicard +0.0021

```

The continuous noise column earns MDI 0.0740 — indistinguishable from two genuinely informative features and 7× the equally-useless binary noise, purely because it offers thousands of candidate split points. Permutation importance on held-out data sends both noise columns to ~0.002 and preserves the real features' ranking.

Listing 2 — Coefficients as importance: scale and collinearity traps

```

"""Listing 2: coefficients as importance -- scale and collinearity traps."""
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler

rng = np.random.default_rng(1)
n = 5000
income_usd = rng.normal(60_000, 15_000, n) # dollars: huge scale
age_years = rng.normal(40, 10, n) # years: small scale
# both matter EQUALLY in standardized terms
z = 1.0 * (income_usd - 60_000) / 15_000 + 1.0 * (age_years - 40) / 10
y = (z + rng.normal(0, 1, n) > 0).astype(int)

X = np.c_[income_usd, age_years]
raw = LogisticRegression(max_iter=5000).fit(X, y)
print("raw coefficients :", np.round(raw.coef_[0], 6),
      "\n-> income looks ~0, age looks big")

Xs = StandardScaler().fit_transform(X)
std = LogisticRegression(max_iter=5000).fit(Xs, y)

```

```

print("scaled coefficients:", np.round(std.coef_[0], 3),
      "-> equal importance revealed")

# collinearity: two near-duplicate features split (or flip) the credit
x1 = rng.normal(size=n)
x2 = x1 + rng.normal(0, 0.05, n) # rho ~ 0.999
y2 = (x1 + rng.normal(0, 1, n) > 0).astype(int)
for i in range(3): # three bootstrap refits
    idx = rng.integers(0, n, n)
    m = LogisticRegression(max_iter=5000).fit(np.c_[x1, x2][idx], y2[idx])
    print(f"bootstrap {i}: coef(x1)={m.coef_[0][0]:+.2f} coef(x2)={m.coef_[0][1]:+.2f}"
          f" sum={m.coef_[0].sum():+.2f}")

```

Output:

```

raw coefficients : [1.16000e-04 1.76346e-01] -> income looks ~0, age looks big
scaled coefficients: [1.732 1.749] -> equal importance revealed
bootstrap 0: coef(x1)=+1.34 coef(x2)=+0.35 sum=+1.69
bootstrap 1: coef(x1)=+2.09 coef(x2)=-0.39 sum=+1.70
bootstrap 2: coef(x1)=+1.88 coef(x2)=-0.03 sum=+1.85

```

Two equally important features differ by a factor of 1,500 in raw coefficient purely because of units; standardization reveals the tie (1.732 vs 1.749). Under near-perfect collinearity, individual coefficients swing wildly across bootstrap refits — one flips sign — while their sum stays a stable ~1.7: only the combined effect is identified.

Listing 3 — PDP and ICE from scratch: averages hide interactions

```

"""Listing 3: partial dependence + ICE from scratch -- averages hide interactions."""
import numpy as np
from sklearn.ensemble import GradientBoostingRegressor

rng = np.random.default_rng(2)
n = 3000
x1 = rng.uniform(-2, 2, n)
x2 = rng.integers(0, 2, n) # binary group flag
x3 = rng.normal(size=n)
# interaction: x1 helps group 1, hurts group 0 -- effects cancel on average
y = np.where(x2 == 1, 2.0 * x1, -2.0 * x1) + 0.5 * x3 + rng.normal(0, 0.3, n)
X = np.c_[x1, x2, x3]
gbm = GradientBoostingRegressor(random_state=0).fit(X, y)
print(f"train R^2: {gbm.score(X, y):.3f}")

def pdp_ice(model, X, feat, grid):
    """Return (pdp, ice): ice[i,g] = f(x_i with feature `feat` set to grid[g])."""
    ice = np.empty((len(X), len(grid)))
    for g, v in enumerate(grid):
        Xmod = X.copy()
        Xmod[:, feat] = v # intervene: set feature for EVERYONE
        ice[:, g] = model.predict(Xmod)
    return ice.mean(axis=0), ice # PDP = average of ICE curves

grid = np.linspace(-2, 2, 9)
pdp, ice = pdp_ice(gbm, X, feat=0, grid=grid)
print("\ngrid      :", np.round(grid, 1))
print("PDP (average)  :", np.round(pdp, 2), "<- looks FLAT: 'x1 has no effect'")

g0, g1 = ice[x2 == 0].mean(axis=0), ice[x2 == 1].mean(axis=0)
print("ICE mean, x2=0  :", np.round(g0, 2), "<- steep negative")
print("ICE mean, x2=1  :", np.round(g1, 2), "<- steep positive")

```



```
slope = np.polyfit(grid, pdp, 1)[0]
s0, s1 = np.polyfit(grid, g0, 1)[0], np.polyfit(grid, g1, 1)[0]
print(f"\nslopes: PDP {slope:+.2f} vs group0 {s0:+.2f} vs group1 {s1:+.2f}")
```

Output:

```
train R^2: 0.939

grid          : [-2.  -1.5 -1.  -0.5  0.   0.5  1.   1.5  2. ]
PDP (average) : [-0.03  0.07  0.05  0.05  0.07  0.07 -0.01 -0.04 -0.01] <- looks
FLAT: 'x1 has no effect'
ICE mean, x2=0 : [ 3.57  2.31  1.35  0.98  0.14 -0.83 -1.67 -2.04 -3.77] <- steep
negative
ICE mean, x2=1 : [-3.72 -2.23 -1.29 -0.91 -0.   1.   1.68  2.   3.83] <- steep
positive

slopes: PDP -0.01 vs group0 -1.68 vs group1 +1.69
```

The feature x_1 has a strong effect on every row — slope -1.68 in one group, $+1.69$ in the other — yet the PDP's fitted slope is -0.01 . An interaction the PDP averages into invisibility is fully exposed the moment the ICE curves are split.

Listing 4 — Exact Shapley values from the definition, verified against KernelSHAP

```
"""Listing 4: exact Shapley values from the definition, verified against KernelSHAP."""
import itertools, math
import numpy as np
from sklearn.ensemble import RandomForestRegressor
import shap

rng = np.random.default_rng(3)
n, d = 2000, 4
X = rng.normal(size=(n, d))
y = 3*X[:, 0] + 2*X[:, 1]*X[:, 2] + rng.normal(0, 0.1, n) # interaction inside
model = RandomForestRegressor(n_estimators=200, random_state=0).fit(X, y)

background = X[rng.choice(n, 100, replace=False)] # reference sample
x = X[0] # instance to explain

def value(S):
    """v(S) = E[f(x_S, X_rest)]: features in S fixed to x, rest drawn from
    background."""
    Xb = background.copy()
    Xb[:, list(S)] = x[list(S)]
    return model.predict(Xb).mean()

def exact_shapley(d):
    phi = np.zeros(d)
    for j in range(d):
        others = [k for k in range(d) if k != j]
        for r in range(d): # all coalition sizes
            for S in itertools.combinations(others, r):
                w = math.factorial(len(S)) * math.factorial(d - len(S) - 1) /
                math.factorial(d)
                phi[j] += w * (value(S + (j,)) - value(S)) # marginal contribution
    return phi

phi = exact_shapley(d)
base = value(()) # E[f(X)]
print("exact Shapley phi :", np.round(phi, 3))
```

```
print(f"base + sum(phi) = {base + phi.sum():.3f} vs f(x) = {model.predict([x])[0]:.3f}"
      "    <- local accuracy (additivity)")

ks = shap.KernelExplainer(model.predict, background)
phi_ks = ks.shap_values(x, nsamples=2000, silent=True)
print("KernelSHAP phi      :", np.round(phi_ks, 3))
print(f"max |exact - kernel| = {np.abs(phi - phi_ks).max():.4f}")
```

Output:

```
exact Shapley phi : [ 5.15 -0.279 -0.298 -0.132]
base + sum(phi) = 5.213 vs f(x) = 5.213    <- local accuracy (additivity)
KernelSHAP phi    : [ 5.15 -0.279 -0.298 -0.132]
max |exact - kernel| = 0.0000
```

Thirty lines implement the game-theoretic definition — every coalition, factorial weights, marginalization over a background sample. The efficiency axiom lands exactly (5.213 = 5.213), and KernelSHAP's regression-based approximation reproduces the exact enumeration to four decimals.

Listing 5 — TreeSHAP on a GBM: local explanation, additivity, global aggregation

```
"""Listing 5: TreeSHAP on a GBM -- local explanations, additivity, global
aggregation."""
import numpy as np
import pandas as pd
import shap
from sklearn.datasets import make_classification
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import train_test_split
from sklearn.inspection import permutation_importance

cols = ["tenure", "monthly_fee", "support_calls", "usage_gb", "age", "contract_len"]
X, y = make_classification(n_samples=6000, n_features=6, n_informative=4,
                          n_redundant=1, random_state=7)
X = pd.DataFrame(X, columns=cols)
Xtr, Xte, ytr, yte = train_test_split(X, y, random_state=0)
gbm = GradientBoostingClassifier(random_state=0).fit(Xtr, ytr)
print(f"test accuracy: {gbm.score(Xte, yte):.3f}")

expl = shap.TreeExplainer(gbm)                                # polynomial-time exact for trees
base_val = float(np.ravel(expl.expected_value)[0])
sv = expl.shap_values(Xte)                                    # (n_test, d) in log-odds space
print(f"shap matrix shape: {sv.shape}, base value (mean log-odds): {base_val:.3f}")

# local: one customer, additivity check in log-odds
i = 5
margin = base_val + sv[i].sum()
print(f"\ncustomer {i}: base {base_val:+.3f} + contributions {sv[i].sum():+.3f}"
      f" = {margin:+.3f} log-odds -> p = {1/(1+np.exp(-margin)):.3f}"
      f" (model says {gbm.predict_proba(Xte.iloc[[i]])[0,1]:.3f}")
contrib = pd.Series(sv[i], index=cols).sort_values(key=np.abs, ascending=False)
for k, v in contrib.items():
    print(f" {k:<13}{v:+.3f}")

# global = aggregate of locals: mean |SHAP| vs permutation importance rankings
gshap = pd.Series(np.abs(sv).mean(axis=0), index=cols).sort_values(ascending=False)
pi = permutation_importance(gbm, Xte, yte, n_repeats=10, random_state=0)
```



```

gperm = pd.Series(pi.importances_mean, index=cols).sort_values(ascending=False)
print("\nglobal mean|SHAP| ranking:", list(gshap.index))
print("permutation ranking      :", list(gperm.index))

```

Output:

```

test accuracy: 0.932
shap matrix shape: (1500, 6), base value (mean log-odds): -0.012

customer 5: base -0.012 + contributions -4.911 = -4.924 log-odds -> p = 0.007 (model
says 0.007)
  contract_len -3.614
  support_calls -0.931
  age          -0.237
  monthly_fee  -0.147
  usage_gb      +0.022
  tenure       -0.005

global mean|SHAP| ranking: ['contract_len', 'monthly_fee', 'support_calls', 'age',
'usage_gb', 'tenure']
permutation ranking      : ['contract_len', 'monthly_fee', 'support_calls', 'age',
'usage_gb', 'tenure']

```

One customer's 0.007 churn probability decomposes exactly: base log-odds -0.012 plus feature contributions -4.911 , dominated by `contract_len` (-3.614). Aggregating |SHAP| across 1,500 rows gives a global ranking that matches permutation importance feature-for-feature.

Listing 6 — LIME from scratch: perturb, weight, fit a local linear model

```

"""Listing 6: LIME from scratch -- perturb, weight by proximity, fit a local linear
model."""
import numpy as np
import pandas as pd
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.linear_model import Ridge
from sklearn.datasets import make_classification
from sklearn.preprocessing import StandardScaler

cols = ["tenure", "monthly_fee", "support_calls", "usage_gb", "age", "contract_len"]
X, y = make_classification(n_samples=6000, n_features=6, n_informative=4,
                          n_redundant=1, random_state=7)
model = GradientBoostingClassifier(random_state=0).fit(X, y)
scaler = StandardScaler().fit(X)

def lime_explain(x, n_samples=5000, kernel_width=1.5, seed=0):
    rng = np.random.default_rng(seed)
    d = len(x)
    # 1) perturb around x in STANDARDIZED space (Gaussian cloud)
    Z_std = scaler.transform([x]).repeat(n_samples, axis=0) \
        + rng.normal(0, 1, (n_samples, d))
    Z = scaler.inverse_transform(Z_std)
    # 2) query the black box on the perturbations
    f = model.predict_proba(Z)[:, 1]
    # 3) proximity kernel: nearby points dominate the local fit
    dist = np.linalg.norm(Z_std - scaler.transform([x]), axis=1)
    w = np.exp(-dist**2 / kernel_width**2)
    # 4) weighted linear surrogate = the explanation
    lin = Ridge(alpha=1.0).fit(Z_std, f, sample_weight=w)
    return lin.coef_, lin.intercept_, lin.score(Z_std, f, sample_weight=w)

```

```

p_all = model.predict_proba(X[:, 1])
x = X[np.argmin(np.abs(p_all - 0.5))]
# explain a BOUNDARY case, not a saturated one
coef, b, r2 = lime_explain(x)
print(f"model p(churn|x) = {model.predict_proba([x])[0,1]:.3f}")
print(f"local surrogate fit R^2 (weighted): {r2:.3f}\n")
expl = pd.Series(coef, index=cols).sort_values(key=np.abs, ascending=False)
print("LIME local coefficients (per +1 std of feature):")
for k, v in expl.items():
    print(f"    {k:<14}{v:+.4f}")

# instability check: rerun with different perturbation seeds
tops = [pd.Series(lime_explain(x, seed=s)[0], index=cols)
        .abs().idxmax() for s in range(5)]
print("\ntop feature across 5 seeds:", tops)

```

Output:

```

model p(churn|x) = 0.499
local surrogate fit R^2 (weighted): 0.623

LIME local coefficients (per +1 std of feature):
contract_len    -0.2778
monthly_fee      +0.1896
age              -0.1462
usage_gb         -0.0466
tenure           +0.0141
support_calls    -0.0058

top feature across 5 seeds: ['contract_len', 'contract_len', 'contract_len',
'contract_len', 'contract_len']

```

Fifteen lines of algorithm: Gaussian perturbations, black-box queries, proximity weights, weighted ridge. At the decision boundary the surrogate reads cleanly (`contract_len` -0.278 per std) and the top feature survives five reruns. Explaining a saturated instance ($p \approx 0.003$) instead yields a near-flat local surface and R^2 of 0.07 — LIME is only as informative as the local gradient.

Listing 7 — Correlated features break importance: credit splitting and impossible points

```

"""Listing 7: correlated features break importance -- credit splitting and impossible
points."""
import numpy as np
import pandas as pd
import shap
from sklearn.ensemble import RandomForestRegressor
from sklearn.inspection import permutation_importance
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(4)
n = 3000
x1 = rng.normal(size=n)
dup = x1 + rng.normal(0, 0.01, n) # near-perfect copy of x1
x3 = rng.normal(size=n)
y = 2.0 * x1 + 1.0 * x3 + rng.normal(0, 0.2, n)

for name, feats in [("without duplicate", {"x1": x1, "x3": x3}),
                    ("with duplicate", {"x1": x1, "x1_dup": dup, "x3": x3})]:
    X = pd.DataFrame(feats)

```

```

Xtr, Xte, ytr, yte = train_test_split(X, y, random_state=0)
rf = RandomForestRegressor(n_estimators=100, random_state=0).fit(Xtr, ytr)
pi = permutation_importance(rf, Xte, yte, n_repeats=10, random_state=0)
sv = shap.TreeExplainer(rf).shap_values(Xte)
ms = np.abs(sv).mean(axis=0)
row = " ".join(f"{c}: perm {p:.3f} shap {s:.3f}"
               for c, p, s in zip(X.columns, pi.importances_mean, ms))
print(f"{name} {row}")

# why permuting correlated features is dangerous: it manufactures impossible rows
X2 = pd.DataFrame({"x1": x1, "x1_dup": dup, "x3": x3})
Xp = X2.copy(); Xp["x1"] = rng.permutation(Xp["x1"].values)
print(f"\ncorr(x1, x1_dup) real data      : {X2['x1'].corr(X2['x1_dup']):+.3f}")
print(f"corr(x1, x1_dup) after permute : {Xp['x1'].corr(Xp['x1_dup']):+.3f}")
" <- evaluates the model on points that cannot exist"

```

Output:

```

without duplicate  x1: perm 1.638 shap 1.659  x3: perm 0.352 shap 0.796
with duplicate    x1: perm 0.478 shap 0.864  x1_dup: perm 0.377 shap 0.797  x3: perm
0.352 shap 0.793

corr(x1, x1_dup) real data      : +1.000
corr(x1, x1_dup) after permute : -0.024  <- evaluates the model on points that cannot
exist

```

Adding a duplicate collapses x_1 's permutation importance $1.638 \rightarrow 0.478$ (the twin covers for it) while SHAP splits the credit $1.659 \rightarrow 0.864 + 0.797$ with the sum conserved. The final lines show why: permutation destroys a correlation of $+1.000$, scoring the model on rows that cannot exist.

Listing 8 — Fairness auditing: unawareness fails, and the audit depends on the ground truth

```

"""Listing 8: fairness auditing -- unawareness fails, and the audit depends on the
ground truth."""
import numpy as np
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(5)
n = 20000
group = rng.integers(0, 2, n)                # protected attribute A (0 / 1)
skill = rng.normal(0, 1, n)                  # true qualification, same
distribution in both groups
zipc = 0.9 * group + rng.normal(0, 0.5, n)   # proxy feature: strongly encodes
group
merit = (skill + rng.normal(0, 0.3, n) > 0.4).astype(int)    # who is ACTUALLY
qualified
y = np.where(rng.random(n) < 0.45 * (1 - group), 0, merit)
# history erased 45% of g0 hires

X = np.c_[skill, zipc]                       # protected attribute NOT used as a
feature
(Xtr, Xte, ytr, yte, gtr, gte,
 mtr, mte) = train_test_split(X, y, group, merit, random_state=0)
clf = GradientBoostingClassifier(random_state=0).fit(Xtr, ytr)  # trained on BIASED
labels
pred = clf.predict(Xte)

```

```

def audit(yhat, ytruth, g, label):
    r0, r1 = yhat[g == 0].mean(), yhat[g == 1].mean()           # selection rates
    tpr = [yhat[(g == k) & (ytruth == 1)].mean() for k in (0, 1)]
    fpr = [yhat[(g == k) & (ytruth == 0)].mean() for k in (0, 1)]
    print(f"{label}")
    print(f"    selection rate    g0 {r0:.3f}    g1 {r1:.3f}    dem-parity diff {r1-r0:+.3f}"
          f"    disparate impact {r0/r1:.2f}")
    print(f"    TPR                g0 {tpr[0]:.3f}    g1 {tpr[1]:.3f}    gap {tpr[1]-tpr[0]:+.3f}")
    print(f"    FPR                g0 {fpr[0]:.3f}    g1 {fpr[1]:.3f}    gap {fpr[1]-fpr[0]:+.3f}")

audit(pred, yte, gte, "audited against HISTORICAL labels (the biased ones):")
print()
audit(pred, mte, gte, "audited against TRUE merit:")

# mitigation: group-specific thresholds to equalize TPR on merit (equal opportunity)
proba = clf.predict_proba(Xte)[: , 1]
thr = {k: np.quantile(proba[(gte == k) & (mte == 1)], 0.20) for k in (0, 1)}
pred_eo = (proba > np.where(gte == 0, thr[0], thr[1])).astype(int)
print(f"\ngroup thresholds g0 {thr[0]:.3f} / g1 {thr[1]:.3f} (aim: TPR ~= 0.80 on merit)")
audit(pred_eo, mte, gte, "after equal-opportunity thresholding, against TRUE merit:")

```

Output:

```

audited against HISTORICAL labels (the biased ones):
selection rate    g0 0.273    g1 0.331    dem-parity diff +0.058    disparate impact 0.82
TPR              g0 0.754    g1 0.835    gap +0.081
FPR              g0 0.161    g1 0.053    gap -0.108

audited against TRUE merit:
selection rate    g0 0.273    g1 0.331    dem-parity diff +0.058    disparate impact 0.82
TPR              g0 0.743    g1 0.835    gap +0.093
FPR              g0 0.023    g1 0.053    gap +0.030

group thresholds g0 0.422 / g1 0.559 (aim: TPR ~= 0.80 on merit)
after equal-opportunity thresholding, against TRUE merit:
selection rate    g0 0.303    g1 0.309    dem-parity diff +0.006    disparate impact 0.98
TPR              g0 0.800    g1 0.793    gap -0.006
FPR              g0 0.038    g1 0.041    gap +0.003

```

Without ever seeing the protected attribute, the model reconstructs historical bias through the zip-code proxy: disparate impact 0.82 and a 9-point TPR gap against truly qualified group-0 candidates. Note the audit against historical labels *inverts* the FPR story (−0.108) — contaminated ground truth contaminates the audit. Group-specific thresholds equalize opportunity (gap −0.006) and, here, repair demographic parity (0.98) as a side effect.

Pitfalls, comparisons and practical tips

Method selection at a glance:

Question	Method	Scope	Cost	Key failure mode
Which features matter overall?	Permutation importance	Global	n_features × repeats evals	Correlated features deflate each other

Question	Method	Scope	Cost	Key failure mode
Which features matter (free)?	Impurity (MDI)	Global	Free with trees	Cardinality bias; training-set artifact
What does this feature contribute given the others?	Drop-column	Global	Retrain per feature	Expensive; correlated twin hides value
What shape is the effect?	PDP + ICE	Global	grid $\times$ n evals	PDP averages away interactions; impossible points
Why this prediction? (principled)	SHAP (TreeSHAP if trees)	Local $\rightarrow$ global	Poly for trees; heavy agnostic	Background choice; correlated credit-splitting
Why this prediction? (fast, any modality)	LIME	Local	perturbations $\times$ evals	Instability; kernel-width arbitrariness
Does behavior differ by group?	Fairness metrics + group SHAP	Global	Cheap given predictions	Ground-truth labels themselves biased

The recurring pitfalls:

- **Trusting MDI rankings.** The default `feature_importances_` inflates high-cardinality features and reflects training-set structure. Habit: recompute with permutation importance on held-out data before presenting any ranking (Listing 1's noise feature outranking real features is the canonical demo).
- **Reading raw coefficients as importance.** Only after standardization, and never individually under collinearity — report the correlation structure or grouped effects instead (Listing 2).
- **Presenting a PDP without ICE.** A flat PDP means "no effect" or "two opposite effects canceling"; only ICE curves distinguish them (Listing 3).
- **Explaining probabilities when SHAP explained log-odds.** TreeSHAP for classifiers decomposes the margin; contributions are additive there and not in probability space. Say which space you're in.
- **Forgetting the background.** SHAP values answer "compared to what?" — a different reference sample changes every number. Choose it to match the question (population average vs a specific cohort) and document it.
- **One-off LIME.** A single LIME run with default kernel width is an anecdote. Rerun across seeds, report stability, and prefer boundary cases where the local surface is informative (Listing 6).
- **Any importance under correlation.** Permutation deflates, SHAP splits, coefficients see-saw. Cluster correlated features and interpret groups (Listing 7).
- **Fairness through unawareness.** Dropping the protected attribute while proxies remain changes nothing except your ability to audit — you often need the attribute *at audit time* to measure the gaps (Listing 8).

- **Auditing against biased labels.** If the historical labels encode the discrimination, equalized odds against those labels certifies the bias. Interrogate the ground truth first.
- **Confusing the model with the world.** Every method here explains *what the model does*, not *what causes what*. SHAP on a proxy feature is evidence about the model's reliance, never causal evidence about reality. Causal claims need causal designs (Chapter 33 returns to this).

Practical workflow for a tabular model: permutation importance for the global ranking (with error bars, on test data) → TreeSHAP beeswarm for direction and shape → ICE for the top features to catch interactions → SHAP force/waterfall plots for individual decisions that need justification → group-wise fairness metrics if decisions touch people. Total cost for a tree ensemble: minutes.

Interview questions and answers

Q1. Distinguish global from local interpretability with an example of each.

Global explains overall model behavior: "income and tenure are the strongest drivers of predicted churn across the portfolio" (permutation importance, mean |SHAP|, PDP). Local explains one prediction: "this customer's 0.92 churn score is driven by three support calls last month and a month-to-month contract" (SHAP values for that row, LIME). They can disagree: a feature can be globally minor but decisive for a subpopulation, and Listing 3's interaction shows a per-row effect that averages to zero globally.

Q2. Why is impurity-based feature importance biased, and toward what?

MDI credits each feature with the impurity reduction of its splits, and features offering more candidate split points — continuous features, high-cardinality categoricals — get more chances to find a spuriously good split, so they accumulate importance even when uninformative. It's also computed on training data, so it reflects overfitting. In Listing 1 a continuous noise column ties two real features by MDI (0.074) while permutation importance on held-out data zeroes it. The interviewer listens for: cardinality bias + training-data artifact + the fix (permutation on test data).

Q3. Explain permutation importance. What are its two main weaknesses?

Shuffle one feature's column on held-out data — destroying its relationship to the target while keeping its marginal distribution — and measure the metric drop; repeat and average. Weakness one: cost, one full evaluation pass per feature per repeat. Weakness two: correlated features — the model recovers the shuffled feature's signal from its correlated twin so the drop understates importance, and the shuffled rows are off-manifold combinations the model never trained on (Listing 7: importance $1.638 \rightarrow 0.478$ after adding a duplicate; correlation $+1.000 \rightarrow -0.024$ after permuting).

Q4. When are linear-model coefficients a valid importance measure?

When features are standardized to a common scale (coefficients have units — per-dollar vs per-decade differences swamp real differences, Listing 2's $1,500\times$ illusion) and features are not strongly collinear (under collinearity only the combined effect is identified; individual coefficients swing and flip sign across refits while their sum stays stable). Also, coefficients measure association given the other features in the model — dropping or adding covariates changes them.

Q5. What is a partial dependence plot and what independence assumption does it make?

PDP of feature j at value v = the average prediction when x_j is set to v for every row, swept over a grid — it shows the shape of the marginal effect. It assumes feature j is independent of the others: the averaging fabricates rows combining $x_j = v$ with covariate values that never co-occur (forcing weight = 45 kg on tall people), so under correlation the curve is built on impossible points and can be arbitrary there.

Q6. Your PDP for a feature is flat. What are the two possible explanations and how do you tell them apart?

Either the feature truly has no effect, or it has strong effects of opposite sign in different subpopulations that cancel in the average. Plot ICE curves: flat ICE everywhere means no effect; ICE curves fanning into crossing bundles means an interaction the PDP erased. Listing 3 stages the second case — PDP slope -0.01 while the two groups have slopes -1.68 and $+1.69$.

Q7. Define the Shapley value and name the four axioms.

The Shapley value of feature j is its marginal contribution to the coalition payout $v(S \cup \{j\}) - v(S)$, averaged over all coalitions S with the weight $|S|!(|F|-|S|-1)!/|F|!$, which equals averaging over all orderings in which features could be revealed. Axioms: efficiency (values sum to $f(x) - E[f(X)]$), symmetry (interchangeable features get equal credit), dummy (a feature that changes no coalition's value gets zero), additivity (values for a sum of models = sum of values). It is the unique attribution satisfying all four.

Q8. What does SHAP's efficiency (local accuracy) property give you that LIME does not?

The SHAP values of one prediction sum exactly to the prediction minus the base value — a complete accounting: you can present "base rate 20%, +30 points from utilization, -8 from tenure, = 42%" and it adds up. LIME's coefficients are slopes of a local surrogate; they satisfy no summation constraint, so they rank and describe direction but never decompose the prediction. Listing 4's check: $\text{base} + \sum \phi = 5.213 = f(x)$ exactly.

Q9. Why is computing exact Shapley values hard in general, and how do KernelSHAP and TreeSHAP each get around it?

The definition sums over all 2^d coalitions, each requiring an expectation over the background — exponential in features. KernelSHAP recasts Shapley values as the solution of a weighted linear regression over sampled coalitions (the Shapley kernel makes the regression solution equal the Shapley value), trading exactness for sampling error; it stays model-agnostic. TreeSHAP exploits tree structure to compute the exact values in polynomial time, $O(T \cdot L \cdot D^2)$ per instance, by tracking how coalitions flow through splits — exact and fast, but trees only.

Q10. For a tree classifier, in what space are TreeSHAP values additive, and why does it matter?

Log-odds (margin) space. The sigmoid is nonlinear, so contributions cannot be additive in probability — a +1 log-odds contribution moves p a lot near 0.5 and almost none near 0.99. Report and plot in log-odds when doing arithmetic; convert only the final sum (Listing 5: base $-0.012 + \Sigma -4.911 = -4.924 \rightarrow \sigma(-4.924) = 0.007$). Presenting SHAP bars in "probability points" that don't sum to the prediction is a common, detectable error.

Q11. How does the choice of background dataset change SHAP values?

SHAP answers "why does $f(x)$ differ from $E[f(X)]$ under this background" — the reference defines both the base value and every conditional expectation $v(S)$. Explaining a loan denial against the population average gives different attributions than against previously approved applicants. There is no universally right background: match it to the contrastive question being asked and hold it fixed across explanations you intend to compare. Small backgrounds also add Monte-Carlo noise in KernelSHAP.

Q12. Walk through the LIME algorithm step by step.

(1) Generate perturbations around the instance: Gaussian noise in standardized feature space for tabular; randomly masking words/superpixels for text/images. (2) Query the black box for each perturbed point. (3) Weight points by a proximity kernel π_x (e.g. $\exp(-d^2/w^2)$) so nearby behavior dominates. (4) Fit a sparse/regularized weighted linear model to the (perturbation, prediction) pairs; its coefficients are the explanation, and the weighted R^2 tells you whether the local linear story is even adequate. Listing 6 is the full algorithm in ~15 lines.

Q13. Why are LIME explanations unstable, and what do you do about it?

Three sources: random perturbations (finite-sample noise), the arbitrary kernel width (too narrow = noisy fit, too wide = no longer local), and off-manifold sampling (perturbations may be unrealistic points where the model is undefined-ish).

Mitigations: many perturbations, rerun across seeds and report whether top features persist (Listing 6 does 5 seeds), tune kernel width to a stable regime, check the surrogate's weighted R^2 before trusting the coefficients, and prefer explaining points where the model isn't saturated.

Q14. SHAP vs LIME — give a decision rule.

Tree ensemble on tabular data: TreeSHAP, no contest — exact, fast, axiomatic. Expensive black box where KernelSHAP is too slow, or text/image inputs where perturbation semantics are natural: LIME (or its descendants). Need a complete decomposition that sums to the prediction, or global aggregation from locals: SHAP. Need only a quick ranked sanity check of one prediction: LIME is fine, run it several times. Both inherit the correlated-features caveat; neither is causal.

Q15. How do you get a global explanation out of a local method like SHAP?

Compute SHAP values for a representative sample and aggregate: mean $|\text{SHAP}|$ per feature = global importance; a beeswarm plot (dots = rows, x = SHAP value, color = feature value) adds direction and nonlinearity; SHAP dependence plots (feature value vs SHAP value) recover PDP-like shapes with interaction coloring. Listing 5 shows mean- $|\text{SHAP}|$ ranking matching permutation importance exactly — and when the two disagree, suspect correlated features or a feature that matters intensely for few rows.

Q16. Two features in your model are near-duplicates. Describe what happens to permutation importance, SHAP, and coefficients.

Permutation: both deflate — shuffle one and the twin covers, so measured drops understate reality ($1.638 \rightarrow 0.478$ in Listing 7). SHAP: credit splits roughly evenly between the twins with the total conserved by efficiency ($1.659 \rightarrow 0.864 + 0.797$). Coefficients: unidentified individually — they trade off against each other across refits, signs can flip, only the sum is stable (Listing 2). Fix: group correlated features (cluster by correlation), interpret at group level, or drop one twin before interpreting.

Q17. What is the difference between explaining the model and explaining the world?

All methods here are descriptions of the fitted function f — which inputs move its output. That is not causality: the model may lean on a proxy (zip code) because it correlates with the true cause (historical discrimination, or income), and SHAP will faithfully report the model's reliance on the proxy. Acting on the world ("change the feature to change the outcome") requires causal assumptions the model never encodes. Saying "SHAP shows X causes Y" in an interview is an instant red flag; "SHAP shows the model relies on X" is correct.

Q18. Define demographic parity, equalized odds, and equal opportunity.

Demographic parity: equal selection rates across groups, $P(\hat{Y}=1|A=a)$ constant in a — ignores the labels entirely. Equalized odds: equal TPR and equal FPR across groups — errors balanced conditional on the true outcome. Equal opportunity: the TPR half only — among truly qualified/positive individuals, equal chance of being selected. Parity suits contexts where the label itself is suspect or aspiration is equal representation; odds/opportunity suit contexts where the label is trusted and you want equal error burden.

Q19. What is the 80% (four-fifths) rule?

A disparate-impact screening threshold from US employment guidelines: the selection rate of the disadvantaged group divided by that of the advantaged group should be at least 0.8; below that, the practice is presumptively discriminatory and needs justification. It's a legal heuristic, not a statistical test — Listing 8's model sits at exactly 0.82, "passing" while still carrying a 9-point TPR gap against qualified group-0 candidates, which is why one metric is never the audit.

Q20. Why doesn't removing the protected attribute from the features make a model fair?

Because correlated proxies (zip code, name, school, purchase history) reconstruct it — Listing 8's model never sees the group yet reaches disparate impact 0.82 through the zip proxy, having learned from labels where 45% of one group's positives were erased. Worse, unawareness removes your ability to *measure* and *correct* gaps: group-aware auditing and mitigation typically need the attribute at evaluation time. The bias lives in the data-generating process, not in one column.

Q21. Calibration within groups and equalized odds — can you have both?

Generally no: the impossibility results (Kleinberg et al. 2016; Chouldechova 2017) show that when base rates differ across groups, a classifier cannot simultaneously be calibrated within each group and have equal TPR and FPR across groups, outside trivial cases (perfect prediction or equal base rates). Practical consequence: fairness is a set of mutually incompatible desiderata; you must argue from the application which criterion binds — and be ready to defend the choice, since optimizing one provably sacrifices another.

Q22. Your fairness audit uses historical outcome labels. What is the risk?

If historical decisions were biased, the labels encode the bias, and metrics conditioned on them are distorted: Listing 8's model shows an FPR gap of -0.108 against historical labels — apparently over-selecting the disadvantaged group — while against true merit the same predictions show a $+0.093$ TPR gap against it. Equalizing odds against poisoned labels certifies discrimination. Defenses: interrogate label provenance, use outcome measures less mediated by past decisions, model the label bias explicitly, or evaluate on cohorts where decisions were randomized/consistent.

Q23. Name the three mitigation families for unfair models with one example each.

Pre-processing: fix the data — reweighing samples, repairing feature distributions, removing proxies (hard: proxies hide in combinations). In-processing: fix the training — add a fairness penalty to the loss, adversarial debiasing where an adversary tries to predict the group from representations. Post-processing: fix the outputs — group-specific decision thresholds; Listing 8 equalizes TPR with per-group thresholds, collapsing the opportunity gap from $+0.093$ to -0.006 without touching the model. Post-processing is cheapest and auditable, but requires the attribute at decision time, which some legal regimes restrict.

Q24. What is a surrogate model in the global sense, and what must you always report with it?

Fit an interpretable model (shallow tree, sparse linear) to the black box's *predictions* — not the labels — and read the surrogate as a description of the black box. Always report fidelity: the surrogate's R^2 /accuracy at mimicking the black box on held-out data. A surrogate with 0.6 fidelity describes 60% of the model's behavior; conclusions from it about the remaining 40% are fiction. Same rule locally: Listing 6 reports the weighted R^2 (0.623) alongside the LIME coefficients.

Q25. A regulator asks why customer 4711's loan was denied. Sketch your answer pipeline.

(1) TreeSHAP (or KernelSHAP) values for that application against a documented reference population; (2) present the top signed contributions in log-odds converted at the end to probability, verifying additivity; (3) translate to reason codes — the standardized adverse-action categories — by mapping the top negative contributors; (4) counterfactual check: verify the cited features actually flip the decision when improved (a reason code that can't change the outcome is not actionable); (5) stability check across the model's retraining history. Interviewer wants: reference-dependence, additivity, actionability, not just "run SHAP".

Q26. Write pseudocode for permutation importance.

Given model f , held-out (X, y) , metric m : `base = m(y, f(X)); for j in features: scores = []; for r in 1..R: Xp = X.copy(); Xp[:, j] = shuffle(Xp[:, j]); scores.append(base - m(y, f(Xp))); importance[j] = mean(scores), std(scores)`. Key details: held-out data (not train), multiple repeats with reported spread, metric must match what you care about (importance under AUC $\neq$ importance under log loss), and correlated features need grouped shuffles.

Q27. Why can two models with identical accuracy have very different SHAP explanations, and what does that imply for trust?

Underdetermination (the "Rashomon effect"): many functions fit the same data equally well while using different features — especially under correlated features, where models can trade one twin for the other freely. Retrain with a different seed and the explanation can shift even as the metric holds. Implication: an explanation describes *this* model instance, not the task; for consequential decisions check explanation stability across retrains/seeds, and treat features whose importance persists across the Rashomon set as more trustworthy than any single ranking.

Q28. You need to justify predictions in a high-stakes setting. Argue interpretable-by-design vs black-box + post-hoc.

For high stakes, the burden of proof favors intrinsic interpretability: a sparse scoring system or shallow tree is its own explanation — exact, stable, contestable — while post-hoc explanations are approximations that can be unfaithful, unstable (LIME), or reference-dependent (SHAP), and can rationalize a flawed model persuasively. The counterargument: on complex signals the accuracy gap costs real outcomes, and TreeSHAP on a GBM is exact-for-the-model and auditable. Defensible synthesis: first *measure* the accuracy gap — on many tabular problems it's near zero (fit the interpretable model, compare honestly); pay for a black box only when the gap is material, and then budget for explanation validation, stability testing, and fairness audits as first-class deliverables.